

TRAINING MODULE

CRUDE DISTILLATION UNIT (CDU)

UNIT: 11

VIETNAM OIL AND GAS CORPORATION (PETROVIETNAM)
DUNG QUAT REFINERY (DQR) PROJECT
DUNG QUAT, VIETNAM



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	CRUDE DISTILLATION UNIT (CDU)	REV: 1	DATE: 18/03/08	

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UNIT: 11

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	TRAINING MODULE		DOC NO: 8474L-011-A5016-0000-001-001	
	CRUDE DISTILLATION UNIT (CDU)	REV: 1	DATE: 18/03/08	

IMPORTANT

THIS TRAINING MODULE HAS BEEN PREPARED BY ARAMIS FOR THE DUNG QUAT REFINERY.

THIS MODULE MUST BE RECOGNIZED AS A TOOL AND GUIDE ONLY. IT WOULD BE IMPOSSIBLE TO ANTICIPATE AND PRESENT ALL POTENTIAL VARIABLES AND PROCESS CONDITIONS THAT OPERATIONAL PERSONNEL MIGHT BE EXPOSED TO.

IT IS IMPERATIVE THAT THE READER ALWAYS AS CERTAIN THAT REFERENCE MATERIALS UTILIZED, WHILE PERFORMING OPERATIONAL DUTIES, CONFORM AT A MINIMUM TO THE LATEST ISSUE OF STANDARD OPERATING PROCEDURES, SAFETY CODES, ENGINEERING STANDARDS, AND GOVERNMENT REGULATIONS.

SOME DESIGN FIGURES MIGHT NOT BE IN LINE DURING THE START-UP OF THE REFINERY.

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TRAINING MODULE

CRUDE DISTILLATION UNIT (CDU)

UNIT: 11

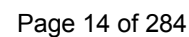
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SECTION 1 : GENERAL DESCRIPTION

The Crude Distillation Unit (CDU) provides primary separation of crude oil feedstocks: Crude oil is preheated against product and pumparound streams before being routed to a fire heater. The primary fractionation is carried out in the main crude column fractionator and associated side stream strippers. Overhead naphtha is further processed in the naphtha stabilizer column. Products are cooled and rundown to intermediate storage or further processing as appropriate. Light gas oil and heavy gas oil streams are vacuum dried prior to rundown.



The CDU is located in the area 1, with the Kerosene Treater Unit (unit 014), in the south of the plant.



Figure 1: 2D Refinery Plot Plan

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1.1. Purpose of the Unit

The objective of the CDU is to provide primary separation of crude oils to produce straight run blendstocks of distillate products (after suitable downstream treatment processes) and feedstocks for other downstream process units:

- LGO sent to storage and LCO HDT
- HGO sent to storage and LCO HDT
- Kerosene sent to Kerosene Treater Unit (KTU) 014
- Full Range Naphta sent to Naphta Hydro Treater (NHT) 012
- Distillation residue sent to storage and Residue Fluid Catalytic Cracker (RFCC) 015



South West View



South East View

Figure 3: Unit 11 3D Drawings

1.2. Basis of Design

1.2.1. Duty of Plant

The unit is designed to operate on two crude oils:

- 6.5 Million Tonnes per Annum of Bach Ho crude oil (sweet case).
- 6.5 Million Tonnes per Annum of Dubai crude oil (sour case).

Furthermore, one operating case is based on a blend of 84.6 wt% of Bach Ho crude oil and 15.4 wt% of Dubai crude oil, such that the Jet A-1 specification for freeze point and density can be achieved when processing the crude blend.

The Crude Distillation Unit consists principally of a Distillation Section which separates the different products from the Crude Oil, and a Crude Oil Preheat and Heat section that raise the temperature of the crude oil up to the necessary condition to carry out the distillation.

In addition, there are other sections that assist the process, such as the Desalter, the Stabilizer, or the Vacuum system.

1.2.2. Feed Characteristics

1.2.2.1. Bach Ho Crude

This crude oil is a light sweet crude oil with an API Gravity 39.2 and sulphur content of 0.03 %. It has a K value of 12.3, which gives it paraffin classification. Bach Ho has a medium yield of naphtha and good yields of middle distillates and vacuum gas oil.

Bach Ho is a high quality refining crude; low in contaminants which is well suited to cracking refineries.

1.2.2.1.1. Distillation Curve

The following table shows the distillation and density curve for Bach Ho crude:

Component Properties			
Fraction (°C)	Wt %	Wt % Cumm.	Density
Lights Ends	2.86	2.86	-
68-93	1.53	4.39	0.6816
93-157	8.43	12.82	0.7460
157-204	7.24	20.06	0.7734
204-260	8.38	28.44	0.7972
260-315	10.21	38.65	0.8160
315-371	12.11	50.76	0.8285
371-427	12.58	63.34	0.8437
427-482	12.84	76.18	0.8539
482-566	9.74	85.92	0.8904
>566	13.81	99.73	0.9313
Loss		0.27	

1.2.2.1.2. Light Ends

The Bach Ho crude light ends content is summarized in the following table:

Component	Wt%
Methane	0.0002
Ethane	0.0031
Propane	0.0327
Isobutane	0.0488
n-butane	0.2122

Component	Wt%
Isopentane	0.3741
n-pentane	0.6270
Cyclopentane	0.0300
2,2-dimethylbutane	0.0243
2,3-dimethylbutane	0.0530
2-methyl-pentane	0.3885
3-methyl-pentane	0.2099
n-hexane	0.8528

1.2.2.2. Dubai Crude

Dubai crude is a sour crude oil with an API Gravity 31.2 and a total sulphur content of 2.1% wt. It has a K value of 11.78, which gives it intermediate classification.

1.2.2.2.1. Distillation Curve

The distillation and density curves for this crude oil are shown in the table below:

Component Properties		
Temperature (°C)	Vol% Cumm.	Density
89	5	0.702
120.4	10	0.741
259.8	30	0.830
372.0	50	0.890
482.2	70	0.946
678.9	90	1.033

1.2.2.2.2. Light ends

The Dubai crude light ends content is summarized in the following table:

Component	Vol%
Ethane	0.001
Propane	0.016
Isobutane	0.15
n-butane	0.59
Isopentane	0.62
n-pentane	0.93

Component	Vol%
Cyclopentane	0.09

1.2.2.2.3. Sulphur content

The total sulphur content in Dubai crude is 2.1%wt with a hydrogen sulphide content less than 0.0001% wt.

The sulphur distillation curve for Dubai crude oil is shown in the table below:

Mid Vol (%)	Total Sulfur (Wt %)
15	0.0004
20	0.0012
25	0.0024
30	0.0048
35	0.0092
40	0.00136
45	0.0184
50	0.0224
55	0.0244
60	0.0252
65	0.0264
70	0.0276
75	0.0296
80	0.0320
85	0.0356
90	0.0392

1.2.3. Product Specifications

1.2.3.1. Product Specifications for operation on 100% Bach Ho feed

The following products specifications shall be meet while unit is operating on 100% Bach Ho feed:

Specification	Value	Test Method
C5 content in LPG rich stream	1.5 mol% max.	ASTM D2163
C4 in Full Range Naphta	0.3 wt% max.	G.C.
Gap between the 5%vol. of Kerosene and 95%vol. of Full Range Naphta	0°Cmin.	ASTM D86
Gap between the 5%vol. of Light Gas Oil	0°Cmin.	ASTM D86

and 95%vol. of Kerosene		
Overlap between the 95%vol. of Light Gas Oil and 5%vol. of heavy Gas Oil	20°C max.	ASTM D86
Heavy Gas Oil flash point	65°C min.	ASTM D93
Kerosene flash point	40°C min.	ASTMD93
Kerosene smoke point	20mm min.	ASTM D1322
Kerosene Density @ 15°C	0.83 kg/l max.	ASTM D1298
Light Gas Oil cetane index	45 min.	ASTM D4737
Light Gas Oil flash point	65°C min.	ASTM D93
Light Gas Oil pour point	0°C max.	ASTM D97
Residue vaporizing below 360°C	10%vol. max.	ASTM D1116 (note 1)

Note 1:

Result from ASTM D1116 test converted to standard atmospheric pressure of 760 mm Hg.

1.2.4. Material Balances

The design of the CDU is based on the following cut points:

Products	TBP cut point (°C)
Full Range Naphta / Kerosene	165
Kerosene / Light Gas Oil	205
Light Gas Oil / Heavy Gas Oil	330
Heavy Gas Oil / Atmospheric residue	370

The resulting flows of these True Boiling Point (TBP) cut points correspond with the normal unit operating case.

1.2.4.1. Bach Ho crude

According to the TBP cut points defined in the above table, the resulting flows for Bach Ho Crude oil are:

Product	Flow (kg/h)	% Wt of feed (water included)
LPG	2181	0.27
Naphta	108314	13.30
Kerosene	51188	6.28
Light Gas Oil	170716	20.96

Product	Flow (kg/h)	% Wt of feed (water included)
Heavy Gas Oil	69822	8.57
Residue	407324	50.01

1.2.4.2. Dubai crude

The design TBP cut points lead to the following flows for Dubai crude oil:

Product	Flow (kg/h)	% Wt of feed (water included)
LPG	7000	0.86
Naphta	130072	16.01
Kerosene	48747	6.00
Light Gas Oil	154839	19.06
Heavy Gas Oil	64933	7.99
Residue	403303	49.64

1.2.4.3. Flexibility cases

The design TBP cut points defined in section 1.3.2 can be modified to attaining product quality requirement. Indeed the unit is designed to operate with different TBP cut point as described in the following sections. In order to change from one case to another, the flow set points should be changed to the value shown in this section in order to achieve the required product quality.

All elements of the CDU have sufficient sizing margins incorporated into the design to achieve the flexibility requirements described below. For affected equipment, this flexibility is considered as an alternative (i.e. not additional) to the design margins.

The stream numbers shown in the following chapters refer to PFD's stream numbers.

1.2.4.3.1. Naphta flexibility cut point

In order to maximize or minimize naphta production, naphtha TBP cut point can be modified in the range of 160 °C to 170 °C. These TBP cut points correspond with minimum and maximum naphtha production cases, respectively.

Maximum naphta production:

This flexibility case's flows, for both crude oils, are summarized in the table below:

MAXIMUM NAPHTHA DUBAI CASE

Stream number	1	2	3	4	5	6	9	10	11	12
Description	CRUDE FEED	MAIN FRACTIONATOR OFF GAS	STABILISER FEED	STABILISER OFF GAS	STABILISER LPG	FULL RANGE NAPHTHA	KEROSENE	LGO	HGO	RESIDUE
DRY FLOWRATE (Kg/h)	812498	0	143000	346	6854	135800	41194	160143	58906	409255
DENSITY (kg/m ³ @ 15 °C)	869		710		562	720	788	836	886	964
DRY FLOWRATE (m ³ /h)	935	0	201.4		12.2	188.6	52.3	191.6	66.5	424.4
MOLECULAR WEIGHT				51						
TEMPERATURE (°C)	50		50	50	50	186.2	40	235.8	318.7	351.4
PRESSURE (kg/cm ² g)	20		1.3	7.4	20.5	8.15	5.5	1.8	2	2
WATER FLOWRATE (kg/h)	0	0	51	2	3	0	0	236	68	275
TOTAL FLOWRATE (kg/h)	812498	0	143051	348	6856	135800	41194	160379	58974	409530

MAXIMUM NAPHTHA BACH HO CASE

Stream number	1	2	3	4	5	6	9	10	11	12
Description	CRUDE FEED	MAIN FRACTIONATOR OFF GAS	STABILISER FEED	STABILISER OFF GAS	STABILISER LPG	FULL RANGE NAPHTHA	KEROSENE	LGO	HGO	RESIDUE
DRY FLOWRATE (Kg/h)	812500	0	119519	1	2157	117360	44119	175175	71582	402105
DENSITY (kg/m ³ @ 15 °C)	828		711		565	714	779	814	842	879
DRY FLOWRATE (m ³ /h)	981	0	168		3.8	164.3	56.7	215	85	457.3
MOLECULAR WEIGHT				49						
TEMPERATURE (°C)	50		50	50	50	191.3	40	241	322	350
PRESSURE (kg/cm ² g)	20		1.3	7.4	20.5	8.15	5.5	1.8	2	2
WATER FLOWRATE (kg/h)	2010	0	41	0	1	0	0	234	73	254
TOTAL FLOWRATE (kg/h)	814510	0	119560	1	2158	117360	44119	175409	71655	402359

Minimum naphta production

The resulting flows for this operating case are shown below (refer to doc. No. 8474L-011-ML-001 for more information):

Stream number	1	2	3	4	5	6	9	10	11	12
Description	CRUDE FEED	FRACTIONATOR OFF GAS	STABILISER FEED	STABILISER OFF GAS	STABILISER LPG	FULL RANGE NAPHTHA	KEROSENE	LGO	HGO	RESIDUE
DRY FLOWRATE (Kg/h)	812498	0	154950	327	6901	124730	52244	160143	58906	409247
DENSITY (kg/m ³ @ 15 °C)	869		705		562	716	784	836	886	964
DRY FLOWRATE (m ³ /h)	935	0	219.7		12.3	174.2	66.6	191.6	66.4	424.4
MOLECULAR WEIGHT				50						
TEMPERATURE (°C)	50		50	50	50	183	40	235	319	352
PRESSURE (kg/cm ² g)	20		1.3	7.4	20.5	8.15	5.5	1.8	2	2
WATER FLOWRATE (Kg/h)	0	0	56	2	3	0	0	236	68	276
TOTAL FLOWRATE (Kg/h)	812498	0	155006	329	6904	124730	52244	160379	58974	409521

MINIMUM NAPHTHA BACH HO CASE

Stream number	1	2	3	4	5	6	9	10	11	12
Description	CRUDE FEED	FRACTIONATOR OFF GAS	STABILISER FEED	STABILISER OFF GAS	STABILISER LPG	FULL RANGE NAPHTHA	KEROSENE	LGO	HGO	RESIDUE
DRY FLOWRATE (Kg/h)	812500	0	135413	1	2188	105233	56225	175175	71581	402096
DENSITY (kg/m ³ @ 15 °C)	828		705		565	709	775	814	842	879
DRY FLOWRATE (m ³ /h)	981	0	192		3.8	148.4	72.5	215.4	85.1	457.6
MOLECULAR WEIGHT				49						
TEMPERATURE (°C)	50		50	50	50	187	40	241	322	350
PRESSURE (kg/cm ² g)	20		1.3	7.4	20.5	8.15	5.5	1.8	2	2
WATER FLOWRATE (Kg/h)	2010	0	47	0	1	0	0	234	73	253
TOTAL FLOWRATE (Kg/h)	814510	0	135460	1	2189	105233	56225	175409	71654	402349

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1.2.4.3.2. Kerosene flexibility cut point

Kerosene normal cut point range of 165 °C to 205 °C can be varied to attaining product requirements. In order to maximize scope for producing kerosene of acceptable quality, the CDU is designed to produce a minimum TBP cut range of 170 °C to 200°C. Additionally, in order to maximize kerosene production, and if permitted by product quality considerations, the CDU is designed to produce a maximum TBP cut range of 160 °C to 210°C.

Maximum Kerosene Production

Kerosene flows for this flexibility case (TBP cut range of 160 °C to 210 °C) are summarized in the table below:

MAXIMUM KEROSENE DUBAI CASE

Stream number	1	2	3	4	5	6	9	10	11	12
Description	CRUDE FEED	MAIN FRACTIONATOR OFF GAS	STABILISER FEED	STABILISER OFF GAS	STABILISER LPG	FULL RANGE NAPHTHA	KEROSENE	LGO	HGO	RESIDUE
DRY FLOWRATE (Kg/h)	812498	0	154950	327	6901	124730	59475	152912	58906	409247
DENSITY (Kg/m ³ @ 15 °C)	869		706		562	716	785	838	886	964
DRY FLOWRATE (m ³ /h)	935	0	219.6		12.3	174.2	75.7	182.6	66.5	424.4
MOLECULAR WEIGHT				50						
TEMPERATURE (°C)	50		50		50	183	40	239	319	352
PRESSURE (Kg/cm ² g)	20		1.3		20.5	8.15	5.5	1.8	2	2
WATER FLOWRATE (Kg/h)	0	0	56	2	3	0	0	226	68	275
TOTAL FLOWRATE (Kg/h)	812498	0	155006	329	6904	124730	59475	153138	58974	409521

MAXIMUM KEROSENE BACH HO CASE

Stream number	1	2	3	4	5	6	9	10	11	12
Description	CRUDE FEED	MAIN FRACTIONATOR OFF GAS	STABILISER FEED	STABILISER OFF GAS	STABILISER LPG	FULL RANGE NAPHTHA	KEROSENE	LGO	HGO	RESIDUE
DRY FLOWRATE (Kg/h)	812500	0	135412	1	2188	105231	62725	169000	71581	401772
DENSITY (Kg/m ³ @ 15 °C)	828		706		565	709	776	815	842	879
DRY FLOWRATE (m ³ /h)	981	0	192		3.8	148.4	80.8	207.5	85.1	457.2
MOLECULAR WEIGHT				49						
TEMPERATURE (°C)	50		50		50	187	40	244	322	350
PRESSURE (Kg/cm ² g)	20		1.3		20.5	8.15	5.5	1.8	2	2
WATER FLOWRATE (Kg/h)	2010	0	47	0	1	0	0	225	73	253
TOTAL FLOWRATE (Kg/h)	814510	0	135459	1	2189	105231	62725	169225	71654	402025

Minimum Kerosene production

The resulting flows for this flexibility case are shown in the following table:

Stream number	1	2	3	4	5	6	9	10	11	12
Description	CRUDE FEED	MAIN FRACTIONATOR OFF GAS	STABILISER FEED	STABILISER OFF GAS	STABILISER LPG	FULL RANGE NAPHTHA	KEROSENE	LGO	HGO	RESIDUE
DRY FLOWRATE (Kg/h)	812498	0	154950	327	6901	124730	59475	152912	58906	409247
DENSITY (kg/m ³ @ 15 °C)	869		706		562	716	785	838	886	964
DRY FLOWRATE (m ³ /h)	935	0	219.6		12.3	174.2	75.7	182.6	66.5	424.4
MOLECULAR WEIGHT				50						
TEMPERATURE (°C)	50		50	50	50	183	40	239	319	352
PRESSURE (kg/cm ² g)	20		1.3	7.4	20.5	8.15	6.5	1.8	2	2
WATER FLOWRATE (Kg/h)	0	0	56	2	3	0	0	226	68	275
TOTAL FLOWRATE (Kg/h)	812498	0	155006	329	6904	124730	59475	153138	58974	409521

MAXIMUM KEROSENE DUBAI CASE

Stream number	1	2	3	4	5	6	9	10	11	12
Description	CRUDE FEED	MAIN FRACTIONATOR OFF GAS	STABILISER FEED	STABILISER OFF GAS	STABILISER LPG	FULL RANGE NAPHTHA	KEROSENE	LGO	HGO	RESIDUE
DRY FLOWRATE (Kg/h)	812500	0	135412	1	2188	105231	62725	169000	71581	401772
DENSITY (kg/m ³ @ 15 °C)	828		706		565	709	776	815	842	879
DRY FLOWRATE (m ³ /h)	981	0	192		3.8	148.4	80.8	207.5	85.1	457.2
MOLECULAR WEIGHT				49						
TEMPERATURE (°C)	50		50	50	50	187	40	244	322	350
PRESSURE (kg/cm ² g)	20		1.3	7.4	20.5	8.15	5.5	1.8	2	2
WATER FLOWRATE (Kg/h)	2010	0	47	0	1	0	0	225	73	253
TOTAL FLOWRATE (Kg/h)	814510	0	135459	1	2189	105231	62725	169225	71654	402025

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1.2.4.3.3. Diesel flexibility cut point:

In order to potentially improve diesel blending flexibility, the CDU is designed to achieve a TBP cut point of 340 °C between the light gas oil and the heavy gas oil products. Also, in order to provide additional flexibility for the campaign production of military diesel, the CDU is designed to achieve a TBP cut point of 320 °C between the light gas oil and heavy gas oil products.

Based on the above, three operating cases have been considered: Maximum light gas oil production (TBP cut range of 200 °C to 340 °C), minimum light gas oil production (TBP cut range of 210 °C to 320 °C) and maximum heavy gas oil production (TBP cut range of 320°C to 370 °C).

Maximum light gas oil production

Maximum light gas oil production is summarized in the table below:

MAXIMUM LGO DUBAI CASE

Stream number	1	2	3	4	5	6	9	10	11	12
Description	CRUDE FEED	MAIN FRACTIONATOR OFF GAS	STABILISER FEED	STABILISER OFF GAS	STABILISER LPG	FULL RANGE NAPHTHA	KEROSENE	LGO	HGO	RESIDUE
DRY FLOWRATE (Kg/h)	812498	0	137637	337	6876	130423	39406	181756	44444	409255
DENSITY (Kg/m ³ @ 15 °C)	869		708		562	718	784	837	890	964
DRY FLOWRATE (m ³ /h)	935	0	194.5		12.2	181.6	50.2	217.2	49.9	424.4
MOLECULAR WEIGHT										
TEMPERATURE (°C)	50		50	50	50	184	40	236	320	352
PRESSURE (Kg/cm ² g)	20		1.3	7.4	20.5	8.15	5.5	1.8	2	2
WATER FLOWRATE (Kg/h)	0	0	50	2	3	0	0	261	51	274
TOTAL FLOWRATE (Kg/h)	812498	0	137687	339	6879	130423	39406	182017	44495	409529

MAXIMUM LGO BACH HO CASE

Stream number	1	2	3	4	5	6	9	10	11	12
Description	CRUDE FEED	MAIN FRACTIONATOR OFF GAS	STABILISER FEED	STABILISER OFF GAS	STABILISER LPG	FULL RANGE NAPHTHA	KEROSENE	LGO	HGO	RESIDUE
DRY FLOWRATE (Kg/h)	812500	0	113425	1	2172	111251	44038	198250	54681	402106
DENSITY (Kg/m ³ @ 15 °C)	828		708		565	712	776	815	845	879
DRY FLOWRATE (m ³ /h)	981	0	160		3.8	156.3	56.7	243.5	64.8	457.6
MOLECULAR WEIGHT										
TEMPERATURE (°C)	50		50	50	50	189	40	241	323	348
PRESSURE (Kg/cm ² g)	20		1.3	7.4	20.5	8.15	5.5	1.8	2	2
WATER FLOWRATE (Kg/h)	2010	0	39	0	1	0	0	256	55	257
TOTAL FLOWRATE (Kg/h)	814510	0	113464	1	2173	111251	44038	198506	54736	402363

Minimum light gas oil production

This flexibility case's flows for both crude oils are shown in the table below:

MINIMUM LGO DUBAI CASE											
Stream number	1	2	3	4	5	6	9	10	11	12	
Description	CRUDE FEED	FRACTIONATOR OFF GAS	STABILISER FEED	STABILISER OFF GAS	STABILISER LPG	FULL RANGE NAPHTHA	KEROSENE	LGO	HGO	RESIDUE	
DRY FLOWRATE (Kg/h)	812498	0	137637	337	6877	130423	53787	137393	74425	409256	
DENSITY (kg/m ³ @ 15 °C)	869		708		562	718	787	835	882	964	
DRY FLOWRATE (m ³ /h)	935	0	194.4		12.2	181.6	68.3	164.7	84.4	424.4	
MOLECULAR WEIGHT											
TEMPERATURE (°C)	50		50	50	50	185	40	235	316	351	
PRESSURE (kg/cm ²)	20		1.3	7.4	20.5	8.15	5.5	1.8	2	2	
WATER FLOWRATE (Kg/h)	0	0	50	2	3	0	0	209	86	275	
TOTAL FLOWRATE (Kg/h)	812498	0	137687	339	6880	130423	53787	137602	74511	409531	

MINIMUM LGO BACH HQ CASE

Stream number	1	2	3	4	5	6	9	10	11	12	
Description	CRUDE FEED	FRACTIONATOR OFF GAS	STABILISER FEED	STABILISER OFF GAS	STABILISER LPG	FULL RANGE NAPHTHA	KEROSENE	LGO	HGO	RESIDUE	
DRY FLOWRATE (Kg/h)	812500	0	113425	1	2173	111251	56388	149338	91244	402105	
DENSITY (kg/m ³ @ 15 °C)	828		708		565	712	778	813	840	879	
DRY FLOWRATE (m ³ /h)	981	0	160.1		3.8	156.2	72.5	183.7	108.6	457.3	
MOLECULAR WEIGHT				49							
TEMPERATURE (°C)	50		50	50	50	189	40	239	319	350	
PRESSURE (kg/cm ²)	20		1.3	7.4	20.5	8.15	5.5	1.8	2	2	
WATER FLOWRATE (Kg/h)	2010	0	39	0	1	0	0	207	93	253	
TOTAL FLOWRATE (Kg/h)	814510	0	113464	1	2174	111251	56388	149545	91337	402358	

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1.2.5. Utility/Power/Chemicals/Catalyst consumption

1.2.5.1. Utility Consumption

The following describes the utility consumption of the CDU based on the Estimated Utility Consumption Documents:

8474L-011-CN-0003-001 100% Bach Ho
8474L-011-CN-0003-002 Bach Ho / Dubai Blend

Only the case 100% Bach Ho is described in the following. For the Bach Ho / Dubai Blend, refer to the abovementioned document.

Note:

(): Intermittent Producer/Consumer

+: Indicates Quantity Produced

-: Indicates Quantity Consumed

1.2.5.1.1. Electrical Power

Item Tag	Description	Electrical Power (kW)			Remark
		Motor Load Rating	Mech. Running Load	Elec. Oper. Load	
A-1101-D-01A	1 st Stage Desalter Transformer A		-30.0		
A-1101-D-01B	1 st Stage Desalter Transformer B		(-30.0)		
A-1101-D-02A	2 nd Stage Desalter Transformer A		-30.0		
A-1101-D-02B	2 nd Stage Desalter Transformer B		(-30.0)		
E-1111	Main Fractionator Condenser		-265.2		Total for 12 drivers
E-1112	Top pumparound cooler		-136.2		Total for 6 drivers
E-1114	Kerosene Air Cooler		-20.4		Total for 2 drivers
E-1116	LGO product cooler		-93.2		Total for 4 drivers

Item Tag	Description	Electrical Power (kW)			Remark
		Motor Load Rating	Mech. Running Load	Elec. Oper. Load	
E-1117	HGO product cooler		-27.0		Total for 2 drivers
E-1121	Stabiliser Condenser [1]		-96		Total for 4 drivers
E-1126	Full Range Naphta Air Cooler		-42.6		Total for 2 drivers
E-1129	Desalter effluent air cooler		-11.4		Total for 3 drivers
E-1133	Tempered water air cooler		-98.0		Total for 4 drivers
P-1101A	Crude booster pump		-761.6		
P-1101B	Crude booster pump		(-761.6)		
P-1102A	Top pumparound pump		-97.3		
P-1102B	Top pumparound pump		(-97.3)		
P-1103A	Kerosene pumparound		-56.7		
P-1103B	Kerosene pumparound		(-56.7)		
P-1104A	LGO pumparound pump		-256.0		
P-1104B	LGO pumparound pump		(-256.0)		
P-1105A	HGO pumparound pump		-49.8		
P-1105B	HGO pumparound pump		(-49.8)		
P-1106A	Residue pump		-629.1		
P-1106B	Residue pump		(-629.1)		
P-1107A	Kerosene Product Pump		-49.5		
P-1107B	Kerosene Product Pump		(-49.5)		
P-1110A	Stabiliser Feed Pump [2]		-95.5		
P-1110B	Stabiliser Feed Pump [2]		(-95.5)		
P-1112A	LGO product pump		-79.3		
P-1112B	LGO product pump		(-79.3)		
P-1113A	HGO product pump		-40.0		
P-1113B	HGO product pump		(-40.0)		
P-1114A	Stabiliser reflux pump [2]		-8.1		
P-1114B	Stabiliser reflux pump [2]		(-8.1)		
P-1115A	Stabiliser LPG pump [2]		-9.7		

Item Tag	Description	Electrical Power (kW)			Remark
		Motor Load Rating	Mech. Running Load	Elec. Oper. Load	
P-1115B	Stabiliser LPG pump [2]		(-9.7)		
P-1118A	Desalter water recycle pump		-27.4		
P-1118B	Desalter water recycle pump		(-27.4)		
P-1119A	Desalter water charge pump		-63.9		
P-1119B	Desalter water charge pump		(-63.9)		
P-1120A	Drier slop oil pump		-8.6		
P-1120B	Drier slop oil pump		(-8.6)		
P-1121A	Ejector condensate pump		-9.3		
P-1121B	Ejector condensate pump		(-9.3)		
P-1122A	Tempered water pump		-35.8		
P-1122B	Tempered water pump		(-35.8)		
P-1123A	Demulsifier injection pump		-0.3		
P-1123B	Demulsifier injection pump		(-0.3)		
P-1124A	Antifoulant injection pump		-0.3		
P-1124B	Antifoulant injection pump		(-0.3)		
P-1125A	Corrosion inhibitor injection pump		-0.3		
P-1125B	Corrosion inhibitor injection pump		(-0.3)		
P-1126A	Neutraliser injection pump		-0.3		
P-1126B	Neutraliser injection pump		(-0.3)		
P-1127A	Closed drain pump		-11.8		
P-1127B	Closed drain pump		(-11.8)		
P-1128	Oily water pump		(-4.9)		
B-1101A	Heater forced draft fan		-60.0		
B-1101B	Heater forced draft fan		(-60.0)		
H-1101	Crude charge heater soot blowers		(-1.3)		[3]
	Misc. electrical users (lighting weld)			-80.0	
	Emergency loads			(-16.0)	
	GENERAL TOTAL		-3188.6	-80.0	

Notes:

[1]: E-1122 motor power consumption reported correspond to controlling case (100% Dubai Case)

[2]: Motor power consumption reported correspond to Dubai case (pump design case)

[3]: Total sootblowers = 12. It is considered one working at the same time

1.2.5.1.2. Steam

Item No.	Description	Steam (T/h)			Cond (T/h)			Losses (T/h)
		HP STM	MP STM	LP STM	HP cond	MP cond	LP cond	
T-1101	Main fractionator [1]			-16.0				16.0
T-1102	Kerosene stripper [1]			(-1.6)				(1.6)
T-1103	LGO stripper [1]			-5.1				5.1
T-1104	HGO stripper [1]			-2.6				2.6
E-1121	Stabiliser reboiler [2]	-9.1				10.6		
J-1101A	1 st stage ejector		-0.3					0.3
J-1101B	1 ST stage ejector		-0.3					0.3
J-1101C	1 ST stage ejector		(-0.3)					(0.3)
J-1102A	2 nd stage ejector		-0.1					0.1
J-1102B	2 nd stage ejector		-0.1					0.1
J-1102C	2 nd stage ejector		(-0.1)					(0.1)
	Steam purging for steam out			(-7.5)				(7.5)
H-1101	Crude Charge heater soot blowers [3]		(-4.5)					(4.5)
H-1101	Crude charge heater snuffing steam			(-11.0)				(11.0)
H-1101	Crude charge heater atomizing steam [4]		(-2.7)					(2.7)
	Steam tracing		-1.0			1.0		

Item No.	Description	Steam (T/h)			Cond (T/h)			Losses (T/h)
		HP STM	MP STM	LP STM	HP cond	MP cond	LP cond	
	Utility stations [5]		(-0.1)					(0.1)
H-1101	Crude charge heater decoking steam		(-18.0)					(18.0)
D-1116	Closed drain steam tracing			(-0.01)		(0.01)		
	TOTAL	-9.1	-0.9	-24.7	0.0	10.6	1.0	24.6

Notes:

[1]: Via H-1101

[2]: E-1121 utilities design requirement correspond to Dubai case, HP steam = 12.89 T/h, HP BFW = 2.09 T/h and MP condensate = 14.98 T/h

[3]: Total soot blowers = 12. It is considered one working at the same time

[4]: Atomizing steam only required for fuel oil firing

[5]: Total estimated number of utility stations is 23. It is considered 2 utility stations working at the same time. Flow estimated.

1.2.5.1.3. Boiler Feed Water

Item No.	Description	BFW (T/h)			Remark
		HP BFW	LP BFW	COLD BFW	
E-1122	Stabiliser reboiler	-1.5			[1]
D-1115	Tempered water drum		(-10.0)		
	TOTAL	-1.5	(-10.0)		

Note:

[1]: E-1121 utilities design requirement correspond to Dubai case, HP steam = 12.89 T/h, HP BFW = 2.09 T/h and MP condensate = 14.98 T/h

1.2.5.1.4. Demineralised Water

Item No.	Description	DW (T/h)	Remark
	Safety showers	(-14.0)	[1]
	TOTAL	(-14.0)	

Note:

[1]: Total estimated number of safety showers is 3. It is considered 2 utility safety shower working at the same time. Estimated flow. Flow reported for safety shower is drinking water.

1.2.5.1.5. Cooling Water

Item No.	Description	Cooling Water		Fresh Water (T/h)	Remarks
		ΔT (°C)	m ³ /h		
E-1115	Kerosene Water cooler	6.0	-43.2		
E-1119	HGO product water cooler	21.0	-42.4		
E-1127	Full range naphta water cooler	6.0	97.5		[1]
E-1130	Vacuum dryer condenser	12.0	-218.2		[2]
E-1131	1 st stage aftercondenser				
E-1132	2 nd stage aftercondenser				
	Miscellaneous pumps (including unit 014 pumps)	15.0	-50.0		
D-1102	2 nd stage desalter			(-40.7)	[3]
D-1107	Drier off gas seal pot			(-0.2)	
P-1128	Oily water lifting pump			-0.2	
D-1117	Decoking KO drum			(-7.0)	
	Utility stations			(-4.0)	[4]
	TOTAL		-451.3	-0.2	

Notes:

[1]: E-1127 cooling water requirement design case is 120.84 m³/h corresponding to Dubai case.

[2]: Cooling water flow for E-1130 is the total for E-1130/31/32

[3]: Via D-1109. Desalter wash water supply is normally stripped sour water via D-1109. As a back-up to stripped sour water, up to 40.7 t/h of make-up service water may be required, subject to an acceptable salt content.

[4]: Total estimated number of utility stations is 23. It is considered 2 utility stations working at the same time. Flow estimated.

1.2.5.1.6. Instrument & Plant Air

Item No.	Description	Inst. Air Nm ³ /h	Plant Air (Nm ³ /h)		Remark
			Contin.	Intermit.	
D-1117	Decoking KO drum			(-256.0)	
	Utility stations			(-400)	[1]
	Instruments & analyser	-193.0			
A-1190	Oil mist generator	-53.0			
	TOTAL	-246.0	0.0	(-400)	

Notes:

[1]: Total estimated number of utility stations is 23. It is considered 2 utility stations working at the same time. Flow estimated.

1.2.5.1.7. Nitrogen - Bach Ho Max Distillates Case

Item No.	Description	Nitrogen (Nm ³ /h)		Remark
		Contin.	Intermit.	
	Nitrogen for purging		(-1134.0)	
D-1115	Tempered water drum		(-600)	
D-1109	Desalter water surge drum		(-70.5)	
	Utility stations		(-400.0)	
	TOTAL	0.0	(-1134.0)	

1.2.5.1.8. Furnace & Boilers

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Item No.	Description	Duty (MW)	Efficiency (%)	Fuel Fired (MW)	Remarks
H-1101	Crude charge heater		85.0	86.1	[1]
	TOTAL			86.1	

Notes:

[1]: H-1101 can be fired by 100% fuel gas, 100% fuel oil or a mix of both. Normal operation is 100% fuel gas.

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1.2.5.2. Chemicals

	Neutralizer	Corrosion Inhibitor	Antifoulant	Demulsifier
Nature	Ammonia solution	-	-	-
Make up procedure	Ammonia vapour sparging through water	Hand filling of liquid by operator	Hand filling of liquid by operator	Hand filling of liquid by operator
Solution Concentration	5 – 10 wt%	According to vendor's recommendation	According to vendor's recommendation	According to vendor's recommendation
Drum volume	3.5 m ³	0.85 m ³	1.2 m ³ (x2 drums)	1.2 m ³
Injection Flowrate	0.008 m ³ /h	0.005 m ³ /h	0.025 m ³ /h	0.025 m ³ /h

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1.3. Glossary of terms and Acronyms

1.3.1. Acronyms

COMPANIES/ORGANISATIONS

DQR	Dung Quat Refinery
	Dung Quat Industrial Zone
DQIZMB	Management Board
EVN	Electricity Authority of Vietnam
FW	Foster Wheeler Energy Limited
MOC	Ministry of Construction
	Ministry of Science, Technology
MOSTE	and Environment
	Ministry of Planning and
MPI	Investment
SRV	Socialist Republic of Vietnam
TPC	Technip Consortium

OTHERS

ACE	Application Control Environment	MC	Marshalling Cabinet
ADAS	Analyser Data Acquisition System	MCB	Main Control Building
ADP	Alarm Display Panel	MCC	Motor Control Center
AER	Application Engineers Room	MCR	Main Control Room
		MCS	
AI	Analyser Indicator	MOV	MOV Control System
		Control System	
AIT	Auto Ignition Temperature	MDF	Main Distribution Frame
AMS	Asset Management System	MIS	Management Information System
ANSI	American National Standards institute	MMS	Machine Monitoring System
APC	Advanced Process Control	MMT	Minimum Maintained Temperature
API	American Petroleum Institute	MOC	Madrid Operating Center
ARU	CRUDE DISTILLATION UNIT (CDU)	MOM	Minutes of Meeting
ASC	Analyser Speciality Contractor	MOV	Motor Operated Valve
ASME	American Society of Mechanical Engineers	MP	Medium Pressure
ASP	Analyser Systems Package	MPT	Minimum Pressurization Temperature
ASTM	American Society of Testing and Materials	MR	Material Requisition
ATM	Asynchronous Transfer Mode	MRR	Marshalling Rack Room
BCS	Blending Control System	MSD	Material Selection Diagram
BEDD	Basic Engineering Design Data	MSDS	Material Safety Data Sheet
BFD	Block Flow Diagram	MTBF	Mean Time Between Failures

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BFW	Boiler Feed Water	MTTR	Mean Time To Repair
BL	Battery Unit	MTO	Material Take-Off
BOM	Bill of Materials	MTPA	Metric Tonnes per Annum
BPC	Blending Properties Control	MVIP	Multi Vendor Interface Program (Honeywell)
BPCD	Barrels per Calendar Day	NACE	National Association of Corrosion Engineers
BPSD	Barrels per Stream Day	NCR	Non Conformance Report
BRC	Blending Ratio Control	NDE	Non Destructive Examination
CAD	Computer Aid Design	NFPA	National Fire Protection Association
CALM	Catenary Anchor Leg Mooring	NHT	Naphtha Hydrotreater (Unit)
CBT	Commercial Bid Tabulation	NIR	Near Infrared Spectroscopy
CCAR	Control Complex Auxiliary Room	NPSH	Net Positive Suction Head
CCC	Central Control Complex	NPV	Net Present Value
CCR	Continuous Catalytic Reformer	NTU	Naphtha Treater Unit
CCTV	Closed Circuit Television	OAS	Oil Accounting System
CD	Chart Datum	OJT	On Job Training
CDU	CRUDE DISTILLATION UNIT (CDU)	OM&S	Oil Movement and Storage Control System
CENELEC	European Committee for Electrotechnical Standardization	OMSA	Oil Movement and Storage automation
CFC	Chlorofluorocarbons	OOS	Operation Override Switch
CFR	Cooperative Fuel Research (Engine)	OPSS	Operations Planning and Scheduling System
C&I	Control and Instrumentation	OSBL	Outside Battery Limit
CMMS	Computerized Maintenance Management System	OTS	Operator Training Simulator
CNU	(Spent) Caustic Neutralization Unit	PABX	Private Automatic Branch Exchange
CPI	Corrugated Plate Interceptor	PAGA	Public Address / General Alarm
CSI	Control Systems Integrator	PCB	Printed Circuit Board
DAF	Dissolved Air Flotation	PFD	Process Flow Diagram
DAU	Data Acquisition Unit	PFM	Path Find Module
DCS	Distributed Control System	PDB	Project Documents Base
DEA	Diethanolamine	PGC	Process Gas Chromatograph (Analysers)
DEIA	Detailed Environmental Impact Assessment	PHD	Plant History Database
DMDS	Dimethyldisulfide	PI	Plant Air
DMS	Document Management System	PIB	Process Interface Building
DNV	Det Nork Veritas	PID	Piping and Instrument Diagram
DPTD	Design, Pressure, Temperature Diagram	PIM	Project Implementation Manual
DQMIS	Dung Quat Management Information System	PKS	Process Knowledge System (Honeywell DCS)
DQRP	Dung Quat Refinery Project	PLEM	Pipeline End Manifold
DVM	Digital Video Manager	PLG	Planning
DWT	Dead Weight Tonnes	PMC	Project Management

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EL	Equipment List	PMI	Consultant
EOR	End of Run	PMT	Positive Material Identification
EDMS	Electronic Document Management System	PO	Project Management Team
EMC	Electromagnetic Compatibility	POC	Purchase Order
EPC	Engineering Procurement, Construction and Commissioning	PP	Paris Operating Center
ERP	Enterprise Resource Planning	PPB	Project Procedure
ES	Ethernet Switch	PPM	Parts per Billion
ESD	Emergency Shut Down	PRU	Parts per Million
ETP	Effluent Treatment Plant	PWHT	Propylene Recovery Unit
ETS	Effluent Treatment System	QA	Post Weld Heat Treatment
EWS	Engineering Work Station	QC	Quality Assurance
FDC	Feed Development Contract	RA	Quality Control
FAP	Fire Alarm Panel	R&D	Risk Analysis
FAT	Factory Acceptance Test	RDBMS	Risk Analysis
FEL	Front End Loading	RFCC	Research and Development
F&G	Fire and Gas System	RFSU	Real Time Database Management System
FIU	Field Interface Unit	RLU	Residue Fluid Catalytic Cracking
FIC	Flow Indicating Controller	ROW	Ready for Start-Up
FM	Factory Mutual (Approval body)	RPMS	Remote Line Unit
FOTC	Fibre Optic Termination Cabinet	RTD	Right of Way
FSC	Fail Safe Controller (Honeywell ESD)	RTDB	Refinery Performance Management System
FTE	Fault Tolerant Ethernet	RTU	Resistance Temperature Detector
GC	Gas Chromatograph	SAT	Real Time Data Base (System)
GFT	Ground Fault	SBT	Remote Terminal Unit
HAZAN	Hazard Analysis Study	SBMS	Site Acceptance Test
HAZOP	Hazard and Operability Study	SCADA	Segregated Ballast Tanks
HDT	Hydrotreater	SCC	Software Bypass Management System
HEI	Heat Exchange Institution	SCE	Supervisory Control and Data Acquisition
HHP	High High Pressure (Steam)	SCR	Satellite Control Complex
HGO	Heavy Gas Oil	SDH	Simulation Control Environment
HIC	Hydrogen Induced Cracking	SE	Satellite Control Room
HP	High Pressure	S&E	Synchronous Digital Hierarchy
HSE	Health, Safety and Environment	SGS	Safety Earth
HVAC	Heating Ventilation Air Conditioning	SOE	Safety & Environmental
IA	Instrument Air	SOR	Safeguarding System
ICAO	International Civil Aviation Organisation	SOW	Sequence of Events
ICE	Instrument Clean Earth	SP	Start of Run
			Scope of Work
			Specification

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ICS	Integrated Control System	SPIR	Spare Parts and interchangeability Record
IIP	Initial Interface Plan	SPM	Single Point Mooring
I/O	Input/Output	SR	Scope of Supply
IP	Institute of Petroleum	SRU	Sulphur Recovery Unit
IPS	Instrumented Protective System	STC	Construction Standard
IRP	Interposing Relay Panel	STD	Design Standard
IRR	Internal Rate of Return	STEL	Short Term Exposure Limit
IS	Intrinsically Safe	SVAC	Shelter Ventilation and Air Conditioning
ISA	Instrument Society of America		System (Analyser houses)
ISE	Intrinsically Safe Earth	SWS	Sour Water Stripping (Unit)
ISBL	Inside Battery Limit	TAS	Terminal Automation System
ISOM	Isomerisation Unit	TBP	True Boiling Point
ITB	Invitation to Bid	TBT	Technical Bid Tabulation
ITP	Inspection and Test Plan	TCF	Temporary Construction Facilities
JB	Junction Box	TCM	Task Control Module
JCC	Jetty Control Complex	TEMA	Tubular Exchanger Manufacturers' Association
JCR	Jetty Control Room	TGIF	Temperature Gauge Indication Facilities (Tankage)
JSD	Job Specification for Design	TLCR	Truck Loading Control Room
JSS	Job Specification for Supply	TLCS	Truck Loading Control System
JVD	Joint Venture Directorate	TN	Transmittal Note
KLOC	Kuala Lumpur Operating Center	TPS	Total Plant Solution (Honeywell)
KTU	Kerosene Treatment Unit	TQM	Total Quality Management
LAN	Local Area Network	TS	Terminal Server
LCO	Light Cycle Oil	TWA	Time Weighted Average
LCOHDT	LCO Hydrotreater	UFD	Utility Flow diagram
LDE	Lead Discipline Engineer	U/G	Underground
LEL	Lower Exposition Limit (F&G, Analysers)	UL	Underwriter Laboratories (Approval body)
LGO	Light Gas Oil	UPS	Uninterruptible Power Supply
LIMS	Laboratory Information Management System	VDU	Visual Display Unit
LIS	Laboratory Information System	VPU	Vendor Package Unit
LLU	Local Line Unit	WABT	Weight Average Bed Temperature
LP	Low Pressure	WBS	Wash Breakdown Structure
LPG	Liquefied Petroleum Gas	WHB	Waste Heat Boiler
LTU LPG	Treater Unit	YOC	Yokohama Operating Center

1.3.2. Glossary

Refer to separate glossary.

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TRAINING MODULE

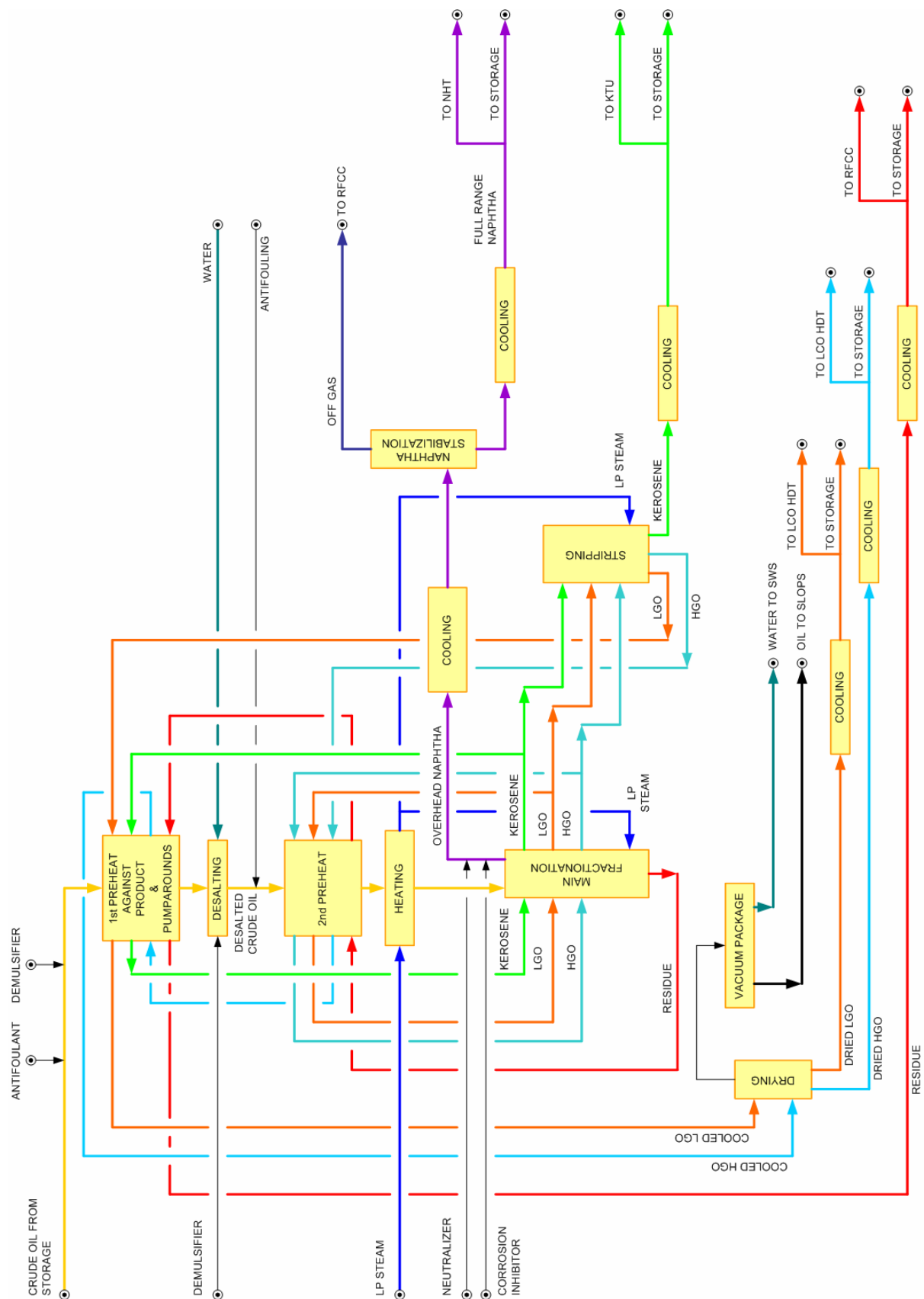
CRUDE DISTILLATION UNIT (CDU)

UNIT: 11

Course Content:

Section 1 - General Description	
Section 2 - Process Flow Description	X
Section 3 - Process Control	
Section 4 - Safeguarding Devices	
Section 5 - Fire & Gas Systems	
Section 6 - Quality Control	
Section 7 - Cause & Effects	
Section 8 - Operating Procedures	
Section 9 - HSE	
Section 10 - Reference Document Index	

SECTION 2 : PROCESS FLOW DESCRIPTION



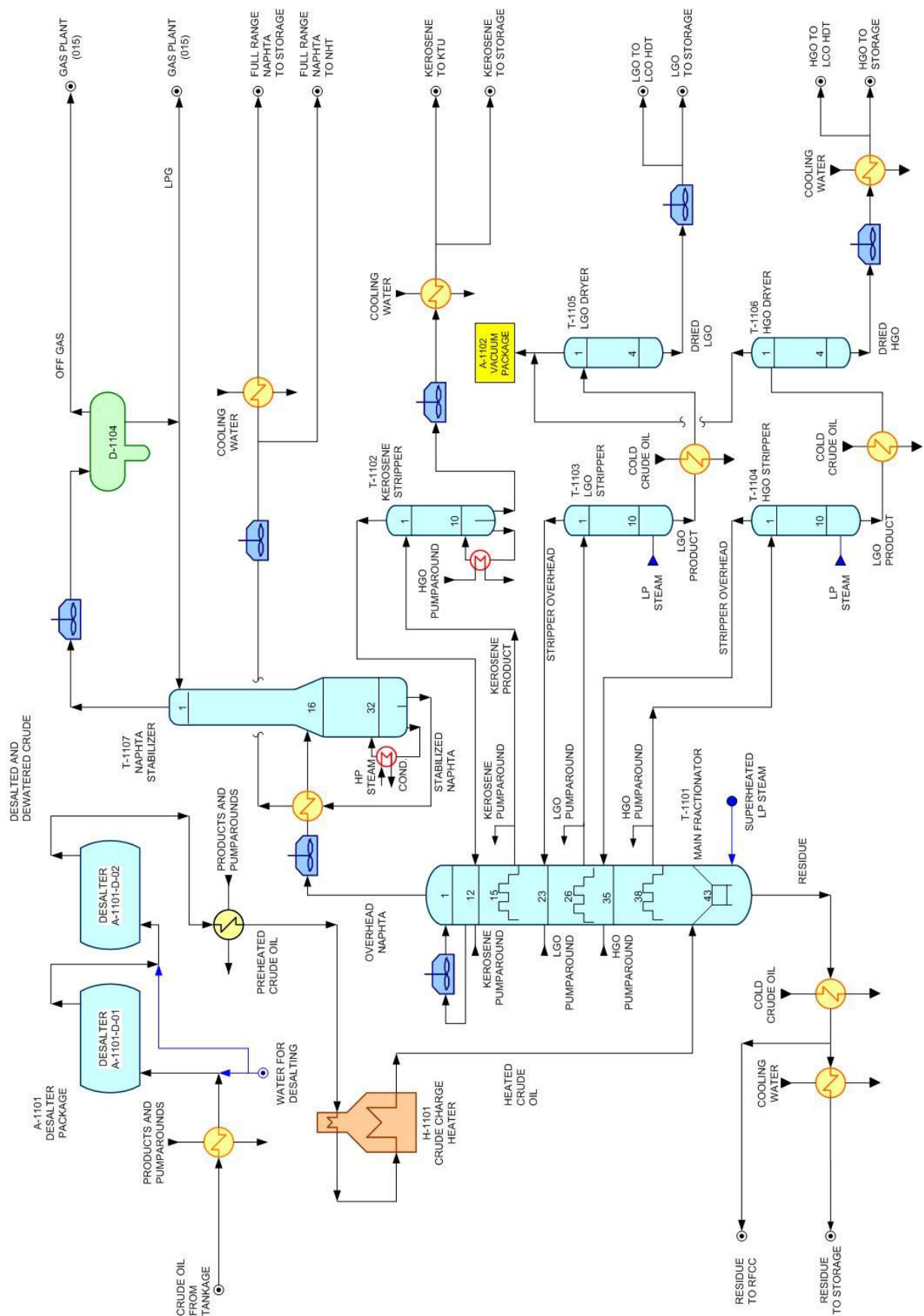


Figure 4: CDU Block Flow Diagram & Process Flow Scheme of Main Streams

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A detailed description of each section of the CDU is shown bellow, sorted from upstream to downstream of crude.

2.1. Crude Preheat

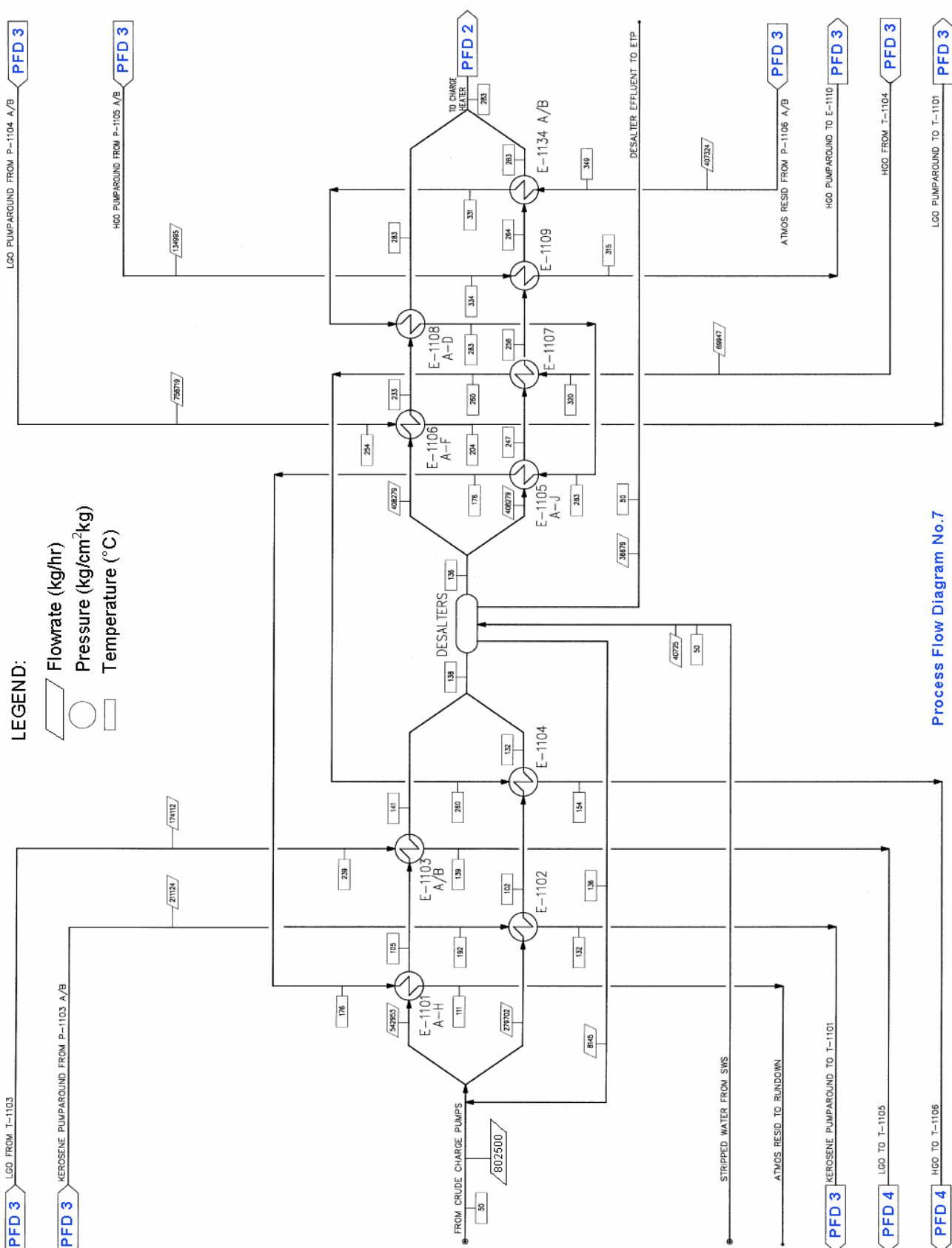


Figure 5: CDU Preheat Simplified PFD

Crude Oil is pumped from storage to the Crude Distillation Unit by the Feed Pumps P-6001A/B/C. At the inlet of the unit, two trains of heat exchangers, (each train has 2 parallel branches) separated by the Desalters, utilize the energy available in the system to raise the crude oil temperature.

The first train, (cold crude preheat) raises the crude temperature from 50°C up to 138-133°C corresponding to Bach-Ho and Dubai cases respectively. The same crude temperature difference is maintained between the parallel branches, by means of a control valve on the inlet of each branch.

Downstream the Desalters, the crude is pumped by the Crude Booster Pump P-1101 A/B to the hot preheat train (second train), which raises the crude oil temperature from 133-131°C up to 283-277°C corresponding to Bach-Ho and Dubai cases respectively. The same crude temperature difference is maintained between the parallel branches, by means of a control valve on the inlet of each branch.

The following tables summarize the equipment involved in the preheat train and the hot fluid used in each heat exchanger:

1 st Preheat Train (Cold Side)	
Train Branch A	Train Branch B
E-1101 A-H*	E-1102
Residue from E-1105	Kerosene pumparound from P-1103
E-1103 A/B	E-1104**
Light Gas Oil from T-1103	Heavy Gas Oil from E-1107

2 nd Preheat Train (Hot side)	
Train Branch A	Train Branch B
E-1105 A-J*	E-1106 A-F Light Gas Oil pumparound from P-1104
Residue from E-1108	
E-1107**	E-1108 A-D*
HGO from T-1104	
E-1109	
HGO pumparound from P-1105	Residue from E-1134
E-1134 A/B*	
Residue from P-1106	

* Heat exchangers in series (E-1101, E-1105, E-1108, E-1134)

** Heat exchangers in series (E-1104, E1107)

2.2. Desalters

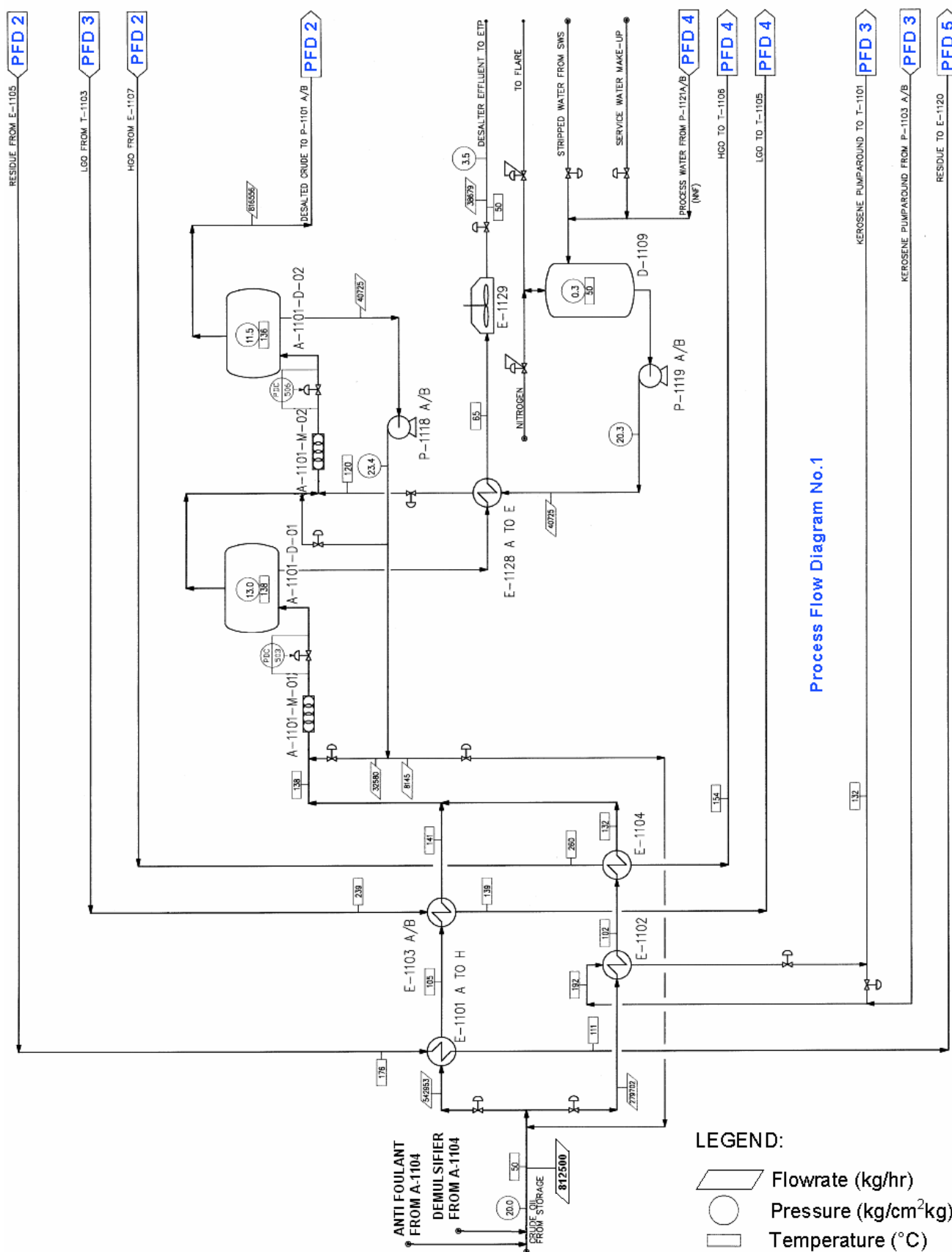


Figure 6: Cold Preheat and Desalters Simplified PFD

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Inorganic salts are removed by emulsifying crude oil with water and separating them in a desalter. The desalting system consists of one train of double desalters (A-1101-D-01/02), which reduces the water extractable soluble salt content to 2.0 ppm wt (max.) and the free water to 0.2% volume (max.) at Desalter operating temperature.

The crude oil containing sediments comes from the cold crude preheat (E-1101 A/H, E-1102, E-1103 A/B and E-1104). The recycled water from the Second Stage Desalter (A-1101-D-02) is injected in this crude inlet. This fluid enters in the first stage static mixer (A-1101-M-01) which is a crude/water disperser, maximising the interfacial surface area for optimal contact between both liquids. Downstream the mixer, the oil/water mix is homogeneously emulsified in the Emulsifying Device 011-PDV-503, upstream the First Stage Desalter (A-1101-D01). This emulsion enters the first stage desalter where it is separated into two phases (crude oil and water) by electrostatic coalescence. The desalted oil floats on the top of the vessel and the salty water decants to the bottom where it is discharged to the ETP (Effluent Treatment Plant).

The crude oil from the first stage (A-1101-D-01) is mixed with the dilution water coming from E-1128 (optionally the water recycled from second stage, pumped by the Desalter Water Recycle Pump P-1118 A/B, may be used for desalting improvement) in the second static mixer (A-1101-M-02) followed by the second stage Emulsifying Device (011-PDV-506).

The degree of emulsion in each stage is adjusted and controlled by differential pressure control loop across each emulsifying device (011-PDIC-503 / 011-PDIC-506).

This emulsion enters the second stage desalter (A-1101-D-02) where the separation is produced by electrostatic coalescence again. The crude oil flows on the top of the vessel while the water leaving from the bottom is recycled upstream the first stage desalter (A-1101-D-01) by P-1118 A/B.

The desalter system is capable of treating the crude with only one stage in operation (any desalter can be bypassed). However, the bypass of the complete system (two desalters bypassed) is not allowed, shutting down the suction valves of P-1101 A/B if both desalter bypass valves are open.

To help the desalter dehydration and salt removal efficiency, and minimize oil content in the water effluent, a demulsifier chemical compound is pumped from Demulsifier Storage Drum (A-1104-D-12) by the Demulsifier Injection Pump (A-1104-P-23 A/B), and injected to both the unit crude feed and upstream the second stage desalter.

Solids present in the crude are accumulated in the desalters' bottom, so a "mud washing" system is periodically used to remove these solids. Mud washing consists of recycling a portion of the desalter water from E-1128 A/E to agitate the accumulated solids so that they are washed out into the effluent water.

Desalter effluent is a combination of the periodically mud wash, produced water that came with the crude, and the wash water resulting from the dilution and salts and other contaminants removal. This desalter effluent is cooled down, first with fresh make-up water in the Desalter Water Exchanger (E-1128 A-E), and afterwards in the Desalter Effluent Air Cooler (E-1129). Finally, it is discharged to ETP (Effluent Treatment Plant).

Desalter Water Charge Pump (P-1119 A/B) pumps the fresh make-up water, from Desalter Water Surge Drum (D-1109) to E-1128 A-E, where is heated up to 120°C, before being injected to the crude outlet from the first stage desalter (A-1101-D-01).

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The source of the D-1109 fresh make-up water could be the Stripped Water System, Service Water or Process Water from P-1121A/B. However, the use of Process water is restricted to the use of sweet crude oil processing.

2.3. Crude Heater

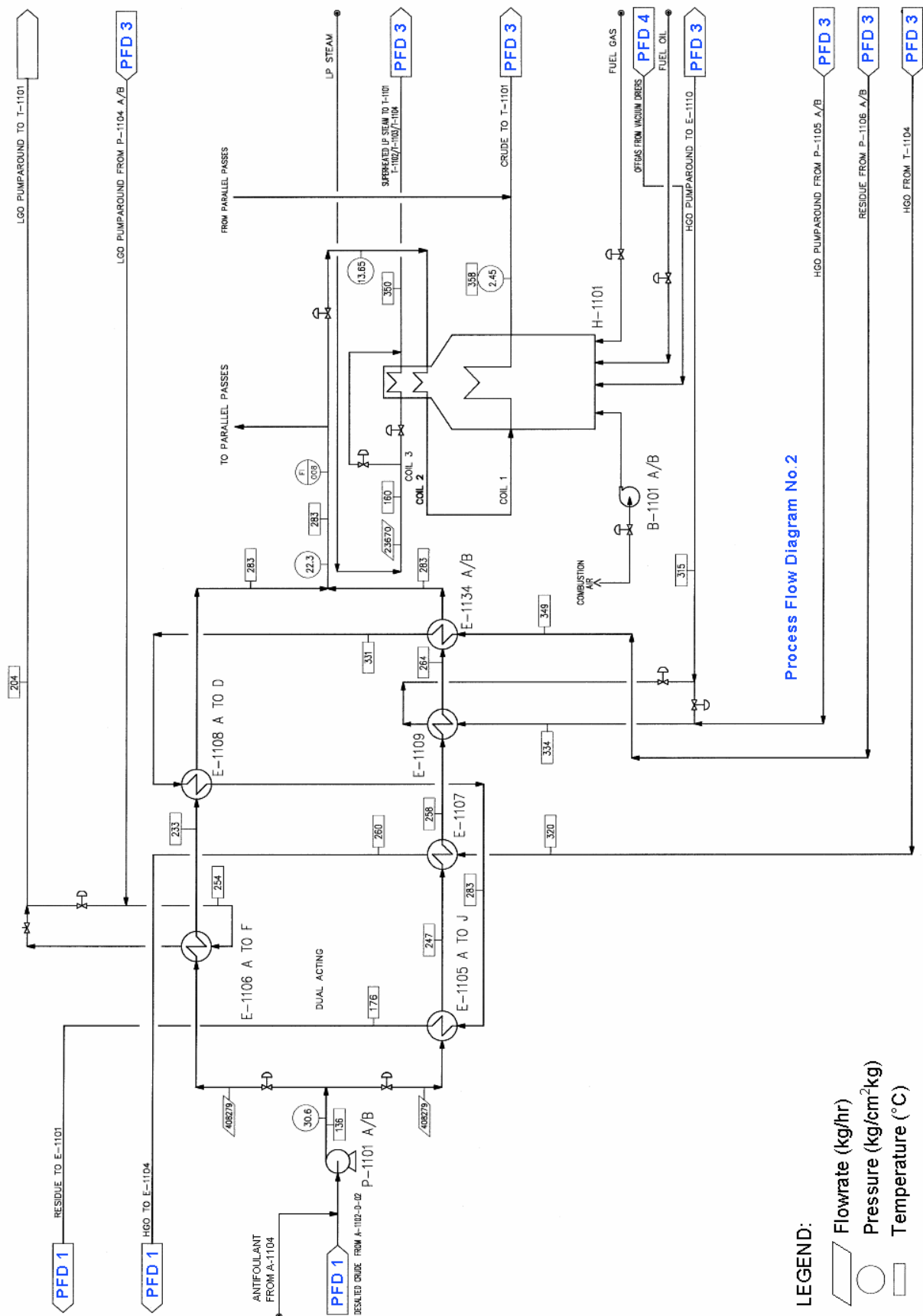


Figure 7: Hot Preheat and Heater Simplified PFD

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In order to raise the temperature of the crude up to the necessary temperature for its distillation (358-364 °C, for Bach Ho and Dubai case respectively) and vaporize part of the crude, a Crude Heater H-1101 is placed after the Hot Crude Preheat trains.

The Crude Heater (H-1101) is a two cells heater with 8 tube-passes, 4 in each cell, and 16 burners, 8 in each cell. The heat exchange occurs in the convection section first and in the radiant section afterwards.

Process flow is divided at entrance to convection section in eight symmetrical passes. After leaving convection section flow is divided such that four passes are directed to cell 1 and the other four are directed to cell 2. Each pass outlet is located at top of radiation section.

3 rows of the convection coil are used to superheat the Low Pressure Steam (LP Steam), which it is used as stripping steam in T-1101 / 1102 / 1103 / 1104.

Both fuel gas and fuel oil can be used as combustible for this heater (100% fuel gas, 100% fuel oil or a mixture of both). Nevertheless, the off gas outlet of D-1106 and D-1107 (See Vacuum Section) is constantly burned in the heater. The air is taken from the atmosphere by 2 Heater Forced Draft Blowers (B-1101 A/B) located in parallel (1 in operation and 1 in standby).

The crude heater is always working with air excess.

2.4. Crude Distillation

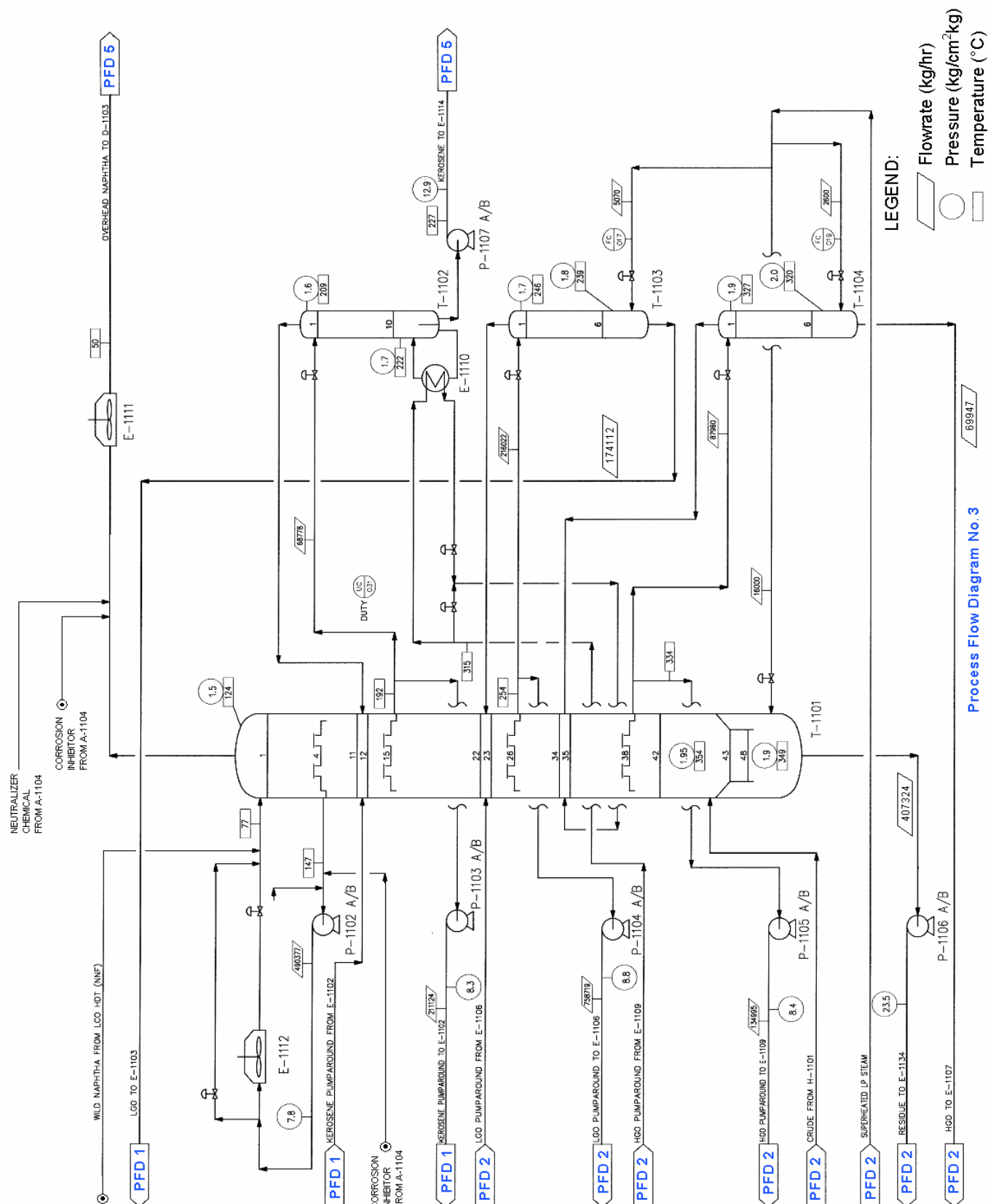


Figure 8: Main Fractionator / Strippers Simplified PFD

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Partially vaporized crude feed enters the Main Fractionator T-1101, in the flash zone where liquid and vapour are separated. Liquid leaving the flash zone is steam stripped to recover light components and discharged from the column as reduced crude. Vapours leaving the flash zone are fractionated into lighter products and three side streams, Heavy Gas Oil (HGO), Light Gas oil (LGO) and Kerosene.

The lighter products (Gas, LPG and Naphtha) from overhead section are condensed and route to the accumulator where naphtha is separate from water and gas, then the naphtha is stabilized in a separate column where LPG is recovered.

The heavier products are obtained by withdrawing portion of the main fractionator's internal reflux and are steam stripped in dedicated side columns (T-1102, T-1103 and T-1104).

The properties of each fraction can be varied as required, but only at the expense of adjacent fraction. The endpoint of sidecut will depend on the quantity withdrawn. Changing the draw-off rate is the way in which sidecuts are kept on endpoint specifications.

Refer to the section 2.1.4 of the operating manual (doc. No. 8474L-011-ML-001) for more information.

To reduce vapour and liquid traffic through the entire column and improve heat recovery and separation efficiency, four intermediate cold reflux or pump around are provided, Top pump around, kerosene pump around, LGO pump around and HGO pump around.

This tower has 48 trays divided in two diameter sections: the first one from tray 1 to 42 with an internal diameter of 6700 mm, and the second one from tray 43 to 48 with a diameter of 4000 mm. The length between tangent lines is 42,850 mm. The feed inlet nozzle is tangential with a distribute pipe and it is located between the trays 42 and 43.

The tower operates in a pressure range of 1.5 (top) to 1.9 (bottom) kg/cm²g, and in a temperature range of 130-124° (top) to 349-354 °C (bottom).

The T-1101 can be divided into 5 sections, described bellow:

- Overhead Section.
- Kerosene Section.
- Light Gas Oil Section.
- Heavy Gas Oil Section.
- Overflash section
- Residue Section.

2.4.1. Overhead Section

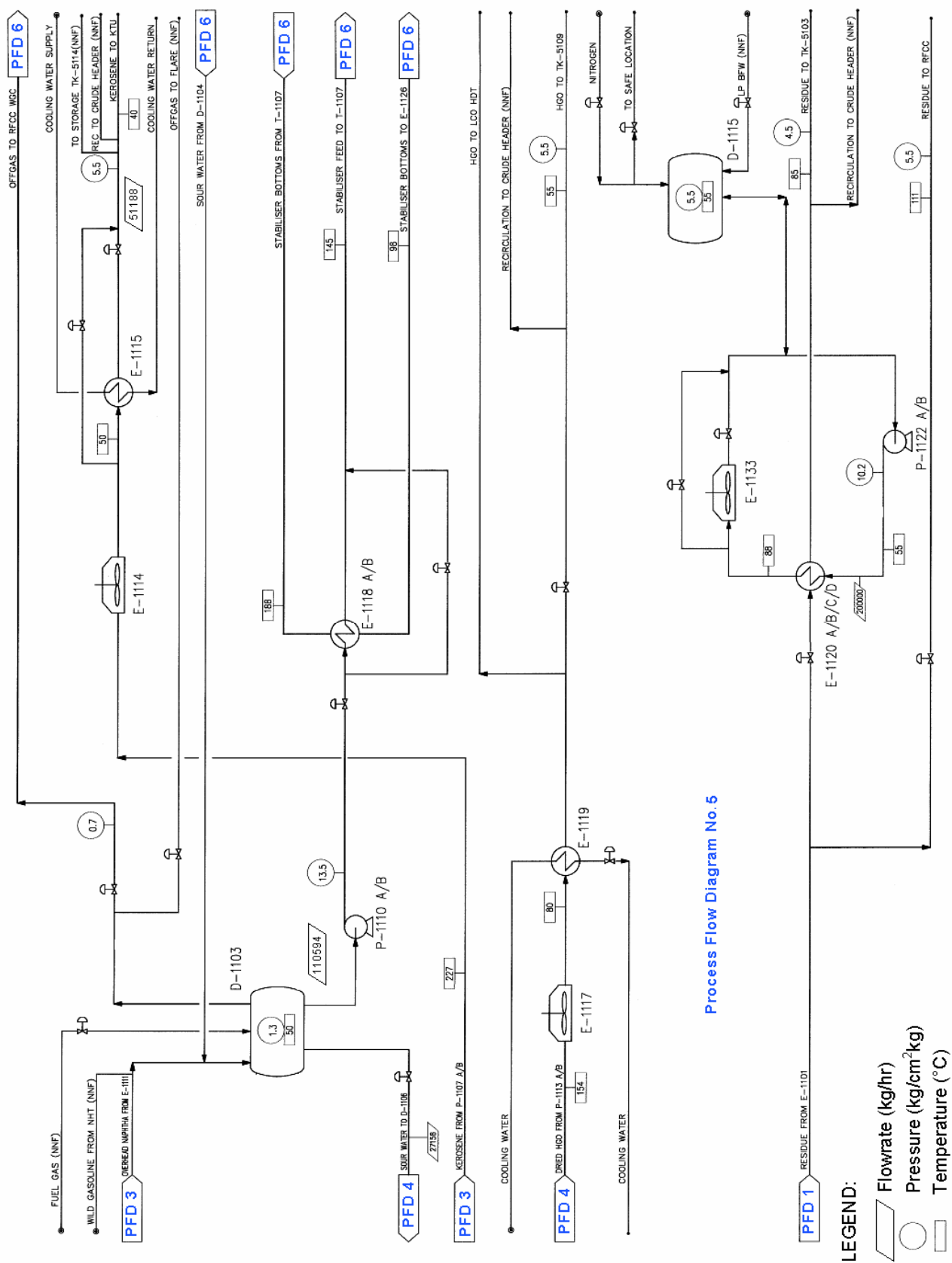


Figure 9: Overhead / Rundown System Simplified PFD

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A top pump-around in the Main Fractionator T-1101 provides reflux to the top section of T-1101 and maintains the overhead temperature of T-1101 overhead. The Top Pump-around Pump (P 1102 A/B) drives the liquid from the tray 4 to E-1112 where the liquid is air cooled, and then routed to the tray 1. The heat removed in Top Pump-around Air Cooler (E-1112) is adjusted to control the overhead temperature (by means of control valves UV-079 and UV-080).

The overhead vapour (124°C), after taking a dose of corrosion inhibitor and neutralizer chemical, condenses totally through the Main Fractionator Condenser E-1111 at 50°C. The outlet from this exchanger gravity flows to the Main Fractionator Accumulator Drum (D-1103).

In D-1103, the water is separated from the unstabilised naphta and drained to D-1106 by the level control valve 011-LV-040. The unstabilised naphta is heated in the Stabilizer Feed/Bottom Exchanger (E-1118 A/B), after it is pumped to the Stabilizer column (T-1107) by means of the Stabilizer Feed Pump (P-1110 A/B). The main fractionator accumulator drum is maintained at the constant pressure of 1.3 kg/cm²g: In case of low pressure, fuel gas is taken into the accumulator in order to increase the pressure. In case of high pressure, off gas from the drum is routed to the RFCC unit. However, in case of overload of the pressure valve to RFCC, off gas is routed to flare.

2.4.2. Kerosene Section

Kerosene is drawn off at tray 15 as a product and to be circulated in the preheat train (E-1102).

Kerosene Pump-around Pump (P-1103 A/B) routes part of the kerosene to the preheat train, specifically to the E-1102. The flow of kerosene pumparound through E-1102 and its bypass is controlled to ensure the separation efficiency in the main fractionator is good enough. Then, the kerosene is routed back to the T-1101, at Tray 12.

The other part of the Kerosene is taken to the Kerosene Stripper (T-1102). This stripper consists of 10 trays and a Kerosene Stripper Reboiler (E-1110), which uses the Heavy Gas Oil (HGO) pump-around as heat source. A facility for stripping steam injection in the bottom of the tower is also available, but it is not necessary in normal conditions.

The top vapour of T-1102 is returned to tray 12 of T-1101. The kerosene product in the bottom is pumped to the Kerosene Air Cooler (E-1114) and afterwards to Kerosene Water Cooler (E-1115), by means of the Kerosene Product Pump (P-1107 A/B), in order to reduce its temperature to 40°C before being sent to the Kerosene Treater Unit (KTU).

2.4.3. Light Gas Oil Section

Light Gas Oil (LGO) is drawn off at tray 26 as a product and for pump-around. Both are used for circulation in the preheat train.

LGO Pump-around Pump (P-1104 A/B) routes part of the LGO to the preheat train, specifically to E-1106 A-F. The flow of LGO pumparound through E-1106 and its bypass is controlled, and so the amount of heat removed, to ensure the separation efficiency in the main fractionator is good enough. Then the LGO is routed back to the T-1101 at tray 23.

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The other part of LGO is taken to the LGO Stripper (T-1103). This stripper consists of 6 trays and an injection of stripping steam in the bottom and on flow control.

The top gas of T-1103 is returned to T-1101 at tray 23. The LGO product in the bottom gravity flows to E-1103 (preheat train) and afterwards to the LGO Dryer (T-1105).

2.4.4. Heavy Gas Oil Section

Heavy Gas Oil (HGO) is drawn off at tray 38 as a product and pump-around. Both are used for circulation in the preheat train.

HGO Pump-around Pump (P-1105 A/B) routes part of the HGO to the preheat train, specifically to E-1109. Then, this HGO is used as the hot fluid in the Kerosene Stripper Reboiler (E-1110). The flow of HGO pumparound through these exchangers and their bypasses is controlled, and so the amount of heat removed, to ensure the separation efficiency in the main fractionator is good enough. Then, the HGO is routed back to the T-1101 at Tray 35.

The other part of HGO leaving tray 38 is taken to the HGO Stripper (T-1104). This stripper consists of 6 trays and an injection of stripping steam in the bottom and on flow control.

The T-1104 top vapour is returned to T-1101 at tray 35. The HGO product in the bottom gravity flows to E-1107 and E-1104 (preheat train) and afterward to the HGO Dryer (T-1106).

Overflash Section

The flash zone is the feed entry point, coming from the heater, located between trays 42 and 43. The heater effluent is feed to the main fractionator column via a tangential nozzle to ensure a good vapour and liquid distribution into the flash zone.

The hot vapour flows up through the tower where it contacts with colder liquid flowing down through the tower.

Liquid from the flash zone flows down over the stripping vapour section where the light components are stripped out.

2.4.5. Residue Section

T-1101 bottom Residue (at 349-354 °C) is pumped to the preheat train by the Residue Pump (P-1106 A/B). In particular, Fractionator residue is pumped to the heat exchangers and following this order: E-1134 A/B, E-1108 A-D, E-1105 A-J, E-1101 A-H (See Crude Preheat).

To strip off any light component that would be otherwise taken out in the residue stream, stripping steam is continuously injected in the bottom of the tower, under flow control.

Part of T-1101 bottoms liquid is routed under flow and level control to the RFCC, while the remaining residue is routed to storage (tank TK-5103) after having been cooled down to 85°C in E-1120 A-D.

Heat is removed from the residue with water in the Residue / Tempered Water Cooler (E-1120 A-D). After that, this water is air cooled in the Tempered Water Air Cooler (E-1133).

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Tempered Water Pump (P-1122 A/B) recycles the water from E-1133, along with any necessary make-up water from the Tempered Water Drum (D-1115), to use it again as the cooling fluid of E-1120 A-B.

2.5. Stabilizer Section

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The unstabilised naphta from D-1103 is preheated in E-1118, before entering the Stabilizer column (T-1107), where the LPG is separated from the stabilized naphta. This tower is a two diameter column (1500 mm in the top and 2600 mm in the bottom), with 32 trays, a Stabilizer Reboiler (E-1121) in the bottom and a top reflux system.

The top vapour flow is partially condensed in the Stabilizer Condenser E-1122 and then it gravity flows to the Stabilizer Reflux Drum (D-1104), where off gas, LPG and water are separated.

Off gas is discharged to the RFCC unit on pressure control. In case of overload, the excess is discharged to the flare system. Water is sent to D-1103 under level control.

Part of the LPG is taken by the Stabilizer Reflux Pump (P-1114 A/B) under flow control and is discharged to the top of T-1107 as reflux.

The other part of the LPG is pumped by the Stabilizer LPG Pump (P-1115 A/B) to Gas Recovery in RFCC. A separate line to LPG off-spec storage is also available.

The liquid in the bottom of T-1107 flows continuously by siphon effect in the Stabilizer Reboiler (E-1121). This heat exchanger uses High Pressure Steam (HP) as hot fluid, which has been previously desuperheated in the Desuperheater (DS-1101) with high pressure Boiling Feed Water (BFW).

The Full Range Naphtha discharged in the bottom is used to preheat the stabilizer feed in the E-1118 A/B. The remained heat of Naphtha is taken off downstream in the Full Range Naphtha Air Cooler (E-1126), and then, in the Full Range Naphtha Water Cooler (E-1127) before being sent to storage.

2.6. Dryers

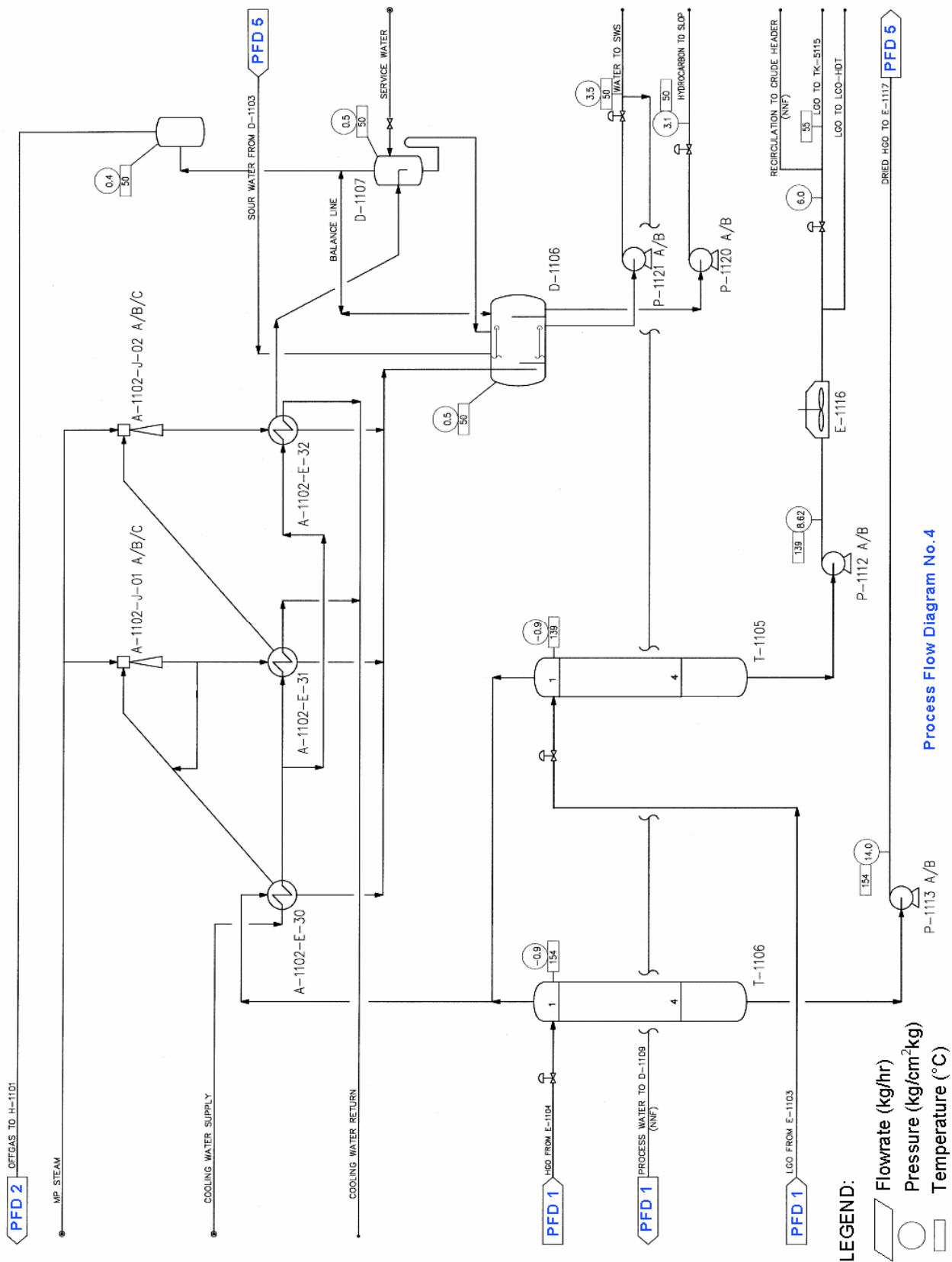


Figure 11: Dryer / Ejector System Simplified PFD

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The LGO and HGO produced by crude distillation are routed to the LGO Dryer T-1105 and HGO Dryer T-1106 respectively. Each column has 4 trays.

The LGO produced in the bottom of T-1105 is pumped to the LGO Product Cooler (E-1116) by the LGO Product Pump (P-1112 A/B). This air cooler reduces the temperature of the LGO to 55°C before being sent to storage (TK-5115) and LCO-HDT under flow control.

The HGO produced in the bottom of T-1106 is pumped to HGO Product Air Cooler (E-1117) by the HGO Product Pump (P-1113 A/B). This exchanger along with the HGO Product Water Cooler (E-1119) reduces the temperature of the HGO to 55°C before being sent to storage (TK-5109) and LCO HDT under flow control.

A reduced operating pressure of -0.9 kg/cm²g is maintained in these towers (T-1105/06) by the Vacuum Package A-1102.

2.7. Vacuum Section

Vacuum system maintains a reduced pressure in the Dryers by Ventury effect. This system consists of a Pre-Condenser (A-1102-E-30), and two stage After-Condenser (A-1102-E-31 and E-1102-E-32). Each After Condenser has a train of three parallel ejectors: A-1102-J-01A/B/C for the first stage and A-1102-J-02 A/B/C for the second stage.

The purpose of the ejectors is to entrain tower overhead vapours and non-condensables, by means of medium pressure motive steam. The purpose of the condensers is to condense as much steam and hydrocarbons as possible. Cooling water is used as cold fluid in the condensers.

Condensate from all condensers is drained to the Drier Oil / Water Separator D-1106 where the water is separated from the hydrocarbon phase. The hydrocarbon phase is pumped to Slop by the Drier Slop Oil Pump (P-1120 A/B), and the water is pumped by the Ejector Condensate Pump (P-1121 A/B) to the Stripped Water System (SWS).

The off-gas is routed to the Drier Off-gas Seal Pot D-1107 before being sent to the burners of H-1101 along with any gas separated in D-1106.

2.8. Chemical Package

Four chemical compounds are continuously used in Crude Distillation Unit. Points of injection and a brief description are shown below:

2.8.1. Neutralizer

The neutralizer maximizes corrosion protection, by means of chlorides neutralization and controlling pH in the naphta as it condenses in the vapour line and overhead condenser.

In this case, a continuous dosage of 5 ppm (0.002 m³/h) of neutralizer is injected to the Fractionator (T-1101) overhead vapour line, upstream of E-1111. The flow will be adjusted manually by the operator.

The chemical is pumped from the Neutralizer Storage Drum (A-1104-D-10) to the injection point by the Neutralizer Injection Pump (A-1104-P-26 A/B).

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2.8.2. Corrosion Inhibitor

The corrosion inhibitor provides excellent carbon steel resistance to acid attack from H₂S, HCl, CO₂, organic acid, SO_x acids and HCN.

In this case there are 2 injection points:

- Fractionator overhead vapours line
- Suction of Top Pump-around P-1102 A/B

The chemical is pumped from the Corrosion Inhibitor Storage Drum (A-1104-D-11) to each injection point by the Corrosion Inhibitor Injection Pump (A-1104-P-25 A/B). A flow meter and a valve are used to regulate the flow injected in each point.

The total corrosion inhibitor dosage to be injected is 2 ppm (0.002 m³/h).

2.8.3. Demulsifier

The demulsifier increases desalter dehydration and salt removal efficiency in order to reduce crude unit corrosion. It helps to maximize crude rates by controlling emulsion buildup at the desalter interface.

In this case, there are two injection points:

- In the crude feed upstream of Cold Preheat Crude Train
- Upstream of second stage desalter, in the suction of P-1101 A/B (this injection is only required during start-up).

The chemical is pumped from the Demulsifier Storage Drum (A-1104-D-12) to each injection point by the Demulsifier Injection Pump (A-1104-P-23 A/B). A flow meter and a valve are used to regulate the flow injected in each point.

The total demulsifier dosage to be injected is 1.5 ppm (0.002 m³/hr).

2.8.4. Antifoulant

The antifoulant reduces process side fouling caused by coke, polymers, sludge, corrosion products, tar, and other particulate matter.

Antifoulant is injected in two points:

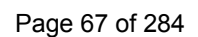
- Feed of Cold Preheat Train
- Feed of Hot Preheat Train in the suction of P-1101 A/B

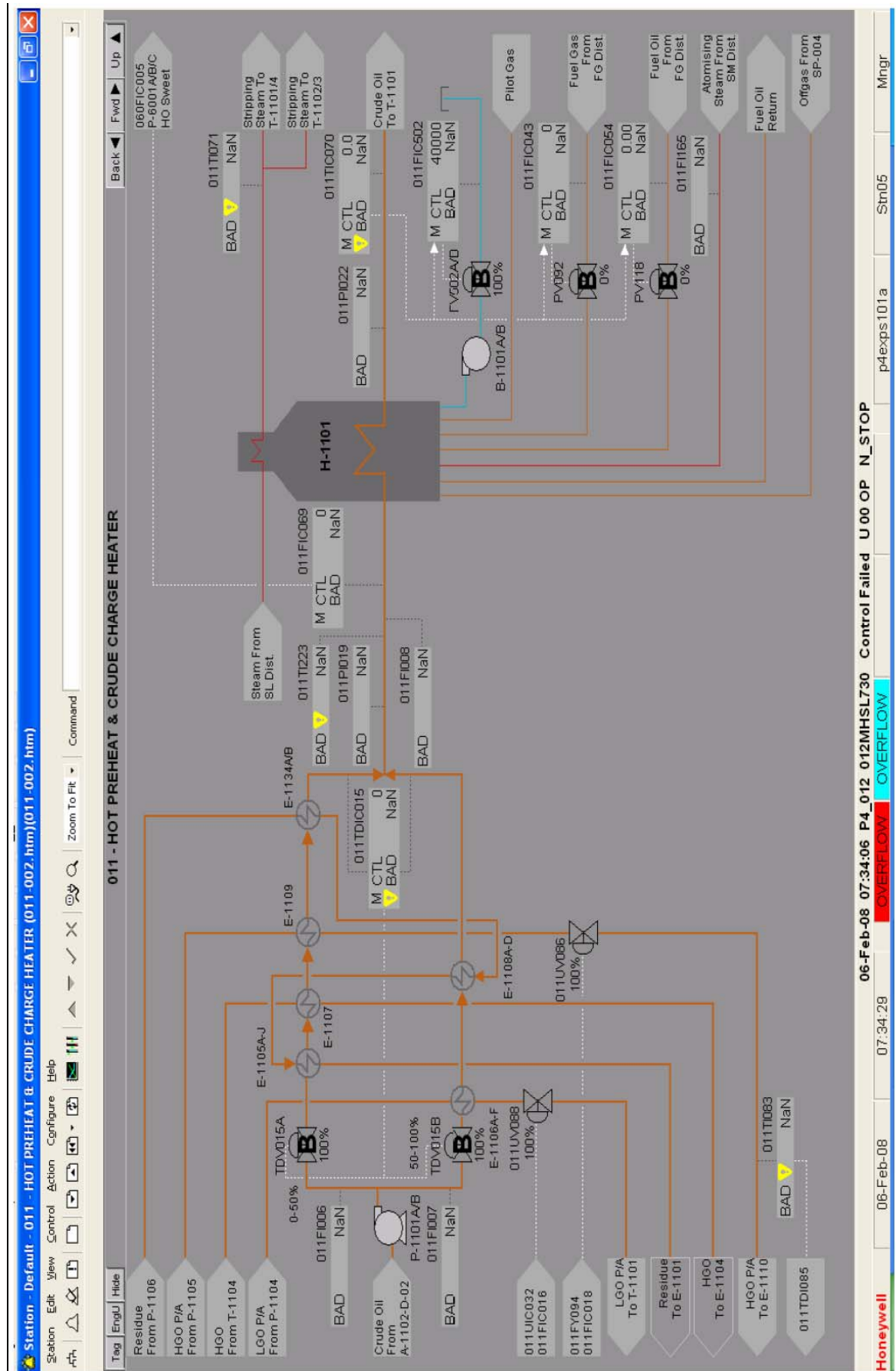
The chemical is pumped from the Antifoulant Storage Drums (A-1104-D-13A/B) to each injection point by the Antifoulant Injection Pumps (A-1104-P-24 A/B). A flow meter and a valve are used to regulate the flow injected in each point.

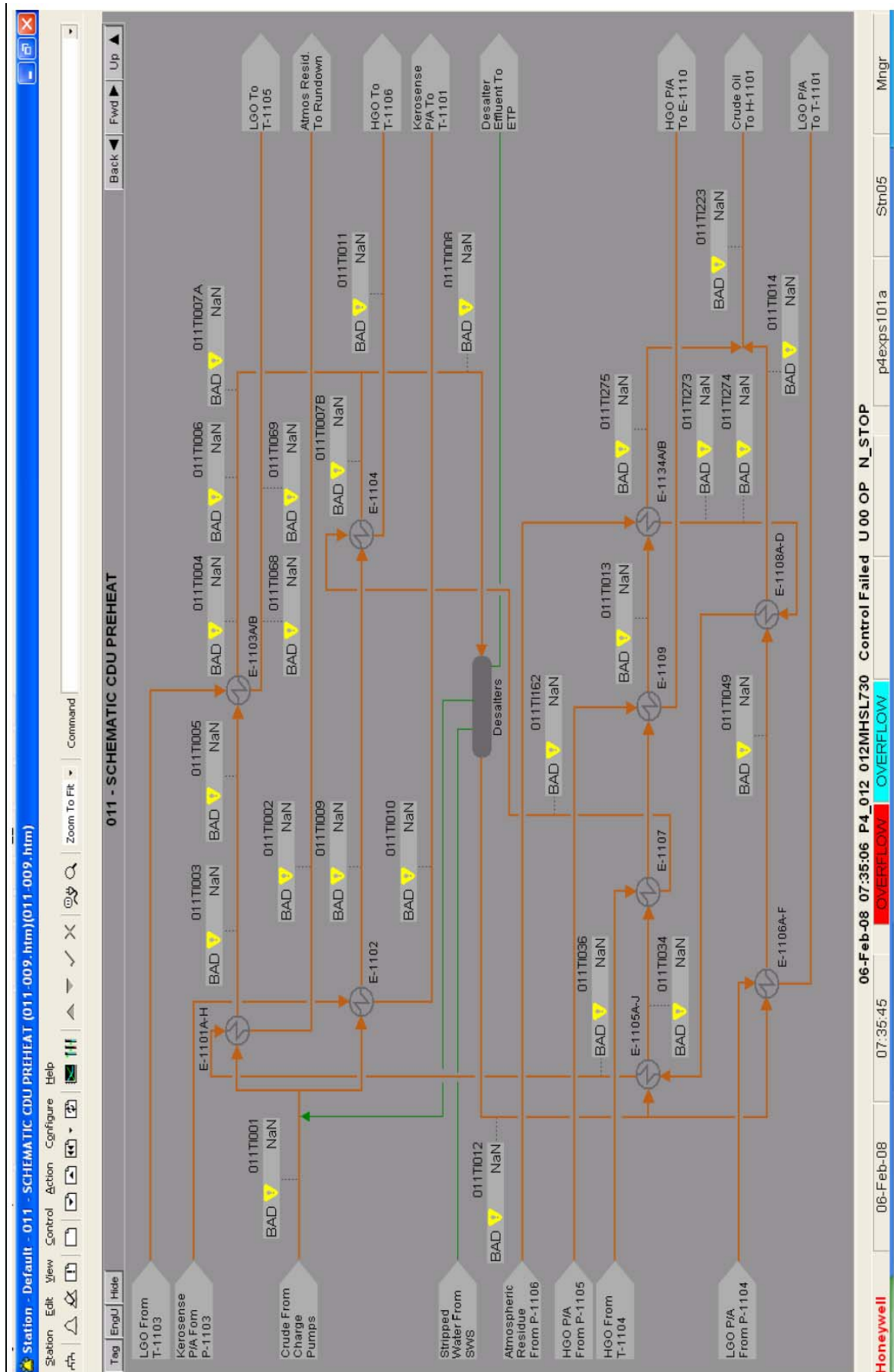
The total antifoulant dosage to be injected is 5 ppm (0.005 m³/h)

2.9. Process Overview with DCS Printouts

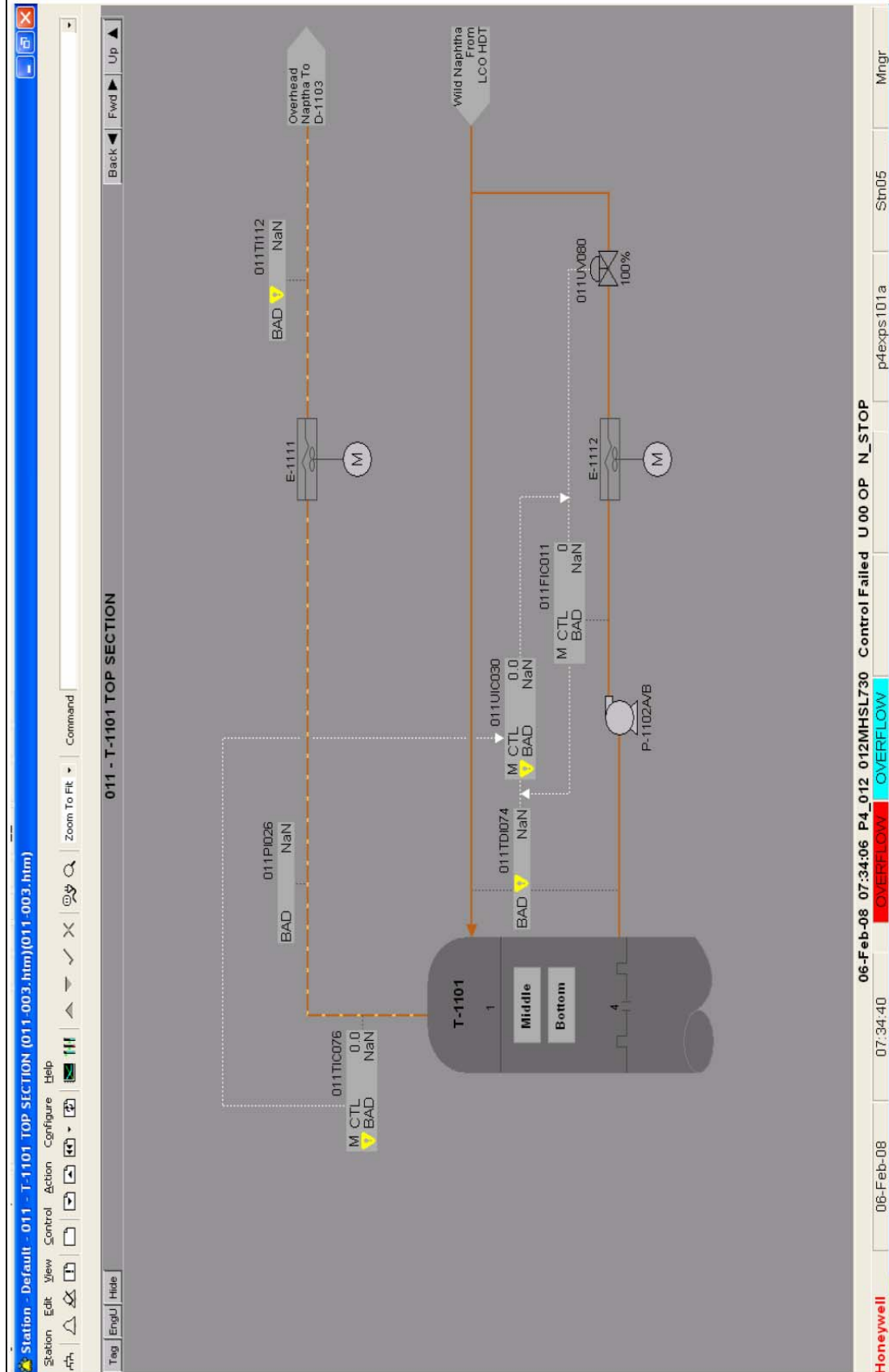
2.9.1. Crude Preheat and desalters

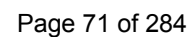






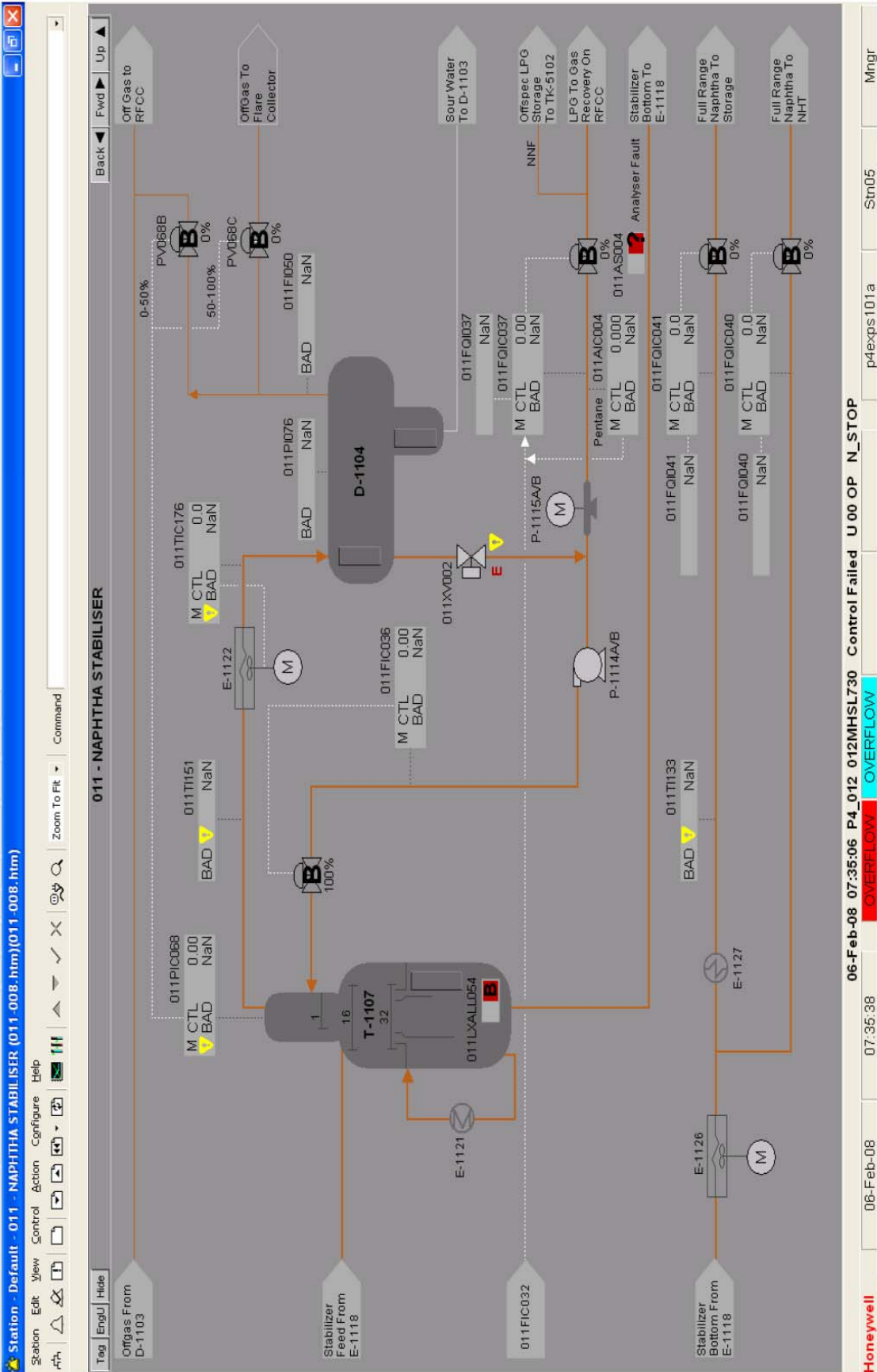
2.9.2. Main Fractionator T-1101







2.9.5. Naphta Stabilizer



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TRAINING MODULE

CRUDE DISTILLATION UNIT (CDU)

UNIT: 11

Course Content:

Section 1 - General Description	
Section 2 - Process Flow Description	
Section 3 - Process Control	X
Section 4 - Safeguarding Devices	
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Section 10 - Reference Document Index	

SECTION 3 : PROCESS CONTROL

For a detailed description of the control loops and the controllers, refer to the process control narrative, 8474L-011-SP-1511-101.

3.1. Control Narrative & Operating parameters

3.1.1. Crude Preheat and desalter

3.1.1.1. Desalter Pressure Control (PIC-144) and Cold preheat temperature control (TDIC-007)

The objective of this system is to control the temperature differential of the two parallel streams leaving the cold preheat train (TDIC-007) whilst minimizing the pressure drop introduced by the control valves.

The pressure control of the desalters can override the differential pressure control.

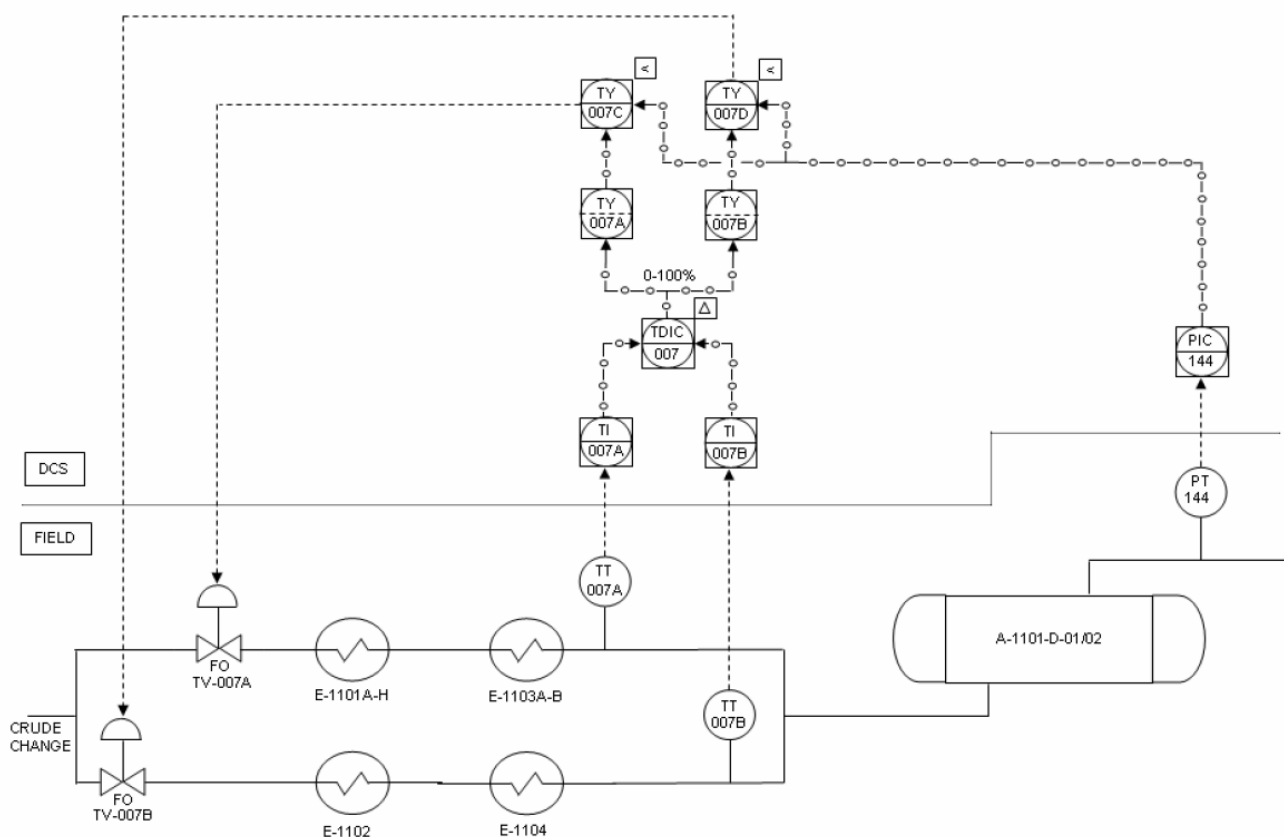


Figure 12: Functional Description of Cold preheat temperature control

Pressure control of the Desalters is achieved by 011-PIC-011 and 011-PIC-144. 011-PIC-144 is provided to give a quick response if the pressure rises above the normal operating range of 011-PIC-011. 011-PIC-144 has a higher setpoint than 011-PIC-011 and it is used to restrict the pressure in the Desalters by closing the cold preheat temperature control valves as described below.

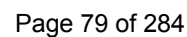
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In order to maintain the same crude temperature difference between the parallel branches, the entering flow to each one is controlled by two control valves, 011-TV-007A and 011-TV-007B, placed at E-1101 and E-1102 inlet, respectively.

If 011-TDIC-007 process variable (ΔT) is higher than the set point (i.e. 011-TI-007A is higher than required or 011-TI-007B is lower than required) then the output from 011-TDIC-007 to 011-TY-007 A/B increases. In the normal operating range of the controller, this output increase closes 011-TV-007B to increase the temperature of crude leaving E-1104 and opens 011-TV-007A to decrease the temperature of the crude leaving E-1103 A/B.

The blocks 011-TY-007A and 011-TY-007B operates as lineal blocks, the outputs of 011-TY-007A and 011-TY-007B pass through the minimum select blocks 011-TY-007C and 011-TY-007D to close the valves if required by pressure controller 011-PIC-144 because of high pressure in desalter A-1101-D-02. In this case, the temperature control is inhibited until the pressure comes down again. This however should be rare.

| The following DCS printout shows the cool crude preheat:



3.1.1.2. Desalters interface level control

The interface level control is one of the most important parameter to operate the desalters.

Both desalters, A-1101-D-01 and A-1101-D-02, are provided with level controls, 011-LIC-503 and 011-LIC-506 respectively. The output signal of 011-LIC-503 is used to regulate the effluent water control valve 011-LV-003. The other controller, 011-LIC-506 is used to regulate the water recirculation control valves 011-FV-005 and 011-FV-004 or, in case the first desalter is bypassed, to regulate the effluent water control valve 011-LV-003 by the selector 011-HS-001.

The control valves for both desalters are placed in the discharge line, so an increase in the vessel level will open the control valves.

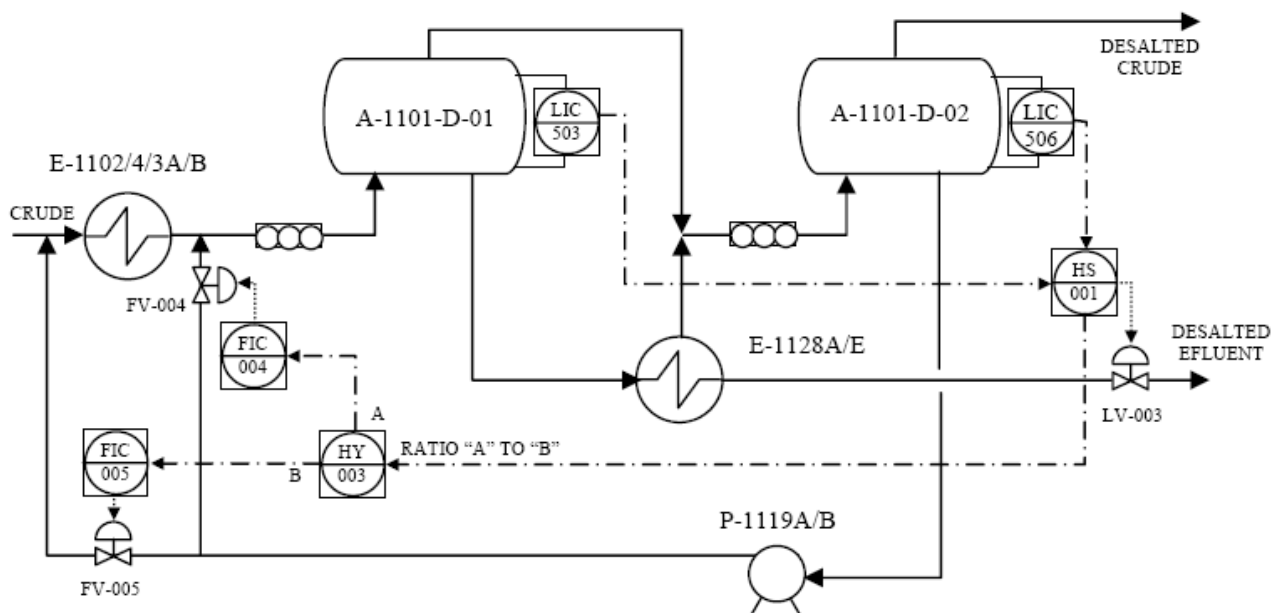


Figure 13: Functional Description of Desalters Interface Level Control

3.1.1.3. Desalter Pressure Control (PIC-011) & CDU Feed Rate (011-FY-092 / 011-HY-002)

The objectives of this system are:

- To control the pressure of the desalters by trimming the CDU feed rate
- To control the crude oil flow rate to the desalters such that the pressure in the Desalters is held constant, setting the flow rate at the crude charge pumps to the same flow as that measured at the fired heaters, 011-FIC-069.
- To maintain the required flow ratio of Bach Ho and Sour Crudes. For this, the operator enters the relation sweet/sour through 011-HY-002 that generates the setpoint of the controllers 060-FIC-005 (Bach Ho) and 060-FIC-012 (sour).

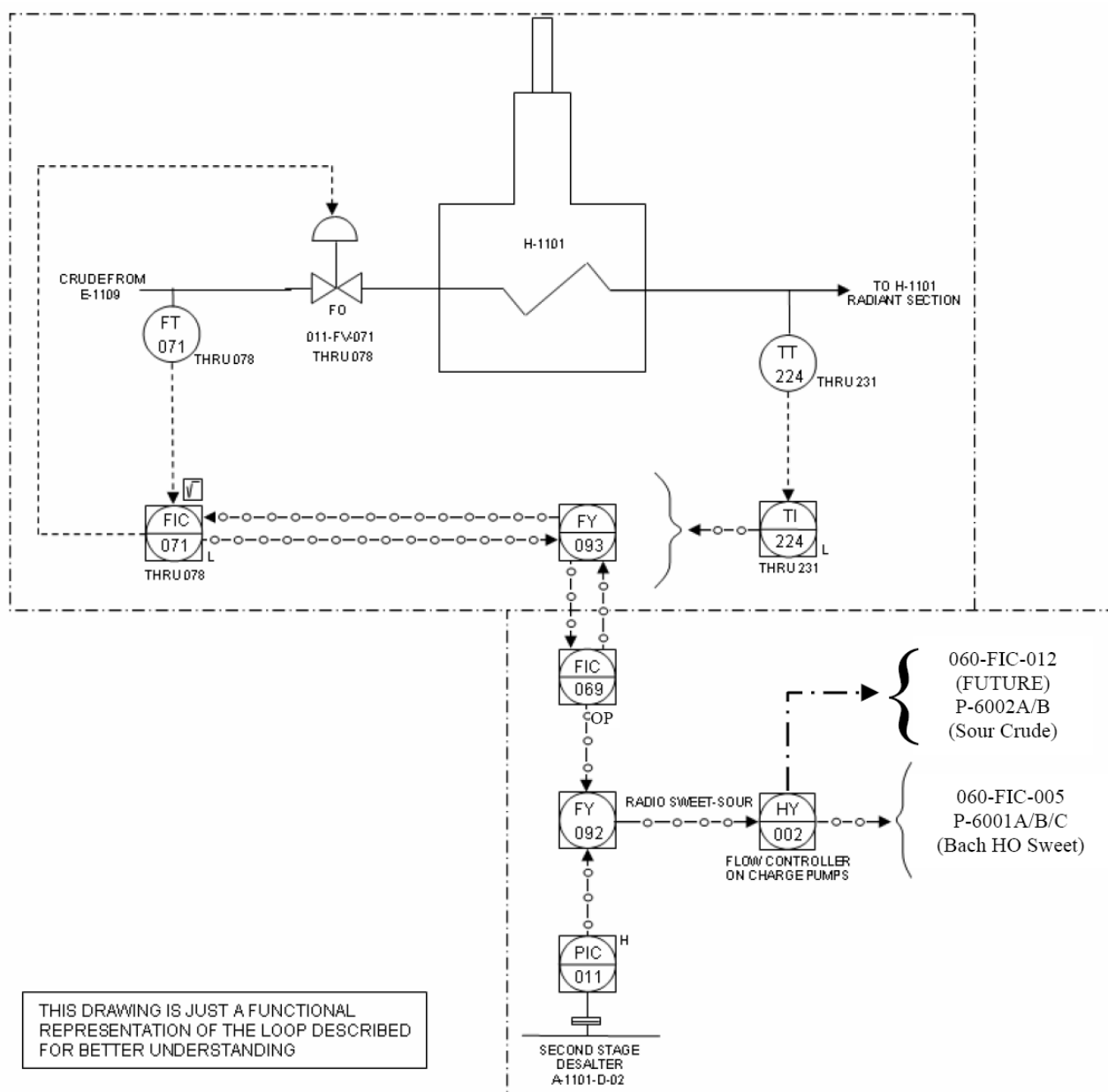


Figure 14: Functional Description of CDU Feed rate Control

Pressure control of the desalters is achieved thru 011-PIC-011, which is used to control the pressure of the desalters by trimming the CDU feed rate, and 011-PIC-144 (as described in the previous section).

The flow rate out of the desalters is set at the heater pass balancing controller (see description below), which sets the flowrate on controller 011-FIC-069. Then in order to overcome flow measurement inaccuracies over a prolonged period and to trim the pressure, a pressure correction is made. This increases the flow if the pressure is too low or decreases the flow if the pressure is too high.

The following DCS printout shows the Desalters:

3.1.1.4. Hot Preheat Temperature Control

The objective of this system is to control the temperature differential of the 2 parallel streams leaving the hot preheat train, whilst minimizing the pressure drop introduced by the control valves.

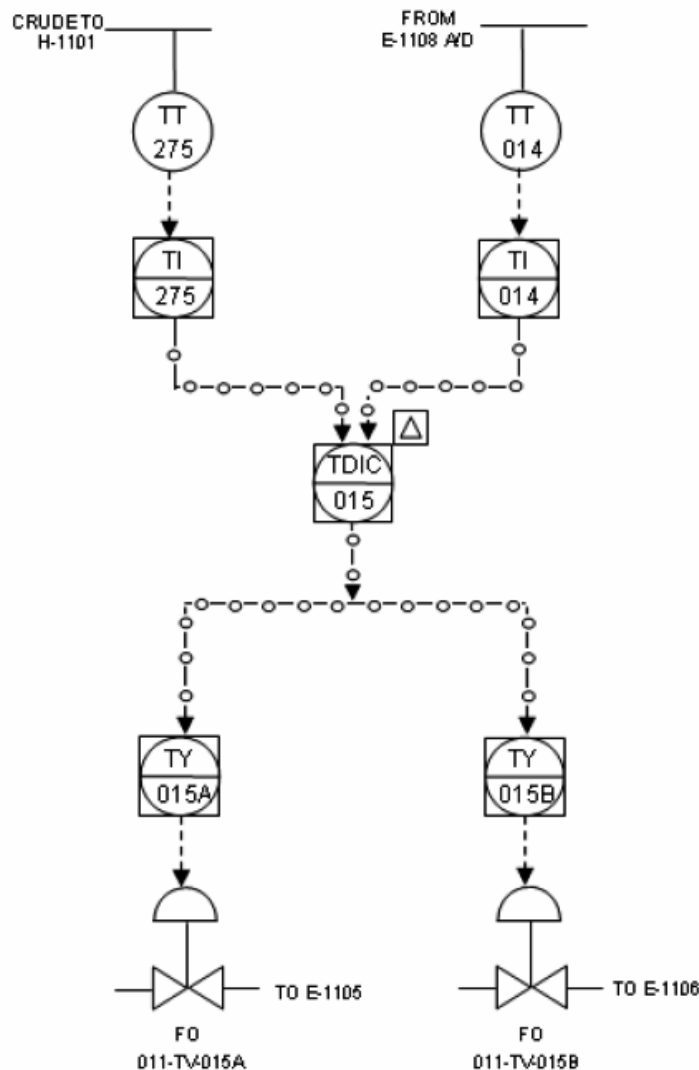


Figure 15: Functional Description of Hot Preheat Temperature Control

In order to maintain the same crude discharge temperature in each parallel branch, the entering flow is controlled by two control valves, 011-TV-015A and 011-TV-015B, placed at E-1105 A-J and E-1106 A-F inlet, respectively.

If 011-TDIC-015 process variable (ΔT) is higher than the set point (i.e. 011-TI-275 is higher than required or 011-TI-014 is lower than required) then the output from 011-TDIC-015 increases. In the normal operating range of the controller, this closes 011-TV-015B to increase the temperature of crude leaving E-1108A-F and opens 011TV-015A to increase the temperature of crude leaving E-1109.

The following DCS printout shows the Hot Crude Preheat:



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3.1.1.5. Heater Pass Balancing (011-FY-093)

Objective:

- To maintain the total flow to the crude charge heater at a desired value set by the operator and to equalize the individual pass outlet temperatures, in order to correct possible hydraulic and heat imbalances.
- To set flow set points for all the passes from a flow weighted average temperature of all passes and the individual pass outlet temperatures
- The total of all the flow setpoints shall equal the desired value set by the operator using 011-FIC-069.
- The fuel rate to the furnace is adjusted to maintain a desired combined outlet temperature as part of the combustion control (as described later).

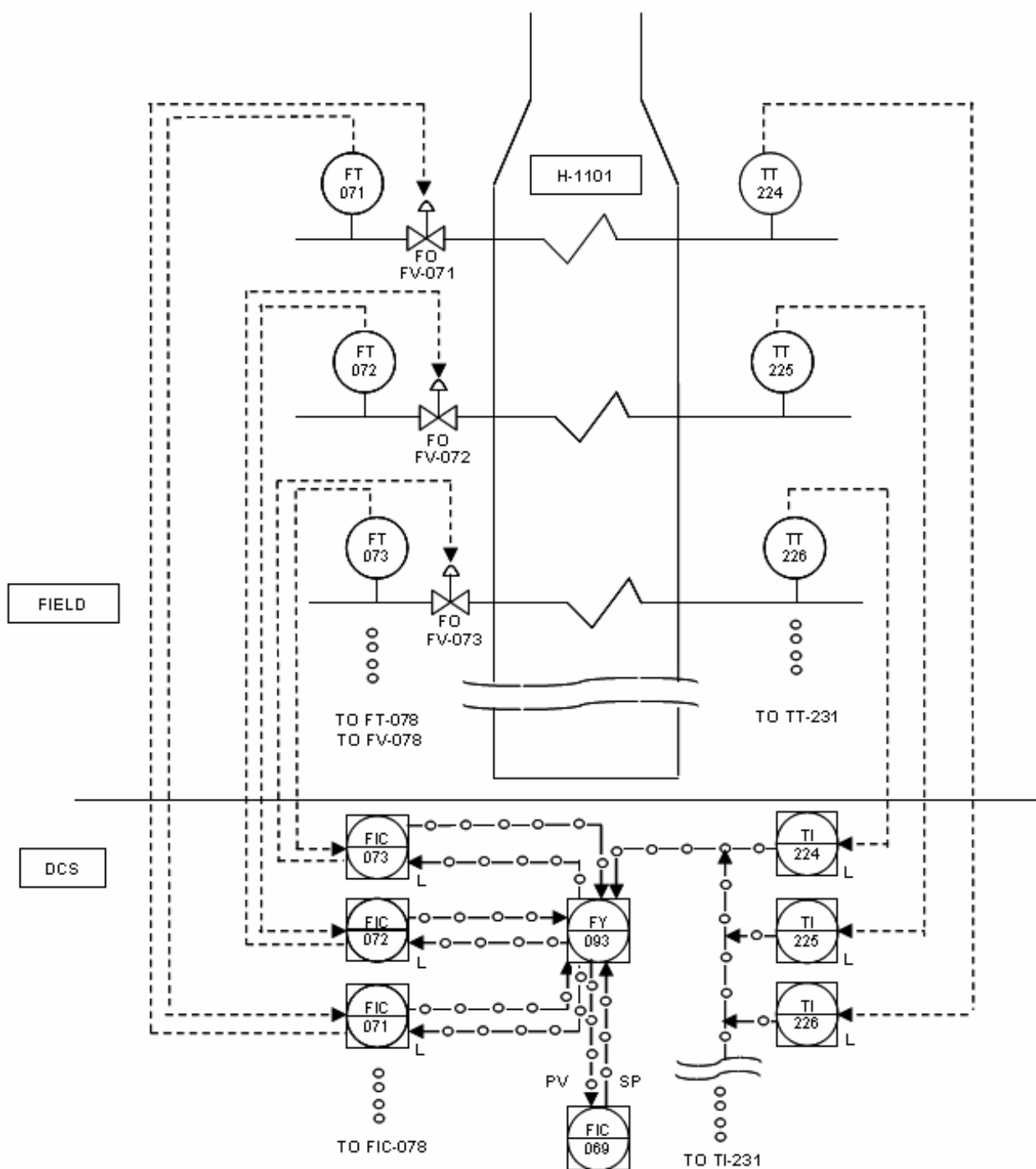


Figure 16: Functional Description of Heater Pass Balancing

For the functional description only 3 heater passes are represented. Actually there are 8 heater passes.

To maintain the total flow of crude charge heater at desired value set by the operator and to equalize the individual pass outlet temperatures, in

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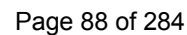
order to correct hydraulic and heat imbalances, the pass balance control receives as input:

- The flow weighted average outlet temperature of all passes. This value is calculated on the 011-FY-093, based on the outlet temperature of each pass given by 011-TT-224 to 231.
- The inlet flowrate (011-FT-071 to 011-FT-078)

These variables are fed to 011-FY-093 where:

- Total flowrate is calculated and used as process variable for 011-FIC-069. The set point of this controller is manually set by operator.
- Output from 011-FIC-069 is divided by the number of passes to calculate flow control action through each pass, that is the setpoint of the controllers 011-FIC-071 thru 011-FIC-093, compensating with outlet temperature of each pass.
- At the same time, 011-FY-093 process value is used by 011-FIC-069 to control the crude charge pumps P-6001 A/B/C and P-6002 A/B via 011-FY-092 and 011-HY-002.

The following DCS printout shows the Crude Charge Heater:



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3.1.2.1. Heat demand

The furnace outlet temperature is controlled by the TIC-070. This controller sets the heat demand and acts on the firing through two selectors relay: a high signal selector (011-FY-502C) and a low signal selector (011-FY-066C). These selector relays ensure that:

- In case of increase of the fuel demand, first the air flow set point is increased and afterwards the fuel set point (high selector function, 011-FY-502C).
- In case of decreasing fuel demand, first the fuel flow set point is decreased and afterwards the air set point (Low selector function, 011-FY-066C)

Considering this configuration, a safe sequence of adjustment of the set point of the fuel gas and air controller is guaranteed.

To permit a certain initial response to load changes, a small negative bias is applied to the total fuel-flow signal to high selector, 011-FY-502B output. Likewise a small positive bias is applied to the air flow to low selector, 011-FY-066B output.

3.1.2.2. High signal selector

The fuel gas flow is measured via 011-FI-044 and is compensated via 011-FY-502A to take into account the change in the fuel gas composition (fuel gas density), which signal is sent from 037-Ai-001 in the fuel gas unit (037).

In the same way, fuel oil flow measurement is calculated from the difference between fuel oil supply, 011-FI-053, and return, 011-FI-055, both compensated by temperature (011-TI-178 and 011-TI-179 respectively), and converted into an equivalent flow of standard fuel gas with calorific value (CV) corrector (011-FY-053C). Calorific value is set as constant by DC operator into 011-FY-053C.

Both flowrates (fuel oil and fuel gas) are added via 011-FY-502B. Furthermore, combustion air flowrate is measured by means of 011-FI-066 and temperature compensated by 011-FY-066A (011-TI-134). 011-XA-502 obtains the air/fuel ratio from the relation between the signal from 011-FY-502B and the air flowrate signal from 011-FY-066A. A low ratio alarm (011-XAL-502) and a low-low air/fuel ratio trip (011-XALL-502) have been considered to protect from unsafe combustion.

Once the total fuel (fuel gas and fuel oil) is measured, this is compared with the heat demand via 011-FY-502C. The higher signal will be the combustion air flow required for the heat demand. This value (corrected by air/fuel ratio) fixes the set point of the combustion air controller (011-FIC-502) that actuates both valves 011-FV-502 A/B. Therefore:

- The maximum value between heat demand and actual fuel flowrate sets the combustion air demand.
- The combustion air required is translated into equivalent fuel gas by means of the air/fuel ratio manually established.

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- When an increase in duty is required, the air flowrate is increased before the fuel flowrate is increased.

Note:

The 011-HIC-077 air/fuel ratio shall be configured with the following data: Minimum engineering unit range 15; the normal setting shall be 17, the operator only can diminish it value of 17 if the excess of oxygene (O₂) is superior to the 3%.

3.1.2.3. Low signal selector (011-FY-066C)

Combustion air flowrate is measured and temperature compensated by means of 011-FI-066 and 011-FY-066A. This value is divided by the air/fuel ratio via 011-FY-066B in order to obtain fuel gas equivalent to airflow measured and enters, along with the heat demand into the low signal selector 011-FY-066C. The outlet of the low signal selector (lowest signal between heat demand and air flowrate) is the inlet to two blocks 011-FY-066D & 011-FY-066E where the set point for fuel gas controller and fuel oil controller respectively are calculated. 011-FY-066D represents the difference between outlet of low signal selector and measured fuel oil flow. This is the set point of the fuel gas controller (011-FIC-043). 011-FY-066E represents the difference between outlet of low signal selector and measured fuel gas flow. This is the set point of the fuel oil controller (011-FIC-054).

The signal from the fuel gas controller (011-FIC-043) and fuel oil controller (011-FIC-054) is sent to a pressure controller (011-PIC-092 and 011-PIC-118 respectively) that actuates the respective valve (011-PV-092 and 011-PV-118).

With this control scheme,

- The set point for the fuel controllers is limited by the available combustion air,
- When a decrease in duty is required, the air flowrate is not decreased until the fuel flowrate is decreased.

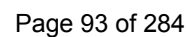
The controllers 011-PIC-092 (fuel gas) and 011-PIC-118 (fuel oil) can operate in two modes: local and cascade. In local mode (during start-up) the set point is fixed by operator. When the normal operation duty is achieved, the controller is switched to cascade mode and the set point for the controllers 011-PIC-092 and 011-PIC-118 are fixed by the desired outlet process temperature.

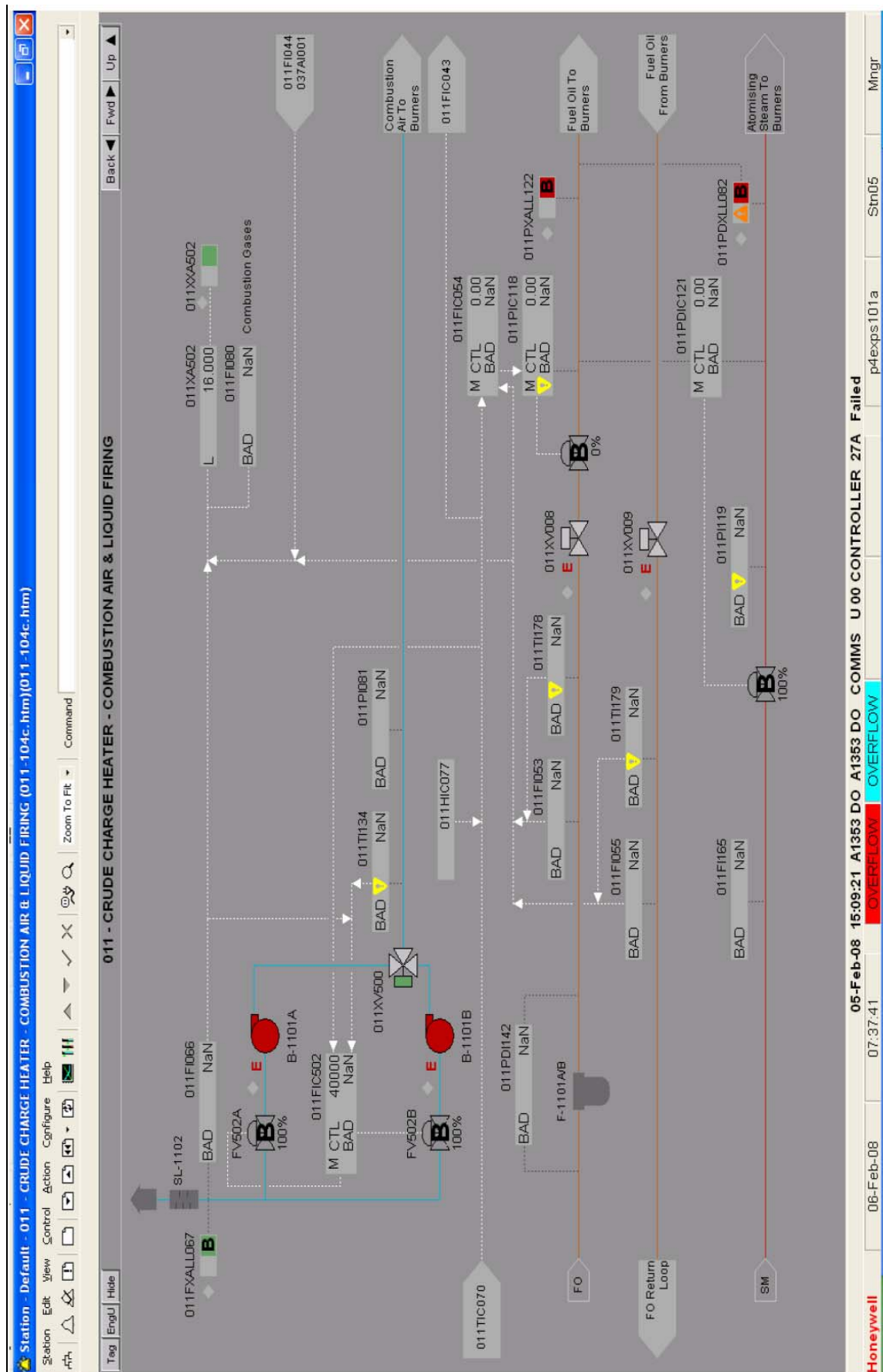
In order to facilitate the start-up (considering the large flexibility required, 1 burner at minimum duty during start-up and 16 burners at design duty in normal operation) and guarantee a minimum pressure for the satisfactory operation at minimum load of burners, a control valve (011-PCV-072) has been installed in parallel to the main fuel gas valve (011-PV-092). This by-pass valve (by-pass regulator) has the following functions:

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- Maintain the minimum pressure required for minimum liberation at burners. This will avoid the low main fuel gas pressure trip due to control or control valve failure.
- By-pass valve will be designed in a range from minimum liberation of one burner to minimum liberation of 16 burners. This provides a better precision during start up that would not be achieved by means of the main control valve that requires a larger pressure design range.

The following DCS printouts show the Charge Heater H-1101 Gas/Liquid Firing:







3.1.3. Superheated low pressure steam temperature

The superheated steam temperature used to feed the strippers is controlled by 011-TIC-063 controller. The objective of this controller is to keep a constant temperature in the superheated steam at heater outlet, by acting on two control valves (011-TV-063A, and 011-TV-063B). 011-TIC-063 compares the desired value (set point fixed by operator) and the measured temperature.

The controller signal is routed to 011-TY-063A or 011-TY-063B, which modify the opening percentage on 011-TV-063A and 011-TV-063B respectively, depending on its value. If the temperature measure is lower than its set point, the controller will tend to close 011-TV-063A and will open 011-TV-063B, in order to increase the steam flow through fired heater.

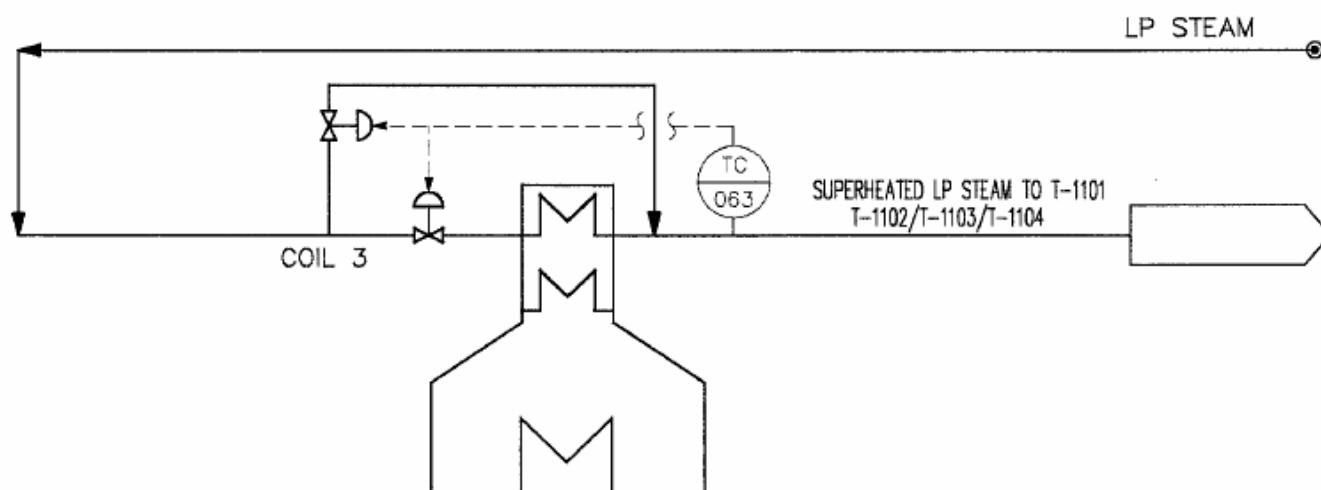


Figure 18: Functional Description of Superheated LP Steam Temperature Control

3.1.4. DS-1101 Desuperheater outlet temperature control

The purpose of this controller is to keep constant the temperature at desuperheater outlet. This is a cascade loop in which the controller 011-PIC-201 output is routed to the 011-TIC-301 set point. The temperature controller acts over 011-TV-301 control valve, in order to ensure the steam is saturated at desuperheater outlet.

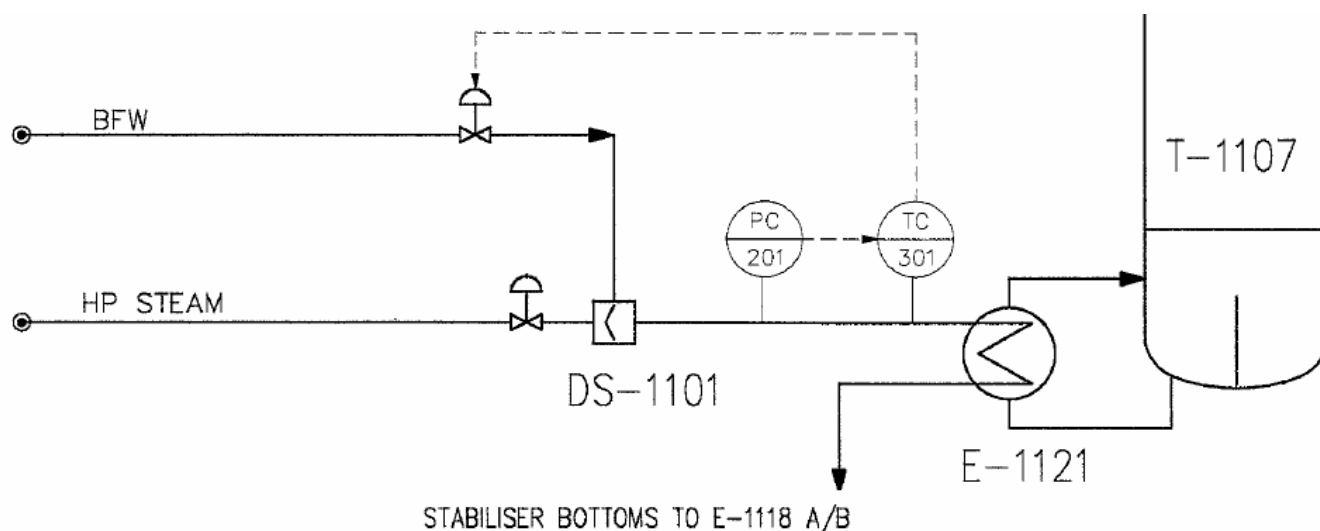


Figure 19: Functional Description of Desuperheater Outlet Temperature Control

3.1.5. Fuel oil/atomizing steam differential pressure controller

The objective of this controller is to ensure that the atomizing steam pressure at burner inlet is 2.1 kg/cm²g higher than the fuel oil pressure at burner inlet. If the pressure difference measured by the controller 011-PDIC-121 is lower than 2.1 kg/cm²g the controller will open the steam control valve 011-PV-121, in order to increase the steam pressure at heater inlet.

3.1.6. Main Fractionator T-1101

Control of the main fractionator is achieved primarily by the removal of heat in four pumparound streams. The top pumparound stream is used to control the tower top temperature by the removal of heat from the top section of the tower. The other pumparound streams are: Kerosene, Light Gas Oil (LGO) and Heavy Gas Oil (HGO).

These pumparound streams control the quality of the side draw products, by altering the temperature profile of the Main Fractionator, T-1101.

3.1.6.1. Overheads Temperature Control (TIC-076)

The top pumparound (TPA) circuit of the main fractionator provides reflux to the top section of T-1101 and maintains the temperature of T-1101 overhead vapour by controlling the amount of heat removed from the TPA circuit. Under normal operation, for a given unit throughput, the flow around the TPA circuit remains constant and the heat duty is controlled by passing more or less flow around E-1122. The top temperature, 011-TIC-076, resets the set point of duty controller 011-UIC-030.

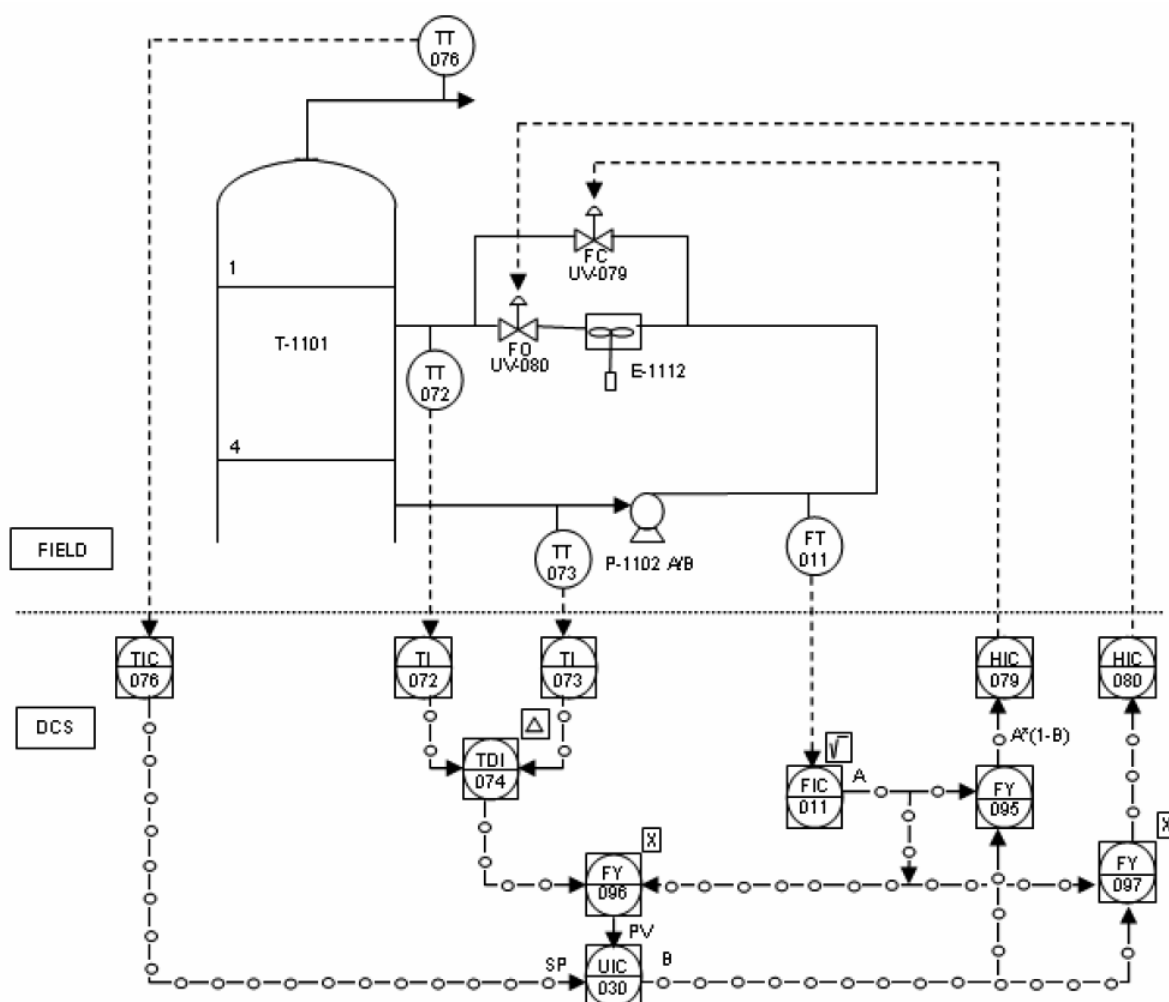


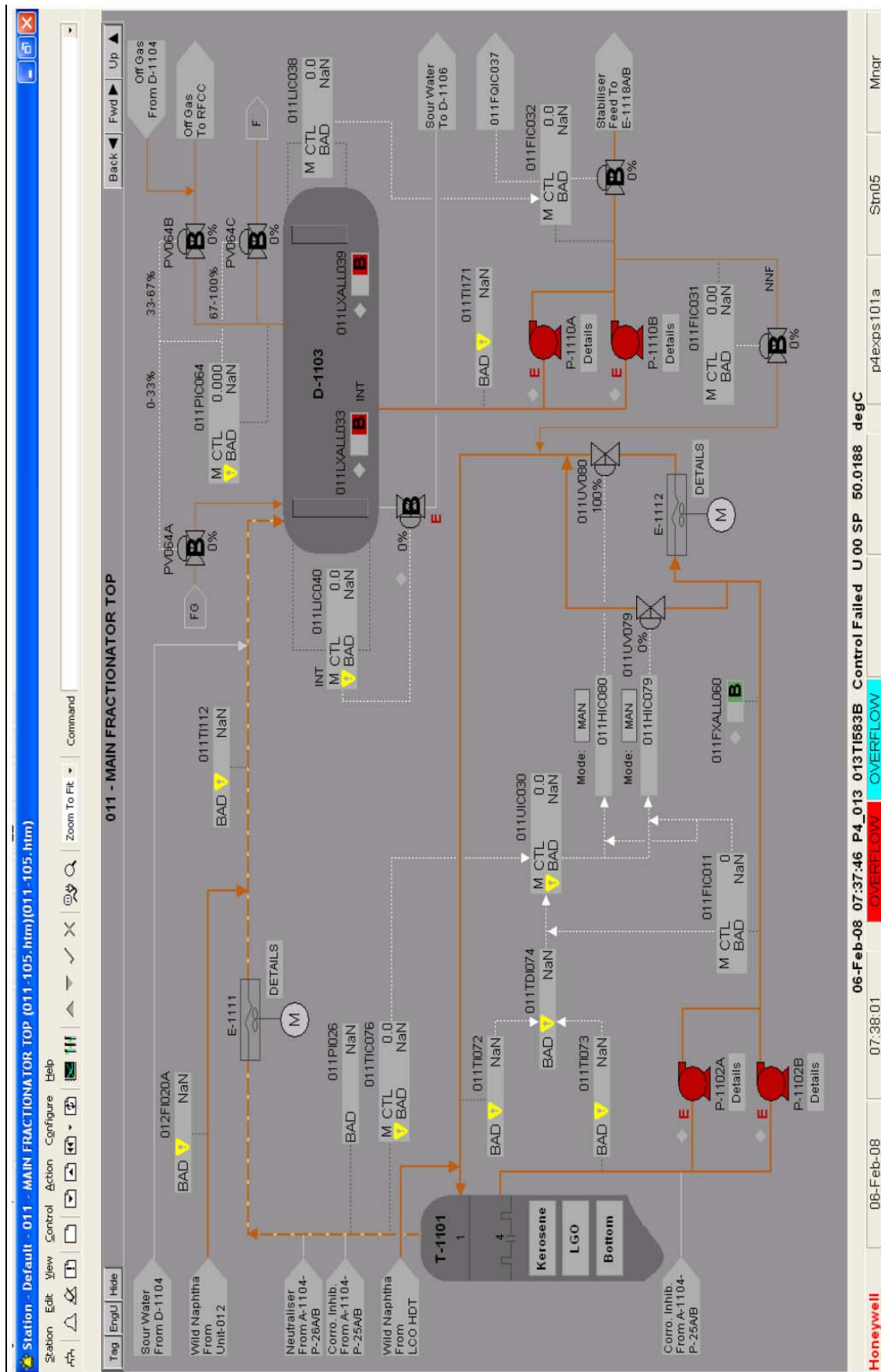
Figure 20: Functional Description of Overheads Temperature Control

Any increase in duty above the setpoint at 011-UIC-030 will produce a decrease in the duty controller output B, which will close valve 011-UV-080 via calculation block 011-FY-097 and hand controller 011-HIC-080 and open valve 011-UV-079 via calculation block 011-FY-095 and controller 011-HIC-079. The result will be to pass less liquid through the exchanger E-1112 and more through bypass valve 011-UV-079, i.e. duty is reduced.

Any increase in flow above the setpoint at controller 011-FIC-001 will produce a decrease in flow controller output A, which will close both valve 011-UV-079 and 011-UV-080 by the same amount via their calculation blocks, 011-FY-095 / 011-FY-097 and hand controllers, 011-HIC-079 / 011-HIC-080, i.e. total flow is reduced.

The flow controller 011-FIC-011, have to have priority above duty controller, 011-UIC-030, to prevent both valve close in temperature or duty failure case. In other hand, the flow controller have to be able to keep the flow in control otherwise duty controller failure for itself or for temperature indicator failure. 011-UIC-030 output shall be limited to 10% - 90%.

The following DCS printout shows the top section of T-1101:



3.1.6.2. Kerosene Pumparound Duty Control (UIC-029)

The objective of Kerosene pumparound is to remove heat from T-1101. In normal operation the flow through the pumparounds circuits remains constant and the heat duty is controlled by passing more or less flow around the heat exchangers.

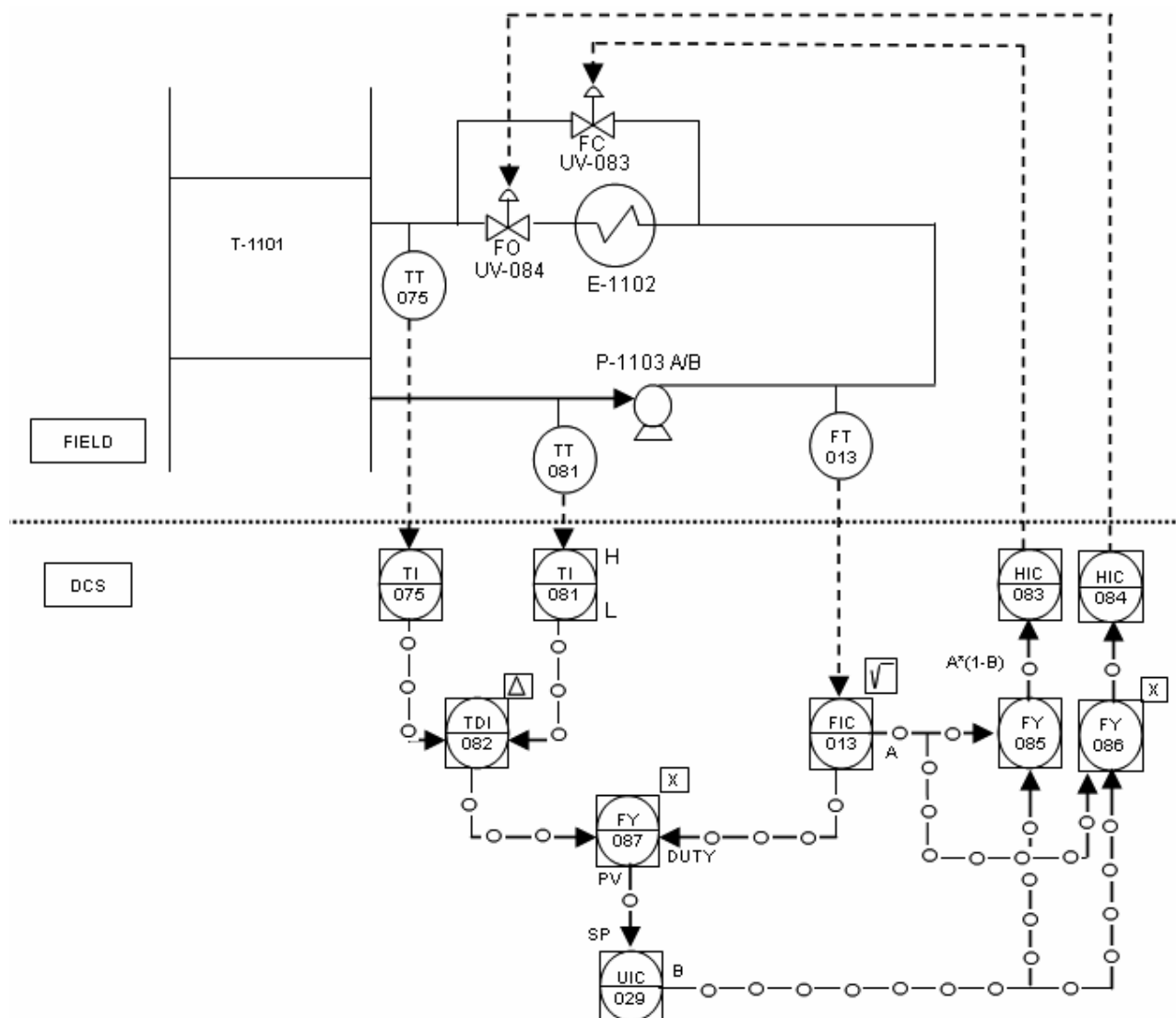


Figure 21: Functional Description of Kerosene Pumparound Duty Control

In order to ensure that separation efficiency in the main fractionator T-1101 is good enough, the amount of heat removed in the exchanger is adjusted by the duty controller 011-UIC-029, using 011-UV-083/084 control valves which adjust the flow through the heat exchanger and its by-pass.

The duty is calculated as the product of the flow rate of the Kerosene pumparound (011-FIC-013), the temperature difference (011-TDI-082) and the average specific heat of the stream.

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The duty controller outputs to calculation blocks 011-FY-085 and 011-FY-086 in order to maintain control by operating the two control valves (011-UV-083 / 011-UV-084) simultaneously in opposite directions.

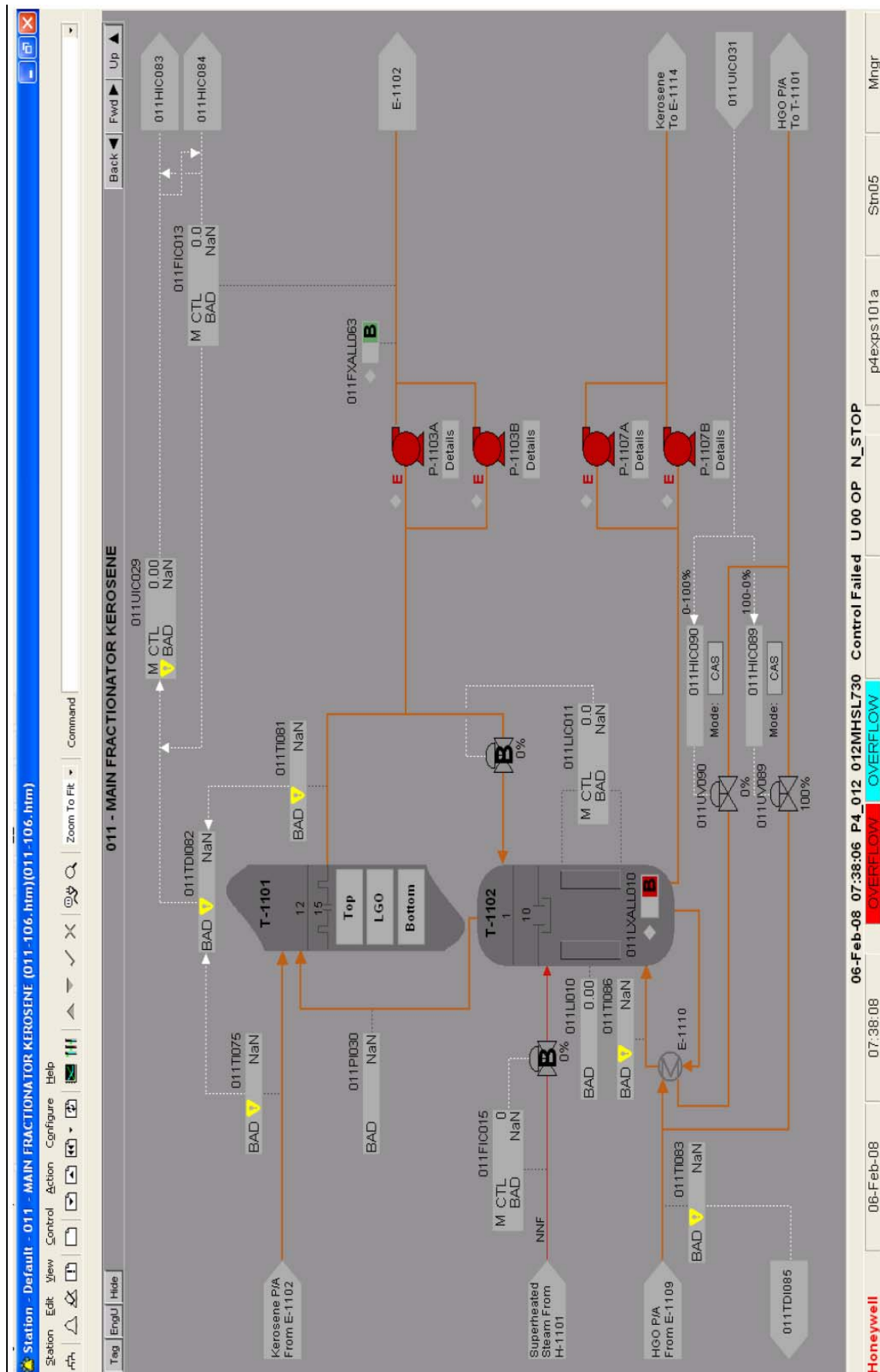
Any increase in duty above the setpoint at 011-UIC-029 will produce a decrease in output B, which will close valve through heat exchanger and open bypass valve. The result will be to pass less liquid through the Heat Exchanger and more through the bypass valve 011-UV-083, i.e. duty is reduced.

The flow controller 011-FIC-013 outputs to calculation blocks 011-FY-085 and 011-FY-086 in order to maintain control by operating the two control valves (011-UV-083 / 011-UV-084) simultaneously in the same direction.

Any increase in flow above the setpoint at 011-FIC-013 will produce a decrease in output A, which will close both valve (through heat exchanger and bypass), i.e. total flow is reduced.

The flow controller 011-FIC-013, have to have priority above duty controller, 011-UIC-029, to prevent both valve (011-UV-083 / 011-UV-084) close in temperature or duty failure case. In other hand, the flow controller have to be able to keep the flow in control otherwise duty controller failure for itself or for temperature indicator failure. Limit 011-UIC-029 output 10%-90%.

The following DCS printout shows the Kerosene Section:



3.1.6.3. LGO Pumparound duty control (UIC-032)

The objective of this system is to control the amount of heat removed from the LGO pumparound stream and to control the flow of the pumparound. In normal operation the flow through the pumparounds circuits remains constant and the heat duty is controlled by passing more or less flow around the head exchangers.

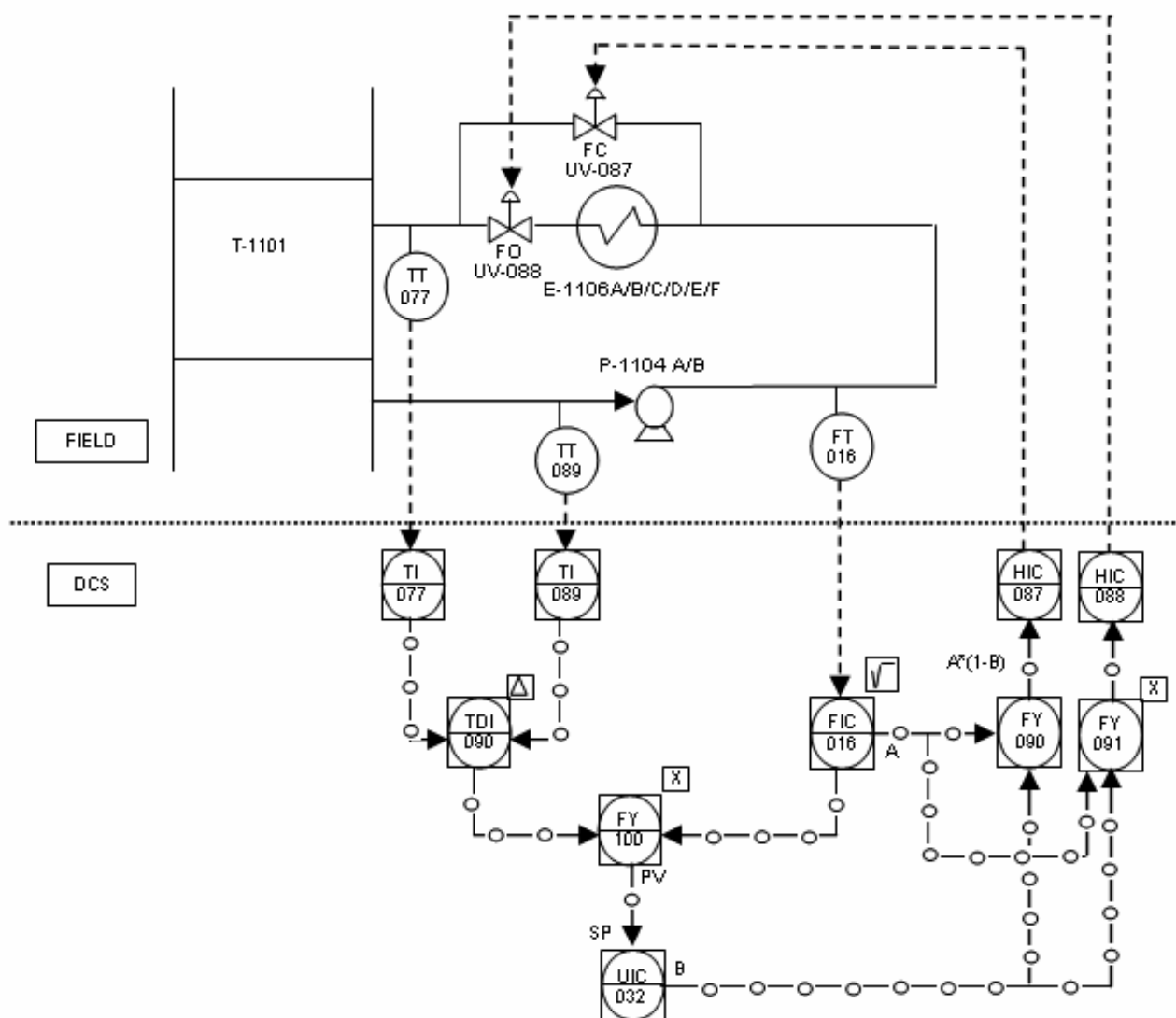


Figure 22: Functional Description of LGO Pumparound Duty Control

In order to ensure that the separation efficiency in the main fractionator is good enough, the amount of heat removed in the exchanger is adjusted by the duty control 011-UIC-032 using 011-UV-087/088 control valves which adjust the flow through the heat exchanger and its by-pass.

Duty controller 011-UIC-032 is used to control the duty removed from the LGO pumparound stream. The duty is calculated as the product of the

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flow rate of the LGO pumparound (011-FIC-016), the temperature difference (011-TDI-090) and the average specific heat of the stream.

The duty controller outputs to calculation blocks 011-FY-090 and 011-FY-091 in order to maintain control by operating the two control valves (011-UV-087 / 011-UV-088) simultaneously in opposite directions.

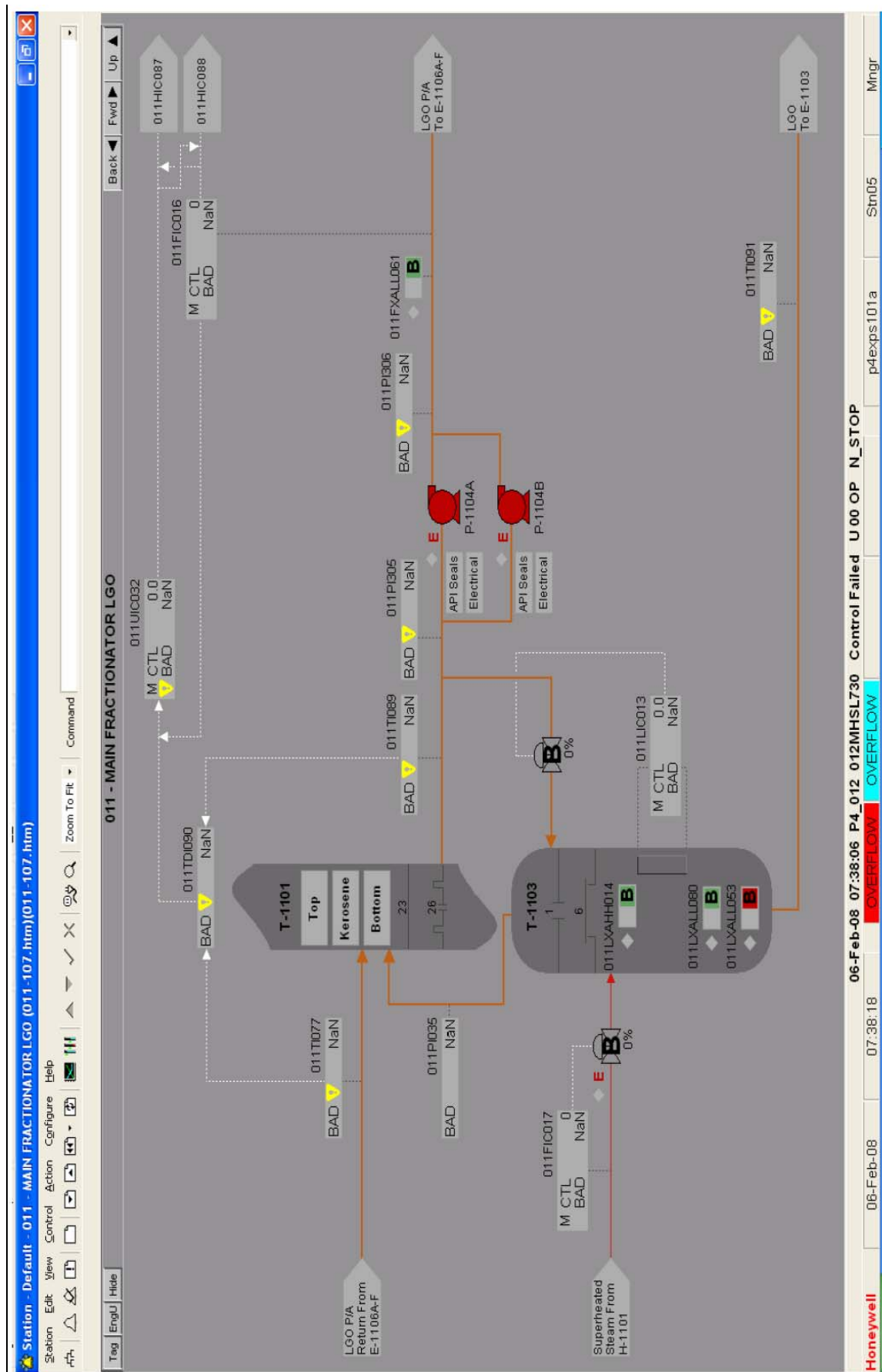
Any increase in duty above the setpoint at 011-UIC-032 will decrease output B, which will close valve through heat exchanger and open bypass valve. The result will be to pass less liquid through the Heat Exchanger and more through the bypass valve 011-UV-087, i.e. duty is reduced.

The flow controller 011-FIC-016 outputs to calculation blocks 011-FY-090 and 011-FY-091 in order to maintain control by operating the 2 control valves (011-UV-087 / 011-UV-088) simultaneously in the same direction.

Any increase in flow above the setpoint at 011-FIC-016 will decrease output A, which will close both valve (through heat exchanger and bypass), i.e. total flow is reduced.

The flow controller 011-FIC-016, have to have priority above duty controller, 011-UIC-032, to prevent both valve (011-UV-087 / 011-UV-088) close in temperature or duty failure case. In other hand, the flow controller have to be able to keep the flow in control otherwise duty controller failure for itself or for temperature indicator failure. 011-UIC-032 output shall be limited to 10%-90%.

The following DCS printout shows the LGO Section in T-1101:



3.1.6.4. HGO Pumparound Duty Control (UIC-031 & UIC-033)

This system control the amount of heat removed from the HGO pumparound in Crude/HGO Exchanger (E-1109) and the Kerosene Stripper Reboiler (E-1110) and control the flow of the pumparound. It is desired that the heat removed is preferentially used to satisfy the Kerosene Stripper Reboiler (E-1110) duty with the excess being used to heat the Crude in the Crude/HGO Pumparound Exchanger (E-1109).

In normal operation the flow through the pumparounds circuits remains constant and the heat duty is controller by passing more or less flow around the head exchangers.

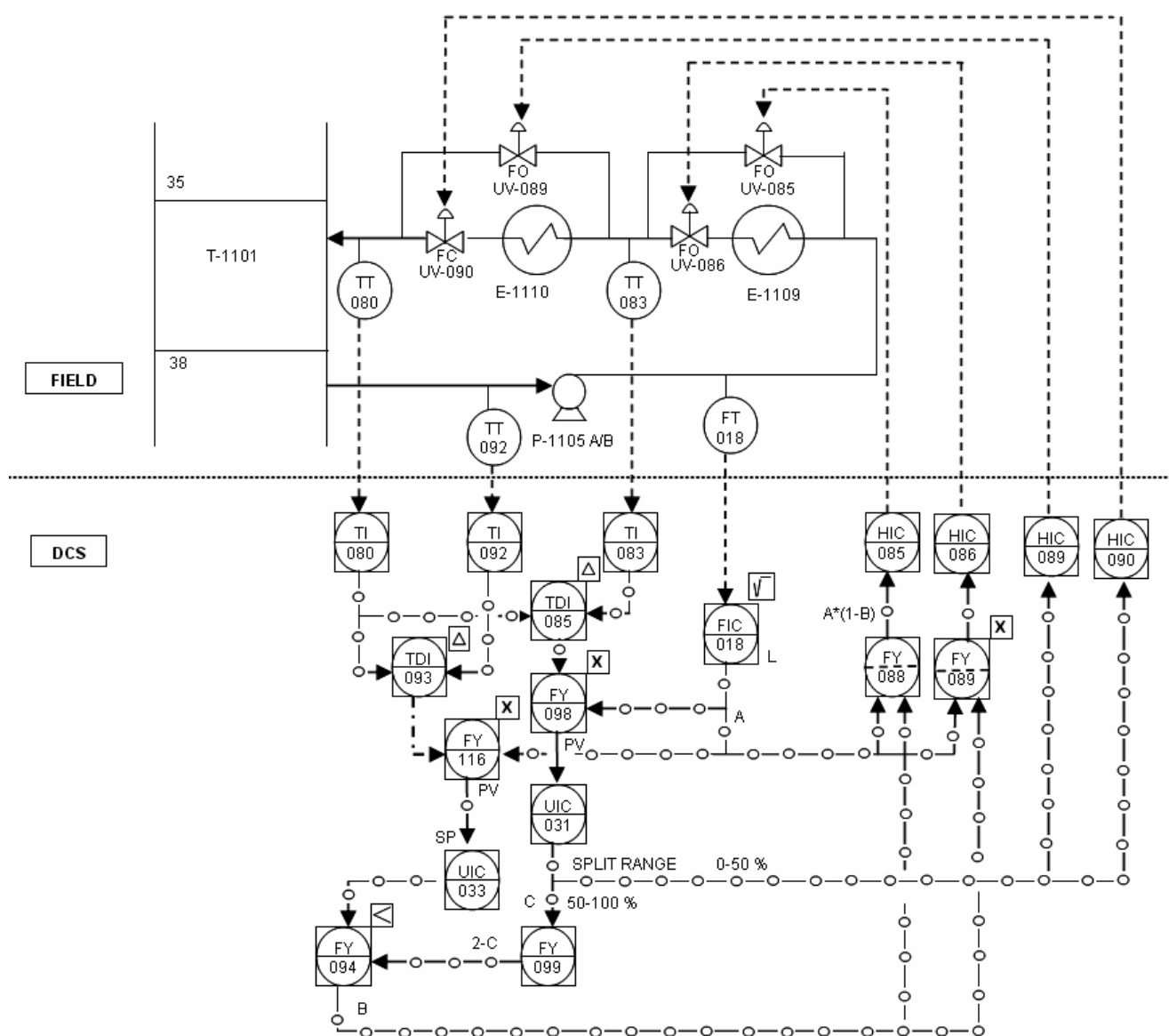


Figure 23: Functional Description of HGO Pumparound Duty Control

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The overall duty removed from the HGO stream is controlled by 011-UIC-033 which effectively controls the duty of E-1109. The kerosene reboiler, E-1110, duty is controlled by 011-UIC-031.

The principle of operation of overall duty control and flow control is the same as for Kerosene and LGO pump rounds.

An HGO stream is drawn from tray 38 and pumped by P-1105 A/B to E-1109, which has a bypass stream. From the recombined streams after E-1109, a slipstream of the HGO stream is diverted to re-boil the bottoms of the Kerosene Stripper (T-1102), via E-1110, before returning to tray 35 of T-1101.

The required heat duty of the HGO in E-1110 is set at duty controller 011-UIC-031 and calculated by 011-FY-098.

UIC-031 is a split range controller:

0-50 %: Increase the heat removed in E-1110 by reducing the flow through bypass valve 011-UV-089 and increasing the flow through E-1110 (valve 011-UV-090). This method is used to ensure that at any one time the total valve opening is 100%. (i.e. at 25% controller output both valves are open 50%)

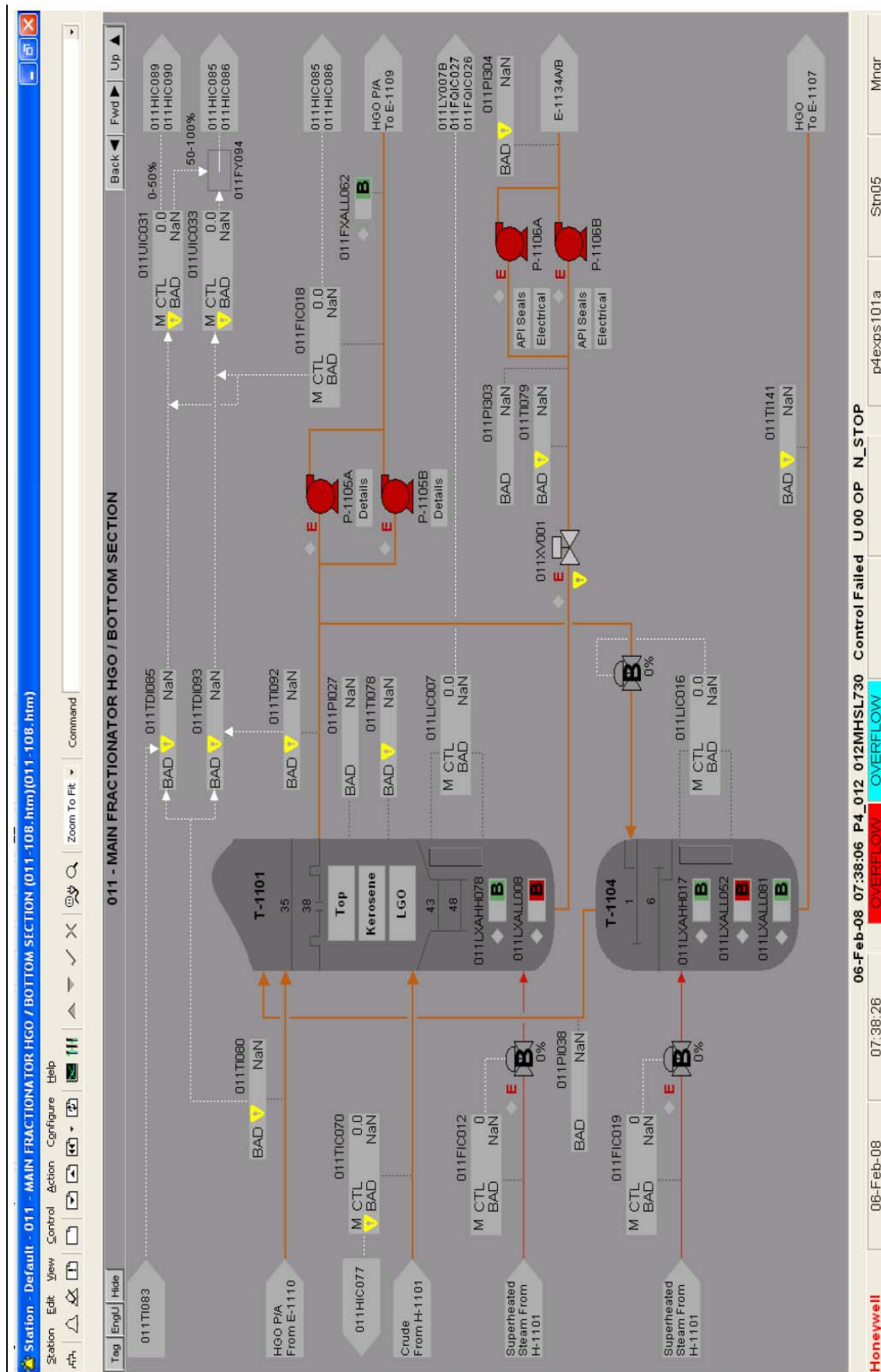
50-100 %: Reduced the actual duty of E-1109 to increase the inlet temperature to E-1110 and increase its duty through the function block, 011-FY-099.

Duty controller 011-UIC-033 is used to control the total duty removed from the HGO pumparound stream by altering the duty of E-1109. The duty is calculated as the product of the flowrate of the HGO pumparound (011-FIC-018), the temperature difference (011-TDI-093) and the average specific heat of the stream. The duty controller outputs to 011-FY-094 which itself outputs the minimum of that input and the other input coming from 011-FY-099. The minimum value is sent to calculation blocks (011-FY-88, 011-FY-89) in order to maintain control by operating the two control valves simultaneously in the same direction.

The flow controller 011-FIC-018 maintains control by operating the two control valves simultaneously in the same direction.

The flow controller, 011-FIC-018, has to have priority above duty controller, 011-UIC-031/033, to prevent both valve closes in temperature or duty failure case. In other hand, the flow controller have to be able to keep the flow in control otherwise duty controller failure for itself or for temperature indicator failure. 011-UIC-031/033 output shall be limited to 10-90%.

The following DCS printout shows the HGO/Bottom Section of T-1101:



3.1.6.5. Main Fractionator Level Control (LIC-007)

Refer to Inter-Units Controls section: Atmospheric Residue from the CDU to the RFCC.

3.1.6.6. Atmospheric Residue Cooling Control (FIC-030, TIC-111 & TIC-168)

The objective of this system is to control the temperature of Atmospheric Residue being sent to storage. It is desired to achieve this by regulating the temperature water supply temperature whilst keeping its flow constant.

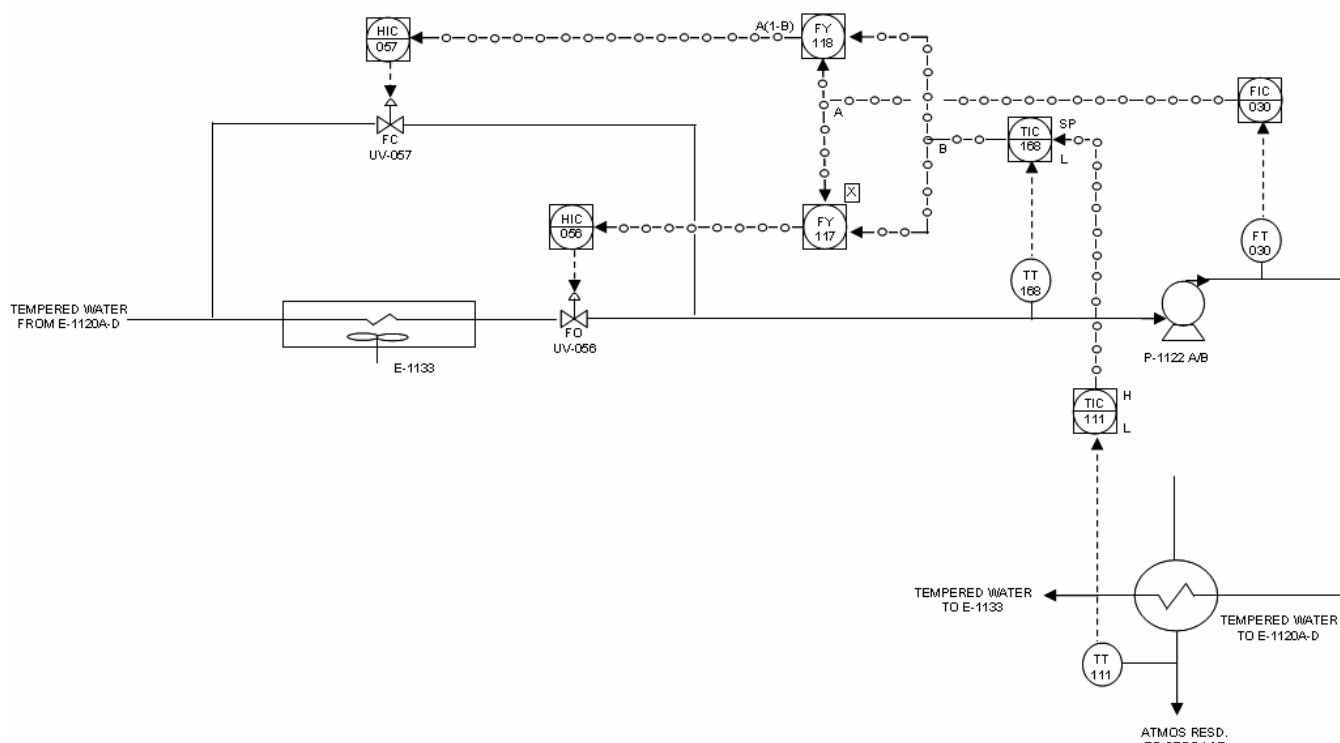


Figure 24: Functional Description of Atmospheric Residue Cooling Control

Any increase in temperature above the set point at 011-TIC-111 will produce a decrease in the set point to 011-TIC-168.

011-TIC-168 shall be provided with a set point low limit to prevent water being delivered too cold to E-1120.

The flow controller 011-FIC-030 acts on both valves, 011-UV-056 (E-1133 outlet) and 011-UV-057 (E-1133 bypass), by the same amount via their calculation blocks and hand controllers 011-HIC-056, 011-HIC-057.

Any increase in temperature above the set point at 011-TIC-168 will produce an increase in FIC-030 output, which will open E-1133 outlet control valve 011-UV-056 and close the bypass valve 011-UV-057. The result will be to pass more liquid through E-1133 and less through the bypass valve, i.e. temperature is reduced.

3.1.6.7. Kerosene Rundown Control (FQIC-033 & TIC-122)

The objective of this system is to control the flowrate and temperature of the kerosene rundown stream.

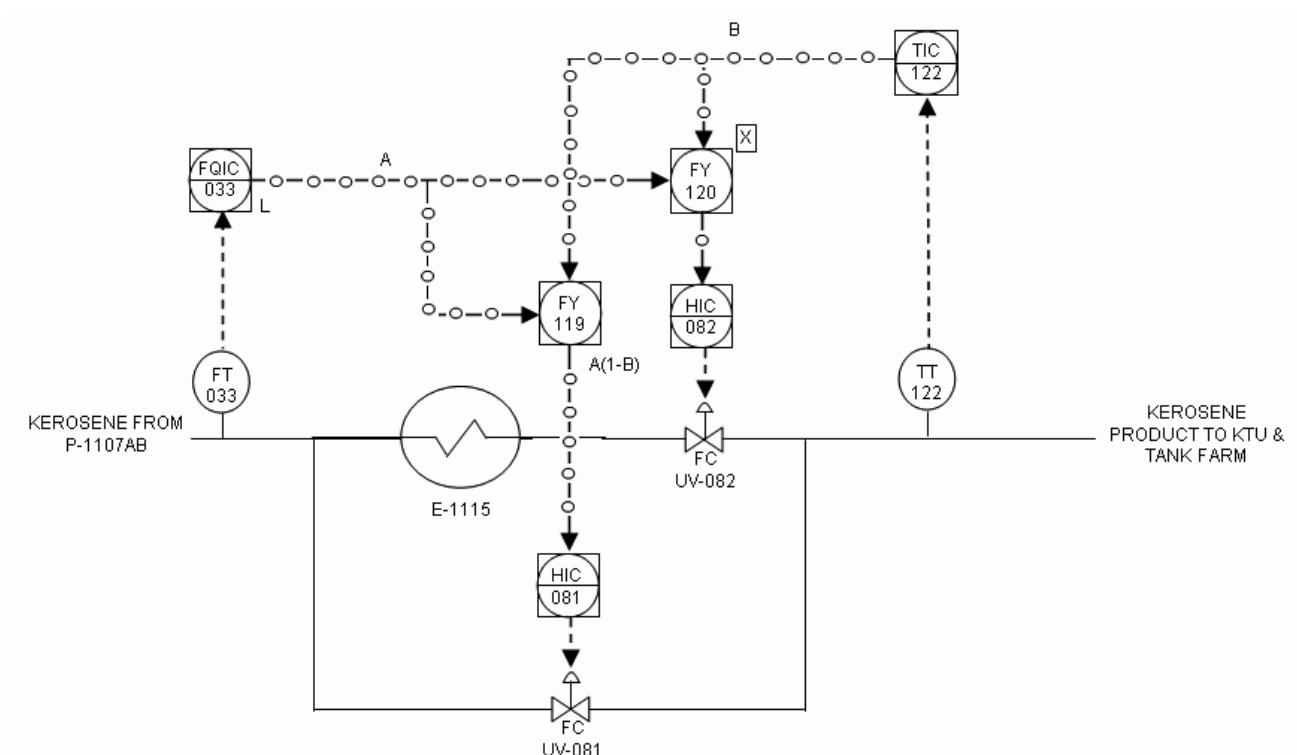
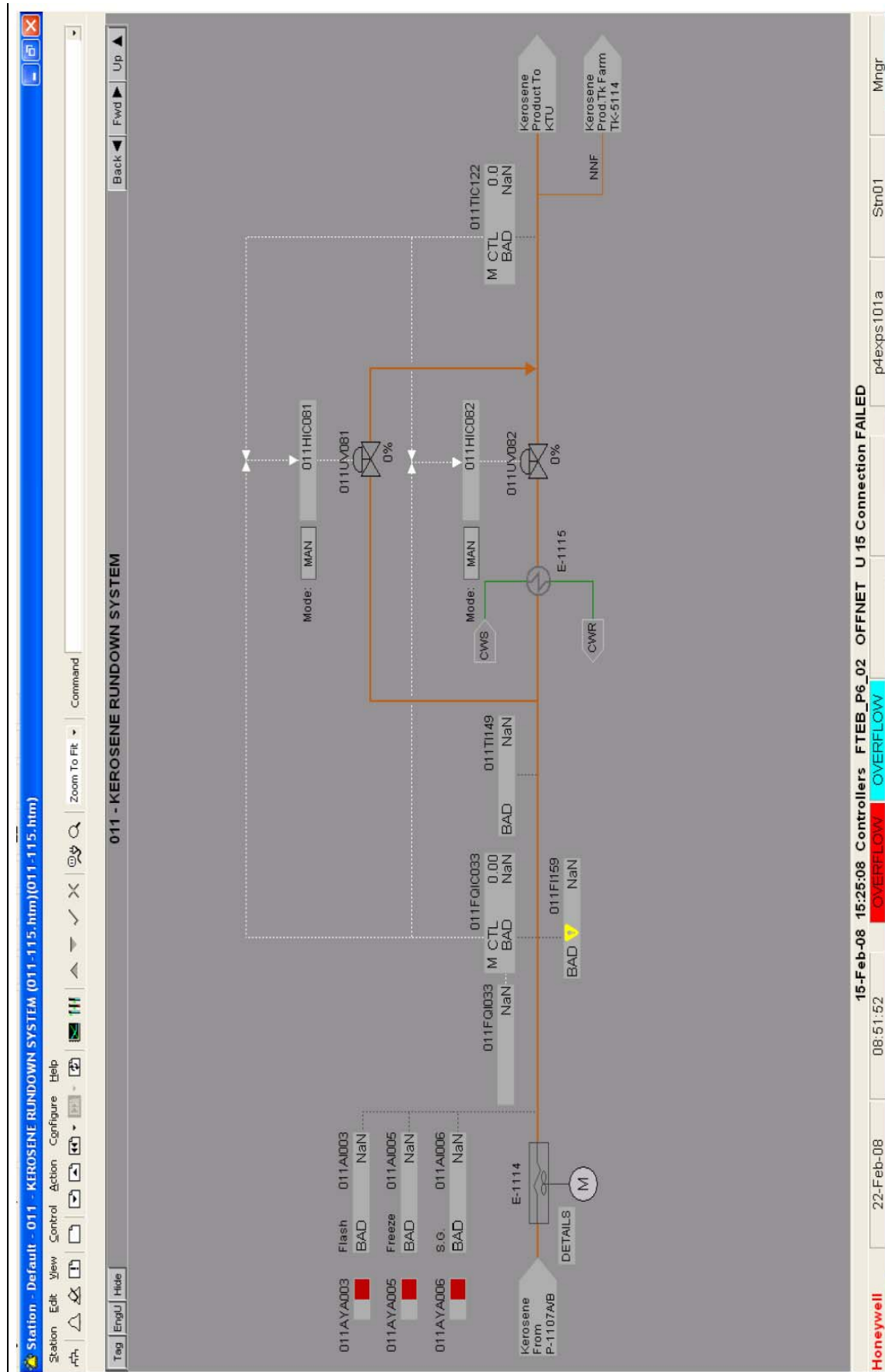


Figure 25: Functional Description of Kerosene Rundown Control

The flow controller (011-FQIC-033) acts both valves, 011-UV-081 (E-1115 bypass) and 011-UV-082 (E-1115 outlet) by the same amount via their calculation blocks (011-FY-119, 011-FY-120) and hand controllers (011-HIC-081, 011-HIC-082).

Any increase in temperature above the set point at 011-TIC-122 will produce an increase in the controller output, which will close valve 011-UV-081 and open valve 011-UV-082. The result will be to pass more liquid through E-1115 and less through the bypass valve 011-UV-081, i.e. temperature is reduced.

The following DCS printout shows the Kerosene Rundown System:



3.1.6.8. HGO/LGO flow controller / LCO HDT Charge control

Refer to Inter Units Control section: LCO HDT Feed Control.

3.1.6.9. T-1101 Accumulator Drum Pressure Control

The objective of this controller is to maintain a constant pressure in the accumulator drum D-1103 by controlling the flow gas in and out of the drum, by a three way split range pressure controller 011-PIC-064:

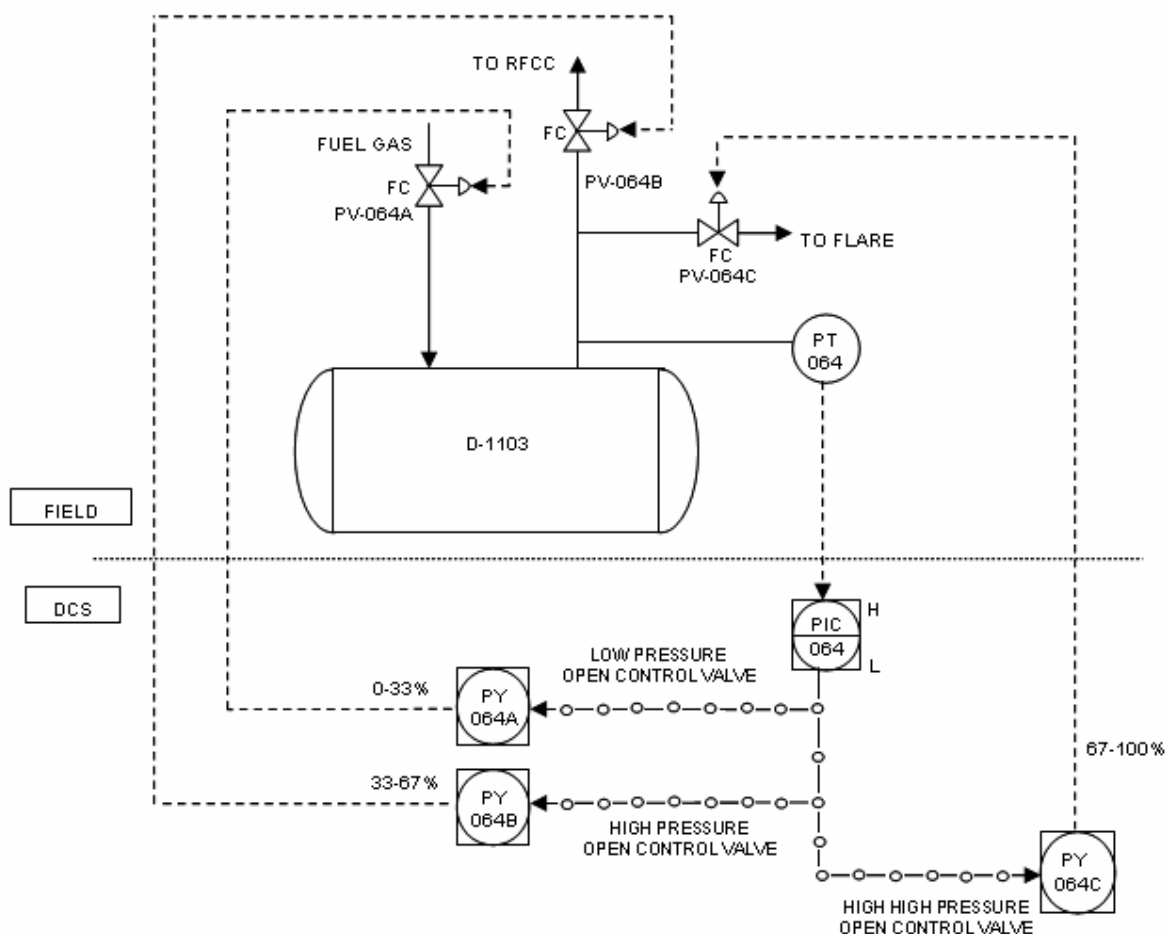


Figure 26: Functional Description T-1101 Accumulator Drum Pressure control

- In the low range fuel gas is brought into the accumulator drum to increase the pressure (PV-064A)
- In the mid range off gas from the drum is routed to the RFCC (PV-064B)
- In the high range the off gas is routed to flare (PV-064C)

3.1.7. Naphta Stream

3.1.7.1. Fractionator T-1101 Overhead Accumulator Drum Level Control (LIC-038)

The objective of this system is to provide a steady stabilizer feed, whilst minimizing any upstream upsets.

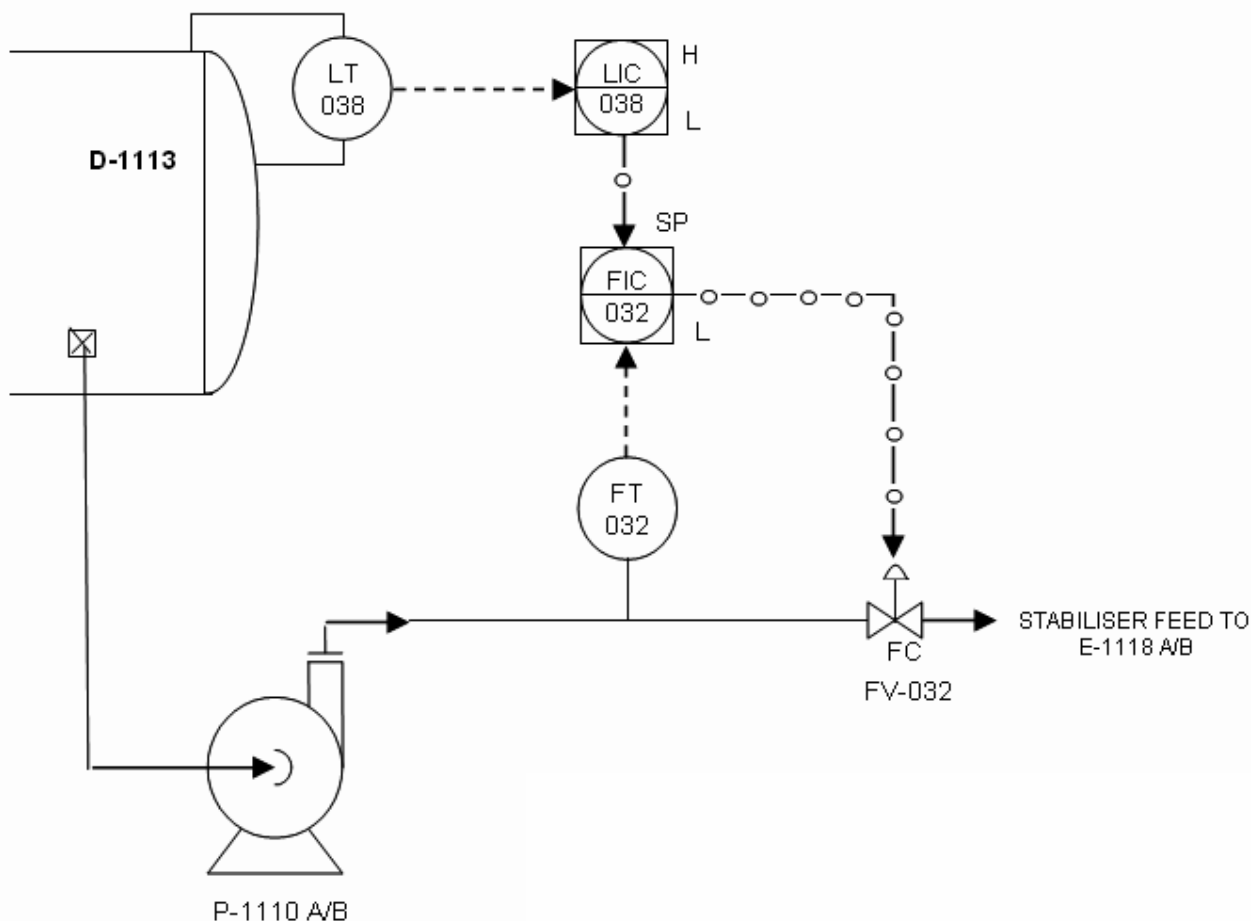


Figure 27: Functional Description of Fractionator T-1101 Overhead Accumulator Drum Level Control

This control is a cascade loop in which the 011-LIC-038 output signal is used as 011-FIC-032 set point, which acts on 011-FV-032 control valve: P-1110 A/B discharge to E-1118 A/B.

This controller will typically react slowly to any changes in level within safe limits, to minimize the rate of change of flow rate to the stabilizer. When the level is outside of the safe limits the controller will act quickly to restore the level back to within the safe limits. The use of such a controller will minimize the effect of upsets being passed on to the stabilizer tower, by taking advantage of the available surge volume in the accumulator drum. 011-LIC-038 will reset 011-FIC-032 to provide smoothed flow to the stabilizer tower T-1107.

3.1.7.2. Naphta Stabilizer Level Control (011-LIC-042)

Refer to Inter-Unit Controls section: Full Range Naphta from the CDU to the NHT.

3.1.7.3. T-1107 Stabilizer feed temperature control

The purpose of this controller is to keep constant the stabilizer feed temperature. The controller 011-TIC-121 output is routed to bypass control valve 011-TV-121. If the feed to stabilizer temperature decreases, the controller will close the control valve, in order to increase the flow through the heat exchanger E-1118 A/B.

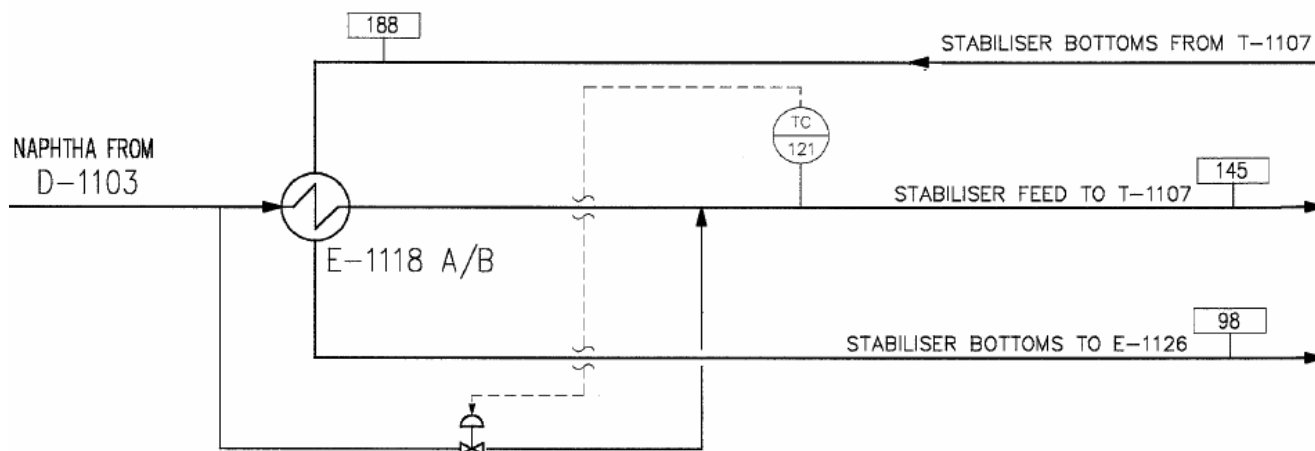


Figure 28: Functional Description of Naphta Stabilizer Feed Temperature Control

3.1.7.4. T-1107 Stabilizer bottom temperature control

The objective of this control is to maintain constant the bottom temperature of the stabilizer. This is a cascade control loop in which the temperature controller 011-TIC-113 output is used as 011-FIC-034 set point, the later acting on 011-FV-036 control valve: Reflux to T-1107

If the bottom temperature decreases the controller will increases the 011-FIC-034 set point, in order to increase the steam flow to the stabilizer reboiler E-1121.

3.1.7.5. Stabilizer Reflux Drum Level Control

The objective is to control D-1104 level by controlling the reflux flow to the Stabilizer whilst minimizing fluctuations in the flow rate.

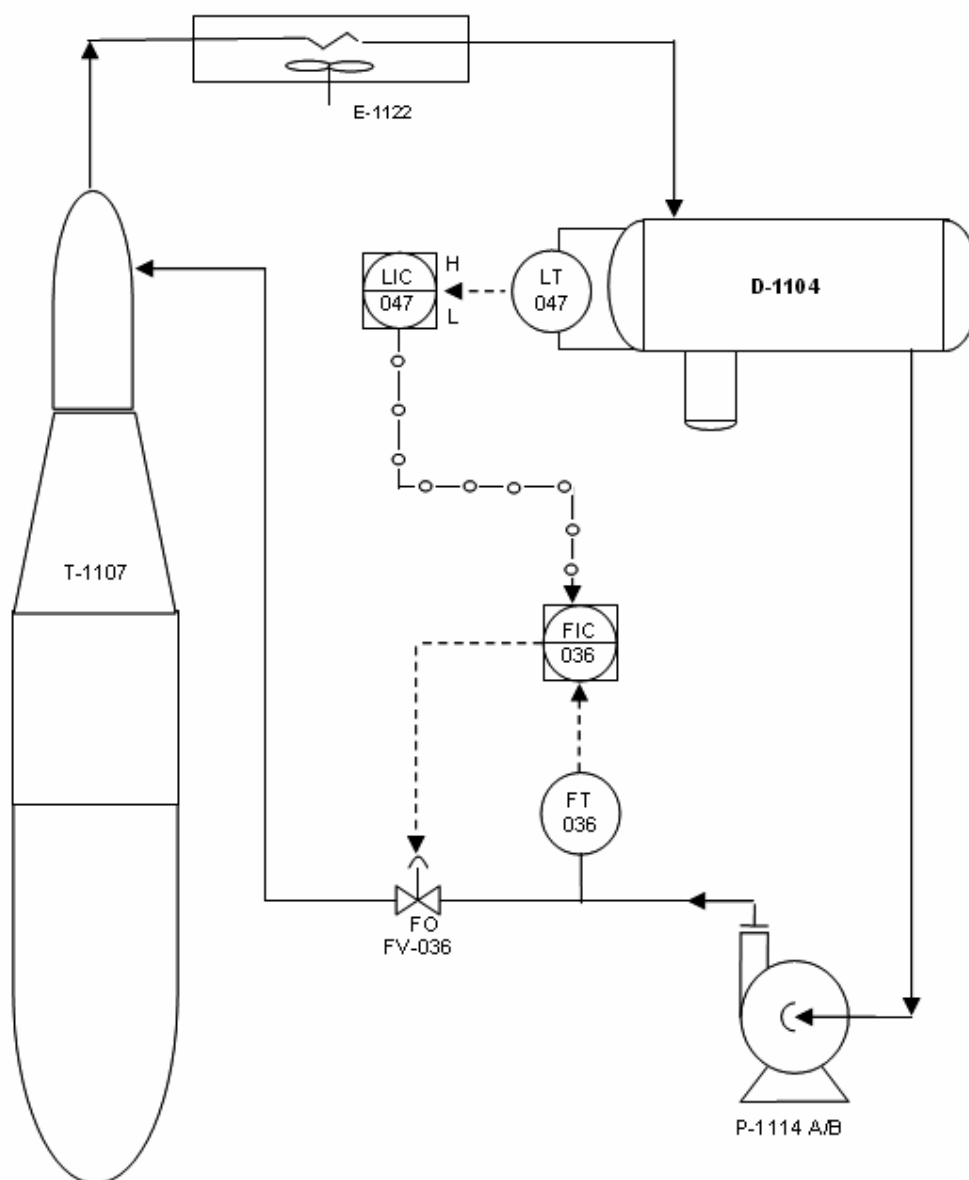


Figure 29: Functional Description of Naphta Stabilizer Reflux Drum Level Control

Stabiliser Reflux Drum level controller (011-LIC-047) shall utilize averaging level control. The algorithm shall be set to take maximum advantage of the drum's holding volume to reduce the effects of any upsets that would otherwise be passed onto the Stabiliser. A typical algorithm will achieve a smooth outflow from the drum when the level deviates within a safe range. The controller will act rapidly to restore an acceptable level when the liquid level exceeds the safe limits. The level is slowly returned to set point to maximize the surge capacity of the drum.

3.1.7.6. LPG Flow Control (011-FQIC-037)

The objective of this system is to control the flow of the LPG stream from the top of the Stabilizer, T-1107, in order to control its separation process.

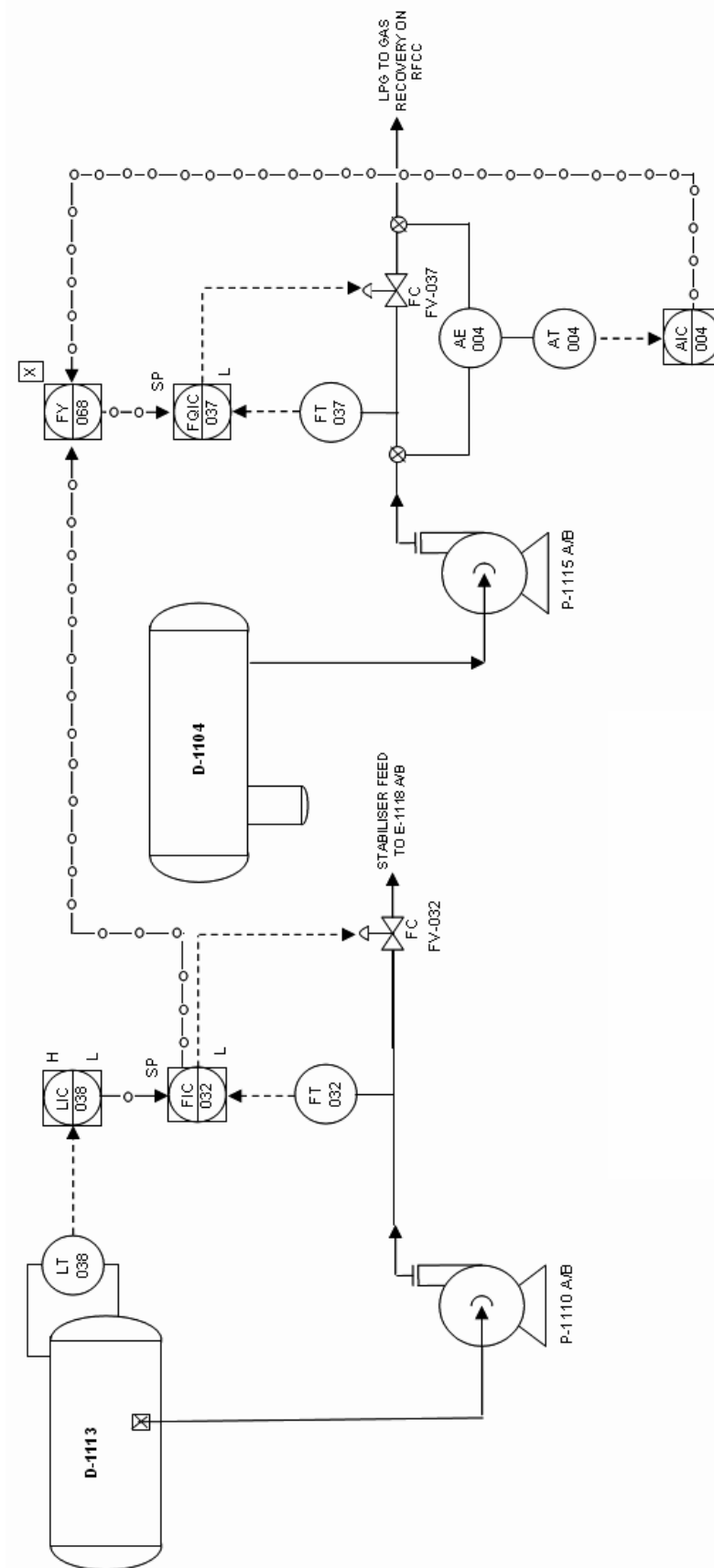


Figure 30: Functional Description of LPG Flow Control

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The flow rate of LPG drawn off from Stabilizer Reflux Drum, D-1104, (011-FQIC-037), is controlled with a calculation set point as a ratio of the flow to the Stabilizer (011-FIC-032) , with a correction for the Pentane content in the LPG stream as measured by 011-AIC-004.

3.1.7.7. Naphta Stabilizer Accumulator Drum Pressure Control

The objective is to control the pressure on the accumulator drum D-1104 by a two way split range pressure controller 011-PIC-068.

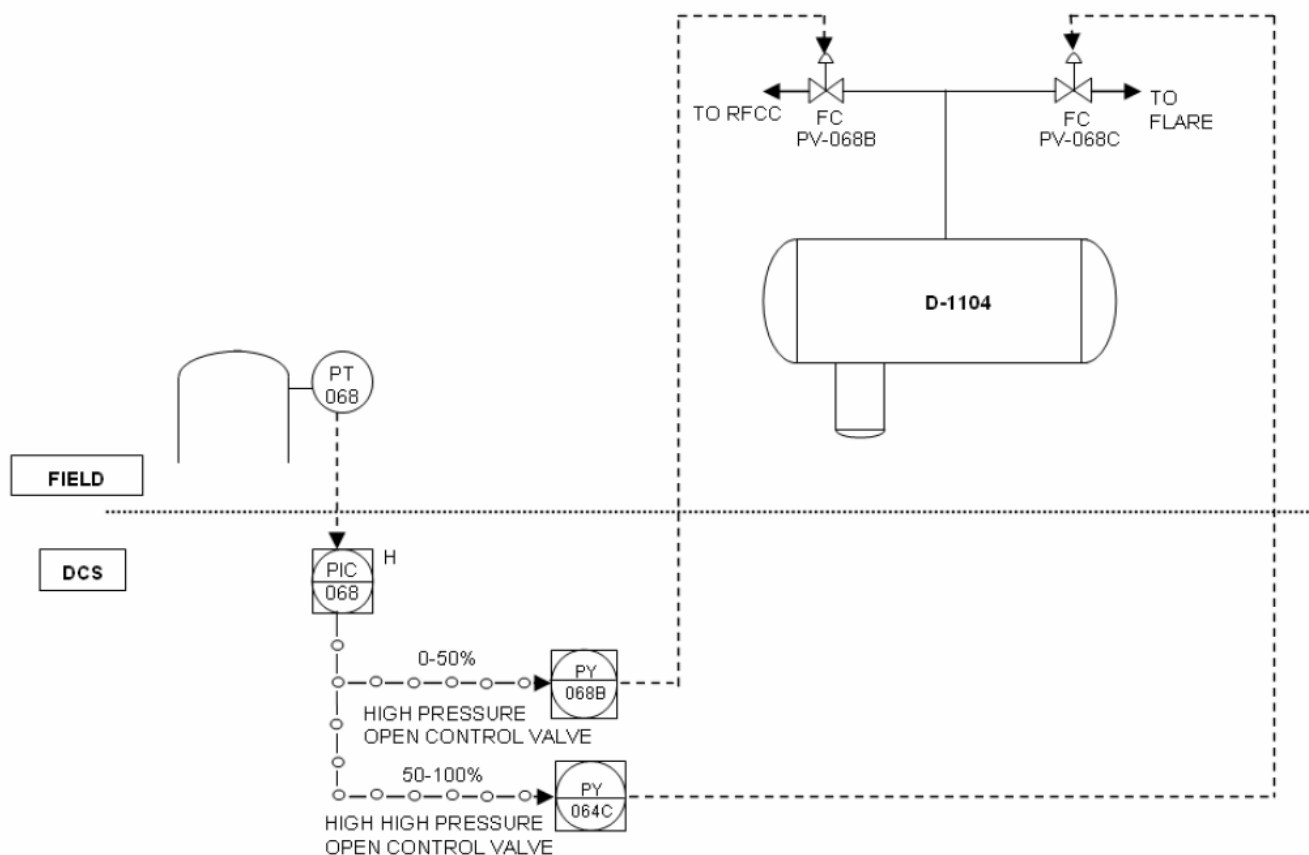
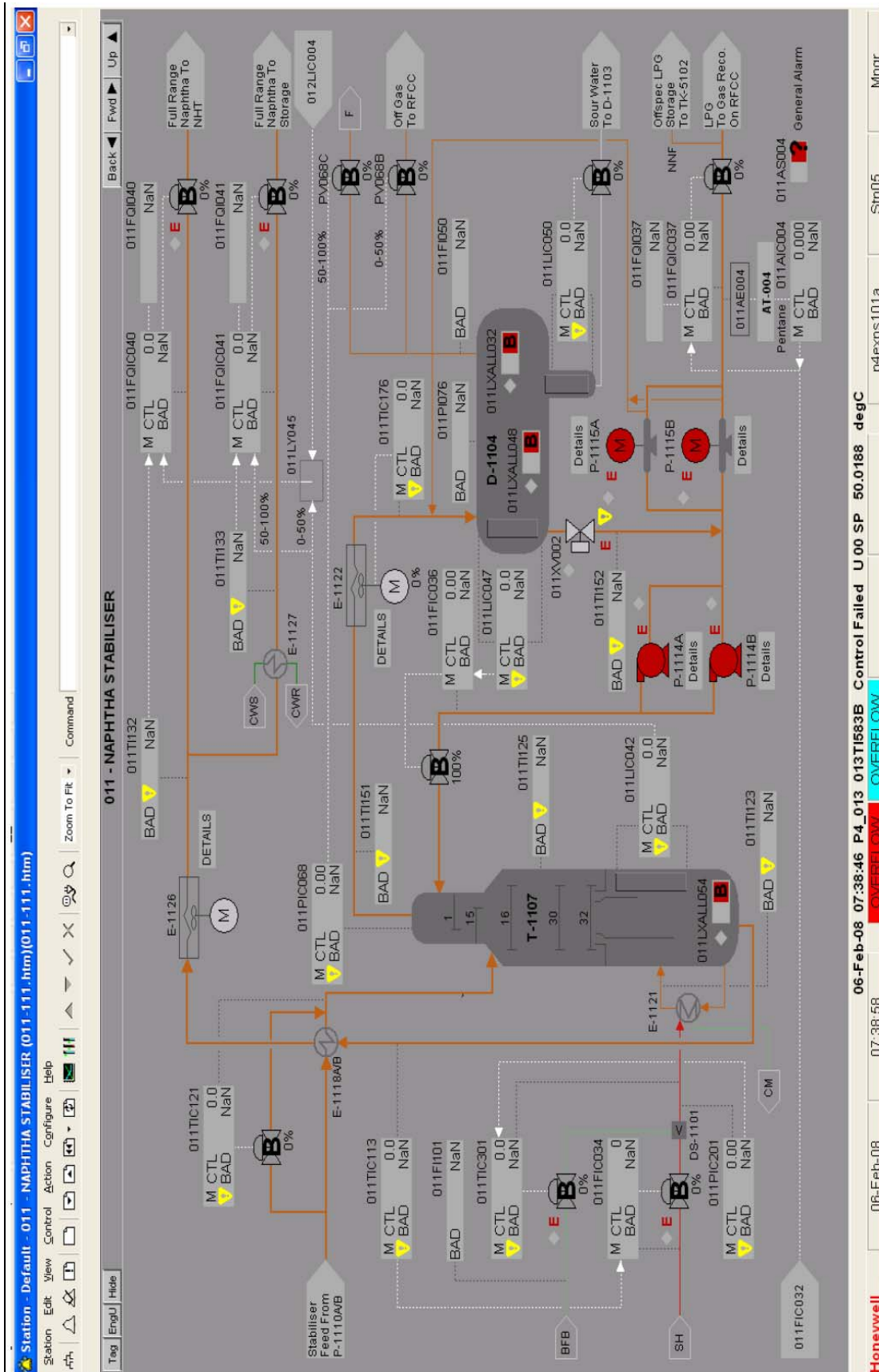


Figure 31: Functional Description of T-1107 Accumulator Drum Pressure Control

In the low range, signal from PIC-068 is selected by low signal selector PY-068B to open PV-068B: the off gas is routed RFCC.

In the high range, signal from PIC-068 is selected by high signal selector PY-068A to open PV-068A: the off gas goes to flare.

3.1.8. DCS printout of the Naphta Stabilizer T-1107



3.1.9. Tempered Water Accumulator Drum Pressure Control

The objective is to control the pressure on the accumulator of residue cooling water drum D-1115 by a two way split range pressure controller 011-PIC-109.

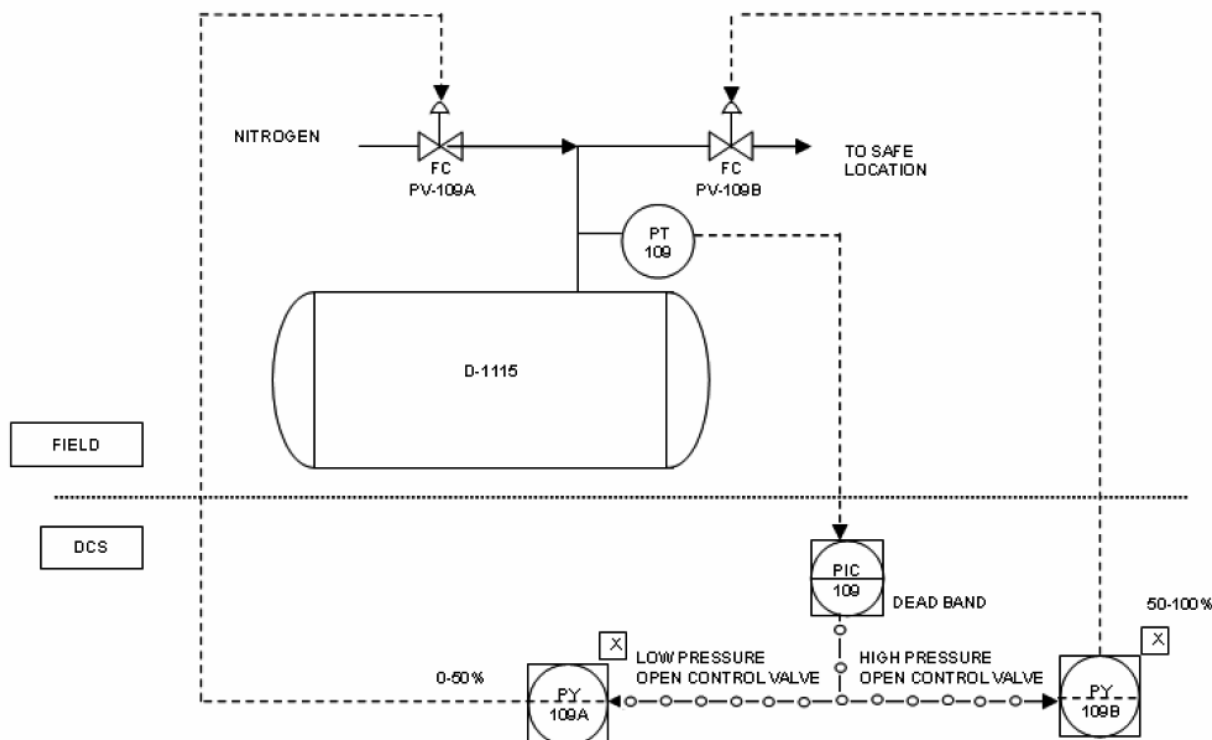


Figure 32: Functional Description of Tempered Water Accumulator Drum Pressure Control

In the low range nitrogen is brought into D-1115 by means of PV-109A.

In the high range nitrogen/vapor flow is vented to safe location by means of PV-109B.

3.1.10. Kerosene/LGO/HGO strippers (T-1102/T-1103/T-1104) and LGO/HGO dryers (T-1105/T-1106) level control

The control loop is the same for all the towers mentioned. The level controller acts over the control valve placed on the hydrocarbon stream feed, so if the level value rises above its set point the controller will close the valve. In the table below the controllers and control valves for each tower are shown.

	Level controller	Control valve
T-1102	011-LIC-011	011-LV-011
T-1103	011-LIC-013	011-LV-013
T-1104	011-LIC-016	011-LV-016
T-1105	011-LIC-019	011-LV-019
T-1106	011-LIC-022	011-LV-022

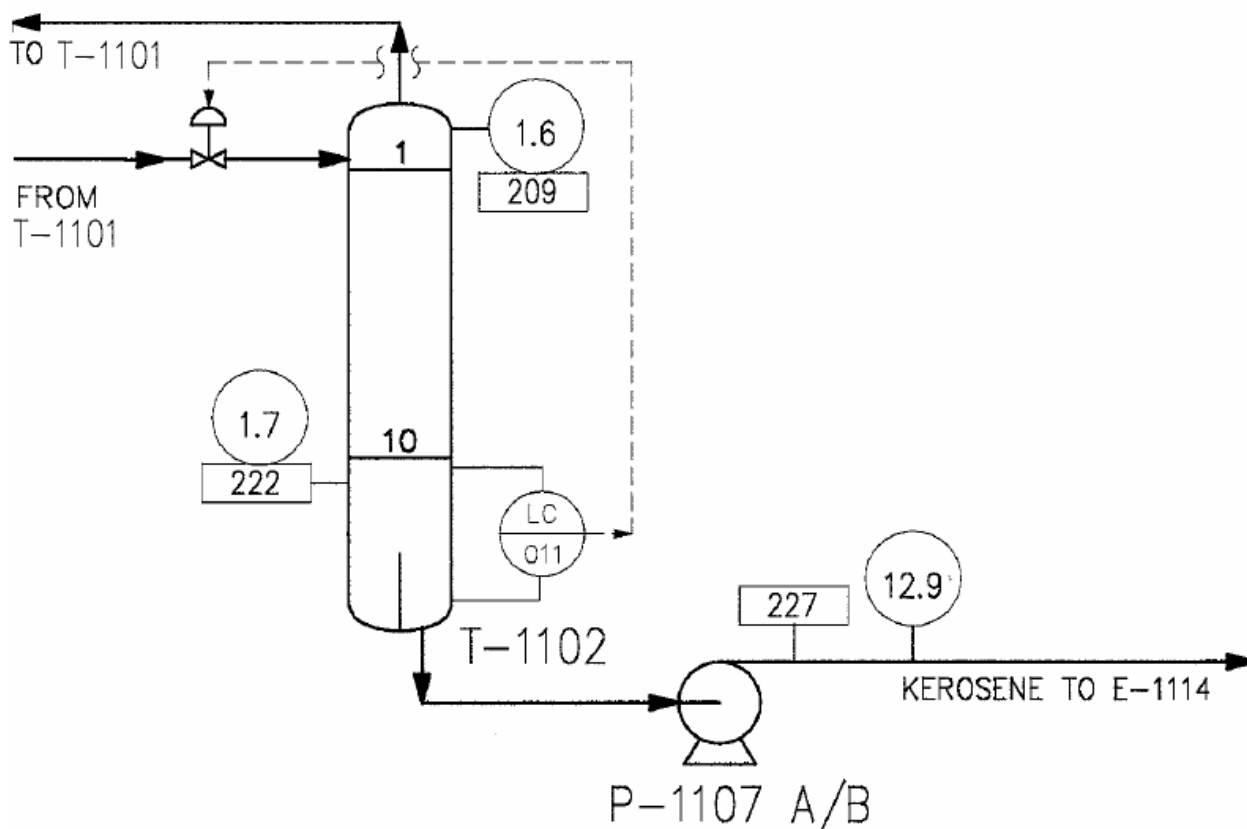
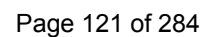


Figure 33: Functional Description of Kerosene Stripper Level Control

The following DCS printout shows the LGO & HGO Driers:



3.1.11. D-1109 level control

The objective of this controller is to maintain a constant level in the desalter water surge drum (D-1109) by controlling the water flow inlet, by a three way split range level controller 011-LIC-025.

Each way of the split range refers to a water source:

- Process water
- Stripped water from SWS
- Service water

If the level is near its high limit the controller will close the three sources by closing the valves 011-FV-025, 011-LV-025A and 011-LV-025B. When the level begins to decrease, the controller will start opening the process water (this water only can be used when the plant is operating with 100% Bach Ho case).

If the level continues decreasing, the split range will try to open the stripped water control valve (011-LV-025A) by 011-LY-025A. This control is routed to the minimum selector 011-LY-025D, which selects the minimum signal (less opening in the control valve) between the column level control from SWS unit and the D-1109 signal.

If the level still decreases the service water will be used by opening 011-LV-025B control valve.

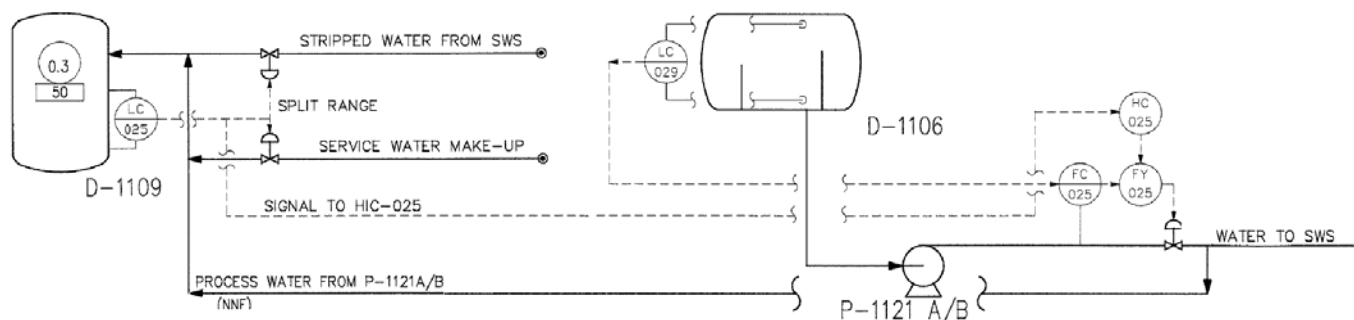


Figure 34: Functional Description of D-1109 Level Control

3.1.12. D-1106 level controls

The objective of water level control is to keep constant the water level inside the vessel. This control is a cascade loop in which the 011-LIC-029 output signal is used as 011-FIC-025 set point. The 011-FIC-025 is routed to 011-FY-025 minimum selector which acts on 011-FV-025 control valve.

The hydrocarbon level is an on/off control, which starts the P-1120 A/B pumps when 011-LIC-028 detects high level and stops when low level is reached.

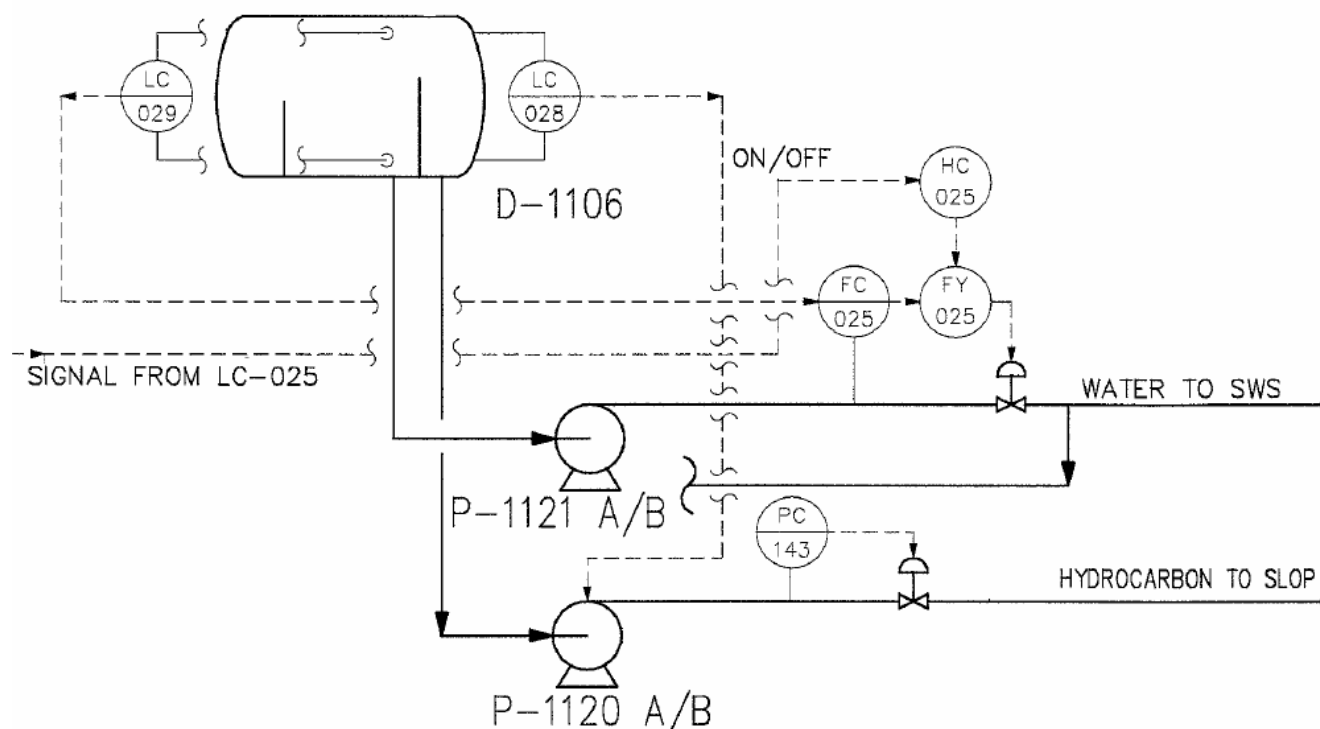


Figure 35: Functional Description of D-1106 Level Control

3.1.13. D-1115 level control

The purpose of this control is to maintain the level inside the vessel. The level controller 011-LIC-070 output is routed to 011-LV-070, which regulate the inlet of low pressure boiler feed water to enter into the vessel.

3.1.14. D-1103 water level control

This control loop is used to maintain constant the water level on D-1103. This controller 011-LIC-040 is routed to 011-LV-040 that is placed on the water outlet line, so if level increases inside the vessel the valve will open.

3.1.15. D-1104 water level control

This control is similar to the one explained in D-1103 water level control: the controller 011-LIC-050 output is routed to 011-LV-050, which is placed on the water outlet line.

3.1.16. P-1120 A/B outlet pressure controller

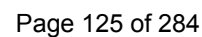
This control loop ensures that the control valve 011-PV-143 is closed before the pump P-1120 A/B starts. As the pressure increases, due to the fact that the pump P-1120 A/B has started, the controller 011-PIC-143 starts to open the control valve 011-PV-143. When the pump is stopped the controller 011-PIC-143 closes the valve 011-PV-143.

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3.1.17. Ejectors inlet pressure control

The pressure inlet at the vacuum package A-1102 is adjusted by the controller 011-PIC-049 which signal is routed to 011-PV-546 control valve. If the pressure at the package inlet is lower than its set point, the control valve will increase the recirculation flow between A-1102-E-31 hydrocarbon inlet and A-1102-J-01 A/B/C hydrocarbon inlet.

The following DCS printout shows the Driers Vacuum System:



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3.1.18. Distributed control system and control centre

For further description of the D.C.S. and the control center, refer to the following documents:

8474L-000-JSS-1511-001 Distribution Control System Technical Specification

8474L-000-JSD-1510-001 Design and Engineering Information for Control and Instrumentation

8474L-000-DW-1510-001 Control System Interconnection Diagram

3.1.19. Inter-Unit Controls & Interfaces

3.1.19.1. Atmospheric Residue from the CDU to the RFCC

Objective

The objective of this system is to control the level in the bottom of the Main Fractionator column and route as much of the outgoing atmospheric residue to the RFCC as is required.

Description

This description should be read in conjunction with the following documents:

CDU P&ID's:

8474L-011-PID-0021-112

8474L-011-PID-0021-125

RFCC P&ID's:

8474L-015-PID-0021-0301

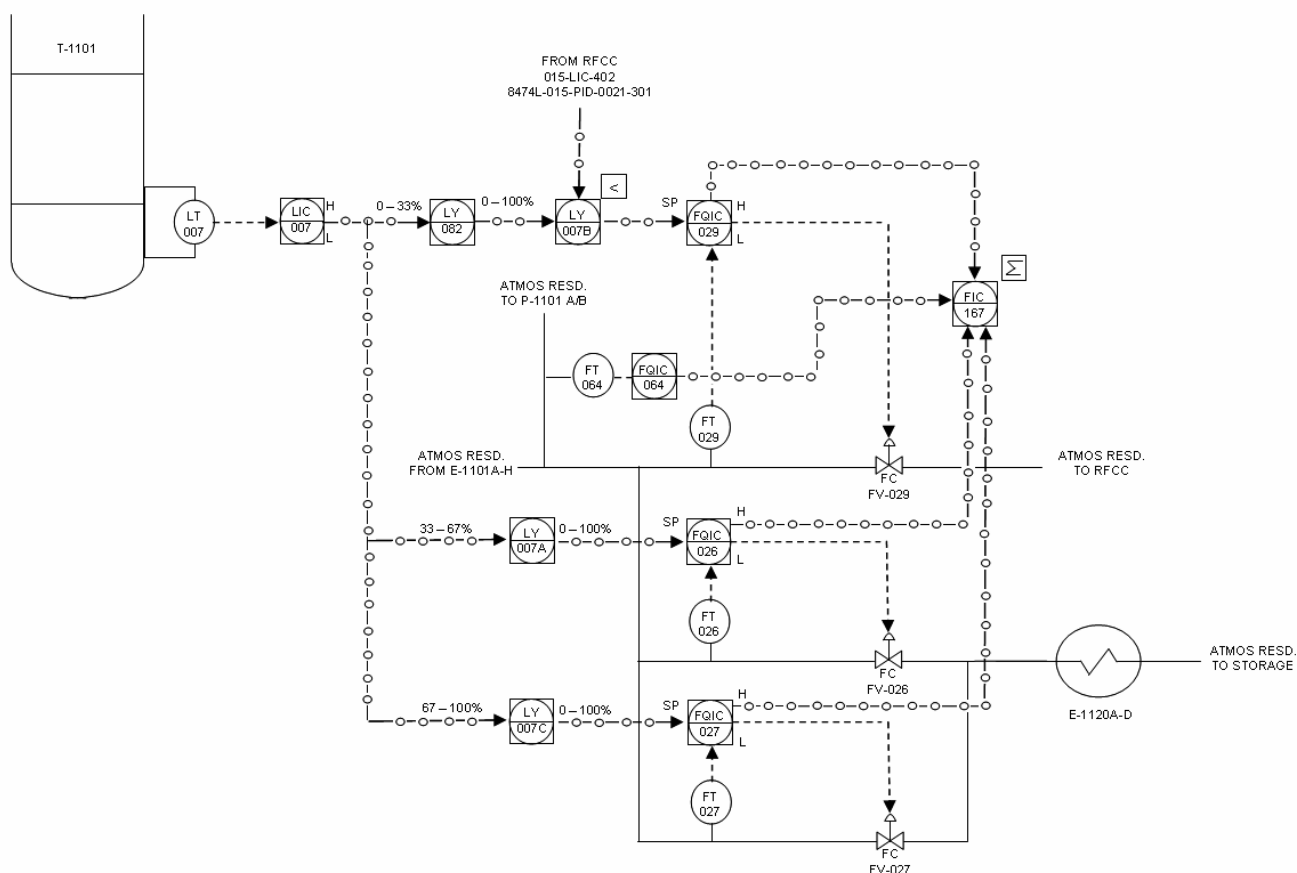


Figure 36: Functional Description of Main Fractionator Level Control

The level in the Main Fractionator, T-1101 is controlled by a typical three way split range level controller 011-LIC-007 (direct acting).

In the low range (0-33 %) the signal from 011-LIC-007 goes to low selector 011-LY-007B, which selects between 011-LIC-007 and the feed requirement of the RFCC, (015-LIC-402), to control the flow to the RFCC.

In the mid range (33-67 %) 011-LIC-007 resets the setpoint of 011-FQIC-026 and in the high range (67-100 %) 011-LIC-007 resets the setpoint of 011-FQIC-027. 011-FQIC-026 and 011-FQIC-027 are parallel controllers used to regulate the flow of residue to storage.

011-FV-026 is a smaller valve than 011-FV-027 and will normally be sufficient to control the amount of residue to storage whilst the RFCC is on-line. In the event that the RFCC is off-line then 011-FQIC-026 will be fully open and 011-FQIC-027 will control the flow to storage. The total flow of atmospheric residue is indicated to operator by signal 011-FIC-167.

The RFCC feed is controlled by a split range level controller, which takes flow preferentially from the CDU by cascading onto 011-FQIC-029 via 011-LY-007B. If there is not enough flow coming from the CDU then flow is taken from storage by acting on a flow controller FIC-402 in the line from storage.

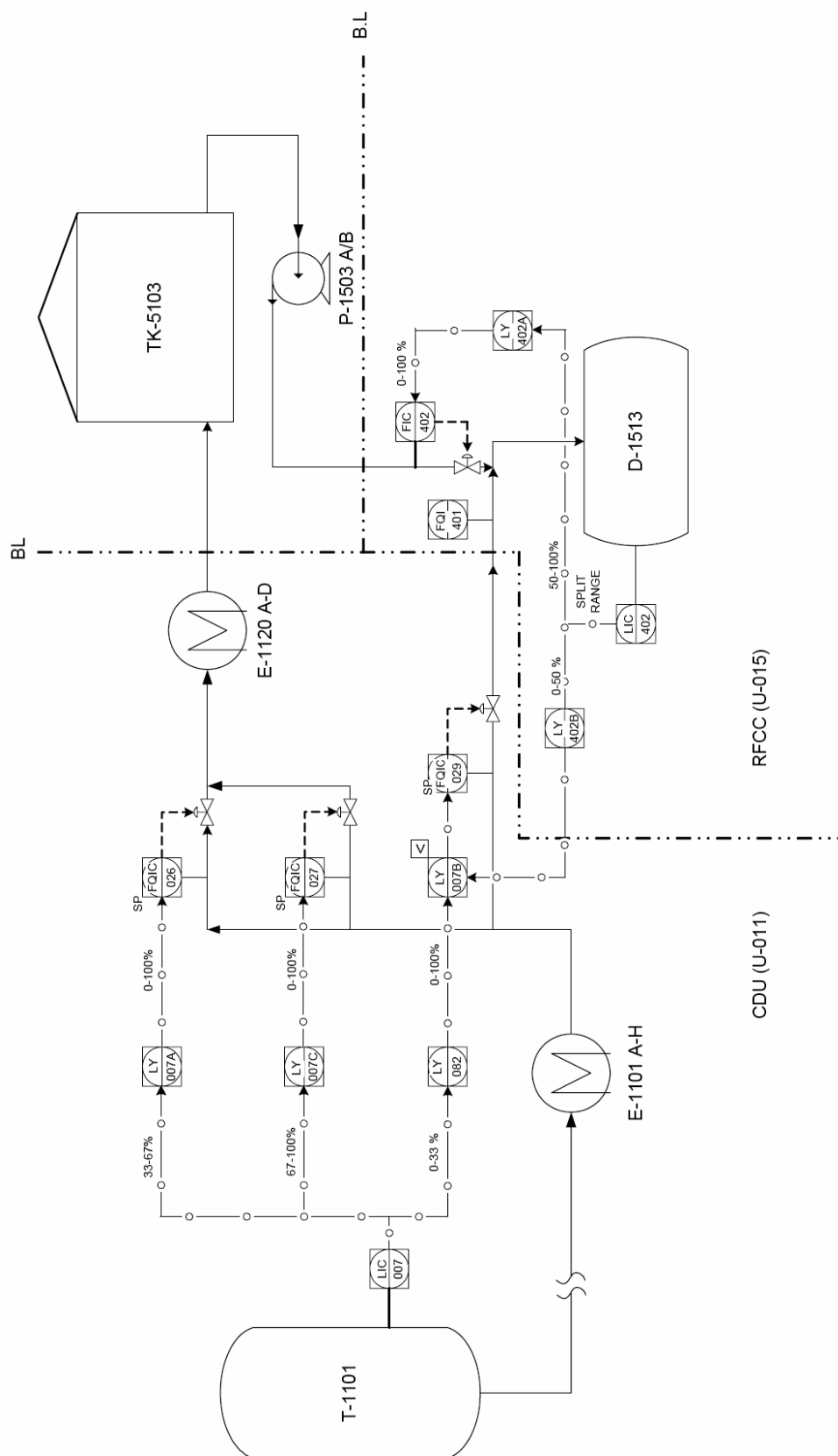
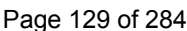


Figure 37: Atmospheric residue from CDU to RFCC



	TRAINING MODULE	DOC NO: 8474L-011-A5016-0000-001-001		
	CRUDE DISTILLATION UNIT (CDU)	REV: 1	DATE: 18/03/08	

3.1.19.2. Full Range Naphta from the CDU to the NHT

Objective

To control the level of the Stabilizer T-1107, and route as much of the outgoing full range naphtha to the NHT as is required

Description

This description should be read in conjunction with the following documents:

CDU P&ID's:

8474L-011-PID-0021-130

8474L-011-PID-0021-132

NHT P&ID's:

8474L-012-PID-0021-011

Control of level in T-1107 is by split range controller 011-LIC-042 (direct acting).

In the low range (0-50%), 011-LIC-042 acts on 011-FQIC-040 through low selector 011-LY-045 to control the flowrate to the NHT.

In the high range (50-100%), 011-LIC-042 acts on the rundown of full range naphtha to storage.

Low selector 011-LY-045 is used to select between T-1107 level controller output and the full range naphtha requirement of the NHT from 012-PIC-001.

The output from 012-PIC-001 (reverse acting) is split range. In the low range naphtha is taken from the CDU via 011-LY-045 and in the high range naphtha is taken from storage by acting on 012-FIC-001.

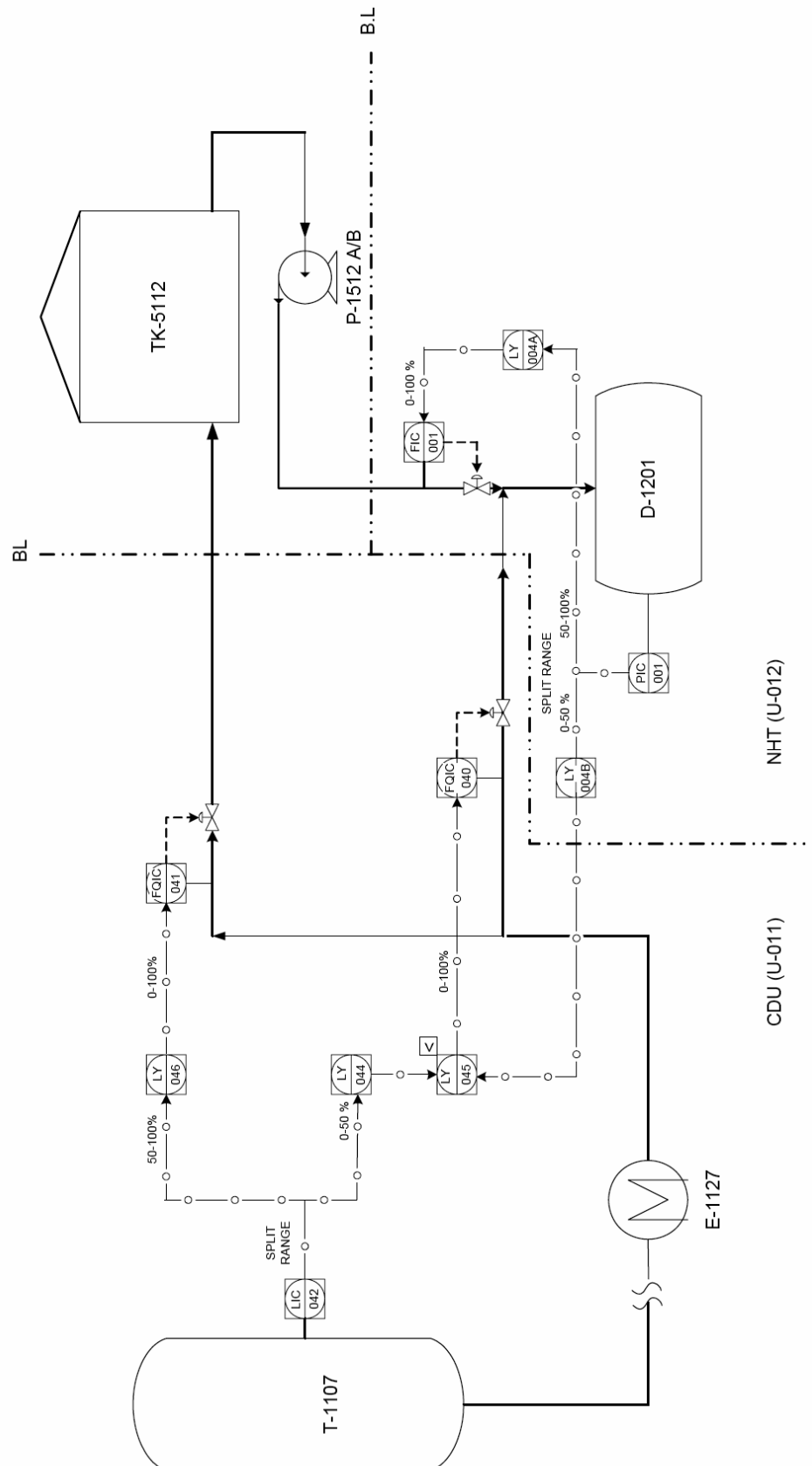


Figure 38: Full range Naphtha from CDU to NHT

	TRAINING MODULE		DOC NO: 8474L-011-A5016-0000-001-001	
	CRUDE DISTILLATION UNIT (CDU)	REV: 1	DATE: 18/03/08	

3.1.19.3. LCO HDT feed control

Objective

The objective of this system is to control the level of the Feed Surge Drum, D-2401, in case of RFCC Max Gasoline Operation, i.e. when feed to LCO HDT Unit, as defined by the Licensor, consists of LCO from RFCC Unit (and potentially LCO Storage), HGO and potentially LGO from CDU Unit.

Description

This description should be read in conjunction with the following documents:

CDU P&ID's:

8474L-011-PID-0021-117

8474L-011-PID-0021-127

LCO HDT P&ID's:

8474L-024-PID-0021-111

RFCC P&ID's:

8474L-015-PID-0021-328

In case of RFCC Max Gasoline Operation, feed to LCO HDT shall consist of:

- LCO from RFCC Unit (flow-controlled within RFCC)
- HGO from CDU Unit, under flow control 024-FIC-002
- Potentially LGO from CDU Unit, under flow control 024-FIC-003
- Potentially LCO from storage tank TK-5111, under flow control 024-FIC-001 which controls the flow rate of LCO from storage that shall be limited to 25% of unit throughput

Selector 024-HS-003 is switched to enable the reverse action of the D-2401 level controller 024-LIC-002 onto controllers 024-FIC-002 and 024-FIC-003, through the calculation module 024-FY-007, therefore adjusting the flow of feeds from CDU Units (HGO and LGO) in the target ratio.

Total LGO flow, from the CDU to both the LCO HDT and Refinery storage, is controlled at a constant value by the action of a flow calculator 011-FY-038 that resets the set point of 011-FQIC-020 by subtracting the flow of LGO to the LCO HDT (024-FIC-003) from the operator input 011-HIC-115. The engineering unit range of 011-HIC-115 shall be 011-FQIC-020 + 024-FIC-003 (0-384 m³/Hr).

Similarly, total HGO flow, from the CDU to both the LCO HDT and Refinery storage, is controlled at a constant value by the action of a flow calculator that resets the set point of 011-FQIC-028 by subtracting the flow of HGO to the LCO HDT (024-FIC-002) from the operator input 011-HIC-007.

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3.1.19.4. Stripped water from the SWS to the CDU

Objective

To control the level of the Sour Water Stripper, T-1801, and route as much of the outgoing sour water to the CDU as it requires.

Description

This description should be read in conjunction with the following documents:

CDU P&ID's:

8474L-011-PID-0021-119

8474L-011-PID-0021-123

8474L-011-PID-0021-124

SWS P&ID's:

8474L-018-PID-0021-113

Control of level in T-1801 is by split range controller 018-LIC-009 (direct acting). In the low range, 018-LIC-009 acts on 011-LV-025A through 018-LY-009A and low selector 011-LY-025D to control the flowrate to the CDU. Low selector 011-LY-025D is used to select between T-1801 level controller output and the requirement for water from the level controller, 011-LIC-025, on the Desalter Water Surge Drum D-1109. In the high range, 018-LIC-009 acts on the control valve through LY-009B in the rundown to effluent treatment.

Control of level in D-1109 is by split range controller 011-LIC-025 (reverse acting). In the low range 011-LIC-025 acts on the control valve in the line from the Drier Oil/Water Separator Drum D-1106, via low select 011-FY-025, which restricts the maximum flow to that set by 011-FIC-025. In the middle range 011-LIC-025 controls the flow from the SWS as described in the paragraph above. In the high range, 011-LIC-025 opens the control valve on the service water supply trough 011-LY-025B.

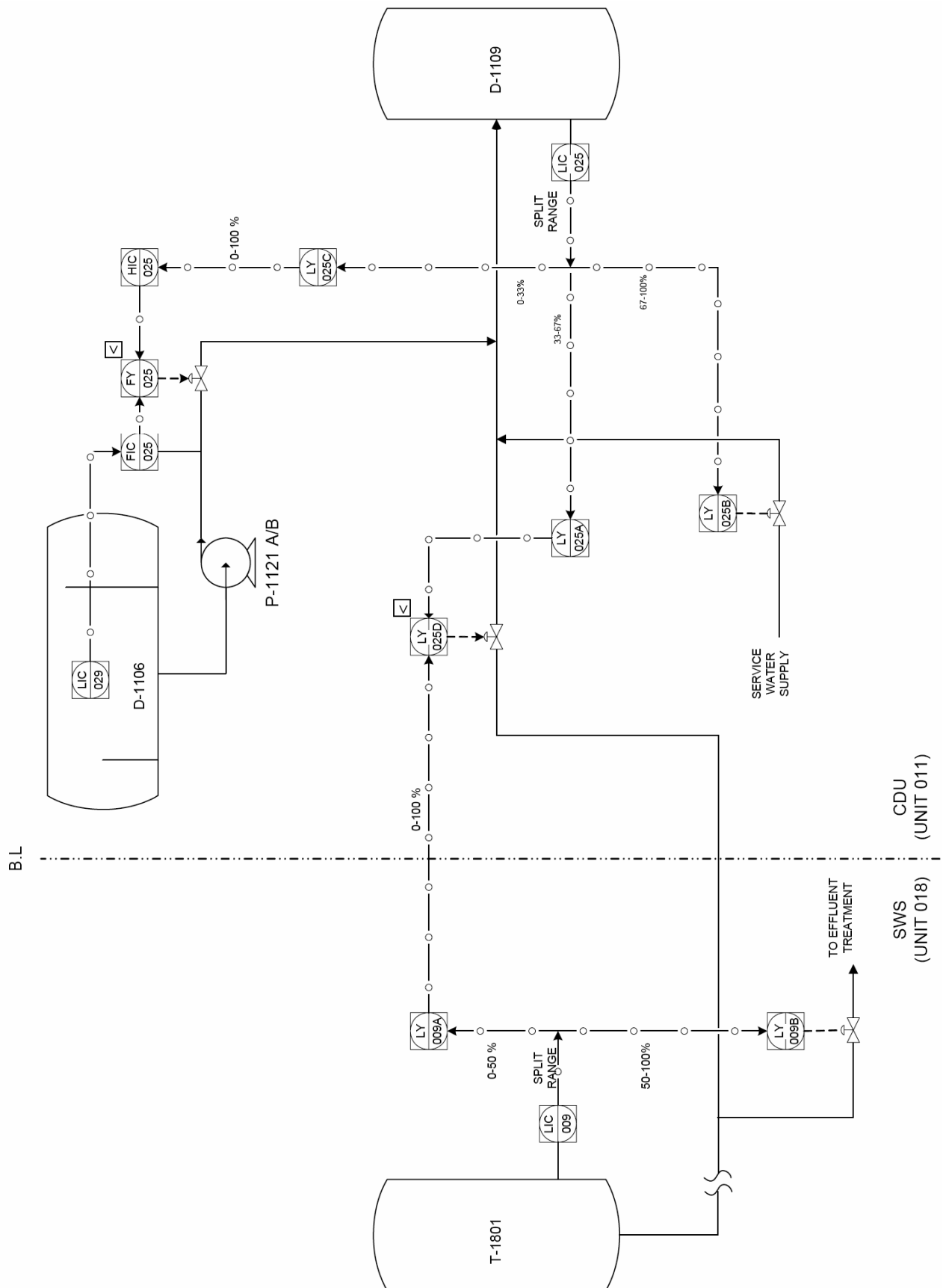


Figure 40: Stripped water from SWS to CDU

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3.1.20. Operating Parameters

For operating conditions of the CDU refer to the following Process Flow Diagrams:

8474L-011-PFD-0010-001	Cold Preheat and Desalters (Bach Ho)
8474L-011-PFD-0010-002	Hot Preheat and Heater (Bach Ho)
8474L-011-PFD-0010-003	Main Fractionators - Strippers (Bach Ho)
8474L-011-PFD-0010-004	Driers - Ejector System (Bach Ho)
8474L-011-PFD-0010-005	Over Head -Rundown System (Bach Ho)
8474L-011-PFD-0010-006	Naphta Stabiliser (Bach Ho)
8474L-011-PFD-0010-007	Schematic -CDU Preheat (Bach Ho)
8474L-011-PFD-0010-008	Mass Balance (Bach Ho)
8474L-011-PFD-0010-009	Cold Preheat and Desalters (Dubai)
8474L-011-PFD-0010-010	Hot Preheat and Heater (Dubai)
8474L-011-PFD-0010-011	Main Fractionators - Strippers (Dubai)
8474L-011-PFD-0010-012	Driers - Ejector System (Dubai)
8474L-011-PFD-0010-013	Over Head -Rundown System (Dubai)
8474L-011-PFD-0010-014	Naphta Stabiliser (Dubai)
8474L-011-PFD-0010-015	Schematic -CDU Preheat (Dubai)
8474L-011-PFD-0010-016	Mass Balance (Dubai)

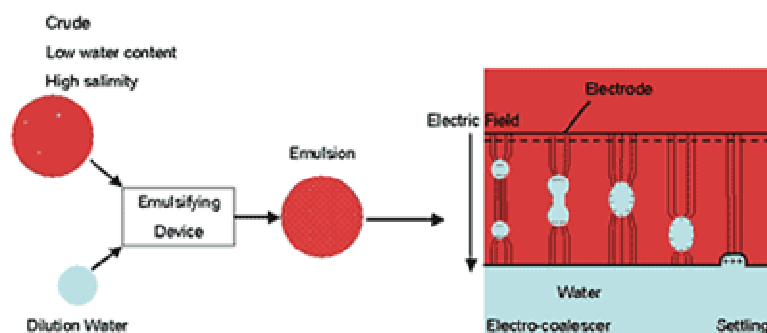
3.2. Instrument List

Refer to attached extracted instrument list of unit 011.

3.3. Main Equipment

3.3.1. Desalters

The desalters package aims at removing the inorganic salts existing in the crude oil fed to the unit by emulsifying crude oil with water and separating them by electrostatic coalescence.



When the oil enters the electric field, the influence of this one causes each water droplet to become an induced dipole.

Consequently water droplets are attracted to one another and they coalesce to create larger droplets.

By gravity, these droplets fall at the bottom of the vessel.

Figure 41: Electrostatic Coalescence Principle

The desalters' package consists of the following main items:

- A-1101-D-01, First stage desalter (including two transformers)
- A-1101-D-02, Second stage desalter (including two transformers)
- A-1101-M-01, First stage static mixer
- A-1102-M-02, Second stage static mixer
- 011-PDV-503, First stage mixing valve
- 011-PDV-506, Second stage mixing valve

The Desalters internals are:

- Oil collector located at the top of the vessel; the treated oil flows through the oil collector
- Oil distributor located at the bottom of the vessel. These distributors distribute the oil across the length of the electrostatic desalter, below energized electrodes grids. This ensures a good distribution of the oil/water mixture as well as a minimum residence time within the electrode area.
- Water collector located at the bottom; water goes out via these collectors
- Electrode grids composed with 2 energized grids and one grounded grid. The grids of electrodes are located just above the vessel centreline in order to generate the electrostatic coalescence in the entire cross-section of the desalter. High voltage electricity for the electrode zone is furnished by power units installed externally on top of the Desalter.

- Mud wash system located at the bottoms; this mud wash system put again in suspension in water the sediments accumulated in the bottom of desalter
- Skimmer system located at the interface oil/water; this system recovers the emulsion at the interface oil/water.

Desalters A-1101 D-01 & A-1101-D-02		
	1 st Stage	2 nd Stage
Contents	Bach Ho or Dubai + produced water	
Max. / Normal / Min. Working Temperature (°C)	150 / 138 / 125 for each stage	
Working Pressure Upstream Emulsifying Device (kg/cm ² g)	1 st Stage: 14.5	2 nd Stage: 11.5
Design Temperature (°C)	175	
Design Pressure (kg/cm ² g)	15	
	Dubai	Bach Ho
Oil Quantity (BOPD)	148000	141000
Normal Salt Content (ppm wt)	56	14
Produced Water Quantity (% vol)	0.4-0.5	-
Fresh Water Quantity (% vol)	4	4
Outlet Salt Content for normal condition	<2	<2
Total Volume (Head Incl.) (m ³)	181	
Diameter of Shell ID (mm)	3600	
Length between Tangent Lines (mm)	16500	
Length for Saddle to Saddle	11000	
Type of Heads	Elliptical 2/1	
Lining	No	
Position	Horizontal	
Interface Level (mm):		
LDHH	1550 from BOV	
LDH	1450 from BOV	
LD (Normal)	1200 from BOV	
LDL	950 from BOV	
LDLL	850 from BOV	
Oil Level	Full	

3.3.2. Columns

3.3.2.1. Main Fractionator T-1101

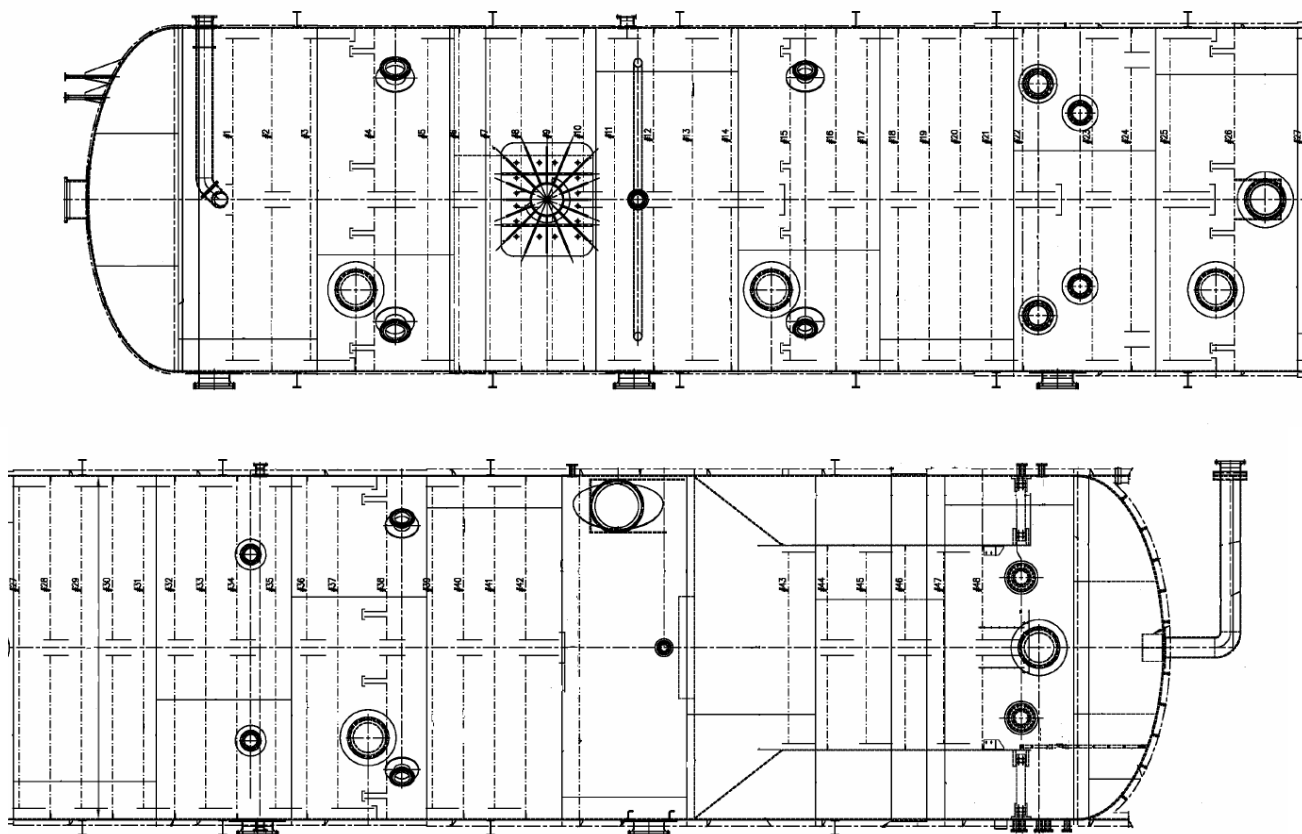


Figure 42: Main Fractionator

Preheated crude oil is distilled in T-1101, which can be divided in 5 main sections:

- Overhead Section with a top pumparound from tray #4 to tray #1
- Kerosene Section: Kerosene is withdrawn at tray #15 as a product and pumparound. Kerosene pumparound is re-injected at tray #12.
- Light Gas Oil Section: LGO is withdrawn at tray #26 as a product and pumparound. LGO pumparound is re-injected at tray #23.
- Heavy Gas Oil Section: HGO is withdrawn at tray #38 as a product and pumparound. HGO pumparound is re-injected at tray #35
- Overflash Section
- Residue Section

T-1101 has 48 trays divided in two diameter sections: the first one from tray 1 to 42 with an internal diameter of 6700 mm, and the second one from tray 43 to 48 with a diameter of 4000 mm. The length between

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tangent lines is 42850 mm. The feed inlet nozzle is tangential with a distribute pipe and it is located between the trays 42 and 43.

The tower operates in a pressure range of 1.5 (top) to 1.9 (bottom) kg/cm²g, and in a temperature range of 130-124° (top) to 349-354 °C (bottom).

Main Fractionator T-1101	
Number of Trays	48
Number of Diameter Sections	2
Section 1	Diameter of 6700 mm from Tray #1 to #42
Secton 2	Diameter of 4000 mm from Tray #43 to #48
Length between Tangent Lines (mm)	42,850
Feed Inlet Nozzle	Tangential with a Distribution Pipe Located between Trays #42 & #43
Top / Bottom Design Pressure (kg/cm ² g)	3.50 / 4.40
Top / Bottom Operating Pressure (kg/cm ² g)	1.50 / 1.98
Top / Bottom Design Temperature (°C)	235 / 385
Top / Bottom Operating Temperature (°C)	130 / 354

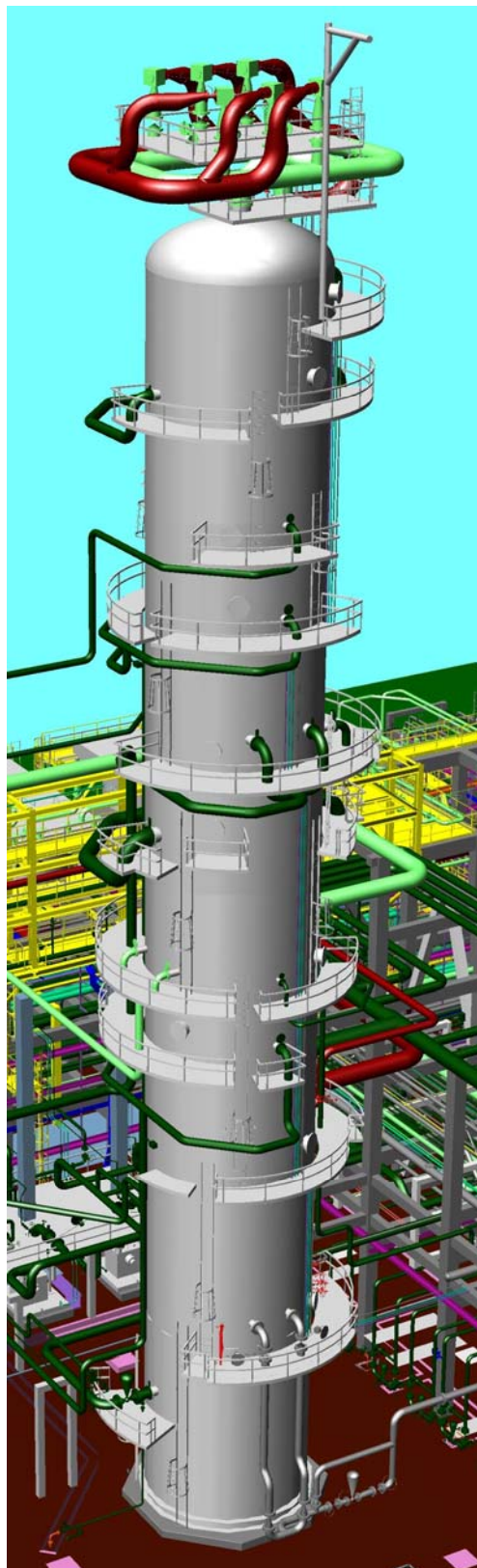


Figure 43: Main Fractionator 3D view

3.3.2.2. Kerosene Stripper

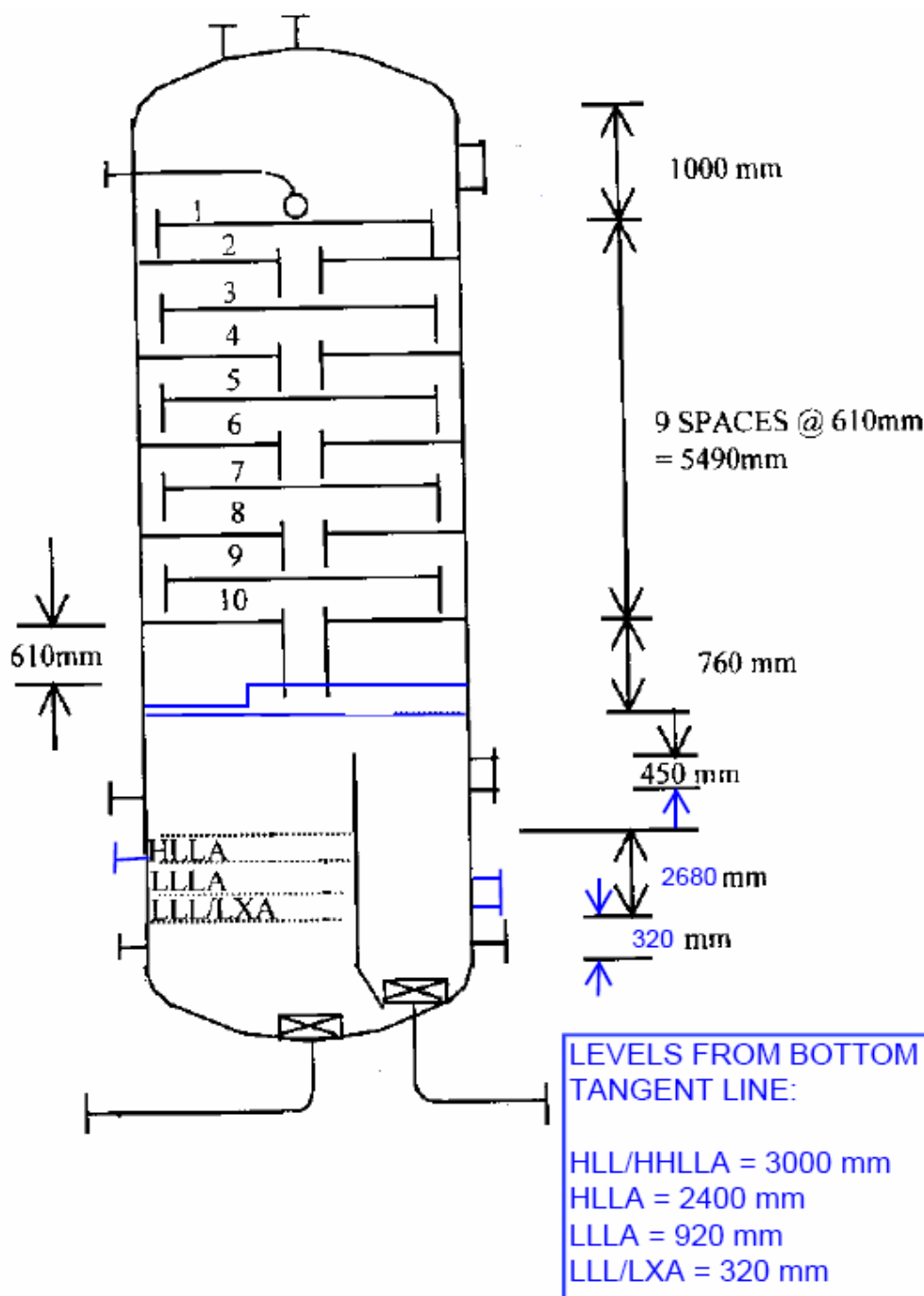


Figure 44: Kerosene Stripper

The function of T-1102 is to purify the kerosene product from tray 15 of T-1101. The Kerosene Stripper (T-1102) consists of 10 trays and a Kerosene Stripper Reboiler (E-1110), which uses the Heavy Gas Oil (HGO) pump-around as heat source. A facility for stripping steam injection in the bottom of the tower is also available.

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Kerosene Stripper T-1102	
Number of Trays	10
Internal Diameter (mm)	1,800
Length between Tangent Lines (mm)	11,150
Feed Inlet Nozzle Location	Tray #1
Top / Bottom Design Pressure (kg/cm ² g)	3.70 / 3.90
Top / Bottom Operating Pressure (kg/cm ² g)	1.60 / 1.70
Top / Bottom Design Temperature (°C)	375
Top / Bottom Operating Temperature (°C)	215 / 234

3.3.2.3. LGO Stripper T-1103

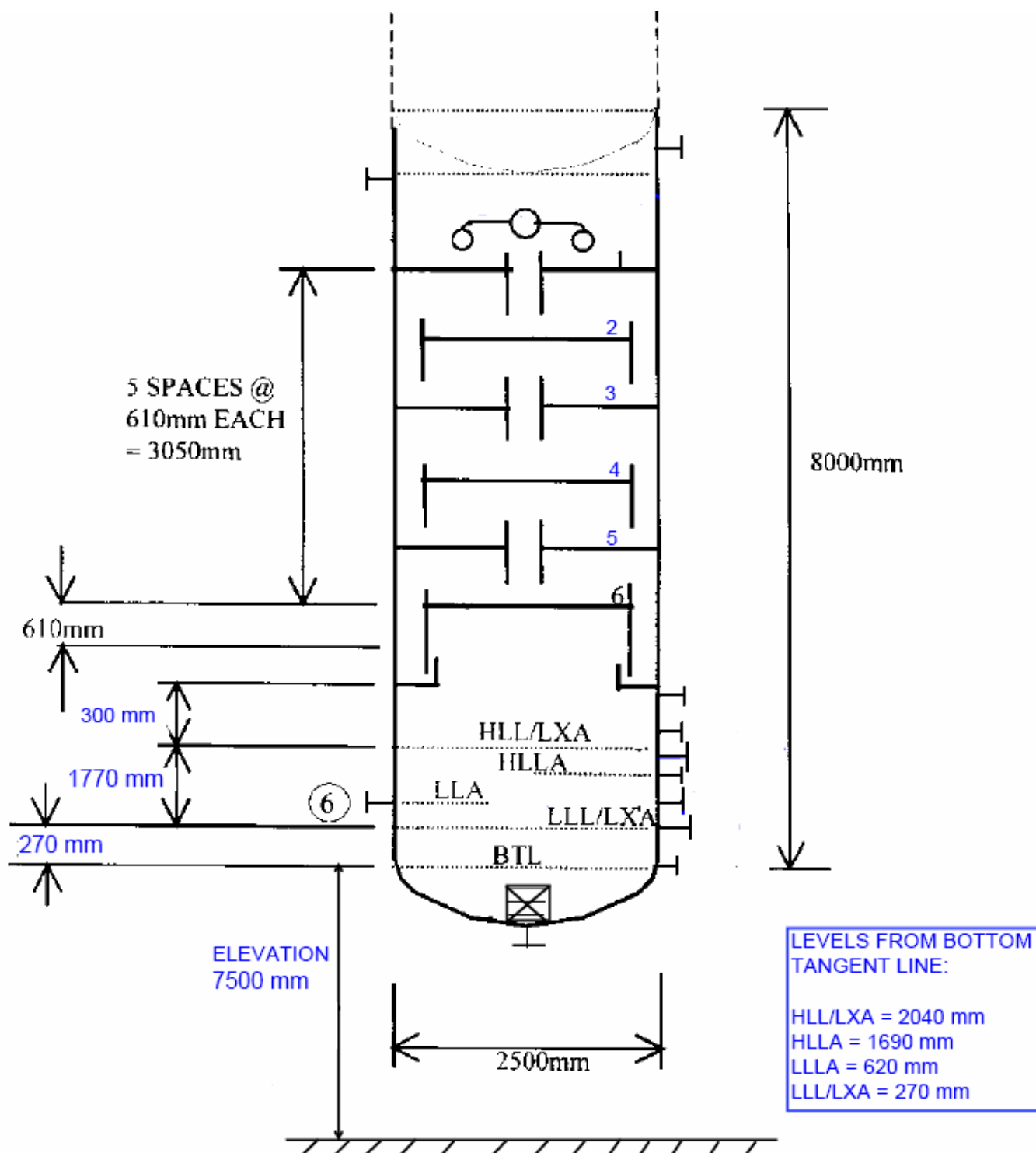


Figure 45: LGO Stripper

The function of T-1103 is to purify LGO product from tray 26 of T-1101. The LGO Stripper (T-1103) consists of 6 trays and an injection of stripping steam at the bottom.

	TRAINING MODULE		DOC NO: 8474L-011-A5016-0000-001-001	
	CRUDE DISTILLATION UNIT (CDU)	REV: 1	DATE: 18/03/08	

LGO Stripper T-1103	
Number of Trays	6
Internal Diameter (mm)	2,500
Length between Tangent Lines (mm)	8,000
Feed Inlet Nozzle Location	Tray #1
Top / Bottom Design Pressure (kg/cm ² g)	3.90 / 4.00
Top / Bottom Operating Pressure (kg/cm ² g)	1.75 / 1.82
Top / Bottom Design Temperature (°C)	375
Top / Bottom Operating Temperature (°C)	251/ 244

3.3.2.4. HGO Stripper T-1104

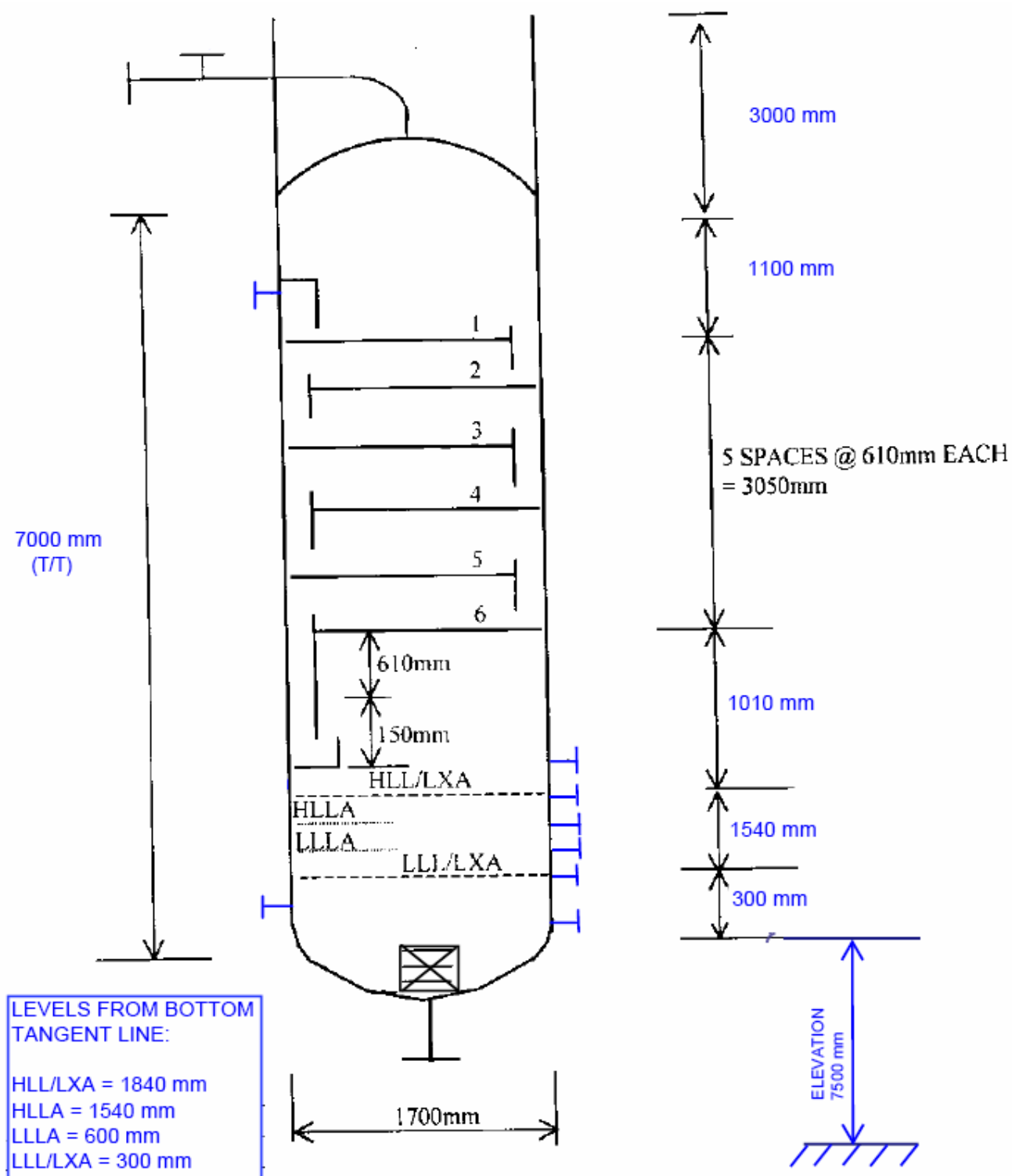


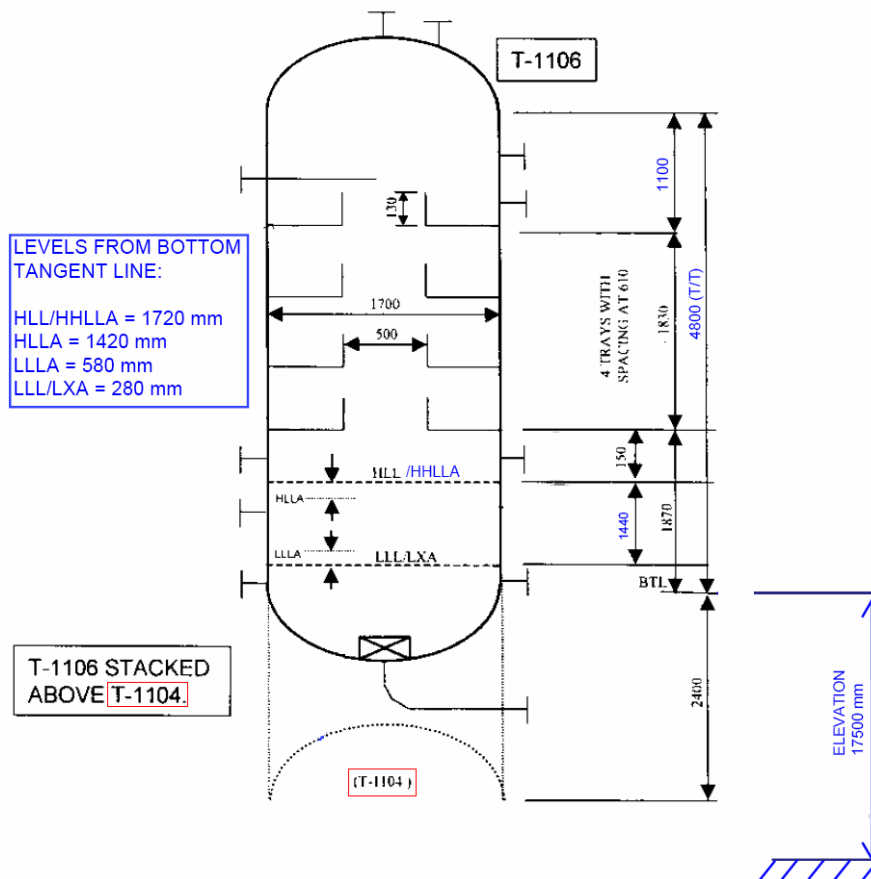
Figure 46: HGO Stripper

The function of T-1104 is to purify the HGO product from tray 38 of T-1101. The HGO Stripper (T-1104) consists of 6 trays and an injection of stripping steam in the bottom.

In T-1105, LGO is vacuum dried before being sent to storage. T-1105 is stacked below T-1102 and above T-1103

LGO Dryer T-1105	
Number of Trays	4
Internal Diameter (mm)	2,500
Length between Tangent Lines (mm)	5,100
Feed Inlet Nozzle Location	Tray #1
Top / Bottom Max. Design Pressure (kg/cm ² g)	3.50 / 3.70
Min. Design Pressure (kg/cm ² g)	Full vacuum
Top / Bottom Operating Pressure (kg/cm ² g)	-0.900
Top / Bottom Design Temperature (°C)	241 / 241
Top / Bottom Operating Temperature (°C)	140 / 140

3.3.2.6. HGO Dryer T-1106



	TRAINING MODULE		DOC NO: 8474L-011-A5016-0000-001-001	
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Figure 48: HGO Dryer

In T-1106, HGO is vacuum dried before being sent to storage. T-1106 is stacked above T-1104

HGO Dryer T-1106	
Number of Trays	4
Internal Diameter (mm)	1,700
Length between Tangent Lines (mm)	4,800
Feed Inlet Nozzle Location	Tray #1
Top / Bottom Max. Design Pressure (kg/cm ² g)	3.50 / 3.70
Min. Design Pressure (kg/cm ² g)	Full vacuum
Top / Bottom Operating Pressure (kg/cm ² g)	-0.900
Top / Bottom Design Temperature (°C)	295 / 295
Top / Bottom Operating Temperature (°C)	160 / 160

3.3.2.7. Naphta Stabilizer T-1107

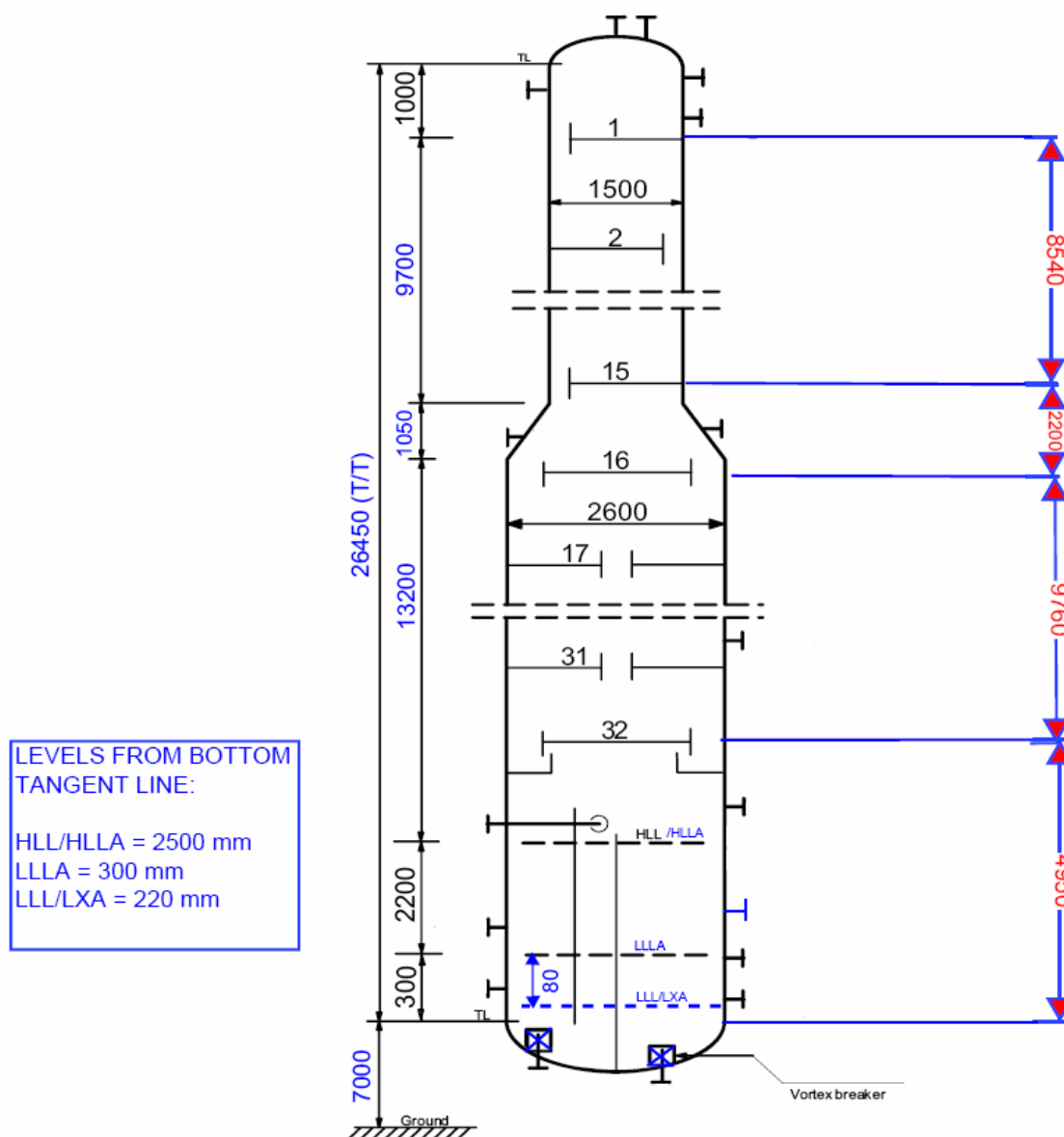


Figure 49: Naphta Stabilizer

Naphta from the main fractionator accumulator drum D-1103 is purified in the stabilizer column. Naphta Product accumulates at the bottom of the stabilizer while the overhead gases (LPG, water vapour and off) are separated in the stabiliser reflux drum. T-1107 has 32 trays divided in two diameter sections, a Stabilizer Reboiler (E-1121) in the bottom, which uses HP steam as heat source, and a top reflux.

Naphta Stabilizer T-1107	
Number of Trays	32
Number of Diameter Sections	2
Section 1	Diameter of 1,500 from Tray #1 to #15

Naphta Stabilizer T-1107	
Section 2	Diameter of 2,600 from Tray #16 to #32
Length between Tangent Lines (mm)	26,450
Feed Inlet Nozzle Location	Above Tray #16
Top / Bottom Design Pressure (kg/cm ² g)	9.60 / 9.85
Top / Bottom Operating Pressure (kg/cm ² g)	7.90 / 8.15
Top / Bottom Design Temperature (°C)	215 / 215
Top / Bottom Operating Temperature (°C)	66 / 191

3.3.3. Vacuum Package A-1102

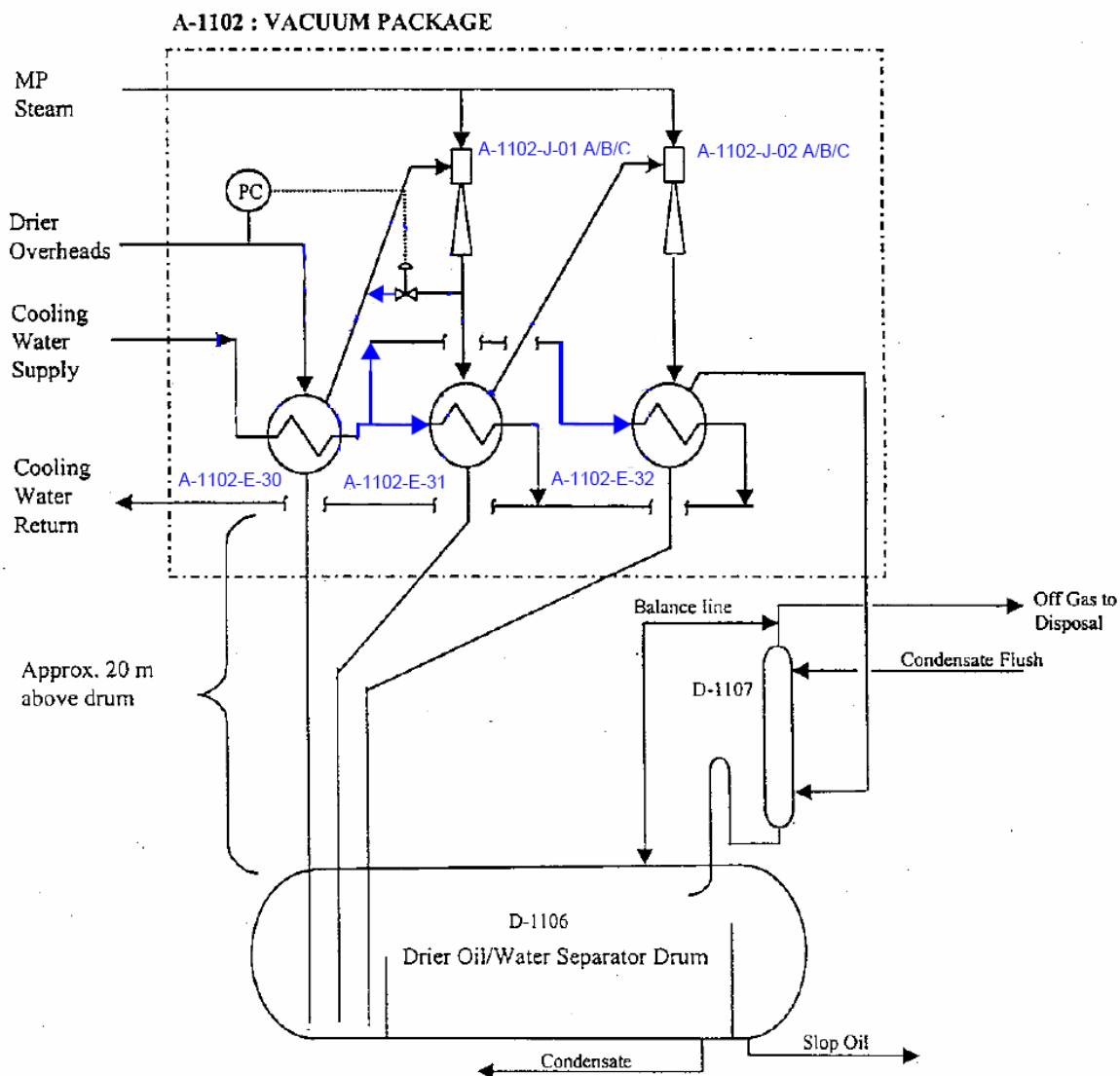


Figure 59: Vacuum Package

The vacuum system maintains a reduced pressure in the Dryers by using steam ejectors, where medium pressure steam is used as motive fluid.

The Vacuum package, A-1102, consists of the following equipment:

- A-1102-J-01 A/B/C First stage ejector
- A-1102-J-02 A/B/C Second stage ejector
- A-1102-E-30 Pre-condenser (Vacuum drier pre-condenser)
- A-1102-E-31 Inter-condenser (First stage after-condenser)
- A-1102-E-32 After-condenser (Second stage after-condenser)
- 011-PV-546 Recycle control valve

Vacuum Package A-1102	
Normal Flow (kg/hr)	3,604
Operating Pressure at Package Inlet (kg/cm ² g)	-0.95
Pressure loss between Drier and Vacuum Package (kg/cm ² g)	0.02
Discharge Pressure in Drier Oil/Water Separator (kg/cm ² g)	0.5
Temperature Vapour from LGO Drier (°C)	140
Temperature Vapour from HGO Drier (°C)	153.5
Operation	Continuous
Number of Ejector Stages	2
Type of Condensers	Surface
Number of Ejectors Elements	3 per Stage

3.3.4. Crude Charge Heater H-1101

Heater H-1101 is a crude heater designed for 83740 kW duty. It consists of a double cell cylindrical radiation heater and a single convection section. Process flow is divided at entrance to convection section in eight symmetrical passes. After leaving convection section flow is divided such that four passes are directed to cell 1 and the other four are directed to cell 2. Each pass outlet is located at top of radiation section.

Additionally, remaining heat from flue gases is used to superheat low pressure steam in the top three rows of convection section.

Radiation section is based on two identical cells of vertical tubes. There are 72 tubes per cell, each of 17.9 m straight length spaced two tube nominal diameters. These tubes are supported on the top and guided at their intermediate part and bottom. Crossovers from convection to radiation section are external and welded.

Convection section located on top of the two radiation cells consists of eighteen rows of eight tubes per row of process coil and three rows of steam superheating

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coil. The three bottom rows of process coil (shock rows) are bare tubes while the fifteen rows left and the three superheating coil rows are finned tubes of $\frac{3}{4}$ " fin height which increases the installed area while the fin height is acceptable to burn Fuel Gas and Fuel Oil.

Each burner includes a self inspirating pilot provided with and ignition rod for automatic ignition and a detection rod for ionisation flame detection. The air intake to the pilot is controlled by means of a venturi that consists of an air door that may be adjusted at field.

Three dampers are located above convection section to enable draft control at the heater. Dampers have been designed with a maximum stop in order to leave always some free area for flue gas pass. Dampers position in case of air or electrical failure is fully open.

This heater has been designed to operate in forced draft mode. For this operation the heater has been provided with two blowers (B-1101 A/B) located in parallel (one in operation and one in spare). Both have been designed for 120% of design air flow. The air flow is controlled with the inlet guide vane of each blower. A diverter (011-XV-500) has been installed to isolate the blower in spare and allow pass for the blower running. The diverter is positioned by means of a local handle and limit switches (011-XZL-500) have been installed to give information of diverter position.

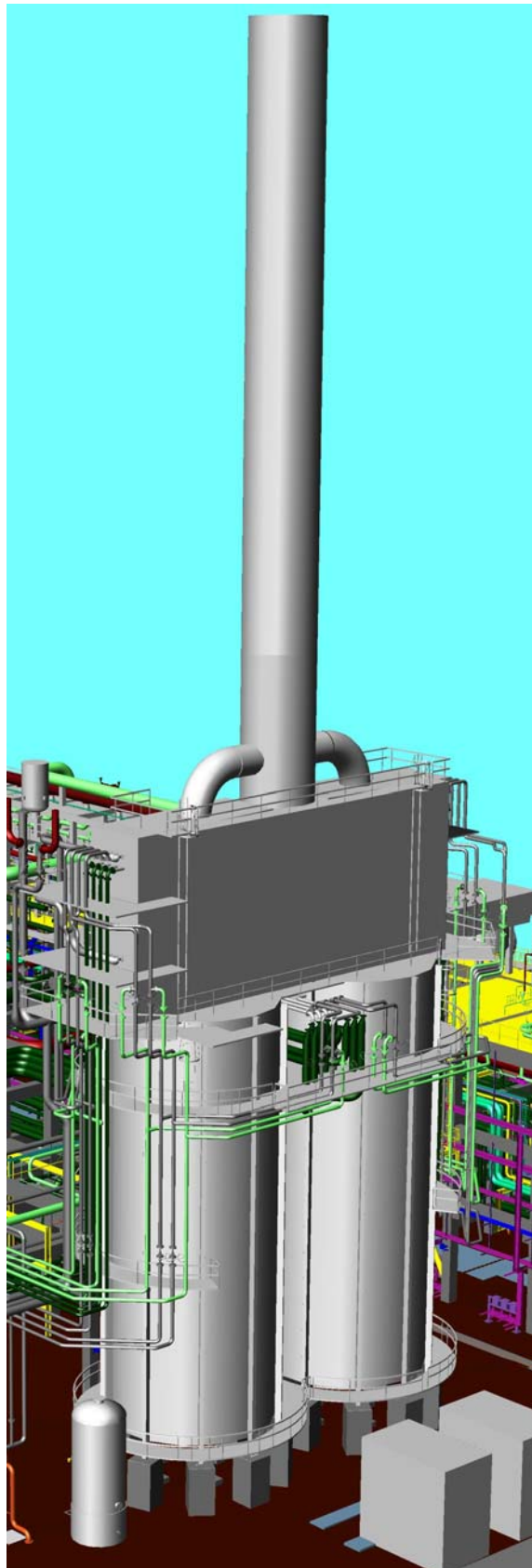


Figure 51: Crude Charge Heater

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Crude Charge Heater H-1101				
Type of Heater	Double Cylinder Radiation + Convection			
Total Heater Design Absorbed Duty (Kw)	83790			
Operating Case	Design (100% Bach Ho)			100% Dubai
Heater Section	RAD	CONV	CONV	RAD/CON
Service	Process		LP SSH	Process
Heat Absorption (KW)	816556		23670	816571
AVG. RAD. Sect. Flux Density, Calc. (W/m ²)	38000			
Inlet Temperature (°C)		283	160	277
Outlet Temperature (°C)	360.4		350	363.5
Inlet Pressure (kg/cm ² g)		13.65	3.3	15.15
Outlet Pressure (kg/cm ² g)	2.45		2.6	2.45
Burner Data				
Number	2x8			
Type	Forced Draft, Dual Firing, Low Nox			
Location / Orientation	Floor / Firing Up			
Heat Released per Burner (KW)	Design: 7069 Normal: 6147 Minimum: 2357 (Fuel Oil) - 1414 (Fuel Gas)			
Pilot Type	Self Inspiring			
Pilot Capacity (Kcal/h)	2000			
Pilot Fuel	Gas			
Ignition Method	Electrical			
Mechanical Design Conditons				
Heater Section	RAD	CONV	CONV	
Service	Crude		SSH	
Tube Orientation	Vertical	Horizontal	Horizontal	
Tube Material	A 335 P9	A 335 P5	A 335 P22	
Tube Outside Diameter (mm)	168.3	168.3	168.3	
Number of Flow Passes	8		4	
Number of Tubes	72/Cell	144	24	

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Crude Charge Heater H-1101				
Number of Tubes per Row		8	8	
Overall Tube Length (mm)	18462	17670	17670	
Number of Bare Tubes	144 Total	24	-	
Total Exposed Surface (m ²)	1405	224	-	
Number of Extended Surface Tubes	-	120	24	
Total Exposed Surface (m ²)	-	6811	1362	
Extended Surface Type	-	Solid Fins		
Extended Surface Material	-	11-13 Cr & Cs	CS	
Tube Layout	Line	Staggered	Staggered	
Tube Supports Location	Top	Ends	Ends	
Tube Support Material	25 Cr 20 Ni	CS	CS	

3.3.5. Rotating Equipment

3.3.5.1. Centrifugal Pumps API 610 Process Heavy Duty

Radially Split, Single Stage Pumps, Double Suction Impeller

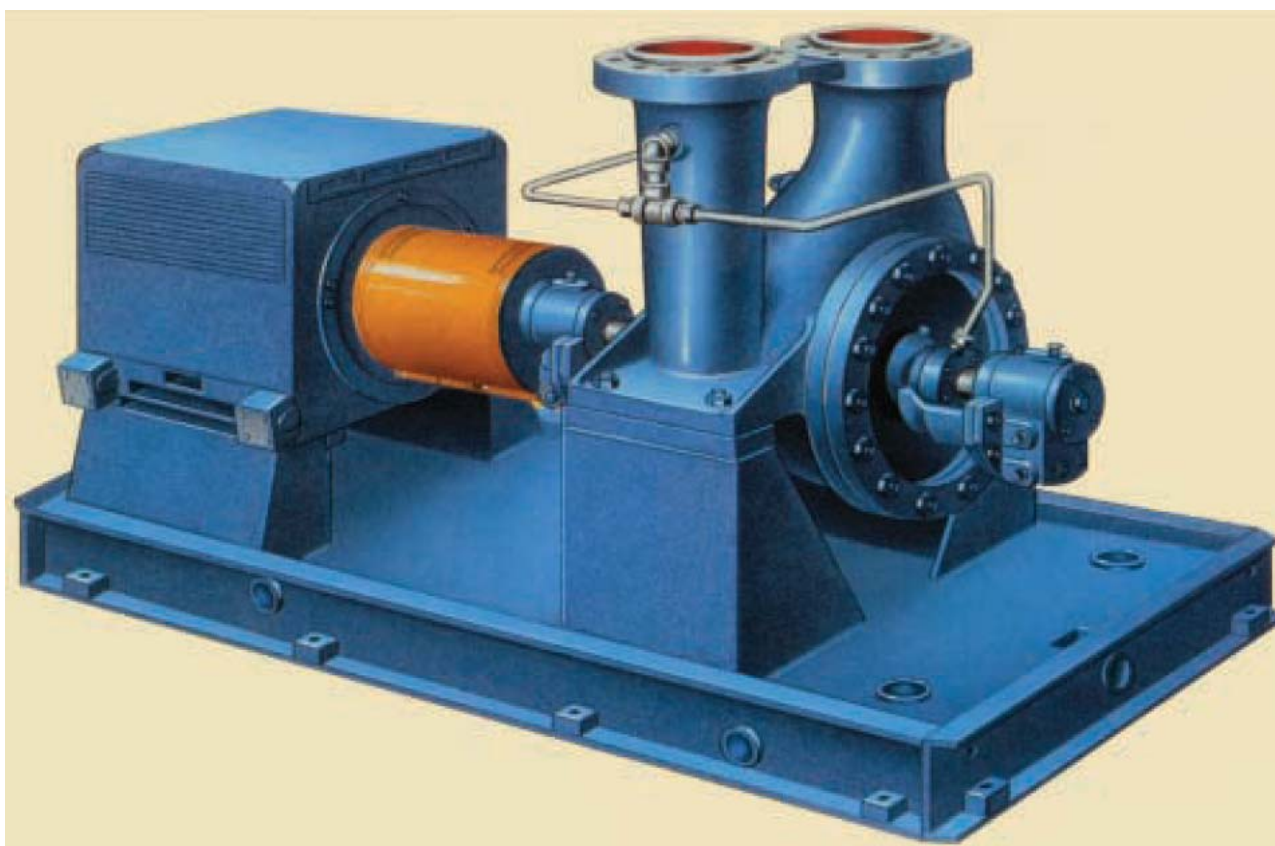


Figure 52: ITT Gould Radially Split, Single Stage Pump

The following process pumps are of the radially split, single stage type:

- Crude Booster Pump P-1101 A/B
- Top Pumparound Pump P-1102 A/B
- LGO Pumparound Pump P-1104 A/B

These are all ITT GOULDS pumps Model 3620 and they have the following design features (vendor commercial information):

- Between bearings design for improved rotor stability at high flows and speeds.
- Double suction impeller to reduced NPSH required at high flows and speeds.
- API-610 full compliance with 10th Edition/ISO 13709.
- Tangential discharge for maximum hydraulic efficiency.
- Finned bearing frame to accommodate high temperatures without need for cooling water.
- Seal chamber meets dimensional requirements of API-610/ISO 13709.
- Bearing cooling for extremely high temperature services.
- Rugged mechanical design: Extra heavy shaft and bearings.

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- Back pull-out construction: Designed for one craft maintenance.
- Dual volute casings to maximize bearing life.
- Centerline support for high temperature stability

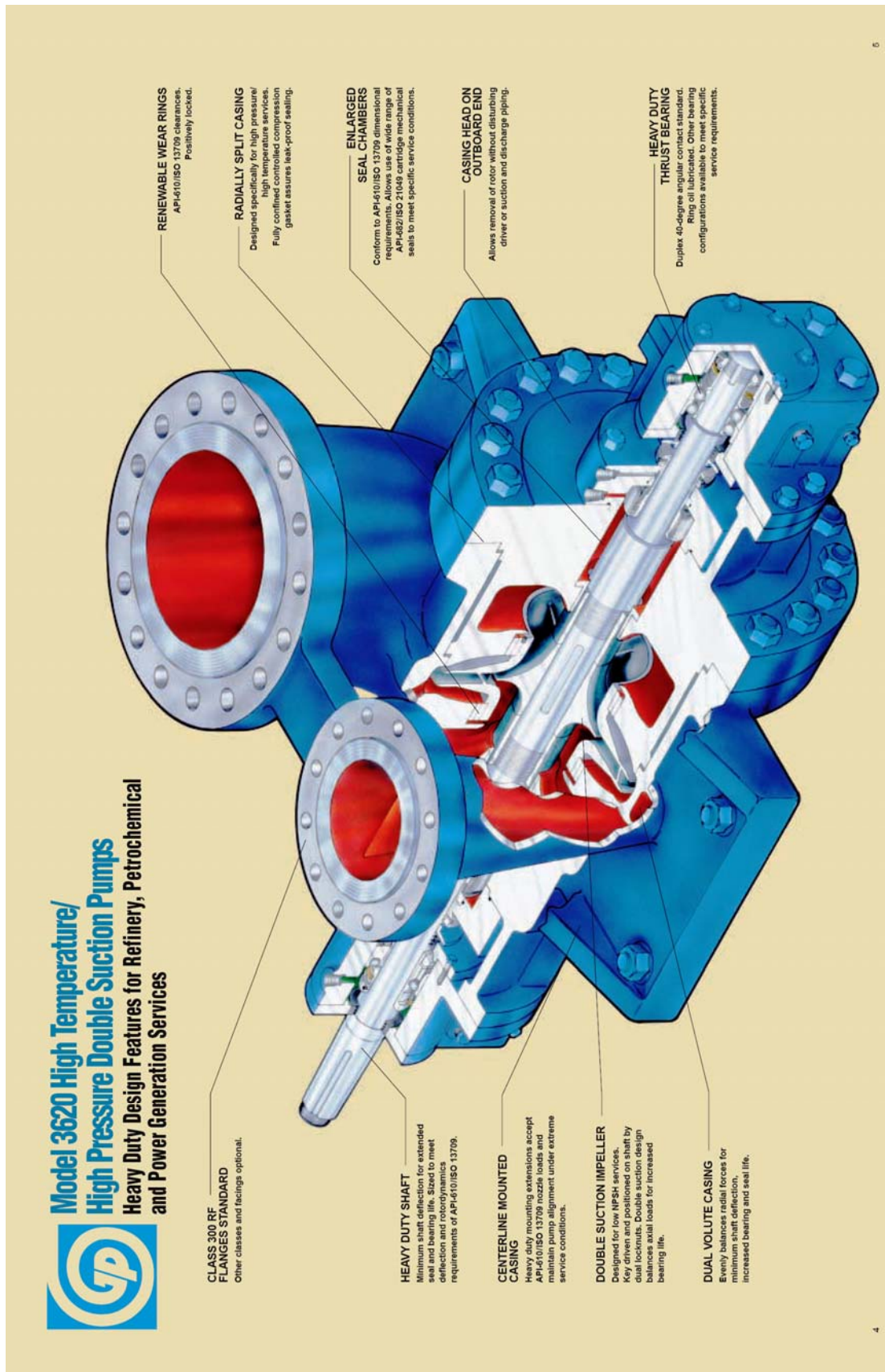


Figure 53: ITT Gould Model 3620 Design

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Radially Split Casing, 2 Stages Between Bearing, Double Suction Impeller

The Residue Pumps P-1106 A/B are of this type. These are Weir Gabbioneta Model DDH pumps having the following main design features (vendor commercial information):

- Double suction 1st. stage impeller, balanced by means of back wear ring on the 2nd stage impeller.
- Ample balancing line from 2nd. stage to suction maintain stuffing box pressures within limits easily acceptable by any type of balanced mechanical seals.
- Centerline casing mounting for high temperature stability.
- Stiff shaft design. Rotor operates below 1st critical speed.
- Deep stuffing boxes designed in accordance to API 682 Std., suitable to accept any type of mechanical or packing seals.
- Labyrinths and deflectors at either bearing housing to maximise protection against dirt.



Figure 54: Weir Gabbioneta Model DDH Pump

Overhung Impeller Centerline Mounted Pumps



Figure 55: ITT Gould Overhung Impeller Centerline Mounted Pump

The following process pumps are of the Overhung Impeller Centerline Mounted type:

- Kerosene Pumparound Pump P-1103 A/B
- HGO Pumparound Pump P-1105 AB
- Kerosene Product Pump P-1107 A/B
- Stabiliser Feed Pump P-1110 A/B
- LGO Product Pump P-1112 A/B
- HGO Product Pump P-1113 A/B
- Desalter Water Recycle Pump P-1118 A/B
- Drier Slop Oil Pump P-1120 A/B
- Ejector Condensate Pump P-1121A/B
- Tempered Water Pump P-1122 A/B

These are all ITT GOULDS pumps Model 3700 and they have the following design features (vendor commercial information):

- API-610 full compliance with 10th Edition/ISO 13709

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- Tangential Discharge for maximum hydraulic efficiency
- Finned bearing frame accommodates high temperatures without need for cooling water
- Seal chamber meets dimensional requirements of API-610/ISO 13709
- Bearing and seal chamber cooling for extremely high temperature services

Bearing Environment:

- o Extra large oil sump provides cooler running bearings
- o Channeled oil lubrication system assures flow-thru of cooled oil to thrust and radial bearings
- o Standard dual oil rings, positively located, provide oil flow to channeled oil lubrication system and prevent oil foaming.

Heat Flinger and Thermal Barrier:

- o Thermal gasket creates heat barrier between seal chamber and bearing frame. Heat flinger dissipates shaft-conducted heat and circulates air to reduce heat build-up

Air Cooling

- o High capacity fan and shroud mounted on ower end effectively reduce bearing frame temperature for cooler running bearings without using cooling water

Water Cooling

- o Finned cooler for maintaining oil/bearing temperature
- Rugged mechanical design: Extra heavy shaft and bearings.
- Back pull-out construction: Designed for one craft maintenance.
- Dual volute casings: 3-inch discharge and larger
- Impeller multiple closed impellers for most casing sizes
- Centerline support high temperature stability

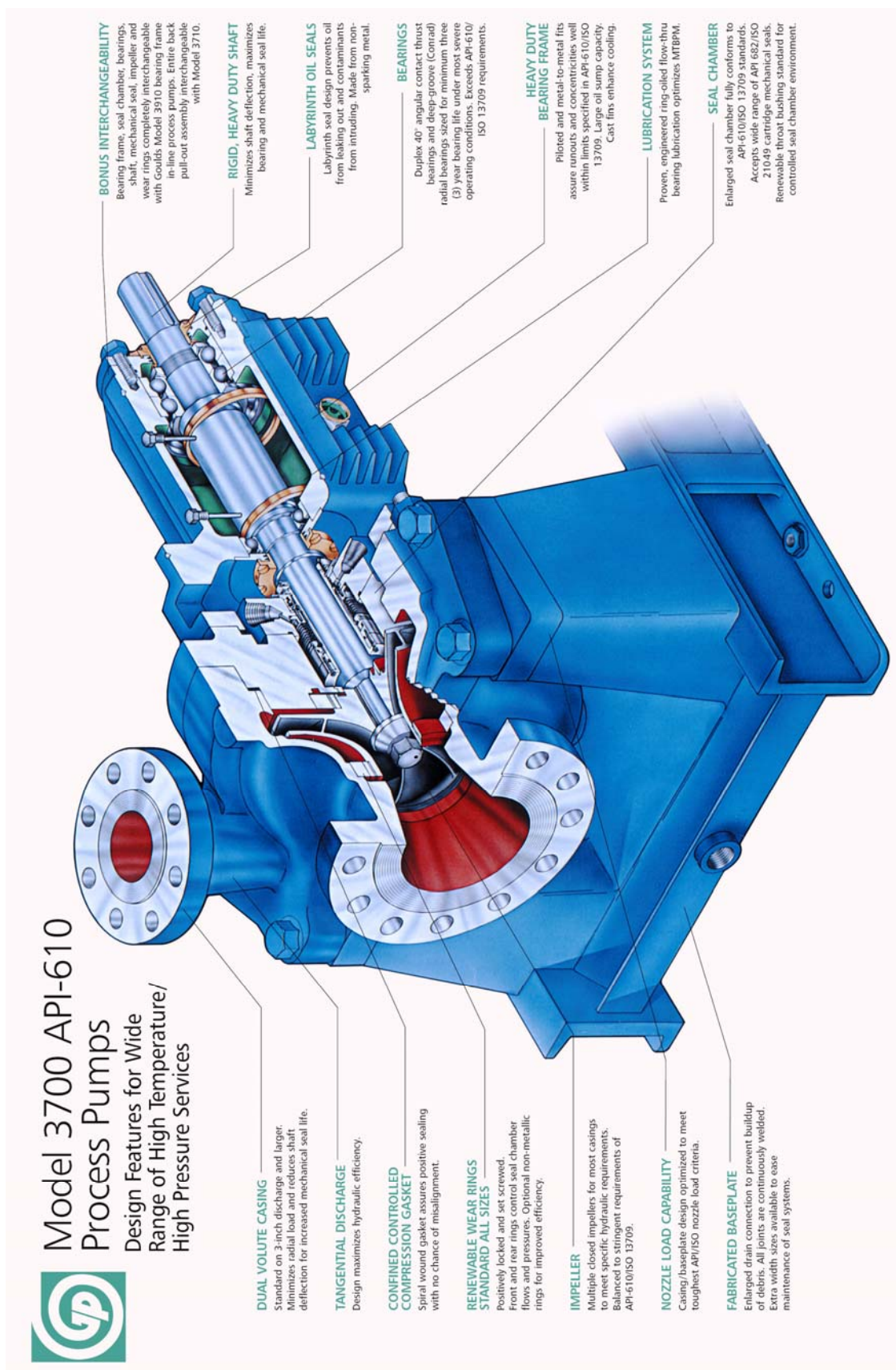


Figure 49: ITT Gould Model 3700 Design

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3.3.5.2. Centrifugal Pumps (High Speed)

The Stabilizer LPG Pump P-1115 A/B pumps are high speed gear driven centrifugal SUNDYNE pumps Model LMV 322-Z.



Figure 50: Sundyne LMV 322

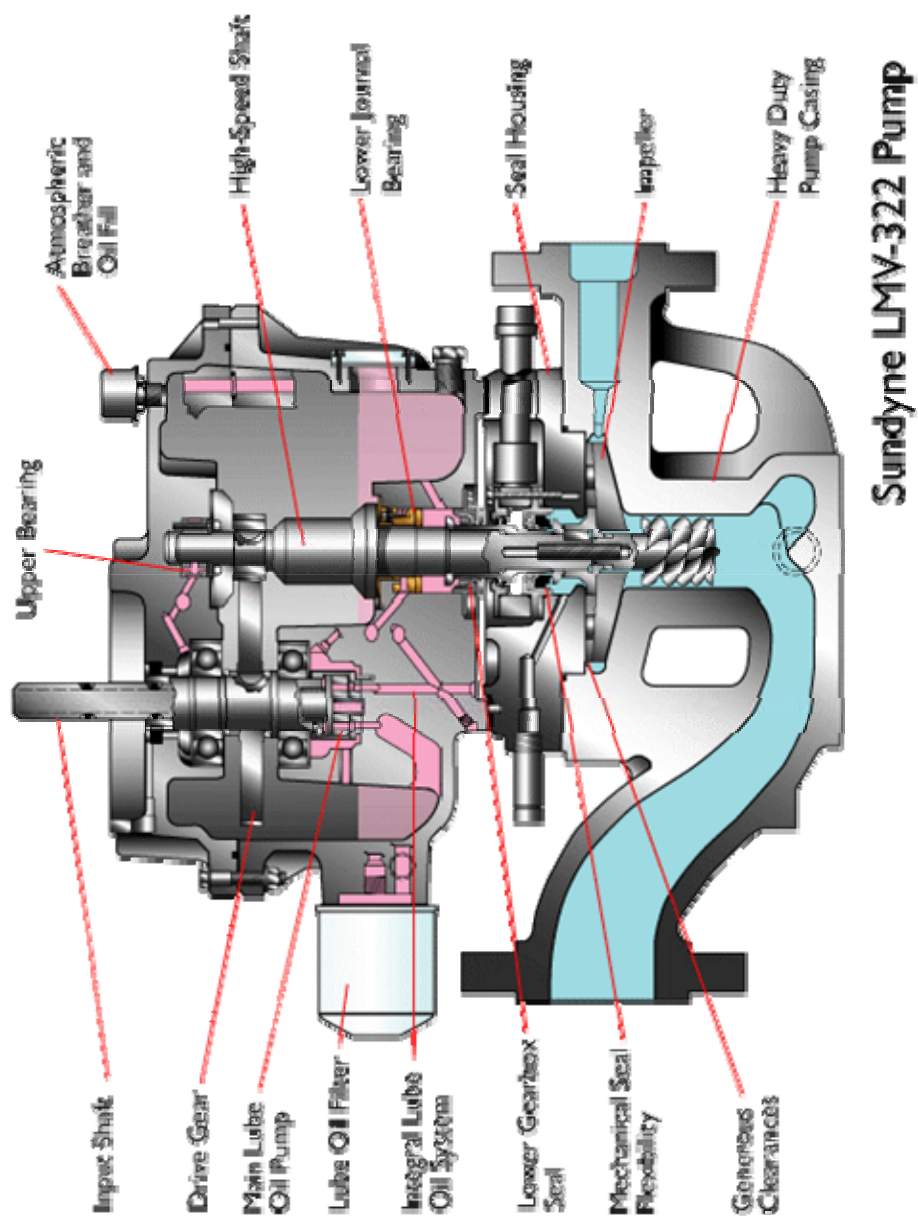


Figure 51: Sundyne LMV 322 Cut Away

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3.3.5.3. Metering Pumps

The following pumps are of the diaphragm metering type:

- Demulsifier Injection Pump A-1104-P-23A/B
- Antifoulant Injection Pump A-1104-P-24 A/B
- Corrossion Inhibitor Pump A-1104-P-25A/B
- Neutralizer Injection Pump A-1104P-26 A/B

These are all Officine Meccaniche Gallaratesi S.p.A. pumps Model DOXA series L, having the following design features:

- manufactured according to API 675 Std
- double diaphragms to avoid leakage due to single diaphragm rupture
- continuous adjustment of plunger stroke length, i.e. of capacity from 0 to 100% while the pump is at rest or running
- metering accuracy better than 1% within the normal operating range of 10÷100% % of capacity
- NSPH required, less than 0.3 kg/cm² abs. at normal operating conditions, for plunger pumps
- crankcase made of cast iron and completely enclosed, with oil splash lubrication
- max. allowable thrust: 2000 N (design 2500 N)
- diaphragms operated through hydraulic circuit with feed and relief valves
- automatic control by means of a pneumatic positioner
- automatic control with positioner controlled by an electrically actuated monophase servomotor
- electric pulses generator and pulsemeter to meter a fixed liquid quantity



Figure 52: Double Diaphragm Metering Pump

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Notes:

No data are available for the following pumps:

- Stabilizer Reflux Pump P-1114 A/B
- Desalter Water Charge Pump P-1119 A/B

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TRAINING MODULE

CRUDE DISTILLATION UNIT (CDU)

UNIT: 11

Course Content:

Section 1 - General Description	
Section 2 - Process Flow Description	
Section 3 - Process Control	
Section 4 - Safeguarding Devices	X
Section 5 - Fire & Gas Systems	
Section 6 - Quality Control	
Section 7 - Cause & Effects	
Section 8 - Operating Procedures	
Section 9 - HSE	
Section 10 - Reference Document Index	

SECTION 4 : SAFEGUARDING DEVICES

4.1. Alarms and Trips

For the CDU alarms & trips, refer to attached list.

4.2. Safeguarding Description

The Emergency Shutdown System (ESD) provides trips and interlocks for preventing or controlling emergency situations which could give rise to hazardous situations leading to injuries to personnel, significant economic loss and/or undue environmental pollution.

Uncontrolled loss of containment is prevented by the provision of pressure safety relief valves and by the ESD system which automatically bring the relevant part of the plant to a safe condition.

A trip or interlock is composed of one or more initiators and one or more actions for preventing hazards. Each trip or interlock is provided with a reset and where necessary an operational override for start-up.

In general if an interlock is invoked, the interlock action is held on until all initiators have returned to the safe state and the interlock reset has been pressed.

In some cases part of the trip action is to set a flow or level controller to manual with its output set to 0% (closed) via the serial link to the DCS from the ESD System. This is strictly not part of the trip action but is done to prevent the operator restarting with the control valve fully open with maximum integral action. In these cases, the operation on the controller is not included in the description of the purpose of the trip but is included in the list of actions.

The following is a description of the interlock system of the CDU.

4.2.1. CDU Emergency Shutdown (011-UX-007)

Purpose: When the ESD button in the Main Control Room (MCR) is pressed, this trip will cause actions and other trips as detailed and ADP alarm is activated.

Initiators:

Initiator Tag No.	Time Delay	P&ID No.	Cause
011-UXHS-007	-	101	CDU Emergency shutdown (ESD)

Actions:

Action No.	Tag	P&ID No	Service
011-UX-001		104	Desalters Protection tripped
011-UX-002		112	T-1101 Inventory Isolation tripped
011-UX-020		113	P-1102 A/B Pump Protection tripped
011-UX-023		103	P-1103 A/B Pump Protection tripped
011-UX-021		115	P-1104 A/B Pump Protection tripped

011-UX-022	116	P-1105 A/B Pump Protection tripped
011-UX-003	115	T-1103 Protection tripped
011-UX-004	116	T-1104 Protection tripped
011-UX-005A	134	H-1101 Heater shutdown tripped
011-UX-026	131	D-1104 Inventory Isolation tripped
011-UX-017	112	T-1101 Protection tripped
011-FV-034	130	Steam to E-1121A/B closed
011-TV-301	130	Boiled feed water to E-1121 closed
011-UXA-007B	101	CDU ESD Auxiliary console (ADP)

Notes:

a) Trip is manually reset by 011-UHSR-007.

The following ESD printout shows the safety logic UX-007:

4.2.2. A-1101-D-01/02 Desalter Protection (11-UX-001)

Purpose: In the event of any of High high pressure in a Desalter or inventory isolation XV-015 is closed or an ESD, the following happen:

- The following pumps are tripped:
 - o Crude charge pumps - P-6001 A/B/C,
 - o Crude booster pumps – P-1101A/B,
 - o Desalter water recycle pumps – P-1118A/B and Desalter water charge pumps – P-1119A/B
- The Fuel gas trip (UX-005B)
- Fuel oil trip (UX-005C)
- Pilot gas trip (UX-005E)
- ADP alarm is activated

Initiators:

Initiator Tag No.	Time Delay	P&ID No.	Cause
011-PXAHH-501/504	-	104	High high pressure in a desalter
ZSC-015	-	105	Inventory isolation valve closed
011-UX-007	-	101	CDU Emergency Shutdown

Actions:

Action Tag No.	P&ID No	Service
P-6001 A/B/C	060-010	060-UX-06 Crude charge pumps tripped
011-UX-008	119	P-1119 A/B Pump Protection tripped
011-UX-011	105	P-1101 A/B Pump Protection tripped
011-UX-018	104	P-1118 A/B Pump Protection tripped
011-UX-005B	134	H-1101 Fuel gas tripped
011-UX-005C	134	H-1101 Fuel oil tripped
011-UX-005E	134	H-1101 Pilot gas tripped

Notes:

- a) Trip is manually reset by 11-UHSR-001
- b) XV-015 reset is done via local push button 11-XHSR-015

The following ESD printout shows the logics UX-001, UX-018, UX-027, UX-028, UX-133, and UX-134:

4.2.3. A-1101-D-01 Transformer TR01B Protection (011-UX-027)

Purpose: In the event of any of the following:

- Low low operating level in the Desalter
- Low low oil operating level in the transformer TR-01B
- High-high oil operating pressure in the transformer TR-01B

Then the transformer TR-01B is stopped.

Initiators:

Initiator Tag No.	Time Delay	P&ID No.	Cause
011-LXALL-502	-	104	Low low operating level in Desalter A-1101-D-01
011-LXALL-519	-	104	Low low oil operating level in transformer TR-01B
011-PXAHH-582	-	104	High high oil pressure in transformer TR-01B

Actions:

Action Tag No.	P&ID No	Service
011-MXS-027	104	TR-01B transformer stopped

Notes:

- a) Trip is manually reset by 11-UHSR-027

4.2.4. A-1101-D-02 Transformer TR01A Protection (011-UX-133)

Purpose: In the event of any of the following:

- Low low operating level in the Desalter
- Low low oil operating level in the transformer TR-01A
- High-high oil operating pressure in the transformer TR-01A

Then the transformer TR-01A is stopped.

Initiators:

Initiator Tag No.	Time Delay	P&ID No.	Cause
011-LXALL-502	-	104	Low low operating level in Desalter A-1101-D-01
011-LXALL-517	-	104	Low low oil operating level in transformer TR-01A
011-PXAHH-580	-	104	High high oil pressure in transformer TR-01A

Actions:

Action Tag No.	P&ID No	Service
011-MXS-133	104	TR-01A transformer stopped

Notes:

- a) Trip is manually reset by 11-UHSR-133

4.2.5. A-1101-D-02 Transformer TR-02B Protection (011-UX-028)

Purpose: In the event of any of the following:

- Low low operating level in the Desalter
- Low low oil operating level in the transformer TR-02B
- High-high oil operating pressure in the transformer TR-02B

Then the transformer TR-02B is stopped.

Initiators:

Initiator Tag No.	Time Delay	P&ID No.	Cause
011-LXALL-505	-	105	Low low operating level in Desalter A-1101-D-02
011-LXALL-520	-	105	Low low oil operating level in transformer TR-02B
011-PXAHH-583	-	105	High high oil pressure in transformer TR-02B

Actions:

Action Tag No.	P&ID No	Service
011-MXS-132	105	TR-02B transformer stopped

Notes:

- a) Trip is manually reset by 11-UHSR-028

4.2.6. A-1101-D-02 Transformer TR-02A Protection (011-UX-134)

Purpose: In the event of any of the following:

- Low low operating level in the Desalter
- Low low oil operating level in the transformer TR-02A
- High-high oil operating pressure in the transformer TR-02A

Then the transformer TR-02A is stopped.

Initiators:

Initiator Tag No.	Time Delay	P&ID No.	Cause
011-LXALL-505	-	105	Low low operating level in Desalter A-1101-D-02
011-LXALL-518	-	105	Low low oil operating level in transformer TR-02A
011-PXAHH-581	-	105	High high oil pressure in transformer TR-02A

Actions:

Action Tag No.	P&ID No	Service
011-MXS-134	105	TR-02A transformer stopped

Notes:

- a) Trip is manually reset by 11-UHSR-134

4.2.7. D-1104 Inventory Isolation (011-UX-026)

Purpose: In the event of any of the following:

- manual initiation of the hand switches in the field
- Initiation from CDU ESD
- Low-low level in D-1104
- XV-002 Inventory isolation close push button remote and local
- XV-002 ESD from local panel

Then, D-1104 bottoms outlet valve XV-002 is closed, isolation valve trip alarm is activated and the discharges pumps are tripped.

Initiators:

Initiator Tag No.	Time Delay	P&ID No.	Cause
011-UXHS-028	-	131	Local initiation pushbutton shutdown
011-UX-007	-	131	CDU emergency shutdown
011-LXALL-032		131	Low low level in D-1104
011-XHSC-002A		131	Inventory isolation close pushbutton (remote)
011-XHSC-002B		131	Inventory isolation close pushbutton (local)
011-XHSC-002B	-	131	XV-002 ESD from local panel

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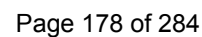
Actions:

Action Tag No.	P&ID No	Service
011-XV-002	131	D-1104 inventory isolation valve
011-UX-006	131	P-1114 A/B & P-1115A/B tripped
011-XXA-002	131	XV-002 trip alarm in local panel
011-MXS-134	105	TR-02A transformed stopped

Notes:

- a) Trip is manually reset by 11-UHSR-026
- b) XV-002 reset is done via local push button 011-XHSR-002

The following ESD printout shows the logics UX-002, UX-003, UX-004, UX-014, UX-017, UX-026, UX-040, UX-041 and UX-042:



4.2.8. H-1101 Heater Protection (011-UX-005)

This section specifies the following trips:

- 011-UX-005A Heater Trip, which shuts down the heater completely
- 011-UX-005B Fuel Gas Trip, which shuts down gas firing
- 011-UX-005C Fuel Oil Trip, which shuts down fuel oil firing
- 011-UX-005D Offgas Trip, which stops offgas firing
- 011-UX-005E Pilot gas Trip, which shuts down pilot gas

4.2.8.1. Heater trip (11-UX-005A)

Purpose: A heater trip is used to shut the heater down on an unsafe condition.

Initiators:

Initiator Tag No.	Time Delay	P&ID No.	Cause
011-PXAHH-017	-	111	High high pressure on crude charge heater
011-LXAHH-079	-	134	High high level in fuel gas K.O. Drum D-1114
011-UX-007	-	134	CDU ESD
011-UXHS-005A	-	134	Local emergency shutdown
011-UXHS-005AA	-	134	Remote emergency shutdown

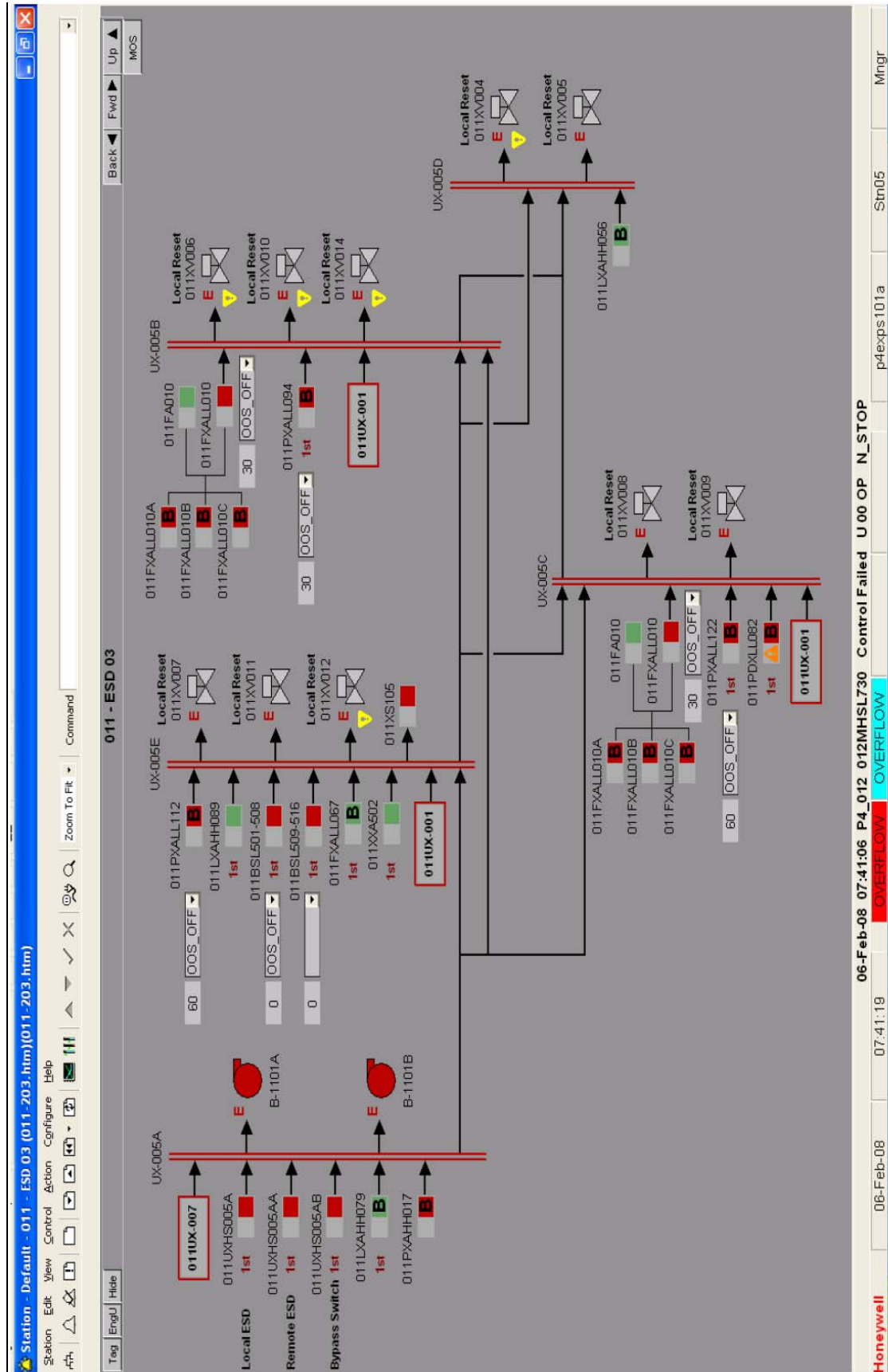
Actions:

Action Tag No.	P&ID No	Service
011-UX-005B	134	Fuel Gas trip initiated
011-UX-005C	134	Fuel Oil trip initiated
011-UX-005E	134	Pilot Gas initiated
011-B-1101A/B	133	Heater Forced Draft Fans A/B stopped
011-UXA-005B	134	Heater ESD auxiliary console (ADP) Annunciator Alarm activated

Notes:

- Trip is manually reset by 011-UHSR-005A
- After trip has been reset, a timer (1.5 minutes) is initiated to override the relevant trip inputs to enable the heater to be brought on-line.

The following ESD printout shows the logics UX-005 A/B/C/D/E:



	TRAINING MODULE		DOC NO: 8474L-011-A5016-0000-001-001	
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4.2.8.2. Fuel Gas trip (11-UX-005B)

Purpose: To shut down the main fuel gas burners and route off gas to atmosphere in the event of the following:

- Low pass feed flow to crude charge heater
- Low-low pressure fuel gas to burner
- 011-UX-005A Heater trip
- 011-UX-005 E Pilot gas trip
- 011-UX-001 Desalter protection trip

Initiators:

Initiator Tag No.	Time Delay	P&ID No.	Cause
011-FXALL-010 A/B/C	-	110	Low pass feed flow to crude charge heater
011-PXALL-094	-	134	Low low pressure fuel gas to burner
011-UX-005A	-	134	H-1101 Heater trip
011-UX-005E	-	134	Pilot gas trip
011-UX-001	-	134	Desalter Protection trip

Actions:

Action Tag No.	P&ID No	Service
011-XV-006	134	Burner fuel gas isolation valve closed
011-XV-010	134	Burner fuel gas isolation valve closed
011-XV-014	134	Burner fuel gas isolation valve opened
011-UX-005D	-	Off gas trip initiated
011-PIC-092	134	Fuel gas controller set to manual and output 0%

Notes:

- Trip is manually reset by 011-UHSR-005B
- XV-006 reset is done via local push button 011-XHSR-006
- XV-010 reset is done via local push button 011-XHSR-016
- XV-014 reset is done via local push button 011-XHSR-014
- After the trip has been reset, a timer (1.5 min) is initiated to override the the fuel gas low low pressure to enable the heater to be brought on line.

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- f) An alarm shall be provided if 011-XV-006 and 011-XV-010 are not closed 5 seconds after the trip is initiated.

4.2.8.3. Fuel Oil trip (11-UX-005C)

Purpose: To shut down the fuel oil burners and route off gas to atmosphere in the event of the following:

- Low pass feed flow to crude charge heater
- Low-low pressure fuel gas to burner
- 011-UX-005A Heater trip
- 011-UX-005 E Pilot gas trip
- 011-UX-001 Desalter protection trip

Initiators:

Initiator Tag No.	Time Delay	P&ID No.	Cause
011-FXALL-010 A/B/C	-	110	Low pass feed flow to crude charge heater
011-PXALL-122	-	135	Low low pressure fuel oil to burners
011-PDXALL-082	-	0024	Low differential pressure on fuel oil steam to burners
011-UX-005A	-	-	Heater trip
011-UX-005E	-	-	Pilot gas trip
011-UX-001	-	0002	Desalter Protection trip

Actions:

Action Tag No.	P&ID No	Service
011-XV-008	135	Fuel oil supply isolation valve closed
011-XV-009	135	Fuel oil supply isolation valve closed
011-UX-005D	-	Off gas trip initiated
011-PIC-118	135	Fuel oil controller set to manual and output 0%

Notes:

- Trip is manually reset by 011-UHSR-005C
- XV-008 reset is done via local push button 011-XHSR-008
- XV-009 reset is done via local push button 011-XHSR-009

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- d) After the trip has been reset, a timer (1.5 min) is initiated to override the fuel oil low low pressure to enable the heater to be brought on line.
- e) An alarm shall be provided if 011-XV-008 and 011-XV-009 are not closed 5 seconds after the trip is initiated.

4.2.8.4. Heater off gas (11-UX-005D)

Purpose: To route Vacuum Package off gas to atmosphere in the event of the following:

- 011-UX-005B Heater fuel gas protection
- 011-UX-005C Heater fuel oil protection
- High-high level on off gas special piping item
- 011-UX-005E Heater Pilot gas trip
- 011-XV-004-004 Close pushbutton from local panel
- 011-XV-005 Close pushbutton from local panel

Initiators:

Initiator Tag No.	Time Delay	P&ID No.	Cause
011-UX-005B	-	134	Heater fuel gas trip
011-UX-005C	-	134	Heater fuel oil trip
011-LXAAH-056		134	High-high level on SP-004 off gas special item
011-UX-005E	-	134	Heater pilot gas trip
011-XV-004	-	134	Close pushbutton from local panel
011-XV-005		134	Close pushbutton from local panel

Actions:

Action Tag No.	P&ID No	Service
011-XV-005	134	Off gas vent to burners closed
011-XV-004	134	Off gas vent to atmosphere opened

Notes:

- a) Trip is manually reset by 011-UHSR-005D
- b) XV-004 reset is done via local push button 011-XHSR-004
- c) XV-005 reset is done via local push button 011-XHSR-005

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- d) The trip can not be reset if both the fuel gas and the fuel oil trips are active.

4.2.8.5. Heater Pilot Gas Trip (11-UX-005E)

Purpose: To shutdown the pilot gas and route off gas to atmosphere in the event of the following:

- Low low pressure on pilot gas to H-1101
- 011-UX-00A Heater trip
- High-high level on pilot gas special piping item
- Lose of flame in more than 4 burners per cell
- 011-XV-007 open pushbutton from local per cell
- 011-XV-011 open pushbutton from local panel
- 011-XV-012 close pushbutton from local panel
- Low-low air/fuel ratio to crude charge heater
- Low-low air flow to heater forced draft fan
- 011-UX-001 Desalters protection trip

Initiators:

Initiator Tag No.	Time Delay	P&ID No.	Cause
011-PXALL-112	-	134	Low-low pressure on pilot gas to H-1101
011-UX-005A	-	134	Heater trip
011-BE-501 to 516		138	Lose of flame in more than 4 burners per cell
011-XV-007	-	134	Open pushbutton from local panel
011-XV-011	-	134	Pen pushbutton from local control panel
011-XV-012	-	134	Close pushbutton from local control panel
011-XXALL-502		134	Low-low air/fuel ratio to crude charge heater
011-FAXLL-067		133	Low-low air flow to heater forced draft fan
011-UX-001		134	Desalters protection trip

Actions:

Action Tag No.	P&ID No	Service
011-XV-007	134	Pilot gas supply closed
011-XV-011	134	Pilot gas supply closed
011-XV-012	134	Pilot vent valve opened
011-UX-005D	134	Heater off gas trip
011-UX-005C	134	Main fuel oil shutdown
011-UX-005B	134	Main fuel gas shutdown
011-UXA-005B	134	Heater ESD auxiliary console (ADP) Annunciator Alarm activated

Notes:

- Trip is manually reset by 011-UHSR-005E
- XV-011 reset is done via local push button 011-XHSR-011
- XV-012 reset is done via local push button 011-XHSR-012
- The trip can not be reset if both the fuel gas and the fuel oil trips are active.
- An alarm shall be provided if XV-007 & XV-011 are not close 5 sec after the trip is initiated.

4.2.8.6. Partial Loss of Flame

Each ionisation detector has been provided with a local and DCS indication of flame on/off and an individual alarm of loss of flame on DCS.

In case an individual pilot flame is lost, an alarm will be shown on DCS, and it is recommended that operator verifies the local panel indication of on/off flame on the affected burner and the absence of pilot flame on the burner itself. Then it is recommended to proceed as follow:

- Manually close the hand valve on the individual pilot line, and re-install the individual pilot blind
- Manually close the hand valve on the individual fuel gas line, off gas line, fuel oil and atomization steam line to burner and re-install the blind on the lines to the burner.

Ionization detectors shall be checked as well as fuel gas pilot supply to this burner etc, in order to determine the cause of the partial loss of flame. The pilot shall not be put into service until the cause has been solved.

If partial loss of flame occurs to the following extension:

- Less than 4 pilot flames confirmation from cell 1
- Less than 4 pilot flames confirmation from cell 2

Then a pilot shutdown will be initiated (see 011-UX-005E).

4.2.9. P-1101 A/B Pump Protection (011-UX-011)

Purpose: In the event of any of the following:

- Desalter Protection trip
- Low low crude flow to E-1105 A-J
- Low low crude flow to E-1106 A-F
- Trip of P-1101A protection from MMS
- Trip of P-1101B protection from MMS
- Desalters inventory isolation valve closed

Then, the Crude Booster Pumps P-1101A/B are shutdown.

Initiators:

Initiator Tag No.	Time Delay	P&ID No.	Cause
011-UX-001	-	105	Desalter protection trip
011-FXALL-058	-	105	Low low flow to E-1105A-J
011-FXALL-059		105	Low low flow to E-1105A-F
011-UXS-091		0041-602	Trip of P-1101A protection from MMS
011-UXS-092		0041-602	Trip of P-1101B protection from MMS
011-XV-015		105	Desalters inventory isolation valve closed

Actions:

Action Tag No.	P&ID No	Service
P-1101A	105	Crude Booster pump A tripped
P-1101B	0003	Crude Booster pump B tripped

Notes:

- a) Trip is manually reset by 11-UHSR-011

4.2.10. P-1102 A/B Top P/A Pumps Protection (011-UX-020)

Purpose: In the event of one of the CDU ESD or a low low flow in P-1102 A/B Pumps, the pumps are stopped

Initiators:

Initiator Tag No.	Time Delay	P&ID No.	Cause
011-UX-007	-	101	CDU ESD
011-FXALL-060	-	113	Low low flow in P-1102 A/B P/A pumps

Actions:

Action Tag No.	P&ID No	Service
P-1102A	113	Top P/A pump stopped
P-1102B	113	Top P/A pump stopped
011-FIC-011	113	Discharge flow controller set to manual and output 0%
011-UXA-020B	113	Top P/A P-1102 A/B auxiliary console

Notes:

- a) Trip is manually reset by 11-UHSR-020

4.2.11. P-1104 A/B Pump Protection (011-UX-021)

Purpose: In the event of any of the following:

- CDU ESD
- A low low flow in the P-1104 A/B discharge line
- Trip protection from MMS

Then, the pumps are stopped and auxiliary console ADP is activated.

Initiators:

Initiator Tag No.	Time Delay	P&ID No.	Cause
011-UX-007	-	101	CDU ESD
011-FXALL-061	-	115	Low low flow in the discharge line from the Kerosene P/A Pumps
011-UXS-093		041-602	P-1104 trip protection from MMS
011-UXS-094		041-602	P-1104B trip protection from MMS

Actions:

Action Tag No.	P&ID No	Service
P-1103A	115	Kerosene P/A Pump stopped
P-1103B	115	Kerosene P/A Pump stopped
011-FIC-013	115	Kerosene flow controller set to manual and output 0%
011-UXA-021	115	P-1104A/B trip auxiliary console ADP

Notes:

- a) Trip is manually reset by 11-UHSR-021

4.2.12. P-1106 A/B Residue Pump Protection (011-UX-025)

Purpose: In the event of any of the following:

- Low low level in the main fractionator
- XV-001 P-1106 A/B suction valve closed
- UX-002 main fractionator inventory isolation trip
- Trip pump protection from MMS

Then, the LGO pump around pumps P-1104A/B are shutdown and auxiliary console ADP is activated.

Initiators:

Initiator Tag No.	Time Delay	P&ID No.	Cause
011-LXALL-008		112	Low low level in T-1101 main fractionator
011-XZSC-001	-	112	XV-001 P-1106 A/B suction valve closed
011-UX-002		109	T-1101 main fractionator inventory isolation trip
011-UXS-095	-	041-602	P-1106A trip pump protection from MMS
011-UXS-096	-	041-602	P-1106B trip pump protection from MMS

Actions:

Action Tag No.	P&ID No	Service
P-1104A	109	LGO Pumparound Pump
P-1104B	109	LGO Pumparound pump
UXA-025B	109	P-1106A/B trip auxiliary console ADP
011-FIC-064	125	Atmospheric residue to P-1101A/B

Action Tag No.	P&ID No	Service
		flow controller set to manual and output 0%
011-FQIC-029	125	Atmospheric residue to RFCC flow controller set to manual and output 0%
011-FQIC-026	125	Atmospheric residue to storage flow controller set to manual and output 0%
011-FQIC-027	125	Atmospheric residue to storage flow controller set to manual and output 0%

Notes:

- a) Trip is manually reset by 11-UHSR-025

4.2.13. P-1107 A/B Kerosene product pump Protection (011-UX-009)

Purpose: In the event of low-low level in the kerosene stripper T-1102, the kerosene product pumps P-1107A/B are shutdown.

Initiators:

Initiator Tag No.	Time Delay	P&ID No.	Cause
011-LXALL-010		114	Low low level in the Kerosene stripper T-1102

Actions:

Action Tag No.	P&ID No	Service
P-1107A	114	Kerosene product pumps P-1107A tripped
P-1107B	114	Kerosene product pumps P-1107B tripped
011-FQIC-033	129	Kerosene from P-1107A/B flow controller set to manual and output 0%

Notes:

- a) Trip is manually reset by 11-UHSR-009

4.2.14. T-1101 Main Fractionator Inventory Isolation (011-UX-002)

Purpose: In the event of any of the following:

- CDU ESD
- Local inventory isolation pushbutton pressed
- XV-001 inventory isolation close pushbutton local pressed

- Low-low level in the main fractionator T-1101
- XV-001 Close pushbutton local
- XV-001 ESD from local panel

Then, the shut off valve in the suction line is closed, the Atmospheric Residue product pumps are stopped (via P-1106A/B Protection), the steam injection to T-1101 is stopped (via T-1101 Protection) and the local panel trip alarm and T-1101 isolated auxiliary console alarms are activated.

Initiators:

Initiator Tag No.	Time Delay	P&ID No.	Cause
011-UX-007	-	101	CDU ESD
011-UXHS-002	-	112	Local inventory isolation pushbutton
011-XHSC-001A	-	112	XV-001 inventory isolation close pushbutton local
011-LXALL-008	-	112	Low-low level in main fractionator column
011-XHSC-001B	-	112	XV-001 close pushbutton local
011-XHS-001B	-	112	ESD from local panel

Actions:

Action Tag No.	P&ID No	Service
011-UX-025	109	P-1106A/B Protection tripped
011-XV-001	112	Atmospheric residue valve closed
011-XV-017	112	T-1101 Protection tripped
011-XXA-001	112	XV-001 Local panel trip alarm activated
011-UXA-002B	112	T-1101 isolated auxiliary console ADP activated

Notes:

- Trip is manually reset by 11-UHSR-002
- XV-001 reset is done via local pushbutton 011-XHSR-001

4.2.15. T-1101 Main Fractionator Protection (011-UX-017)

Purpose: In the event of any of the following:

- High high level in the Main fractionator T-1101
- ESD trip signal

- T-1101 Inventory Isolation trip

Then, the superheated steam injection is stopped by closing the control valve.

Initiators:

Initiator Tag No.	Time Delay	P&ID No.	Cause
011-UX-007	-	101	CDU ESD trip
011-LXAHH-78	-	112	High high level in T-1101
011-UX-002		112	T-1101 Inventory Isolation trip

Actions:

Action Tag No.	P&ID No	Service
011-FSY-012	112	Steam flow control valve to T-1101
011-UXA-017B	112	T-1101 steam closed auxiliary console ADP

Notes:

- a) Trip is manually reset by 11-UHSR-017

4.2.16. T-1103 LGO Stripper protection (011-UX-003)

Purpose: In the event of any of the following:

- High high level in the LGO Stripper T-1103
- ESD trip signal

Then, the superheated steam injection is stopped by closing the control valve.

Initiators:

Initiator Tag No.	Time Delay	P&ID No.	Cause
011-UX-007	-	101	CDU ESD trip
011-LXAHH-014	-	115	High high level in T-1103

Actions:

Action Tag No.	P&ID No	Service
011-FSY-017	115	Steam flow control valve to T-1103 close
011-UXA-003B	115	T-1103 auxiliary console ADP

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Notes:

- a) Trip is manually reset by 11-UHSR-003

4.2.17. T-1104 LGO Stripper protection (011-UX-004)

Purpose: In the event of any of the following:

- High high level in the LGO Stripper T-1104
- ESD trip signal

Then, the superheated steam injection is stopped by closing the control valve.

Initiators:

Initiator Tag No.	Time Delay	P&ID No.	Cause
011-UX-007	-	101	CDU ESD trip
011-LXAHH-017	-	116	High high level in T-1104

Actions:

Action Tag No.	P&ID No	Service
011-FV-019	116	Steam flow control valve close
011-UXA-004B	116	T-1103 auxiliary console ADP

Notes:

- a) Trip is manually reset by 11-UHSR-004

4.2.18. T-1106 inlet valve closed (011-UX-040)

Purpose: In the event of low-low level in the HGO stripper T-1104, the inlet valve to T-1106 HGO Dryer is closed.

Initiators:

Initiator Tag No.	Time Delay	P&ID No.	Cause
011-LXALL-052	-	116	Low low level in T-1104

Actions:

Action Tag No.	P&ID No	Service
LV-022	118	Superheated steam flow control valve closed

Notes:

- a) Trip is manually reset by 11-UHSR-040

4.2.19. T-1105 inlet valve closed (011-UX-041)

Purpose: In the event of low-low level in the HGO stripper T-1103, the inlet valve to T-1105 HGO Dryer is closed.

Initiators:

Initiator Tag No.	Time Delay	P&ID No.	Cause
011-LXALL-053	-	115	Low low level in T-1103

Actions:

Action Tag No.	P&ID No	Service
LV-019	119	Superheated steam flow control valve closed

Notes:

- a) Trip is manually reset by 11-UHSR-041

4.2.20. FV-040/041 full range naphta outlet close valves (011-UX-042)

Purpose: In the event of low-low level in the stabilizer T-1107, the Full range Naphta outlet valves to NHT and storage are closed.

Initiators:

Initiator Tag No.	Time Delay	P&ID No.	Cause
011-LXALL-054	-	115	Low low level in T-1107

Actions:

Action Tag No.	P&ID No	Service
FV-040	132	Full range naphta flow control valve to NHT closed
FV-041	132	Full range naphta flow control valve to storage closed

Notes:

- a) Trip is manually reset by 11-UHSR-042

4.2.21. P-1103A/B Kerosene P/A pump protection (011-UX-023)

Purpose: In the event of any of the following:

- CDU ESD trip signal
- Low-low flow in P-1103A/B discharge

Then, the Kerosene P/A pumps are shutdown.

Initiators:

Initiator Tag No.	Time Delay	P&ID No.	Cause
011-UX-007	-	103	CDU ESD trip
011-FXALL-063	-	103	Low-low flow in P-1103A/B discharge

Actions:

Action Tag No.	P&ID No	Service
P-1103 A	103	Kerosene P/A P-1103A tripped
P-1103 B	103	Kerosene P/A P-1103A tripped
FIC-013	103	LGO to storage flow control set to manual and output 0%

Notes:

- a) Trip is manually reset by 11-UHSR-023

4.2.22. P-1105A/B HGO P/A pump protection (011-UX-022)

Purpose: In the event of any of the following:

- CDU ESD trip signal
- Low-low flow in P-1105A/B HGO P/A discharge

Then, the HGO P/A pumps are shutdown.

Initiators:

Initiator Tag No.	Time Delay	P&ID No.	Cause
011-UX-007	-	103	CDU ESD trip
011-FXALL-062	-	116	Low-low flow in P-1105A/B discharge

Actions:

Action Tag No.	P&ID No	Service
P-1105 A	116	HGO P/A P-1105A tripped
P-1105 B	116	HGO P/A P-1105B tripped
FIC-018	116	LGO flow control set to manual and output 0%

Notes:

- a) Trip is manually reset by 11-UHSR-022

4.2.23. P-1119A/B Desalter charge pumps protection (011-UX-008)

Purpose: When there is a low low level in the Desalter water surge drum D-1109 or on a trip signal from the Desalter Protection, the Desalter water pumps are stopped.

Initiators:

Initiator Tag No.	Time Delay	P&ID No.	Cause
011-LXALL-026	-	119	Low low level in D-1109
011-UX-001	-	119	Desalter protection trip

Actions:

Action Tag No.	P&ID No	Service
P-1119A	119	Desalter Water pump P-1119 A tripped
P-1119B	119	Desalter Water pump P-1119 B tripped
011-FIC-021	104	Desalter water flow controller set to manual and output 0%

Notes:

- a) Trip is manually reset by 11-UHSR-008

4.2.24. P-1118 A/B Desalter water recycle pumps protection (011-UX-018)

Purpose: In the event of:

- UX-001 Desalter Protection trip
- Low-low outlet flow from P-1118 A/B to FIC-004 and FIC-005

Then, the Desalter Water Recycle Pumps P-1118 A/B are shutdown.

Initiators:

Initiator Tag No.	Time Delay	P&ID No.	Cause
011-UX-001	-	119	Desalter protection trip
011-FXALL-056/057	-	104	Low low flow to A-1101-D-01 and low low flow to E-1101/E-1102

Actions:

Action Tag No.	P&ID No	Service
P-1118A	104	Desalter Water Recycle Pump tripped
P-1118B	104	Desalter Water Recycle Pump tripped
011-FIC-004	104	Flow controller on line to A-1101-D-01 set to Manual with 0% output

Action Tag No.	P&ID No	Service
011-FIC-005	104	Flow controller on line to E-1101 and E-1102 set to Manual with 0% output

Notes:

- a) Trip is manually reset by 11-UHSR-018

4.2.25. P-1114 A/B Stabilizer reflux and P-1115 A/B stabilizer LPG pumps protection (011-UX-006)

Purpose: In the event of:

- D-1104 Inventory Isolation (UX-026)
- Low low level in the stabilizer accumulator drum D-1104
- Pump suction isolation valve not open

Then, all running pumps are stopped.

Initiators:

Initiator Tag No.	Time Delay	P&ID No.	Cause
011-UX-026	-	131	D-1104 Inventory Isolation trip
011-LXALL-048	-	131	Low low level in D-1104
011-XZSC-002	-	131	Valve 011-XV-002 not open

Actions:

Action Tag No.	P&ID No	Service
P-1114A	130	Stabilizer Reflux Pump stopped
P-1114B	130	Stabilizer Reflux Pump stopped
P-1115A	131	Stabilizer LPG Pump stopped
P-1115B	131	Stabilizer LPG Pump stopped
011-FIC-036	131	Stabilizer Reflux flow controller set to manual and output of 0%
011-FQIC-037	131	LPG product to RFCC flow controller set to manual and output of 0%
011-LIC-050	131	Sour water controller set to manual and output of 0%

Notes:

- a) Trip is manually reset by 11-UHSR-006

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4.2.26. Main Fractionator accumulator drum water outlet valve LV-040 close (011-UX-014)

Purpose: when there is a low low level in the Overhead accumulator D-1103, the LV-040 valve is closed.

Initiators:

Initiator Tag No.	Time Delay	P&ID No.	Cause
011-LXALL-033	-	128	Low low level in D-1103

Actions:

Action Tag No.	P&ID No	Service
LV-040	128	Water D-1103 outlet valve close

Notes:

- a) Trip is manually reset by 11-UHSR-014

4.2.27. P-1113A/B HGO product pumps protection (011-UX-010)

Purpose: When there is a low low level in the HGO Drier T-1106 (or if T-1106 is by-passed, there is a low low level in the HGO Stripper T-1104), the HGO product pumps are stopped.

Initiators:

Initiator Tag No.	Time Delay	P&ID No.	Cause
011-LXALL-023 OR 011-LXALL-081 as selected by 011-UHS-010	-	118 116	Low low level in T-1106 Low low level in T-1104

Actions:

Action Tag No.	P&ID No.	Service
P-1113A	118	HGO product pump stopped
P-1113B	118	HGO product pump stopped
011-FQIC-028	127	HGO flow controller set to manual and output 0%

Notes:

- a) Trip is manually reset by 11-UHSR-010

4.2.28. P-1112A/B LGO product pumps protection (011-UX-012)

Purpose: When there is a low low level in the LGO Drier T-1105 (or if T-1105 is by-passed, there is a low low level in the LGO Stripper), the LGO product pumps are stopped.

Initiators:

Initiator Tag No.	Time Delay	P&ID No.	Cause
011-LXALL-020 OR 011-LXALL-080 as selected by 011-UHS-112	-	117 115	Low low level in T-1105 Low low level in T-1103

Actions:

Action Tag No.	P&ID No	Service
P-1112A	117	LGO product pump stopped
P-1112B	117	LGO product pump stopped
011-FQIC-020	117	LGO flow controller set to manual and output 0%

Notes:

- a) Trip is manually reset by 11-UHSR-012

4.2.29. P-1113A/B Stabilizer feed pumps protection (011-UX-013)

Purpose: When there is a low low level in the main fractionator accumulator drum D-1103, the stabilizer feed pumps are stopped.

Initiators:

Initiator Tag No.	Time Delay	P&ID No.	Cause
011-LXALL-039	-	128	Low low level in D-1103

Actions:

Action Tag No.	P&ID No	Service
P-1110A	128	Stabilizer feed pump stopped
P-1110B	128	Stabilizer feed pump stopped
011-FIC-032	128	Stabilizer feed flow controller set to manual and output 0%

Notes:

- a) Trip is manually reset by 11-UHSR-013

4.2.30. P-1120A/B Drier slop pumps protection (011-UX-015)

Purpose: When there is a low low level in the drier accumulator drum D-1106 slop oil partition, the slop pumps are stopped.

Initiators:

Initiator Tag No.	Time Delay	P&ID No.	Cause
011-LXALL-030	-	123	Low low level in D-1106 (slop)

Actions:

Action Tag No.	P&ID No.	Service
P-1120A	124	Slop pump stopped
P-1120B	124	Slop pump stopped

Notes:

- a) Trip is manually reset by 11-UHSR-015.

4.2.31. P-1121A/B Ejector condensate pumps protection (011-UX-016)

Purpose: When there is a low low level in the drier accumulator drum D-1106 water partition, the slop pumps are stopped.

Initiators:

Initiator Tag No.	Time Delay	P&ID No.	Cause
011-LXALL-031	-	123	Low low level in D-1106 (water)

Actions:

Action Tag No.	P&ID No.	Service
P-1121A	124	Water pump stopped
P-1121B	124	Water pump stopped
011-FIC-025	124	Condensate flow controller set to manual and output 0%

Notes:

- a) Trip is manually reset by 11-UHSR-016.

4.2.32. P-1121A/ Tempered water pumps protection (011-UX-024)

Purpose: When there is a low low level in the tempered water drum D-1115, the tempered water pumps are stopped.

Initiators:

Initiator Tag No.	Time Delay	P&ID No.	Cause
011-LXALL-069	-	126	Low low level in D-1115 (water)

Actions:

Action Tag No.	P&ID No	Service
P-1122A	126	Tempered Water pump stopped
P-1122B	126	Tempered Water pump stopped
011-FIC-030	126	Tempered Water flow controller set to manual and output 0%

Notes:

- a) Trip is manually reset by 11-UHSR-024.

The following paragraphs describe a series of scenarios where operator intervention is required to solve, or reduce, the contingency which may drives to an undesired plant performance, or, if prolonged, to a more severe problem.

4.2.33. Heat exchanger tube rupture

Although this is a very unlikely scenario, operator monitoring and foreseen sampling allows an early detection and therefore minimize the impact downstream the affected heat exchanger. Once the leak has been detected, the Heat exchanger should be by-passed, the flow rates reduced (when required), put under maintenance in order to repair or isolate the tube or tubes.

On the CDU unit, the rupture of a tube affects mainly the products, therefore, the performance of the specified analysis for each stream will allow detecting any major leak. For further details refer to Quality Control section.

4.2.34. Chemical injection packages, A-1104

A failure on the antifoulant or demulsifying packages will lead to a bad performance of the unit. Loss or reduction of demulsifier implies foam formation within the desalters and excessive water carryover to exchanger bank and column. Reduction or loss of antifoulant leads to a fouling of the heat exchanger train, temperature loss on crude feed to desalters and to increasing temperature on product rundowns.

Impact on desalters' performance may be detected via the following level alarms: 011-LAH/LAL-503, 506, 501 and 504.

Operator will therefore supervise both chemical injection packages: pumps performance and settings, level on reservoirs, etc. in order to ensure the proper operation.

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4.2.35. Desalter Package A-1101

Level within desalter vessels is maintained via level control 011-LIC-503 and LIC-506, which controls the interphase between the crude and the water phases. In case of failure of this controller (for instance: wrong level reading), 011-LV-003, “water to ETP” control valve will move to full opened or closed position.

- In case of valve full closed, 011-LAH-501 and 011-LAH-504 and 011-FQAL-051 will warn the operator about this scenario. If this situation is not soon restored, water will go through the crude outlet into the system causing salt deposition on the pre-heat train and tray damage in the T-1101. Therefore, operator action is required: 011-LIC-503 shall be put into manual mode and the correct set point to keep the valve on the required position will be given by the operator, or, if there is not a satisfying reply from the system, bypass valve shall be opened to re-establish the level in the desalters.
- In the case of valve full opened, 011-LAL-501 and 011-LAL-504 will warn the operator. If this situation is not soon restored, crude will not be properly desalted and a crude leak to the ETP will happen.

Due to the size of the desalter vessels, A-1101-D-01 and D-02, the available time, from the moment on any (high or low) level alarms are activated on the DCS screen to the moment when crude (or water) leaks, is enough to allow an operator to confirm the situation at field and then to proceed with the corrective actions (estimated time: 20 mins).

Additionally, a full trouble shooting guide is included on the desalter package “Operating and Maintenance manual” – 8474L-011-A5016-0814-001-001.

4.2.36. Kerosene pump around and rundown systems

In section 7.3 of the process control narrative, 8474L-011-SP-1511-101, the kerosene pump around control is described. Its purpose is to remove a certain amount of heat from the T-1101. This system is controlled via 011-UIC-029.

A failure on this controller, which is very unlikely, implies that the kerosene product may run out of spec.

011-AE/AT-003/005 and 006 on-line analyzers have been foreseen to warn the operator about any deviation from the product specifications: specific gravity, freeze point and flash points (this analyzer includes and alarm on low level) are monitored.

Additionally, the continuous supervision of the system parameters and the performance of the tests described on the Quality Control Section will allow the operator to detect and correct any deviation from the product specifications.

4.2.37. LGO pump-around and rundown systems

LGO pump-around control loop is described on the section 7.4 of the process control narrative. The amount of heat removed from the LGO stream and the flow of run down product is controlled via the 011-UIC-032.

As for the case of the Kerosene system, the failure of this logic leads to an off-spec LGO production.

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The continuous monitoring of the system parameters and the performance of the tests foreseen on the Quality Control Section will allow the operator to detect and correct any deviation from the product specifications.

If required, in case of a big deviation from the product spec, the unit feed rate shall be reduced and the product diverted to the Heavy Slops in order to allow the operator solving the causes of the deviation.

4.2.38. HGO pump around and rundown systems

As explained on process control narrative, 8474L-011-SP-1511-101 section 7.5, the pump around flow and the amount of duty removed on both the HGO pump around and on the Kerosene Reboiler E-1110 is controlled by 011-UIC-031 and 033.

Although this is a very unlikely scenario, the failure on any of these logic controllers leads to wrong operation of the unit, where products will run off spec.

The continuous monitoring of the system parameters and the performance of the tests foreseen on the Quality Control Section will allow the operator to detect and correct any deviation from the product specifications.

As explained on the previous points, in case of a big deviation from the expected values, the unit feed rate should be reduced in order to minimize the off-spec products' flow and then the production diverted to the Heavy Slops.

4.2.39. T-1101 accumulator drum, D-1103, Interphase controller (011-LIC-040) failure

Level control philosophy is described on the section 3.

A failure on the 011-LIC-040 drives to full open or closed the 011-LV-040, which maintain the Water/crude interphase level in D-1103. In both cases, system warns the operator who has to react in order to correct the situation.

- In case of 011-LV-040 drives open. This scenario implies that the interphase level is lost with a risk of Hydrocarbon carryover to the D-1106. In this case, 011-LXALL-033 will first alarm about the loss of level and then close the 011-LV-040 via solenoid valve. This solenoid valve is activated by the ESD Interlock UX-014. The field operator has to verify that 011-LV-040 is closed. Once the interphase level is re-established on the vessel, operator will maintain it on the expected values by means of the globe by-pass valve and the 011-LG-037 until the problem on 011-LIC-040 is solved.
- In case of 011-LV-040 drives closed. This scenario implies the risk of water carryover to the stabilizer, T-1107. In this case, 011-LZL/LZAL-040 will warn the operator about the failure of the controller and the closure of the control valve. Field operator, by means of the by-pass control valve will restore the interphase level until the problem on 011-LIC-040 is solved.

4.2.40. T-1107 reflux drum, D-1104, Water boot level controller (011-LIC-050) failure

For level control description refer to section 3

In case of 011-LIC-050 failure, 011-LV-050 may go to full open position with the risk of water level loss and LPG carryover from the D-1104 to the D-1103. An open position alarm, 011-LZASO-050 will warn the operator about this scenario.

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Although pressure control system on D-1103 (011-PIC-064) can manage the pressure rise, field operator should isolate control valve to cut the LPG leak to the flare system, and then maintain the required level on the water boot by means of the by-pass globe valve. This valve is accessible from the local level indicator: 011-LG-051.

4.2.41. D-1106 oil level, 011-LIC-028 fails on P-1120 A/B starting

Purpose of 011-LIC-028 is to start the Oil drain pump P-1120 A (or B) when oil reaches a high level. This loop is described in the section 10.10 of the control narrative 8474L-011-SP-1511-101.

If 011-LIC-028 fails, oil level continues to rise lasting around 25 minutes to fill completely the vessel from the High liquid level. In any case, 011-LIAHH-030 will warn to the operator that high liquid level has been passed. Once it has been confirmed that the P-1120 A (or B) is not running, operator will proceed to put the pump into manual mode, via 011-MHS-043A (or 011-MHS-044A for the P-1120B). Following, Oil drain pump must be started by local push button.

4.2.42. Crude Charge Heater H-1101

Refer to vendor document 8474L-011-A3501-0110-001-002 for a complete description of the safeguarding actions for H-1101.

4.3. Safeguarding Equipment

4.3.1. Pressure Safety Devices

Over pressuring of the equipment occurs in many ways. The basic cause of power pressure is imbalance in heat and material flow in the equipment or piping. Pressure safety devices have been installed to protect and/or section against over pressure.

Here is the list of the pressure safety valves found in the CDU:

TAG No.	Description	Type	Location	PID: 8474L-011-PID-0021 -	Setting (kg/cm ² g)
011-TSV-313	Relief Valve (Thermal)	RV	RELIEF FROM E-1106D TO T-1101	108	37.9
011-PSV-020	Relief Valve (Spring)	RV	LGO PRODUCT FROM T-1105 TO FLARE	117	3.5
011-TSV-305	Relief Valve (Thermal)	RV	RELIEF FROM E-1103A TO T-1101	102	37.8
011-TSV-306	Relief Valve (Thermal)	RV	RELIEF FROM E-1103B TO T-1101	102	37.8
011-TSV-307	Relief Valve (Thermal)	RV	RELIEF FROM E-1104 TO T-1101	103	37.8
011-PSV-004	Relief Valve (Pilot)	RV	RELIEF FROM A-1101-D-01 TO T-1101	104	13.8

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TAG No.	Description	Type	Location	PID: 8474L-011-PID-0021 -	Setting (kg/cm ² g)
011-PSV-005	Relief Valve (Pilot)	RV	RELIEF FROM A-1101-D-01 TO T-1101	104	13.8
011-TSV-311	Relief Valve (Thermal)	RV	RELIEF FROM E-1109 TO T-1101	107	37.7
011-TSV-310	Relief Valve (Thermal)	RV	RELIEF FROM E-1107 TO T-1101	107	37.7
011-TSV-312	Relief Valve (Thermal)	RV	RELIEF FROM E-1106B TO T-1101	108	37.9
011-TSV-314	Relief Valve (Thermal)	RV	RELIEF FROM E-1108D TO T-1101	109	37.9
011-PSV-085	Relief Valve (Spring)	RV	SUPHTR STM FROM H-1101 TO T-1101	110	6.3
011-PSV-086	Relief Valve (Spring)	RV	SUPHTR STM FROM H-1101 TO T-1101	110	6.3
011-PSV-161	Relief Valve (Spring)	RV	OVRHD VAP FROM T-1101 TO FLARE	112	3.5
011-PSV-166	Relief Valve (Spring)	RV	OVRHD VAP FROM T-1101 TO FLARE	112	3.5
011-PSV-106	Relief Valve (Spring)	RV	HGO FROM T-1106 TO FLARE	118	3.5
011-PSV-107	Relief Valve (Spring)	RV	HGO FROM T-1106 TO FLARE	118	3.5
011-PSV-021	Relief Valve (Spring)	RV	LGO PRODUCT FROM T-1105 TO FLARE	117	3.5
011-PSV-237	Relief Valve (Spring)	RV	OVRHD RELIEF FROM T-1107 TO FLARE	130	9.6
011-PSV-238	Relief Valve (Spring)	RV	OVRHD RELIEF FROM T-1107 TO FLARE	130	9.6
011-PSV-131	Relief Valve (Spring)	RV	FUEL GAS FROM D-1114 TO FLARE	134	6.7
011-PSV-132	Relief Valve (Spring)	RV	FUEL GAS FROM D-1114 TO FLARE	134	6.7
011-PSV-555	Relief Valve (Spring)	RV	DISCHARGE PUMP A-1104-P-26-B	136	8.2
011-PSV-557	Relief Valve (Spring)	RV	DISCHARGE PUMP A-1104-P-25-B	136	8.3
011-PSV-558	Relief Valve (Spring)	RV	DISCHARGE PUMP A-1104-P-25-A	136	8.3
011-PSV-561	Relief Valve (Spring)	RV	DISCHARGE PUMP A-1104-P-23-A	137	22.0
011-PSV-562	Relief Valve (Spring)	RV	DISCHARGE PUMP A-1104-P-23-B	137	22.0
011-PSV-563	Relief Valve (Spring)	RV	DISCHARGE PUMP A-1104-P-24-A	137	22.0
011-PSV-564	Relief Valve (Spring)	RV	DISCHARGE PUMP A-1104-P-24-B	137	22.0
011-TSV	Relief Valve	RV	SW FROM E-1128D TO	120	38.9

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TAG No.	Description	Type	Location	PID: 8474L-011-PID-0021 -	Setting (kg/cm ² g)
-316	(Thermal)		OW		
011-TSV-317	Relief Valve (Thermal)	RV	SW FROM E-1128A TO OW	120	38.9
011-TSV-315	Relief Valve (Thermal)	RV	RELIEF FROM E-1108C TO T-1101	109	37.9
011-TSV-319	Relief Valve (Thermal)	RV	TPW RELIEF FROM E-1120C	125	22.7
011-TSV-320	Relief Valve (Thermal)	RV	TPW RELIEF FROM E-1120D	125	22.7
011-TSV-308	Relief Valve (Thermal)	RV	RELIEF FROM E-1105A TO T-1101	106	37.8
011-TSV-309	Relief Valve (Thermal)	RV	RELIEF FROM E-1105F TO T-1101	106	37.8
011-PSV-168	Relief Valve (Spring)	RV	PROCESS VAPOUR RELIEF FROM D-1115	126	6.9
011-PSV-169	Relief Valve (Spring)	RV	PROCESS VAPOUR RELIEF FROM D-1115	126	6.9
011-PSV-170	Relief Valve (Spring)	RV	PV FROM D-1106 TO FLARE	123	11
011-PSV-171	Relief Valve (Spring)	RV	PV FROM D-1106 TO FLARE	123	11
011-PSV-172	Relief Valve (Pilot)	RV	RELIEF FROM A-1101-D-02 TO T-1101	105	13.8
011-PSV-173	Relief Valve (Pilot)	RV	RELIEF FROM A-1101-D-02 TO T-1101	105	13.8
011-PSV-175	Relief Valve (Spring)	RV	HP STEAM RELIEF	130	27.2
011-PSV-176	Relief Valve (Spring)	RV	HP STEAM RELIEF	130	27.2
011-PSV-556	Relief Valve (Spring)	RV	DISCHARGE PUMP A-1104-P-26-A	136	8.2
011-TSV-318	Relief Valve (Thermal)	RV	CWR FROM A-1102-E-31/32	122	6.9
011-TSV-321	Relief Valve (Thermal)	RV	CWR RELIEF FROM E-1119	127	10.8
011-TSV-323	Relief Valve (Thermal)	RV	COOLING WATER RETURN FROM E-1127	132	9.1
011-PSV-239	Relief Valve (Spring)	RV	DESALTER WATER SURGE DRUM D-1109	119	2.7
011-TSV-322	Relief Valve (Thermal)	RV	CWR RELIEF FROM E-1115	129	11.9
011-PSV-165	Relief Valve (Spring)	RV	OVRHD VAP FROM T-1101 TO FLARE	112	3.5
011-PSV-162	Relief Valve (Spring)	RV	OVRHD VAP FROM T-1101 TO FLARE	112	3.5
011-PSV-163	Relief Valve (Spring)	RV	OVRHD VAP FROM T-1101 TO FLARE	112	3.5

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TAG No.	Description	Type	Location	PID: 8474L-011-PID-0021 -	Setting (kg/cm ² g)
011-PSV-164	Relief Valve (Spring)	RV	OVRHD VAP FROM T-1101 TO FLARE	112	3.5
011-PSV-195	Relief Valve (Spring)	RV	RELIEF FROM E-1128A/B/C TO OW	120	15.3
011-PSV-196	Relief Valve (Spring)	RV	RELIEF FROM E-1128A/B/C TO OW	120	15.3
011-PSV-197	Relief Valve (Spring)	RV	RELIEF FROM E-1128D/E TO OW	120	15.3
011-PSV-198	Relief Valve (Spring)	RV	RELIEF FROM E-1128D/E TO OW	120	15.3
011-PSV-199	Relief Valve (Spring)	RV	PILOT GAS SPECIAL PIPING SP-003	134	6.7
011-PSV-240	Relief Valve (Spring)	RV	DESALTER WATER SURGE DRUM D-1109	119	2.7
011-TSV-244	Relief Valve (Thermal)	RV	RELIEF FROM E-1134A TO T-1101	107	37.7
011-TSV-245	Relief Valve (Thermal)	RV	RELIEF FROM E-1134B TO T-1101	107	37.7
011-TSV-613	Relief Valve (Thermal)	RV	API SEAL PLAN PUMP P-1115A	8474L-011-PID-0041-601	TBA
011-TSV-614	Relief Valve (Thermal)	RV	API SEAL PLAN PUMP P-1115B	8474L-011-PID-0041-601	TBA
011-TSV-615	Relief Valve (Thermal)	RV	API SEAL PLAN PUMP P-1101A	8474L-011-PID-0041-602	10.0
011-TSV-616	Relief Valve (Thermal)	RV	API SEAL PLAN PUMP P-1101A	8474L-011-PID-0041-602	10.0
011-TSV-617	Relief Valve (Thermal)	RV	API SEAL PLAN PUMP P-1101B	8474L-011-PID-0041-602	10.0
011-TSV-618	Relief Valve (Thermal)	RV	API SEAL PLAN PUMP P-1101B	8474L-011-PID-0041-602	10.0
011-TSV-647	Relief Valve (Thermal)	RV	API SEAL PLAN PUMP P-1102A	8474L-011-PID-0041-604	TBA
011-TSV-649	Relief Valve (Thermal)	RV	API SEAL PLAN PUMP P-1102B	8474L-011-PID-0041-604	TBA
011-TSV-648	Relief Valve (Thermal)	RV	API SEAL PLAN PUMP P-1102A	8474L-011-PID-0041-604	TBA
011-TSV-650	Relief Valve (Thermal)	RV	API SEAL PLAN PUMP P-1102B	8474L-011-PID-0041-	TBA

TAG No.	Description	Type	Location	PID: 8474L-011-PID-0021 -	Setting (kg/cm ² g)
				604	
011-TSV-636	Relief Valve (Thermal)	RV	API SEAL PLAN PUMP P-1103A	8474L-011-PID-0041-603	10.0
011-TSV-638	Relief Valve (Thermal)	RV	API SEAL PLAN PUMP P-1103B	8474L-011-PID-0041-603	10.0
011-TSV-619	Relief Valve (Thermal)	RV	API SEAL PLAN PUMP P-1104A	8474L-011-PID-0041-602	10.0
011-TSV-620	Relief Valve (Thermal)	RV	API SEAL PLAN PUMP P-1104A	8474L-011-PID-0041-602	10.0
011-TSV-621	Relief Valve (Thermal)	RV	API SEAL PLAN PUMP P-1104B	8474L-011-PID-0041-602	10.0
011-TSV-622	Relief Valve (Thermal)	RV	API SEAL PLAN PUMP P-1104B	8474L-011-PID-0041-602	10.0
011-TSV-627	Relief Valve (Thermal)	RV	API SEAL PLAN PUMP P-1105A	8474L-011-PID-0041-602	10.0
011-TSV-628	Relief Valve (Thermal)	RV	API SEAL PLAN PUMP P-1105B	8474L-011-PID-0041-602	10.0
011-TSV-623	Relief Valve (Thermal)	RV	API SEAL PLAN PUMP P-1106A	8474L-011-PID-0041-602	10.0
011-TSV-624	Relief Valve (Thermal)	RV	API SEAL PLAN PUMP P-1106A	8474L-011-PID-0041-602	10.0
011-TSV-625	Relief Valve (Thermal)	RV	API SEAL PLAN PUMP P-1106B	8474L-011-PID-0041-602	10.0
011-TSV-626	Relief Valve (Thermal)	RV	API SEAL PLAN PUMP P-1106B	8474L-011-PID-0041-602	10.0
011-TSV-629	Relief Valve (Thermal)	RV	API SEAL PLAN PUMP P-1107A	8474L-011-PID-0041-602	10.0
011-TSV-630	Relief Valve (Thermal)	RV	API SEAL PLAN PUMP P-1107B	8474L-011-PID-0041-602	10.0
011-TSV-640	Relief Valve (Thermal)	RV	API SEAL PLAN PUMP P-1112A	8474L-011-PID-0041-603	10.0
011-TSV	Relief Valve	RV	API SEAL PLAN PUMP P-	8474L-011-	10.0

TAG No.	Description	Type	Location	PID: 8474L-011-PID-0021 -	Setting (kg/cm ² g)
-642	(Thermal)		1112B	PID-0041-603	
011-TSV-644	Relief Valve (Thermal)	RV	API SEAL PLAN PUMP P-1113A	8474L-011-PID-0041-603	10.0
011-TSV-646	Relief Valve (Thermal)	RV	API SEAL PLAN PUMP P-1113B	8474L-011-PID-0041-603	10.0
011-TSV-633	Relief Valve (Thermal)	RV	API SEAL PLAN PUMP P-1118A	8474L-011-PID-0041-602	10.0
011-TSV-634	Relief Valve (Thermal)	RV	API SEAL PLAN PUMP P-1118B	8474L-011-PID-0041-602	10.0
011-TSV-651	Relief Valve (Thermal)	RV	API SEAL PLAN PUMP P-1127A	8474L-011-PID-0041-602	100
011-TSV-652	Relief Valve (Thermal)	RV	API SEAL PLAN PUMP P-1127B	8474L-011-PID-0041-602	10.0
011-PSV-684	Relief Valve (Spring)	RV	AUXILIARY OIL MIST GENERATOR	8474L-011-PID-0031-595	TBA
011-PSV-685	Relief Valve (Spring)	RV	MAIN OIL MIST GENERATOR	8474L-011-PID-0031-595	TBA
011-TSV-661	Relief Valve (Thermal)	RV	BULK OIL RESERVOIR	8474L-011-PID-0031-595	TBA
011-PSV-696	Relief Valve (Spring)	RV	A-1104-P26B	136	TBA
011-PSV-698	Relief Valve (Spring)	RV	A-1104-P26A	136	TBA
011-PSV-700	Relief Valve (Spring)	RV	A-1104-P25B	136	TBA
011-PSV-702	Relief Valve (Spring)	RV	A-1104-P25A	136	TBA
011-PSV-704	Relief Valve (Spring)	RV	A-1104-P23B	137	TBA
011-PSV-706	Relief Valve (Spring)	RV	A-1104-P23A	137	TBA
011-PSV-694	Relief Valve (Spring)	RV	A-1104-P24A	137	TBA
011-PSV-692	Relief Valve (Spring)	RV	A-1104-P24B	137	TBA
011-PSV-710	Relief Valve (Pilot)	RV	API SEAL PLAN PUMP P-1115A	8474L-011-PID-0041-	TBA

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				605	
011-PSV-712	Relief Valve (Pilot)	RV	API SEAL PLAN PUMP P-1115B	8474L-011-PID-0041-605	TBA

For the fluid discharge parameters through each RV for a particular event, refer to the Flare Discharge Summary: 8474L-011-NM-0006-001.

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TRAINING MODULE

CRUDE DISTILLATION UNIT (CDU)

UNIT: 11

Course Content:

Section 1 - General Description	
Section 2 - Process Flow Description	
Section 3 - Process Control	
Section 4 - Safeguarding Devices	
Section 5 - Fire & Gas Systems	X
Section 6 - Quality Control	
Section 7 - Cause & Effects	
Section 8 - Operating Procedures	
Section 9 - HSE	
Section 10 - Reference Document Index	

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SECTION 5 : FIRE & GAS SYSTEMS

5.1. Fire & Gas detection

Refer to the following documents:

8474-011-DW-1950-001 Fire and Gas Detectors Layout

8474L-011-DW-1960-001 Escape Routes Layout

5.1.1. Fire & Gas Detectors Layout

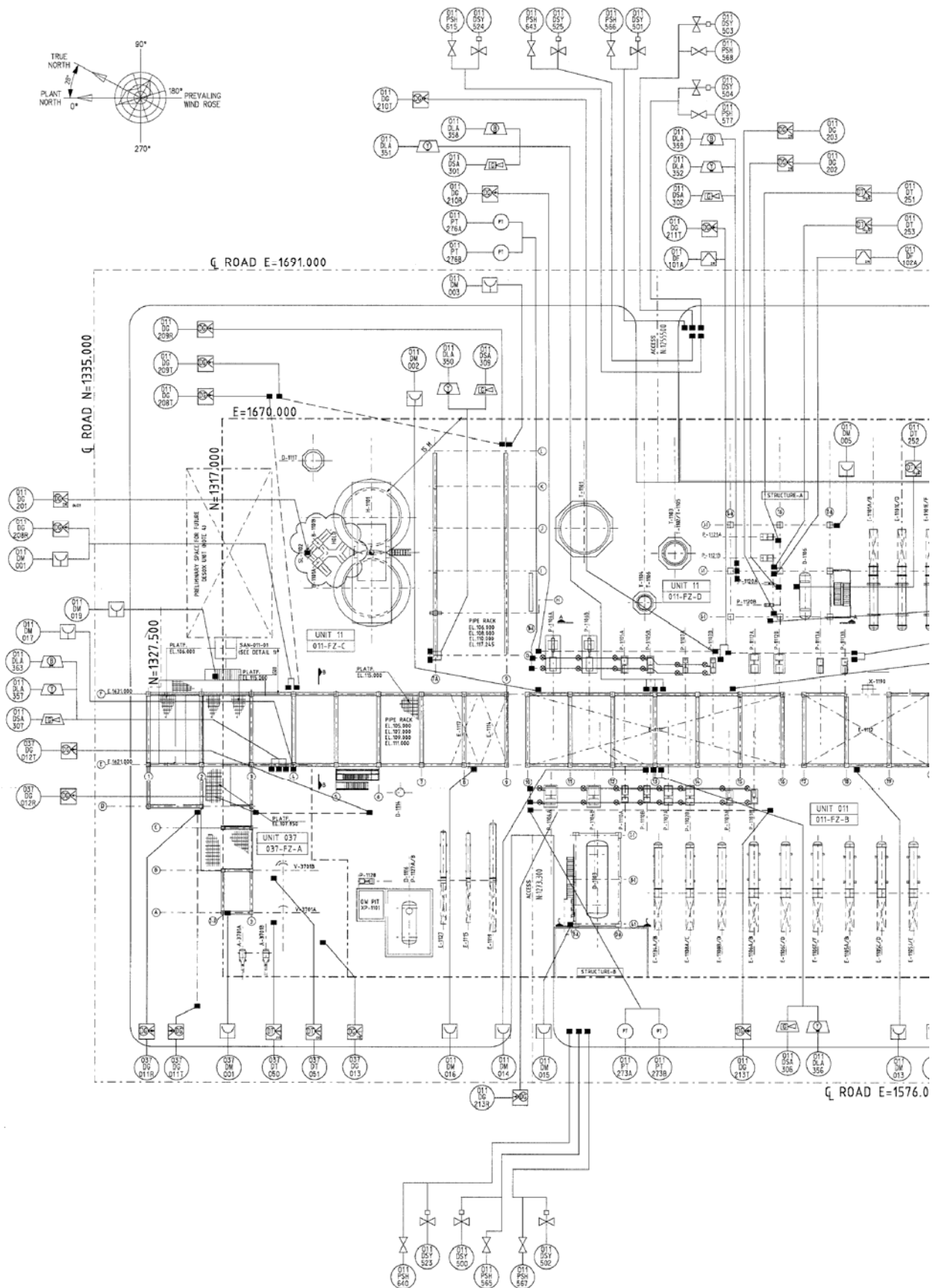


Figure 53: Fire & Gas Detectors Layout (1/2)

LEGEND:	DESCRIPTION	F&G DEVICE TAG LEGEND	ELEVATION	REMARKS
	MANUAL CALL POINT	DM	1.2 M (ABOVE GRADE/PLATFORM)	MOUNTED ON POLE
	IR FLAME DETECTOR	DF	2.5M MIN, 4.0M MAX (ABOVE GRADE)	MOUNTED ON POLE
	POINT TYPE TOXIC GAS DETECTOR (H ₂ S)	DT	MIN 0.5 M (ABOVE GRADE/PLATFORM)	MOUNTED ON POLE OR SUITABLE PLANT STRUCTURE
	POINT TYPE TOXIC GAS DETECTOR (CO)	DT	MIN 0.5 M (ABOVE GRADE/PLATFORM)	MOUNTED ON POLE OR SUITABLE PLANT STRUCTURE
	POINT TYPE FLAMMABLE GAS DETECTOR (INFRA RED)	DG	MIN 0.5 M (ABOVE GRADE/PLATFORM)	MOUNTED ON POLE OR SUITABLE PLANT STRUCTURE
	VISUAL ALARM TOXIC, GAS (BLUE)	DLA	MAX 4.0 M (ABOVE GRADE)	MOUNTED ON POLE OR SUITABLE PLANT STRUCTURE OR WALL.
	VISUAL ALARM FLAMMABLE GAS (YELLOW)	DLA	MAX 4.0 M (ABOVE GRADE)	MOUNTED ON POLE OR SUITABLE PLANT STRUCTURE OR WALL.
	VISUAL ALARM-GENERAL (RED)	DLA		WALL MOUNTED OUTSIDE OF ANALYSER HOUSE
	GAS AUDIBLE ALARM	DSA	MAX 4.0 M (ABOVE GRADE)	MOUNTED ON POLE OR SUITABLE PLANT STRUCTURE OR WALL.
	OPEN PATH FLAMMABLE GAS DETECTOR (INFRA RED)	DG	0.6 M ABOVE THE HIGHEST PUMP	MOUNTED ON POLE OR SUITABLE PLANT STRUCTURE AT A HEIGHT OF 0.6 M ABOVE THE HIGHEST PUMP IN A PUMP ROW.
	POINT TYPE FLAMMABLE GAS DETECTOR (INFRA RED DUCT MOUNTED)	DG	DUCT MOUNTED	AIR INTAKE DUCT
	POINT TYPE TOXIC GAS DETECTOR (H ₂ S) (DUCT MOUNTED)	DT	DUCT MOUNTED	AIR INTAKE DUCT
	PRESSURE TRANSMITTER	PT	BOX MOUNTED	
	POINT TYPE IONISATION SMOKE DETECTOR	DS	MOUNTED AT CEILING	
	POINT TYPE FLAMMABLE GAS DETECTOR (CATALYTIC)	DG	MIN 0.5 M (ABOVE GRADE)	INSIDE ANALYSER SHELTER
	PNEUMATIC HEAT DETECTION LINE			FITTED IN PUMPS
	SOLENOID 3-WAY VALVE	DSY	DELUGE VALVE MOUNTED	
	PRESSURE SWITCH	PSH	DELUGE VALVE MOUNTED	

Figure 55: Summary of Symbols and Elevations

FLAME DETECTORS TAGS	EQUIPMENT PROTECTED
014-DF-011-A 014-DF-011-B	P-1406A/B
011-DF-101-A 011-DF-101-B	P-1103A/B, P-1112A/B, P-1113A/B
011-DF-102-A 011-DF-102-B	P-1120A/B
011-DF-103-A 011-DF-103-B	P-1101A/B
011-DF-104-A 011-DF-104-B	P-1114A/B, P-1115A/B

Figure 56: Flame Detectors Protection

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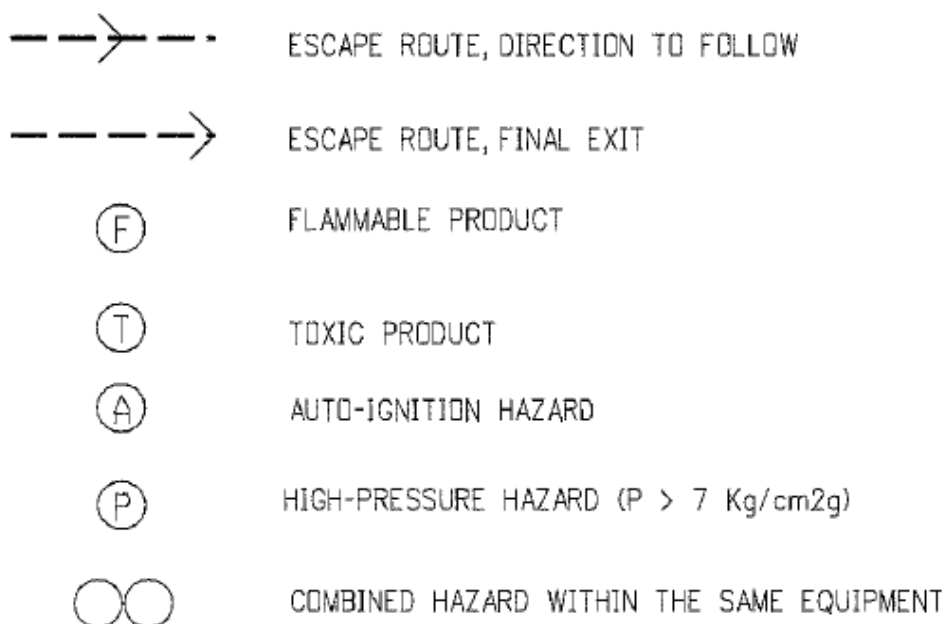


Figure 58: Legend for Escape Route Layout

5.2. Fire Protection

Refer to the following documents:

8474L-011-DW-1933-001 Safety Equipment Layout

8474L-011-DW-1933-002 Fire Protection Layout

5.2.1. Safety Equipment Layout

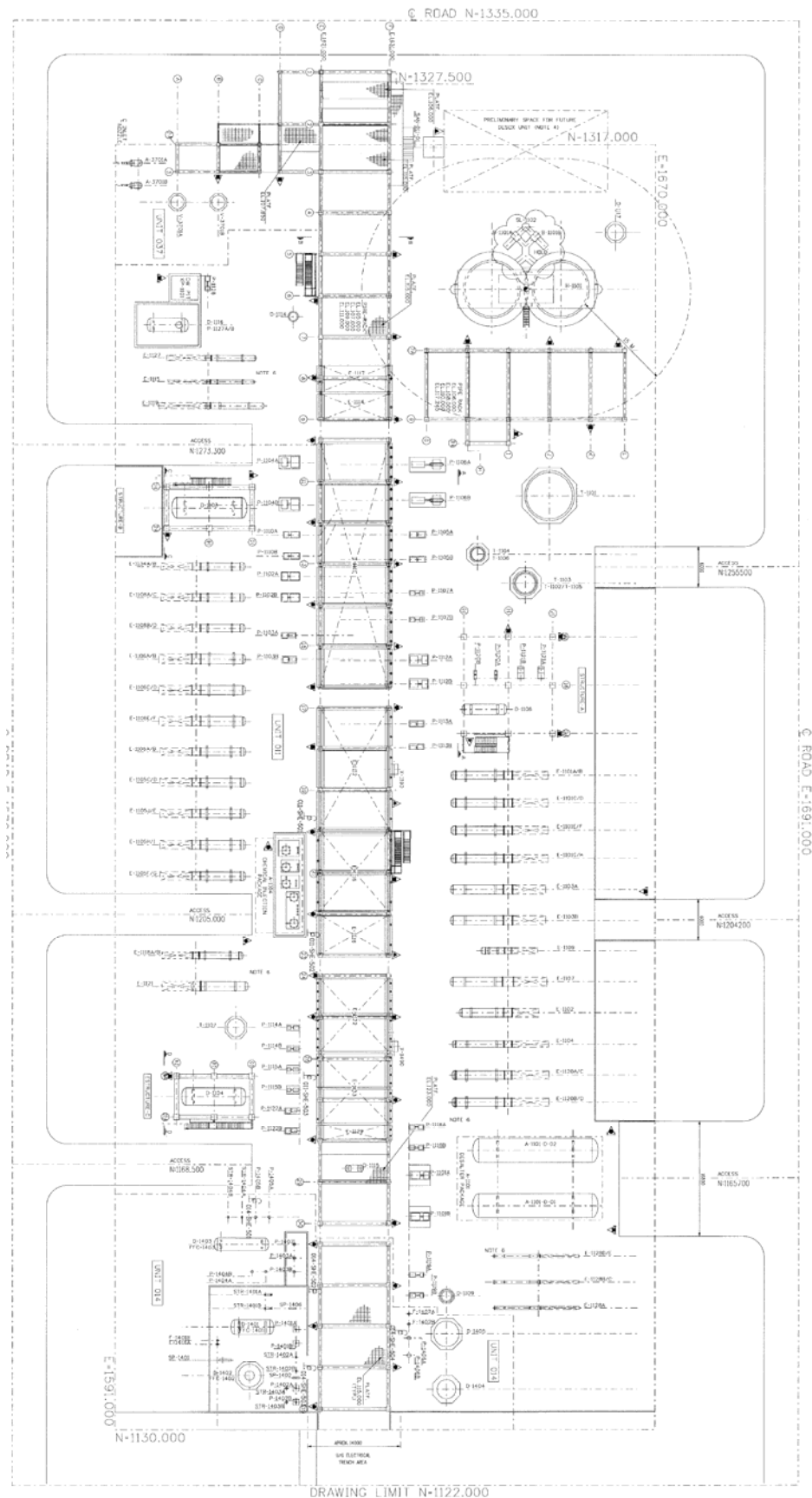


Figure 59: Safety Equipment Layout (1/2)

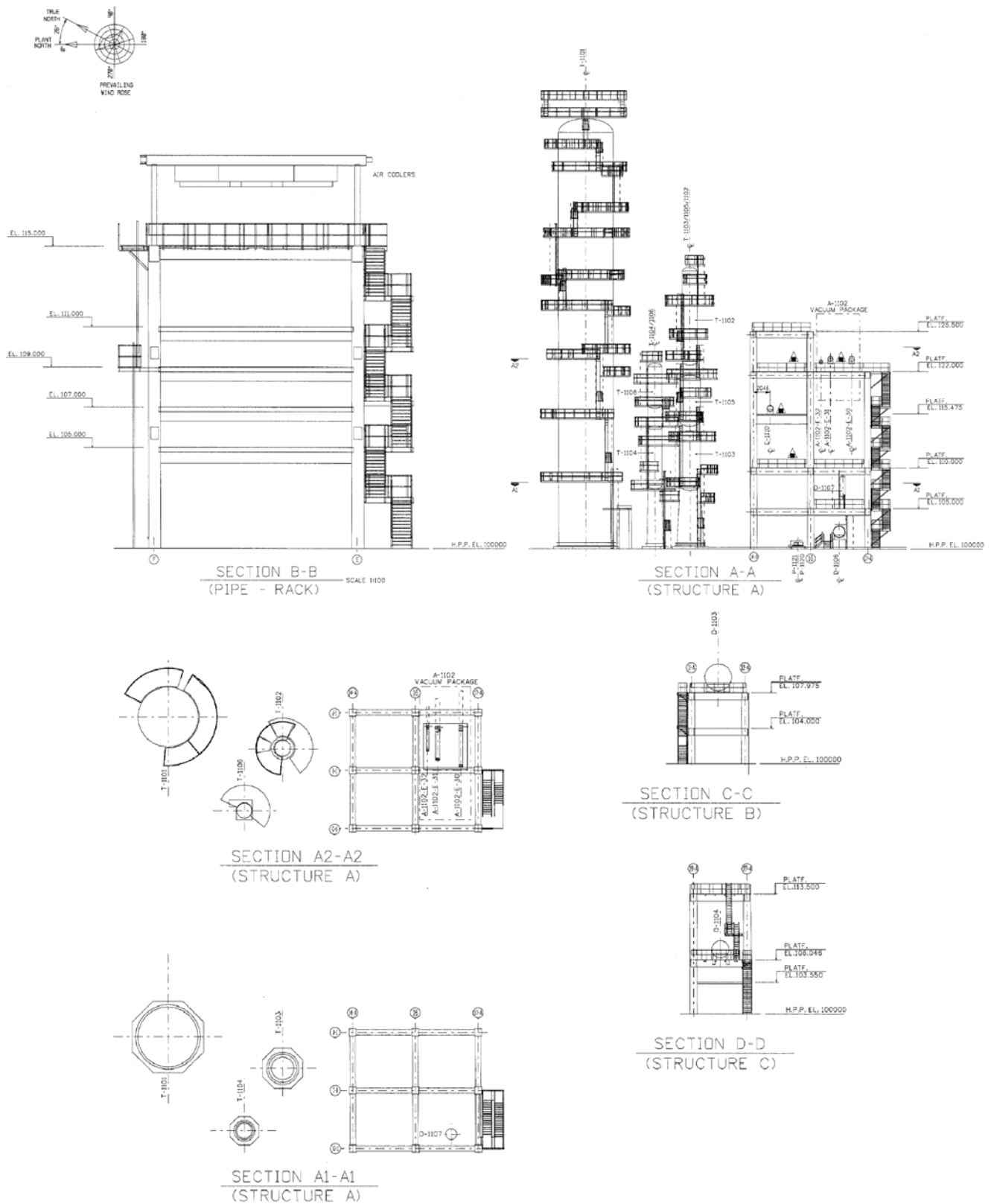


Figure 60: Safety Equipment Layout (2/2)

	TRAINING MODULE		DOC NO: 8474L-011-A5016-0000-001-001	
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SAFETY SHOWER & EYE WASH



9 KG DRY PORTABLE EXTINGUISHER



5 KG CARBON DIOXIDE PORTABLE EXTINGUISHER



9 LITRE FOAM PORTABLE EXTINGUISHER



50 KG POWDER TROLLEY UNITS



100 LITRE WHEELED FOAM EXTINGUISHER

Figure 61: Safety Equipment Legend

5.2.2. Fire Protection Layout





	TRAINING MODULE CRUDE DISTILLATION UNIT (CDU)	DOC NO: 8474L-011-A5016-0000-001-001 REV: 1	DATE: 18/03/08	
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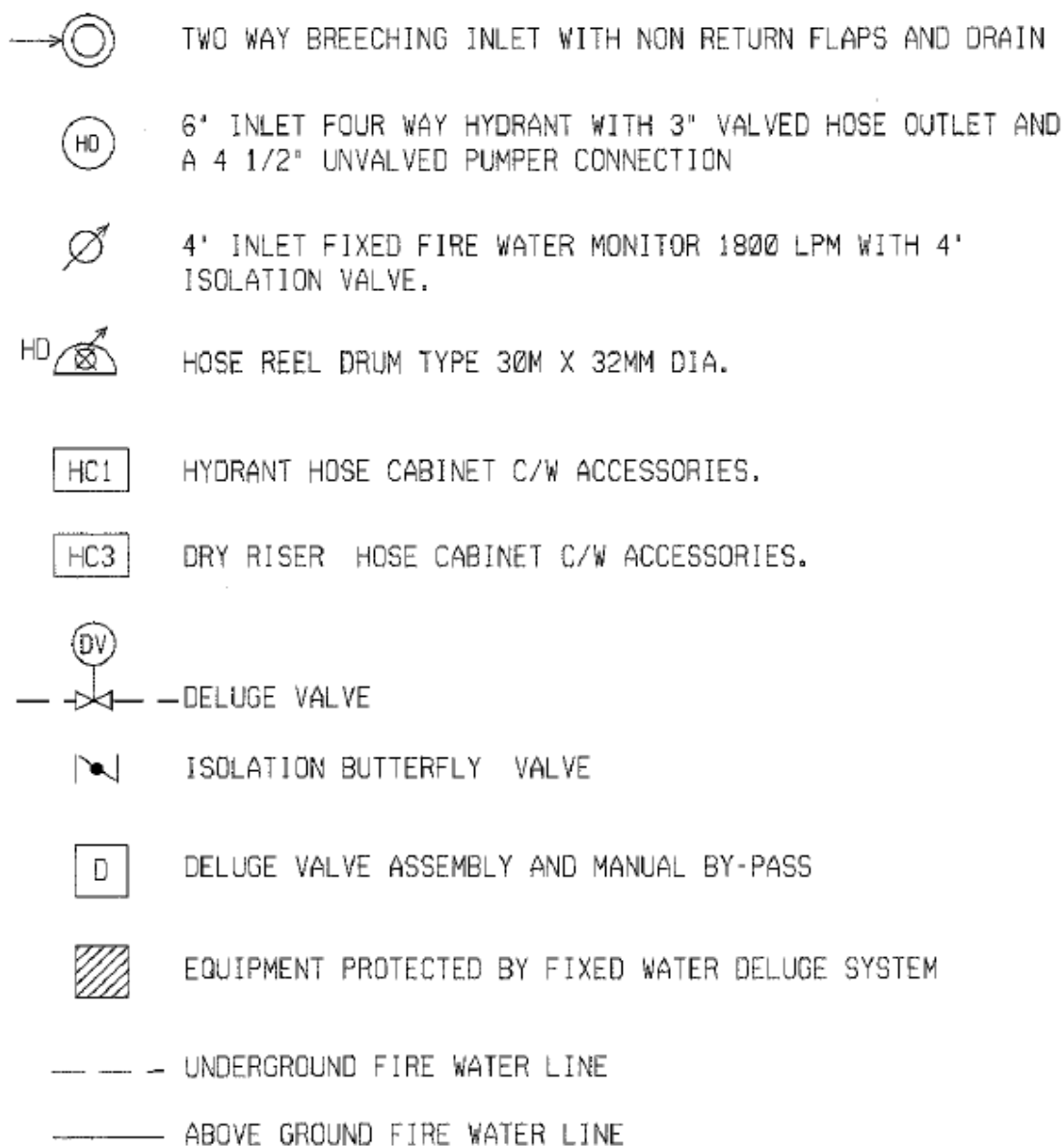
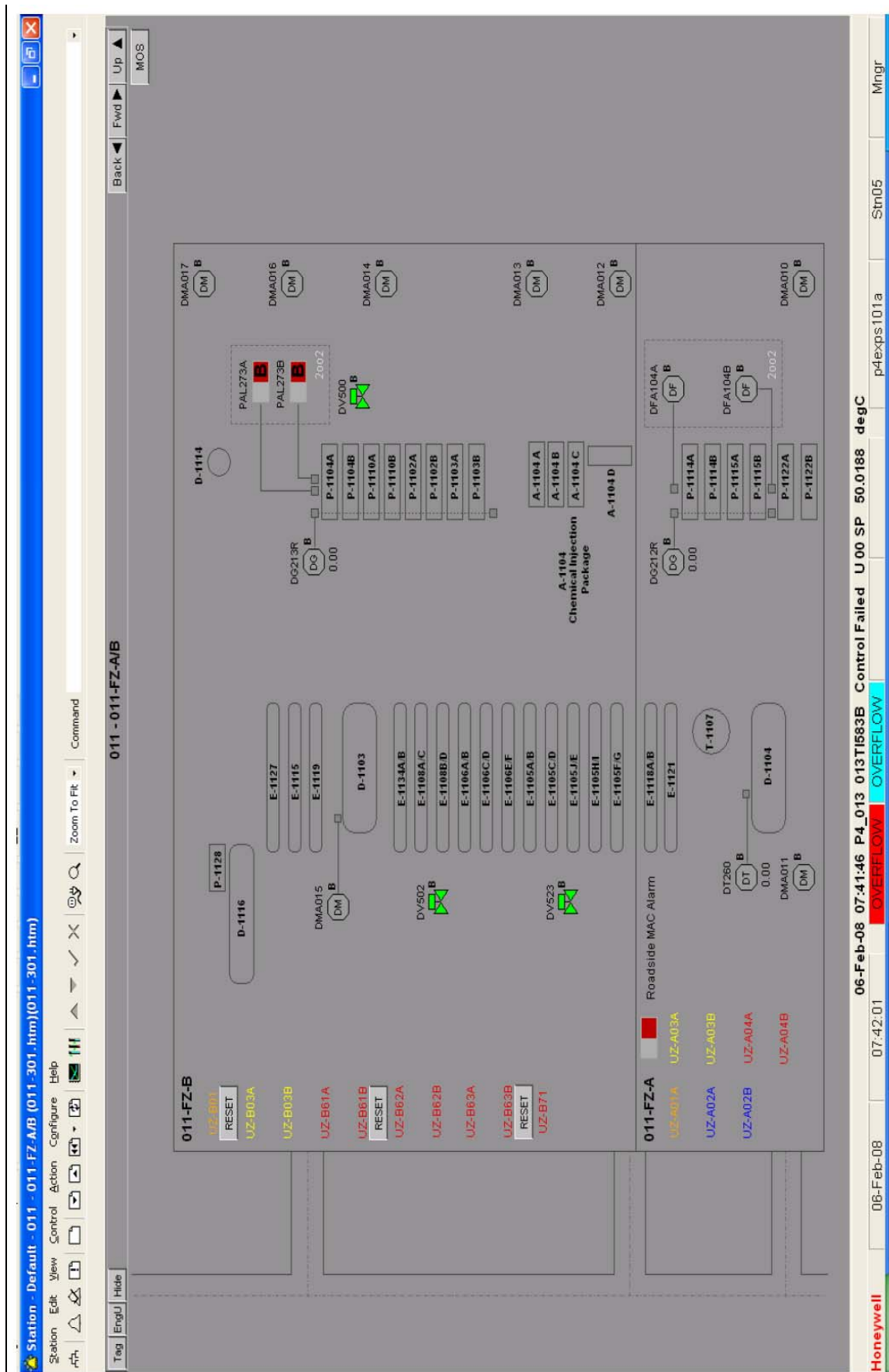
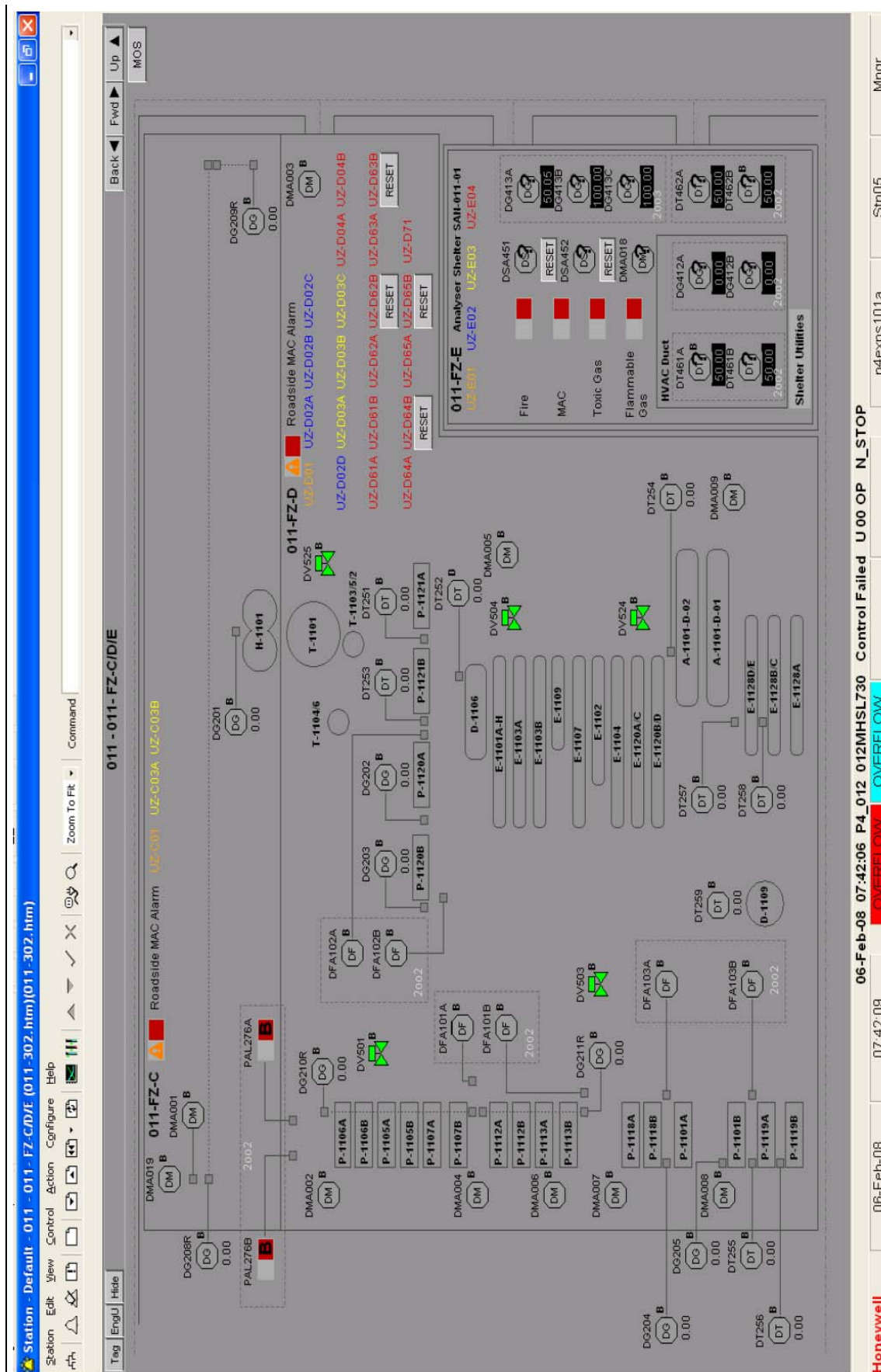


Figure 64: Fire Protection Layout Legend

The following Fire & Gas printouts displays the F&G system layout in the CDU and shows the alarms, detectors and F&G protection system located in this area:





	TRAINING MODULE		DOC NO: 8474L-011-A5016-0000-001-001	
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TRAINING MODULE

CRUDE DISTILLATION UNIT (CDU)

UNIT: 11

Course Content:

Section 1 - General Description	
Section 2 - Process Flow Description	
Section 3 - Process Control	
Section 4 - Safeguarding Devices	
Section 5 - Fire & Gas Systems	
Section 6 - Quality Control	X
Section 7 - Cause & Effects	
Section 8 - Operating Procedures	
Section 9 - HSE	
Section 10 - Reference Document Index	

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SECTION 6 : QUALITY CONTROL

In order to ensure that the products resulting from the operation of this unit are on spec, the unit facilities are provided with sample connections to analyse the process streams. Continuous monitoring of the plant performance by analysing the product streams also helps detect and prevent from major problems.

The first subsection displays tables listing the sample connections, the associated process stream, the properties to be analyzed along with methodology and the frequency of operation.

In addition, on-line analysers sample process streams on a continuous basis to make sure products are on specs and to trip the process in case of deviation. The second subsection describes the on-line analysers of this unit.

6.1. Sampling Analysis

The following table describes the sampling connections in the CDU.

Notes for the following tables:

(1): At sampling line connection to process

(2): at 50°C

(3): Refer to P&ID symbology 8474L-000-PID-0090-006 and 017

(4): Process P&ID / Detail P&ID

	TRAINING MODULE		DOC NO: 8474L-011-A5016-0000-001-001		
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ITEM TAG	P&ID No. (4)	SAMPLING SYSTEM TYPE(3)	SAMPLING SERVICE (Fluid)	PHASE (Gas/Liquid)	OPERATING CONDITIONS (1)		DESIGN CONDITIONS (1)		LIQUID VISCOSITY (cP) (2)	Line No	Analysed value	Methods applied	Frequency
					Pressure (kg/cm ² g)	Temperature (°C)	Pressure (kg/cm ² g)	Temperature (°C)					
001	120/301	SC-3B	DESALTER EFFL. WATER	LIQ	8.3	50	15.2	75	0.55	4*-DEW-110268	Oil content	ASTM D3921 or ASTM D7066	At request
											pH	ASTM D1283	At request
											Hydrogen sulfide	ASTM D4658 or UOP 683	At request
											Chloride	UOP 456	At request
											Total suspended solids	APHA 2540D	At request
001	132/301	SC-18	FULL RANGE NAPHTHA	LIQ	4.55	40	10.5	65	0.5	8*-PL-110085	Distillation curve	ASTM D0086	1/D
											C4+ content	ASTM D5134 or UOP551	1/D
001A	140/309	SCC-6	HC+STEAM	GAS	0.0	535	0.5	560	N/A	6*-PV-110523	CO2 & CO content	ASTM D6522	At request
001B	140/309	SCC-6	HC+STEAM	GAS	0.0	535	0.5	560	N/A	6*-PV-110524	CO2 & CO content	ASTM D6522	At request
001C	140/309	SCC-6	HC+STEAM	GAS	0.0	535	0.5	560	N/A	6*-PV-110525	CO2 & CO content	ASTM D6522	At request
001D	140/309	SCC-6	HC+STEAM	GAS	0.0	535	0.5	560	N/A	6*-PV-110526	CO2 & CO content	ASTM D6522	At request
001E	140/309	SCC-6	HC+STEAM	GAS	0.0	535	0.5	560	N/A	6*-PV-110527	CO2 & CO content	ASTM D6522	At request
001F	140/309	SCC-6	HC+STEAM	GAS	0.0	535	0.5	560	N/A	6*-PV-110528	CO2 & CO content	ASTM D6522	At request
001G	140/309	SCC-6	HC+STEAM	GAS	0.0	535	0.5	560	N/A	6*-PV-110529	CO2 & CO content	ASTM D6522	At request
001H	140/309	SCC-6	HC+STEAM	GAS	0.0	535	0.5	560	N/A	6*-PV-110530	CO2 & CO content	ASTM D6522	At request
002A	141/309	SCC-6	HC+STEAM	GAS	0.0	535	0.5	560	N/A	6*-PV-110515	CO2 & CO content	ASTM D6522	At request
002B	141/309	SCC-6	HC+STEAM	GAS	0.0	535	0.5	560	N/A	6*-PV-110516	CO2 & CO content	ASTM D6522	At request
002C	141/309	SCC-6	HC+STEAM	GAS	0.0	535	0.5	560	N/A	6*-PV-110517	CO2 & CO content	ASTM D6522	At request
002D	141/309	SCC-6	HC+STEAM	GAS	0.0	535	0.5	560	N/A	6*-PV-110518	CO2 & CO content	ASTM D6522	At request
002E	141/309	SCC-6	HC+STEAM	GAS	0.0	535	0.5	560	N/A	6*-PV-110519	CO2 & CO content	ASTM D6522	At request
002F	141/309	SCC-6	HC+STEAM	GAS	0.0	535	0.5	560	N/A	6*-PV-110520	CO2 & CO content	ASTM D6522	At request
002G	141/309	SCC-6	HC+STEAM	GAS	0.0	535	0.5	560	N/A	6*-PV-110521	CO2 & CO content	ASTM D6522	At request
002H	141/309	SCC-6	HC+STEAM	GAS	0.0	535	0.5	560	N/A	6*-PV-110522	CO2 & CO content	ASTM D6522	At request
001	131/309	SC-12	LPG	LIQ	23.4	50	28.2	75	0.12	3*-PL-110097	C2 to C4 composition	UOP 536 or ASTM D2163	1/D
											C5+ content	UOP 536 or ASTM D2163	1/D
001	110/309	SC-20	FUEL GAS	GAS	0.0	400	1.25	425	N/A	4*-VT-110493	O2, CO, CO2, SO2, H2S, Nox content	ASTM D6522 by external laboratory	At request
											VOC content	ASTM D3685 or	
											Dust content	ISO9096	At request

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ITEM TAG	P&ID No. (4)	SAMPLING SYSTEM TYPE(3)	SAMPLING SERVICE (Fluid)	PHASE (Gas/Liquid)	OPERATING CONDITIONS (1)		DESIGN CONDITIONS (1)		LIQUID VISCOSITY (cP) (2)	Line No	Analysed value	Methods applied	Frequency
					Pressure (kg/cm ² g)	Temperature (°C)	Pressure (kg/cm ² g)	Temperature (°C)					
001	201/301	SCC-5B	CRUDE OIL	LIQ	20.0	50	39.0	180	4.7	16"-PL-110001	Boiling Range Distribution	ASTM D5307	At request
											API gravity	ASTM D1298 or ASTM D4052	1/D
											Kinematic viscosity	ASTM D0445	At request
											Reid Pressure vapour	ASTM D0323	At request
											Total Sulphur	ASTM D4294	1/D
											Sediment and Water	ASTM D4007	1/D
											Pour Point	ASTM D0097	1/D
											Salt Content as NaCl	ASTM D9230	1/D
											Ash Content	ASTM D0482	At request
											Conradson Carbon Residue	ASTM D0189	At request
											Metal content,	ASTM D5863	At request
											Vanadium		
											Nickel		
002	104/301	SCC-5B	CRUDE OIL	LIQ	13	138	39	241	4.7	16"-PL-110012	Salt Content as Cl-	ASTM D9230	1/D
											Water content	ASTM D4928	At request
											Water and sediment content	ASTM D4007	At request
											Boiling Range Distribution	ASTM D5307	At request
											API gravity	ASTM D1298 or ASTM D4052	1/D
											Kinematic viscosity	ASTM D0445	At request
											Reid Pressure vapour	ASTM D0323	At request
											Total Sulphur	ASTM D4294	At request
											Pour Point	ASTM D0097	At request
											Ash Content	ASTM D0482	At request
											Conradson Carbon Residue	ASTM D0189	At request
											Metal content,	ASTM D5863	At request
											Vanadium		
											Nickel		

	TRAINING MODULE		DOC NO: 8474L-011-A5016-0000-001-001	
	CRUDE DISTILLATION UNIT (CDU)	REV: 1	DATE: 18/03/08	

ITEM TAG	P&ID No. (4)	SAMPLING SYSTEM TYPE(3)	SAMPLING SERVICE (Fluid)	PHASE (Gas/Liquid)	OPERATING CONDITIONS (1)		DESIGN CONDITIONS (1)		LIQUID VISCOSITY (cP) (2)	Line No	Analysed value	Methods applied	Frequency
					Pressure (kg/cm ² g)	Temperature (°C)	Pressure (kg/cm ² g)	Temperature (°C)					
003	105/301	SCC-5B	DESALTED CRUDE	LIQ	11.5	138	39.0	241	4.7	16"-PL-110015	Boiling Range Distribution	ASTM D5307	At request
											API gravity	ASTM D1298 or ASTM D4052	1/D
											Kinematic viscosity	ASTM D0445	At request
											Reid Pressure vapour	ASTM D0323	At request
											Total Sulphur	ASTM D4294	At request
											Sediment and Water	ASTM D4007	1/D
											Pour Point	ASTM D0097	At request
											Salt Content as NaCl	ASTM D3230	1/D
											Ash Content	ASTM D0482	At request
											Conradson Carbon Residue	ASTM D0189	At request
											Metal content, Vanadium	ASTM D5863	At request
											Nickel		
											Water content	ASTM D4928	1/D
004	201/301	SCC-5B	ATMOSPHERIC RESIDUE	LIQ	12.8	127	34.1	191	340	10"-PL-110057	API gravity	ASTM D1298 or ASTM D4052	1/D
											Flash point	ASTM D0093	1/D
											Residue vaporizing below 360 °C	ASTM D1160	1/D
											Total Sulphur	ASTM D4294	At request
											Pour Point	ASTM D0097	1/D
											Cloud Point		
005	125/301	SCC-5A	ATMOSPHERIC RESIDUE	LIQ	4.5	85	34.1	127	340	10"-PL-110061	API gravity	ASTM D1298 or ASTM D4052	1/D
											Flash point	ASTM D0093	1/D
											Residue vaporizing below 360 °C	ASTM D1160	1/D
											Total Sulphur	ASTM D4294	At request
											Pour Point	ASTM D0097	1/D
											Cloud Point		

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ITEM TAG	P&ID No. (4)	SAMPLING SYSTEM TYPE(3)	SAMPLING SERVICE (Fluid)	PHASE (Gas/Liquid)	OPERATING CONDITIONS (1)		DESIGN CONDITIONS (1)		LIQUID VISCOSITY (cP) (2)	Line No	Analysed value	Methods applied	Frequency
					Pressure (kg/cm ² g)	Temperature (°C)	Pressure (kg/cm ² g)	Temperature (°C)					
007	128/301	SCC-5A	SOUR WATER	LIQ	2.1	50	4.4	75	0.55	4"-SOW-110288	pH	ASTM D1293	At request
											Chlorides	UOP 456	At request
											Sulphides	ASTM D4658 or UOP 883	At request
											Ammonia	ASTM D14268 or UOP 740	At request
											Total dissolved salts	ASTM D1125	At request
											Metals	ISO 10359 (ICVN 6665)	At request
008	129/301	SCC-5B	KEROSENE	LIQ	9.0	40	17.4	65	0.82	8"-K-110250	Flash point	ASTM D0093	3/D
											Smoke point	ASTM D1322	At request
											API gravity	ASTM D1298 or ASTM D4052	1/D
											Freeze point	ASTM D2386	3/D
											Distillation curve	ASTM D0086	3/D
											Total Sulphur	ASTM D4294	At request
009	202/301	SCC-5A	FULL RANGE NAPHTHA	LIQ	4.0	50	10.5	100	0.5	8"-PL-110086	Distillation curve	ASTM D0086	3/D
											C4- content	UOP 551 or ASTM D51134	3/D
											API gravity	ASTM D1298 or ASTM D4052	1/D
											Total Sulphur	ASTM D4294	3/D
011	117/301	SCC-5A	LGO	LIQ	7.9	55	14.8	153	1.8	8"-LGO-110221	Distillation curve	ASTM D0086	3/D
											Flash point	ASTM D0093	3/D
											Pour Point	ASTM D0097	3/D
											API gravity	ASTM D1298 or ASTM D4052	1/D
											Cetane index	ASTM D0976 or ASTM D4737	3/D
											Colour	ASTM D1500	3/D
											Total Sulphur	ASTM D4294	1/D
											Water content	ASTM D6304	At request

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ITEM TAG	P&ID No. (4)	SAMPLING SYSTEM TYPE(S)	SAMPLING SERVICE (Fluid)	PHASE (Gas/Liquid)	OPERATING CONDITIONS (1)		DESIGN CONDITIONS (1)		LIQUID VISCOSITY (cP) (2)	Line No	Analysed value	Methods applied	Frequency
					Pressure (kg/cm ² g)	Temperature (°C)	Pressure (kg/cm ² g)	Temperature (°C)					
012	104/301	SCC-5B	DESALTER EFFL WATER	LIQ	23.9	136	39.0	180	0.55	4"-DEW-110275	pH	ASTM D1293	At request
											Hydrogen Sulfide	ASTM D4658 or UOP 883	At request
											Ammonia	ASTM D1426B or UOP 740	At request
											Chlorides	UOP 458	At request
											Iron	ISO 10359	At request
013	104/301	SCC-5B	CRUDE OIL	LIQ	14.5	138	15.0	175	4.7	1"-VT-110405	Water & Emulsion		Once a shift
											Water and sediment content	ASTM D4007	1/D
											Chlorides	ASTM D3230	1/D
014	105/301	SCC-5B	CRUDE OIL	LIQ	10.5	136	15.0	175	4.7	1"-VT-110406	Water & Emulsion		Once a shift
											Water and sediment content	ASTM D4007	1/D
											Chlorides	ASTM D3230	1/D
015	104/301	SCC-5B	DESALTER EFFL WATER	LIQ	13.0	138	15.2	172	0.55	4"-DEW-110265	pH	ASTM D1293	At request
											Hydrogen Sulfide	ASTM D4658 or UOP 883	At request
											Ammonia	ASTM D1426B or UOP 740	At request
											Chlorides	UOP 458	At request
											Iron	ISO 10359	At request
016	104/301	SCC-5B	SERVICE WATER	LIQ	16.0	120	39.0	145	0.55	4"-SW-116811	pH	ASTM D1293	At request
											Hydrogen Sulfide	ASTM D4658 or UOP 883	At request
											Ammonia	ASTM D1426B or UOP 740	At request
											Chlorides	UOP 458	At request
											Iron	ISO 10359	At request

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Definition of frequency terms:

x/D = Number of analysis per day

x/W = Number of analysis per week

At request = As required, which is an occasionally analysis for trouble shooting identification, process survey, etc.

6.2. On-line analyzers

The on line analyzers in the CDU unit are describe in the table below:

Analyzer tag	Service	Physical property analyzed	Normal value of analyzed component	P&ID
AE/AT-001	Flue gas from crude charge heater H-1011 radiant section	Oxygen content in flue gas	0-10 % vol	111
AE/AT-002	Oil in water	Oil content	0-1 % vol.	124
AE/AT-003	Kerosene product from E-1114 to KTU	Flash point	38-51 °C	124
AE/AT-004	LPG to gas recovery on RFCC	C ₅ + (pentane or heavier) content	1.5 % mol	131
AE/AT-005	Kerosene product from E-1114 to KTU	Freeze Point	-47 °C	124
AE/AT-006	Kerosene product from E-1114 to KTU	Specific Gravity @ 15 °C	0.775-0.840	124

For further details refer to the vendor data sheets:

- 8474L-011-SP-1561-001 for Pentane analyzer, 011-AE/AT-004
- 8474L-011-SP-1562-011 for Oxygen analyzer, 011-AE/AT-001
- 8474L-011-SP-1563-004 for Specific gravity analyzer, 011-AE/AT-006
- 8474L-011-SP-1563-006 for Flash Point analyzer, 011-AE/AT-003
- 8474L-011-SP-1563-018 for Freeze Point analyzer, 011-AE/AT-005
- 8474L-011-SP-1564-007 for Oil in Water analyzer, 011-AE/AT-002

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TRAINING MODULE

CRUDE DISTILLATION UNIT (CDU)

UNIT: 11

Course Content:

Section 1 - General Description	
Section 2 - Process Flow Description	
Section 3 - Process Control	
Section 4 - Safeguarding Devices	
Section 5 - Fire & Gas Systems	
Section 6 - Quality Control	
Section 7 - Cause & Effects	X
Section 8 - Operating Procedures	
Section 9 - HSE	
Section 10 - Reference Document Index	

SECTION 7 : CAUSES AND EFFECT

7.1. Cause & Effect Matrix:

Refer to the following documents:

- Cause and Effect Chart, 8474L-011-DW-1514-201
- H-1101 safeguarding narratives, 8474L-011-A3501-0110-001-002, issued by Heater Vendor
- A-1101 desalter package operation and maintenance manual, chapter 4, 8474L-011-A5016-0814-001-001 issued by desalter vendor

7.1.1. Examples from Cause and Effect Chart 8474L-011-DW-1514-201

To better understand how the Cause & Effect Matrix shall be read, an example is given, which details the information that can be extracted from a sheet of the Cause & Effect Diagram. It shall be read in conjunction with the corresponding sheet of the Cause & Effect Diagram.

7.1.1.1. Sheet 2: A-1101-D01 / D02 Desalter Protection : UX-001

The following causes trigger the logic UX-001:

Cause No.:	Description of the Event:	Equipment Involved:	Event Detected / Activated by:	Located in PID:
1	CDU Emergency Shutdown from the other logic UX-007		-	104
2	High High pressure on 1 st Stage Desalter	A-1101-D-01	PXAHH-501	104
3	High High pressure on 2 nd Stage Desalter	A-1101-D-02	PXAHH-504	105
4	Inventory isolation valve close push button	XV-015	XHSC-015A	105
6	Open Push button local	XV-015	XHSO-015B	105
7	Close push button local	XV-015	XHSC-015B	105
8	ESD local	XV-015	XHS-015B	105

Depending on the triggering cause, the activation of the logic sequence UX-001 leads to the following effects:

Effect No.:	Description of the Effect:	Triggered by Cause No.	Equipment Involved:	Effect Executed by:	Located in PID:
6	Activate (A) crude charge pumps P-6001 A/B/C trip UX-061	1, 2, 3, 4, 7, 8	P-6001 A/B/C	-	104
10	Activate (A) Desalter water charge pump protection UX-008	1, 2, 3, 4, 7, 8	P-1119 A/B	-	119
11	Activate (A) crude booster pump protection UX-011	1, 2, 3, 4, 7, 8	P-1101 A/B	-	105
13	Activate Desalter water recycle pump protection UX-018	1, 2, 3, 4, 7, 8	P-1118 A/B	-	104
14	Close (C) isolation inventory valve	1, 2, 3, 4, 7, 8	XV-015	XSY-015	105
	Open (O) isolation inventory valve	6	XV-015	XSY-015	105
15	Activate (A) isolation inventory valve trip alarm in local panel	1, 4, 7, 8	XV-015	XXA-015	105
16	Activate (A) Heater fuel gas protection UX-005 B	1, 2, 3, 4, 7, 8	H-1101	-	-
17	Activate (A) Heater fuel oil protection UX-005C	1, 2, 3, 4, 7, 8	H-1101	-	-
18	Activate (A) heater pilot gas protection UX-005E	1, 2, 3, 4, 7, 8	H-1101	-	-
26	Activate (A) Desalter A-1101-D-01/02 ESD – Auxiliary console (ADP)	1, 2, 3, 4, 7, 8	-	UXA-001B	104

Maintenance Override Switches (MOS) and Operation Override Switches (OOS) associated with the instrument triggering the trip:

- MOS for PXAHH-501: PHS-501A
- MOS for PXAHH-504: PHS-504A

Resets:

- From DCS: Push button UHSR-001 (see PID 104) resets effects 6, 10, 11, 13, 15, 16, 17, 18, 26
- From Local Panel: XHSR-015 (see PID 105) resets effect No. 26.

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TRAINING MODULE

CRUDE DISTILLATION UNIT (CDU)

UNIT: 11

Course Content:

Section 1 - General Description	
Section 2 - Process Flow Description	
Section 3 - Process Control	
Section 4 - Safeguarding Devices	
Section 5 - Fire & Gas Systems	
Section 6 - Quality Control	
Section 7 - Cause & Effects	
Section 8 - Operating Procedures	X
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SECTION 8 : OPERATING PRACTICES

8.1. Normal Operation

8.1.1. Operating conditions

The following is a narrative of the main process conditions of the CDU in steady state and for the design case 100% Bach Ho.

For a complete description of the Heat and material Balance & Operating Condition of the unit 019, refer to the following process flow diagrams:

8474L-011-PFD-0010-001	Cold Preheat and Desalters (Bach Ho)
8474L-011-PFD-0010-002	Hot Preheat and Heater (Bach Ho)
8474L-011-PFD-0010-003	Main Fractionators - Strippers (Bach Ho)
8474L-011-PFD-0010-004	Driers - Ejector System (Bach Ho)
8474L-011-PFD-0010-005	Over Head -Rundown System (Bach Ho)
8474L-011-PFD-0010-006	Naphta Stabiliser (Bach Ho)
8474L-011-PFD-0010-007	Schematic -CDU Preheat (Bach Ho)
8474L-011-PFD-0010-008	Mass Balance (Bach Ho)
8474L-011-PFD-0010-009	Cold Preheat and Desalters (Dubai)
8474L-011-PFD-0010-010	Hot Preheat and Heater (Dubai)
8474L-011-PFD-0010-011	Main Fractionators - Strippers (Dubai)
8474L-011-PFD-0010-012	Driers - Ejector System (Dubai)
8474L-011-PFD-0010-013	Over Head -Rundown System (Dubai)
8474L-011-PFD-0010-014	Naphta Stabiliser (Dubai)
8474L-011-PFD-0010-015	Schematic -CDU Preheat (Dubai)
8474L-011-PFD-0010-016	Mass Balance (Dubai)

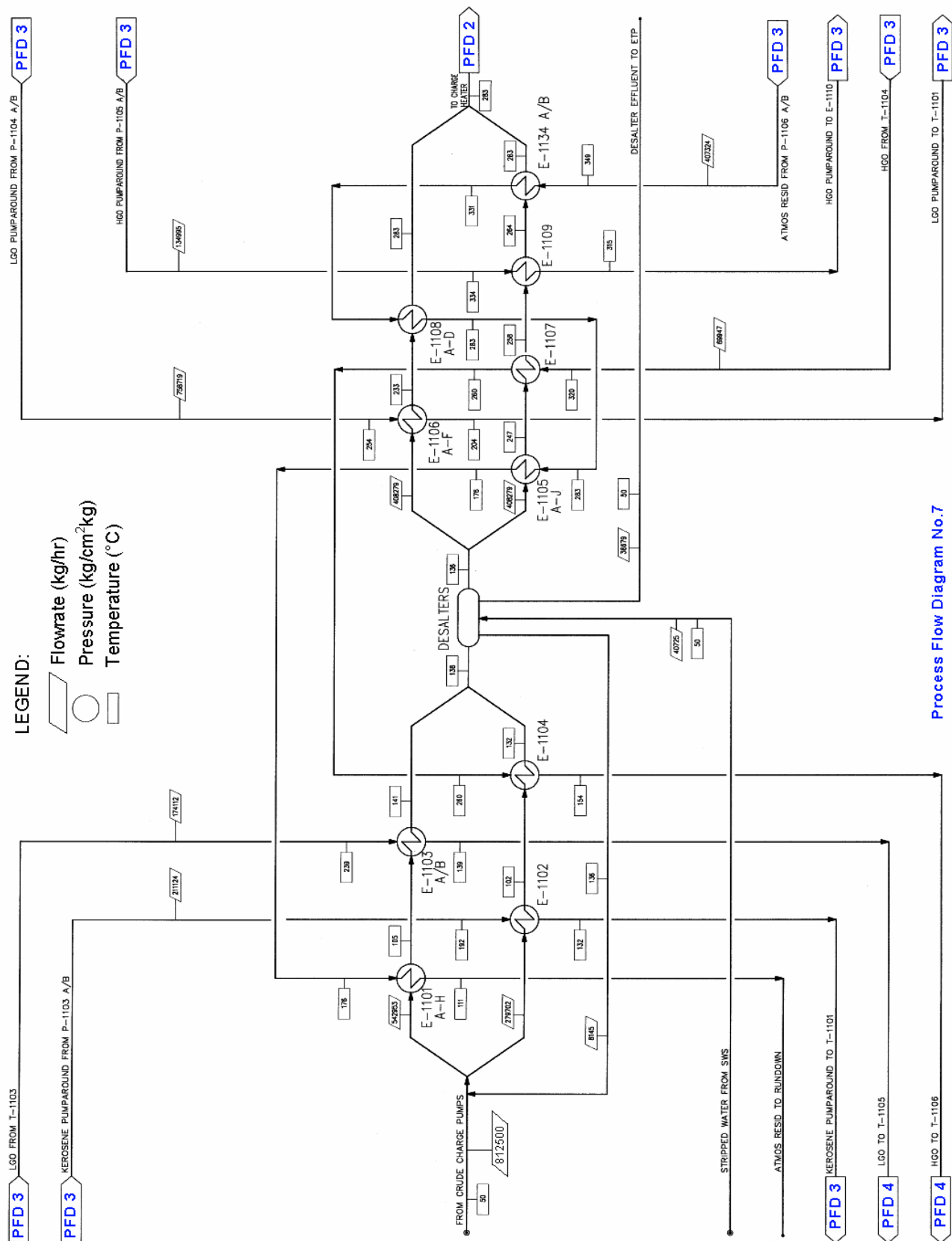


Figure 5: CDU Preheat Simplified PFD

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812,500 kg/hr of crude Oil are pumped from storage to the Crude Distillation Unit at 20 kg/cm²g and 50°C by the Feed Pumps P-6001A/B/C and are mixed with 8145 kg/hr of separated water from the 2nd stage desalter.

The first train of heat exchangers, (cold crude preheat) raises the crude temperature from 50°C up to 138°C. Crude oil emulsified with 32580 kg/hr of water (pumped from the 2nd stage desalter via P-1118 A/B) flows through the 1st desalter. The separated crude oil is mixed with 40725 kg/hr of stripped water pumped from D-1109 by P-1119 A/B at 20.3 kg/cm²g and heated to 120°C in E-1128. After being emulsified, the crude oil flows through the 2nd stage desalter.

Downstream the Desalters, 816556 kg/hr of crude are pumped at 30.6 kg/cm²g and 136°C by the Crude Booster Pump P-1101 A/B to the hot preheat train (second train), which raises the crude oil temperature up to 283°C.

38679 kg/hr of salty water are discharged to the ETP (Effluent Treatment Plant) after being cooled down in E-1128A-E and E-1129 to 50°C.

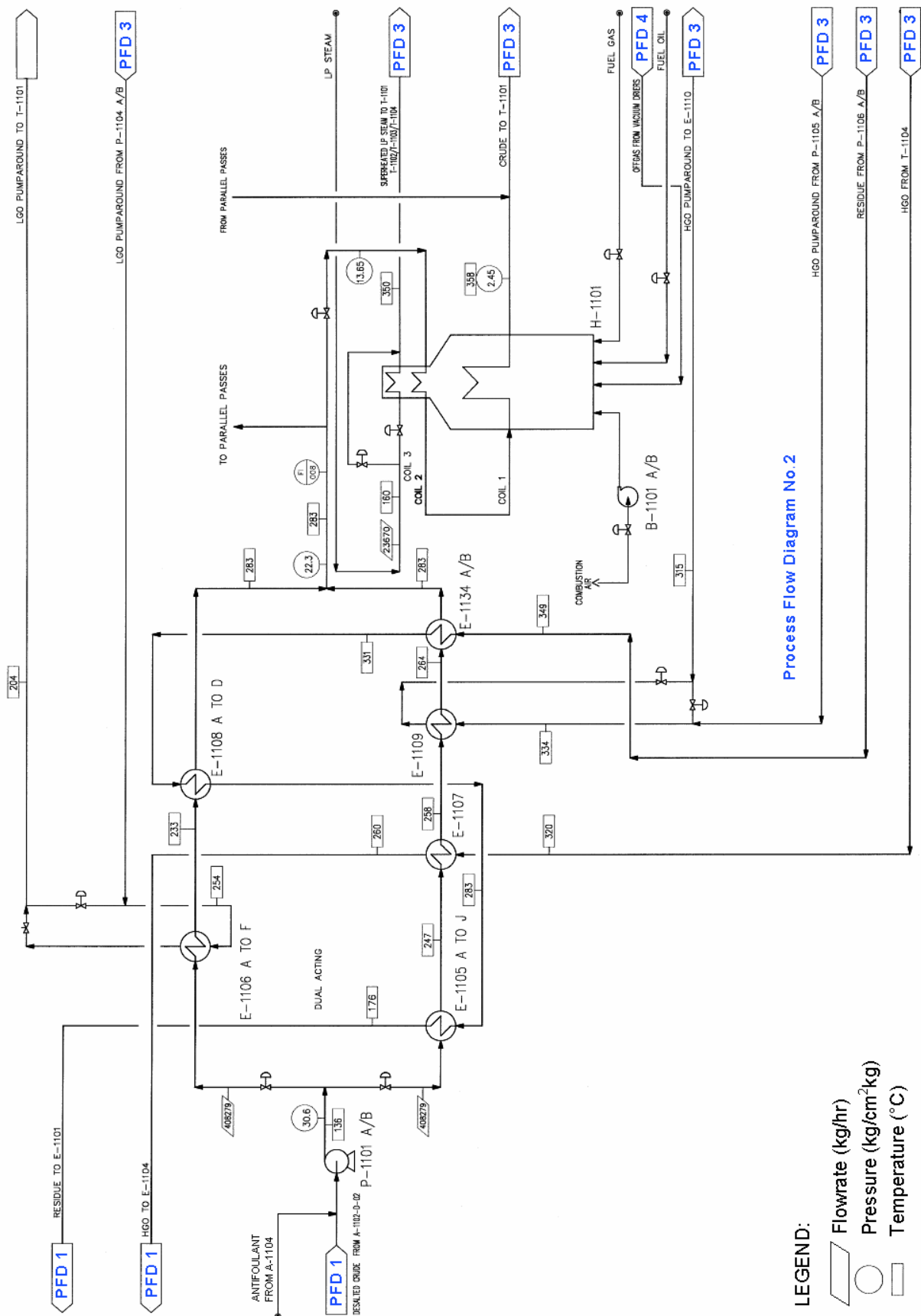


Figure 7: Hot Preheat and Heater Simplified PFD

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Desalted crude oil at 283°C and 22.3 kg/cm²g flows through the convection section and then the radiation sections of heater H-1101, where it is heated to 358°C.

Meanwhile, 23670 kg/hr of LP steam at 160°C is superheated to 350°C in the conection

Partially vaporized crude feed enters the Main Fractionator T-1101 in the flash zone between the trays 42 and 43.

The tower operates in a pressure range of 1.5 (top) to 1.9 (bottom) kg/cm²g, and in a temperature range of 124°C (top) to 349°C (bottom).

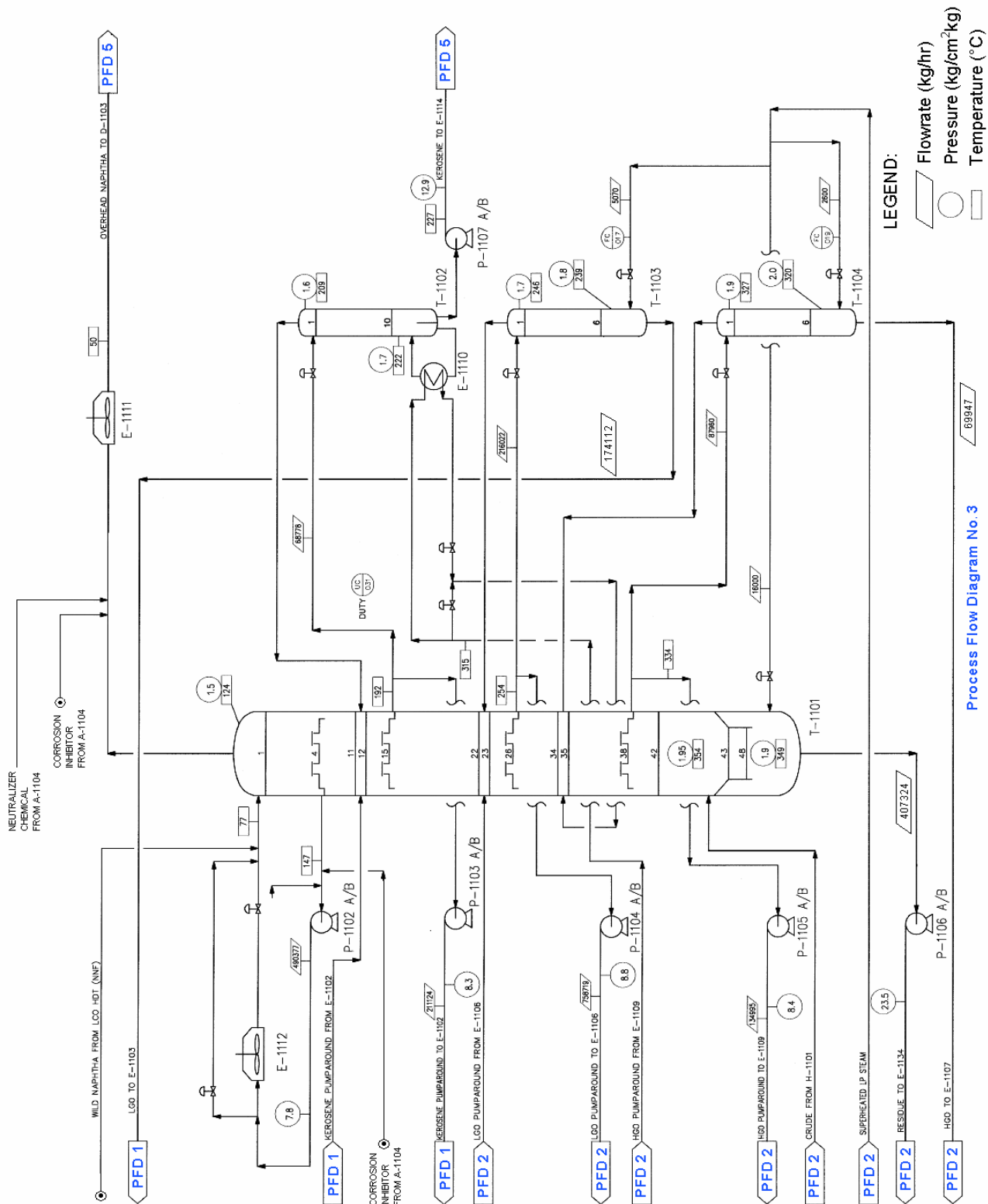


Figure 8: Main Fractionator Simplified PFD

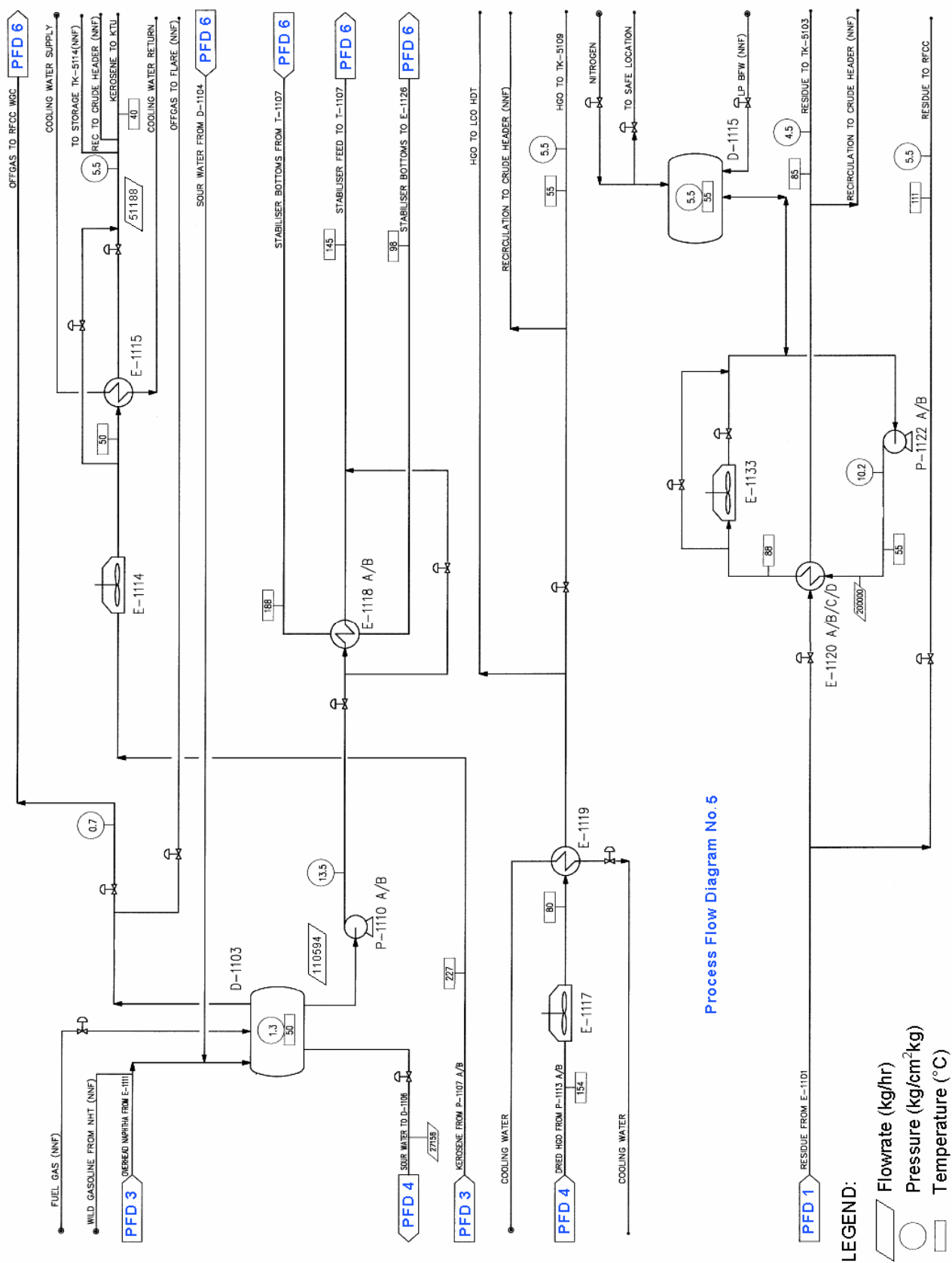


Figure 9: Overhead / Rundown System Simplified PFD

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A top pump-around in the Main Fractionator T-1101 provides reflux to the top section of T-1101 and maintains the temperature of T-1101 overhead. The Top Pump-around Pump (P 1102 A/B) drives 490377 kg/hr of liquid at 147°C from the tray 4 to E-1112 where the liquid is air cooled to 77°C, and then routed to the tray 1.

The overhead vapour (124°C), after taking a dose of corrosion inhibitor and neutralizer chemical, condenses totally through the Main Fractionator Condenser E-1111 at 50°C. The outlet from this exchanger gravity flows to the Main Fractionator Accumulator Drum (D-1103), operated at 1.3 kg/cm²g and 50°C.

In D-1103, 27158 kg/hr of sour water are separated from the condensate and drained to D-1106. The condensate is pumped at 13.5 kg/cm²g by P-1110 A/B to the Stabilizer Feed/Bottom Exchanger (E-1118 A/B) where it is heated to 145°C, and then to the Stabilizer column (T-1107).

Kerosene at 192°C is drawn off at tray 15 as a product and to be circulated in the preheat train (E-1102).

Kerosene Pump-around Pump (P-1103 A/B) routes 211124 kg/hr of the kerosene to the E-1102 in the preheat train, where the kerosene is cooled down to 132°C before being routed back to the tray #12 of the main fractionator.

The other part of the Kerosene, 68778 kg/hr, is taken to the Kerosene Stripper (T-1102). This stripper operates in a pressure range of 1.6 (top) to 1.7 (bottom) kg/cm²g, and in a temperature range of 209°C (top) to 222°C (bottom).

The top gas of T-1102 is returned to tray 12 of T-1101. 51188 kg/hr of kerosene product in the bottom is pumped to the Kerosene Air Cooler (E-1114) and afterwards to Kerosene Water Cooler (E-1115) where its temperature is reduced to 40°C before being sent to the Kerosene Treater Unit (014).

Light Gas Oil (LGO) at 254°C is drawn off at tray 26 as a product and for pump-around. Both are used for circulation in the preheat train.

LGO Pump-around Pump (P-1104 A/B) routes 758719 kg/hr of the LGO to the preheat train, specifically to E-1106 A-F, where it is cooled down to 204°C before being routed back to the tray #23 of the main fractionator.

The other part of LGO, 216022 kg/hr, is taken to the LGO Stripper (T-1103). This stripper operates in a pressure range of 1.7 (top) to 1.8 (bottom) kg/cm²g, and in a temperature range of 246°C (top) to 239°C (bottom). 5070 kg/hr of superheated steam are used as stripping medium in T-1103.

The top gas of T-1103 is returned to T-1101 at tray 23. The LGO product in the bottom gravity flows to E-1103 (preheat train) where it is cooled down to 139°C and afterwards to the LGO Dryer (T-1105).

Heavy Gas Oil (HGO) is drawn off at 334°C at tray 38 as a product and pump-around. Both are used for circulation in the preheat train.

HGO Pump-around Pump (P-1105 A/B) routes 134995 kg/hr of the HGO to the preheat train, specifically to E-1109, where it is cooled down to 315°C. Then, this

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HGO is used as the hot fluid in the Kerosene Stripper Reboiler (E-1110). Finally, the HGO is routed back to the T-1101 at Tray #35.

The other part of HGO, 87960 kg/hr is taken to the HGO Stripper (T-1104). This stripper operates in a pressure range of 1.9 (top) to 2.0 (bottom) kg/cm²g, and in a temperature range of 327°C (top) to 320°C (bottom). 2600 kg/hr of superheated steam are used as stripping medium in T-1104.

The T-1104 top gas is returned to T-1101 at tray 35. The HGO product in the bottom gravity flows to E-1107 and E-1104 (preheat train), where it is cooled down to 156°C and afterward to the HGO Dryer (T-1106).

407324 kg/hr of T-1101 bottom Residue (at 349°C) are pumped to the preheat train by the Residue Pump (P-1106 A/B) at 23.5 kg/cm²g. In particular, Fractionator residue is pumped to the heat exchangers and following this order: E-1134 A/B, E-1108 A-D, E-1105 A-J, E-1101 A-H (See Crude Preheat), where it exits at 111°C.

To strip off any light component that would be otherwise taken out in the residue stream, 16000 kg/hr of stripping steam are continuously injected in the bottom of the tower.

Part of T-1101 bottoms liquid is routed to the RFCC at 111°C and 5.5 kg/cm²g, while the remaining residue is routed to storage (tank TK-5103) at 4.5 kg/cm²g after having been cooled down to 85°C in E-1120 A-D.

Heat is removed from the residue with 200000 kg/hr of water at 55°C in the Residue / Tempered Water Cooler (E-1120 A-D). After that, this water at 88°C is air cooled in the Tempered Water Air Cooler (E-1133).

Tempered Water Pump (P-1122 A/B) recycles the water from E-1133, along with any necessary make-up water from the Tempered Water Drum (D-1115), to use it again as the cooling fluid of E-1120 A-B.

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110594 kg/hr of preheated condensate from D-1103 enters the Stabilizer column (T-1107). This tower operates in a pressure range of 7.9 (top) to 8.15 (bottom) kg/cm²g, and in a temperature range of 66°C (top) to 188°C (bottom).

The top vapour flow is partially condensed in the Stabilizer Condenser E-1122 and then it gravity flows to the Stabilizer Reflux Drum (D-1104), where off gas, LPG and water are separated.

Off gas is discharged to the RFCC unit at 0.7 kg/cm²g and 50°C. Water is sent to D-1103.

Part of the LPG, 15090 kg/cm²g, is taken by the Stabilizer Reflux Pump (P-1114 A/B) under flow control and is discharged to the top of T-1107 as reflux.

The other part of the LPG, 2181 kg/hr, is pumped by the Stabilizer LPG Pump (P-1115 A/B) at 23.9 kg/cm²g and 50°C to Gas Recovery in RFCC.

The liquid in the bottom of T-1107 flows continuously by siphon effect in the Stabilizer Reboiler (E-1121). This heat exchanger uses High Pressure Steam (HP) as hot fluid, which has been previously desuperheated in the Desuperheater (DS-1101) with high pressure Boiling Feed Water (BFW).

108314 kg/hr of Full Range Naphtha discharged in the bottom is used to preheat the stabilizer feed in the E-1118 A/B. The remained heat of Naphtha is taken off downstream in the Full Range Naphtha Air Cooler (E-1126), and then, in the Full Range Naphtha Water Cooler (E-1127) before storage, where the naphta is sent at 40°C and 3.5 kg/cm²g. Part of the full range naphta is also diverted to the Naphta Treater Unit (017), at 50 °C nad 4.0 kg/cm²g, upstream E-1127.

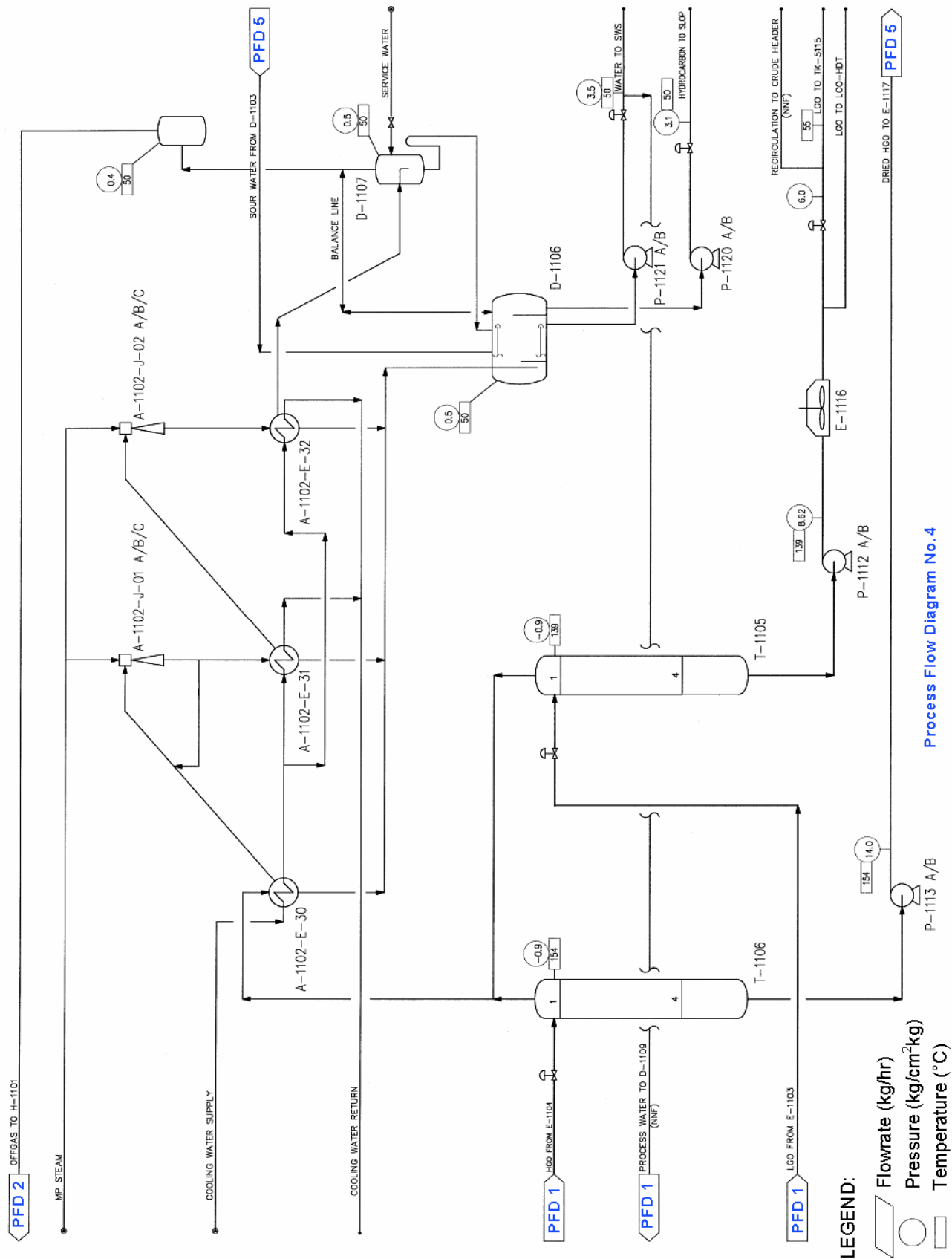


Figure 11: Dryer / Ejector System Simplified PFD

The LGO and HGO produced by crude distillation are routed to the LGO Dryer T-1105 and HGO Dryer T-1106 respectively.

The LGO dryer operates at -0.9 kg/cm²g of column top pressure and 139°C. The LGO produced in the bottom of T-1105 is pumped to the LGO Product Cooler (E-1116) by the LGO Product Pump (P-1112 A/B) at 8.62 kg/cm²g. This air cooler reduces the temperature of the LGO to 55°C before being sent to storage (TK-5115) and LCO-HDT at 6 kg/cm²g.

The HGO dryer operates at -0.9 kg/cm²g of column top pressure and 154°C. The HGO produced in the bottom of T-1106 is pumped to HGO Product Air Cooler (E-1117) by the HGO Product Pump (P-1113 A/B) at 14.0 kg/cm²g. This exchanger along with the HGO Product Water Cooler (E-1119) reduces the temperature of the HGO to 55°C before being sent to storage (TK-5109) and LCO HDT under flow control.

Vacuum system maintains a reduced pressure in the Dryers by Ventury effect. This system consists of a Pre-Condenser (A-1102-E-30), and two stage After-Condenser (A-1102-E-31 and E-1102-E-32). Each After Condenser has a train of three parallel ejectors: A-1102-J-01A/B/C for the first stage and A-1102-J-02 A/B/C for the second stage.

The purpose of the ejectors is to entrain tower overhead vapours and non-condensables, by means of medium pressure motive steam. The purpose of the condensers is to condense as much steam and hydrocarbons as possible. Cooling water is used as cold fluid in the condensers.

Condensate from all condensers is drained to the Drier Oil / Water Separator D-1106 where the water is separated from the hydrocarbon phase. The hydrocarbon phase is pumped to Slop by the Drier Slop Oil Pump (P-1120 A/B) at 3.1 kg/cm²g and 50°C, and the water is pumped by the Ejector Condensate Pump (P-1121 A/B) to the Stripped Water System (SWS) at 3.5 kg/cm²g and 50°C

The off-gas is routed to the Drier Off-gas Seal Pot D-1107 before being sent to the burners of H-1101 along with any gas separated in D-1106.

8.1.2. Alternative Operation

The plant is designed to operate with 100 % Bach Ho case or 100 % Dubai case. Furthermore, the unit is design to operate with different TBP cut point (see section 1, Flexibility cases). In order to change from one case to another, the flow set points' should be change to the value shown in section 1, Flexibility cases.

When the plant is operating with the minimum naphtha flexibility case for both crude oils, a Naphtha recirculation from P-1110 to T-1101 head via line PL-110080 is required, in order to avoid water condensation in T-1101 top trays. The amount of naphtha re-circulated varies from one case to another.

Case	Flow (kg/h)
Bach Ho	28000
Dubai	23000

The pumparound flows and duties vary from Bach Ho to Dubai operation case, but inside the same feed from one flexibility case to another, the same value should be used. The flows and duties are represented in the following table:

	Bach Ho Case		Dubai Case	
	Flow (kg/h)	Duty (kW)	Flow (kg/h)	Duty (kW)
Kerosene Pumparound	211,124	9,000	211,110	9,000
LGO Pumparound	758,719	290,065	758,676	31,000
HGO Pumparound	134,995	4,925	134,963	4,925

Top P/A duty is controlled by T-1101 overhead temperature (011-TIC-076), in order to adjust the overhead flowrate to the values required to each flexibility cases.

As a guideline the next table shows estimated overhead temperatures for each flexibility cases. These values must be adjusted in field.

	Bach Ho (°C)	Dubai (°C)
Design	123.6	123.7
Max. Naphta	129.6	127.5
Min. Naphta	126.2	123.3
Max. Kerosene	126.1	123.6
Min. Kerosene	129.3	127.2
Max. LGO	125.4	123.7
Min. LGO	125.7	124.2
Max. HGO	125.7	123.9

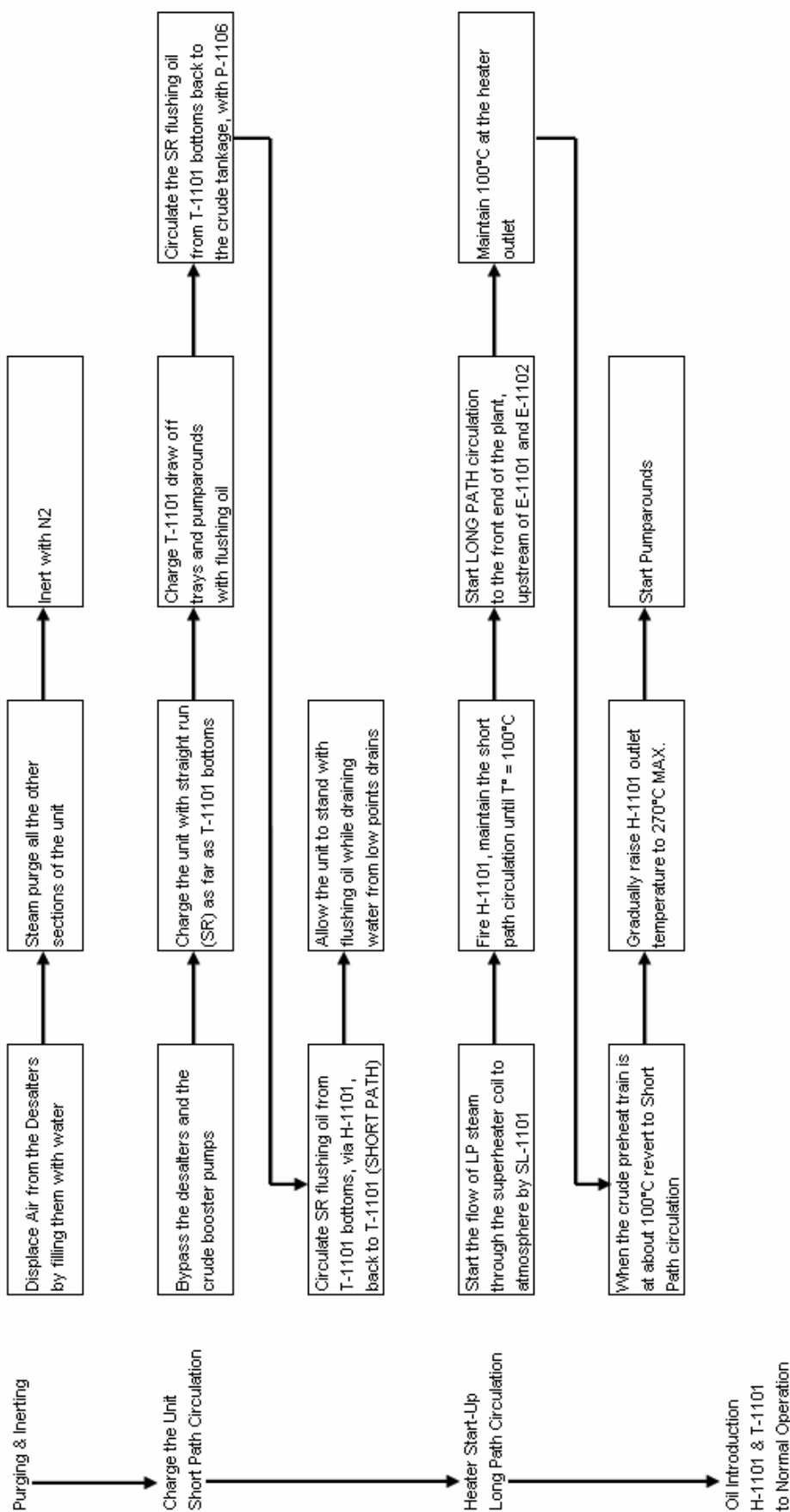
It must be noted that if the plant is operating with a mix of Bach Ho and Dubai case, the process water from P-1121 can not be used to feed D-1109 as it will contain H₂S traces.

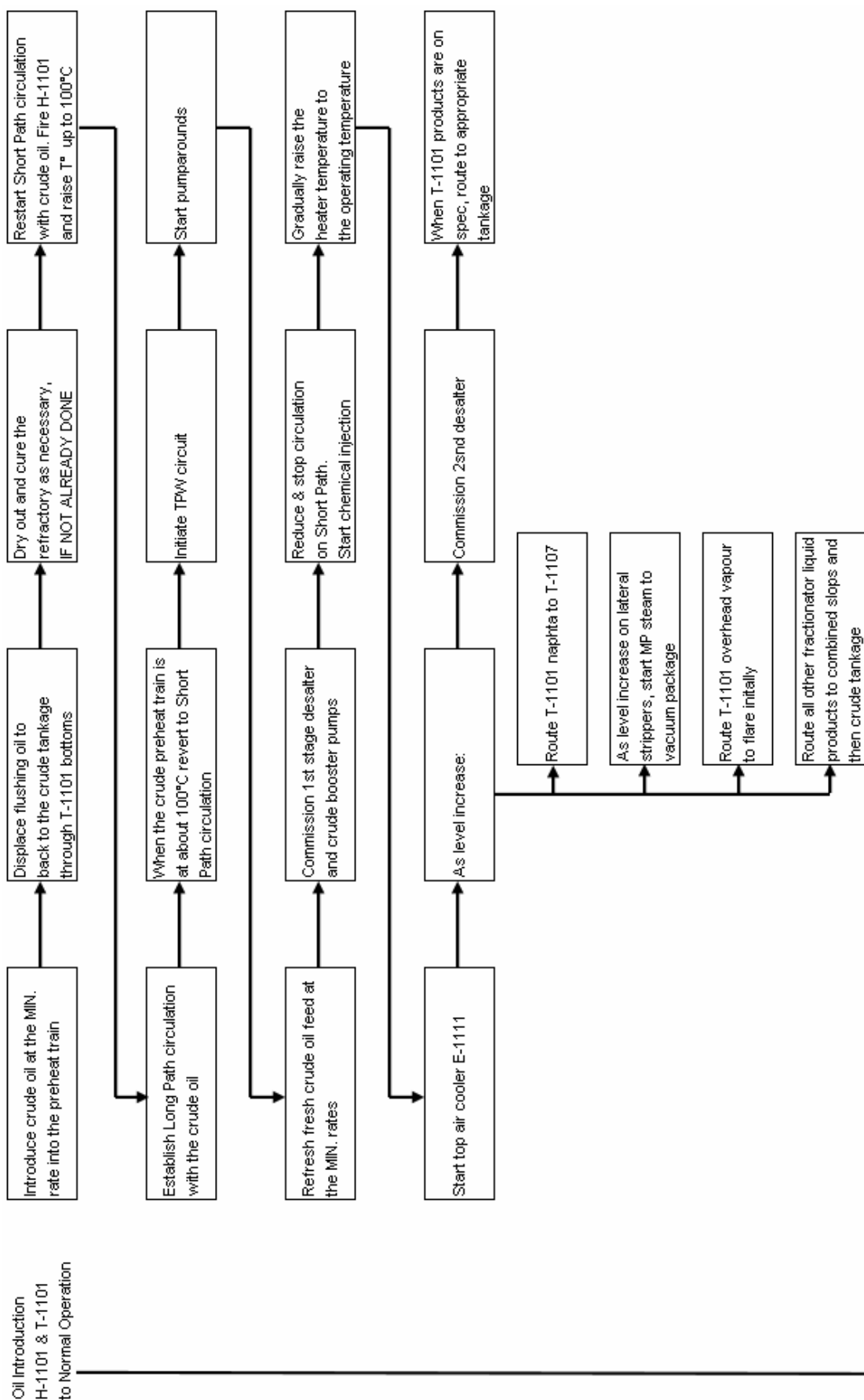
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8.2. Start-up Procedure

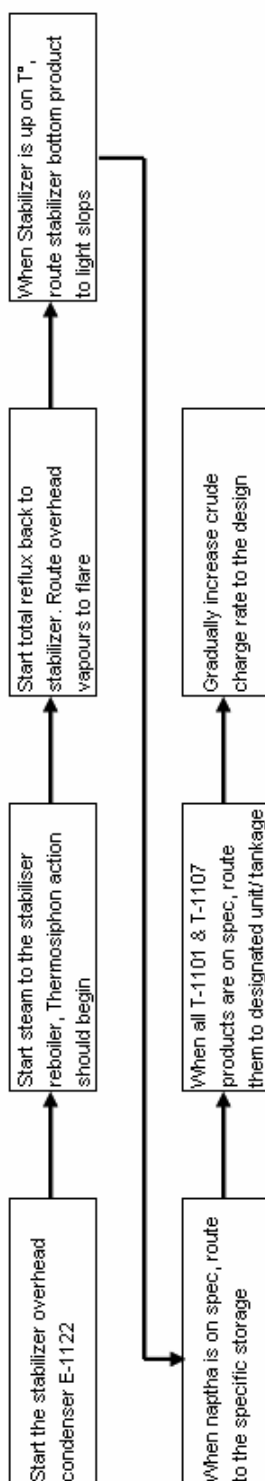
For a detailed description of the start-up procedure, refer to specific procedures into the CDU Operating Manual 8474L-011-ML-001-A.

8.2.1. Summary of Start-up





Stabilizer Start-Up

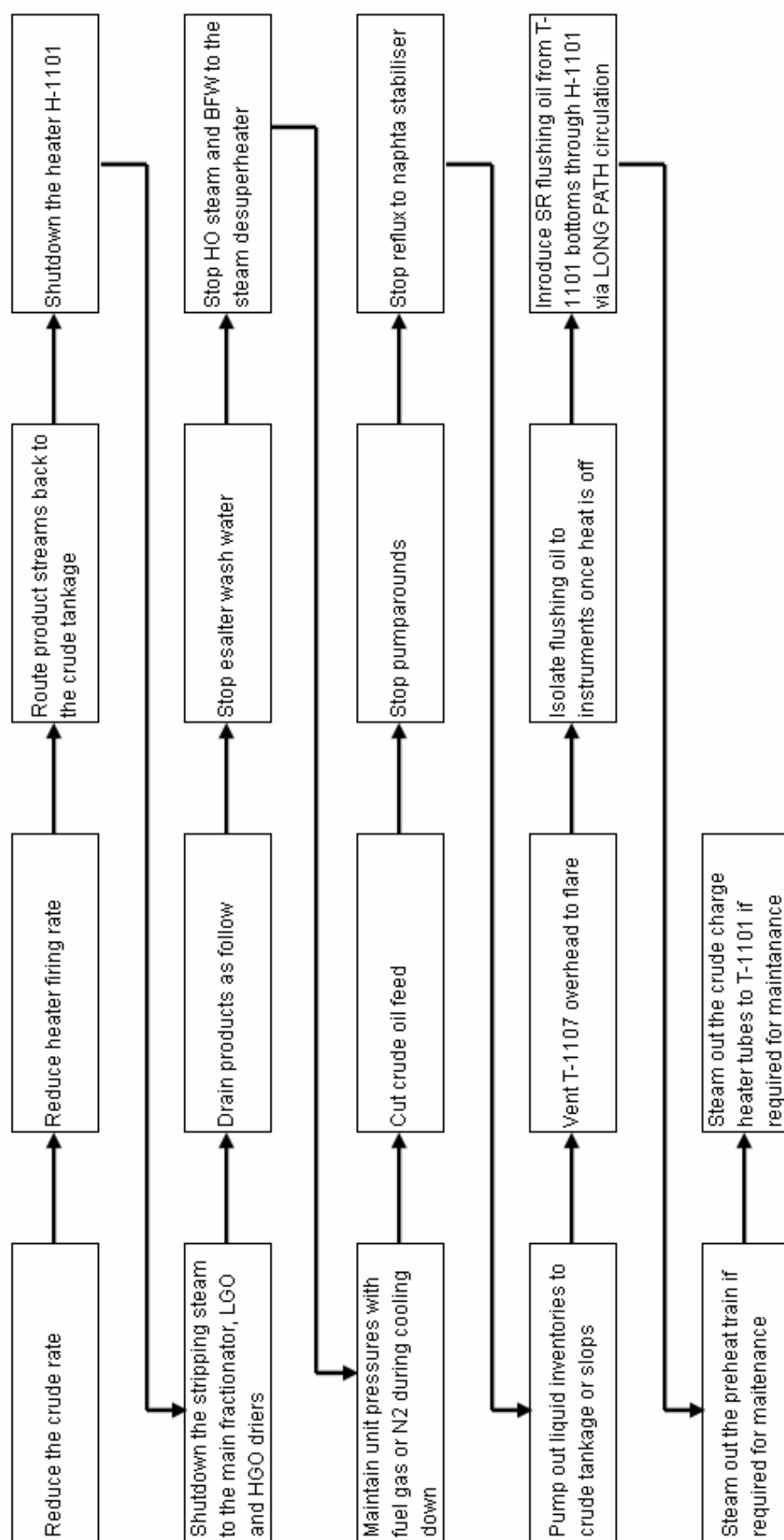


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8.3. Shutdown Procedures

For a detailed description of the shutdown procedure, refer to specific procedures into the CDU Operating Manual 8474L-011-ML-001-A.

8.3.1. Normal Unit shutdown



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8.3.2. Heater Tubes De-coking

Heater decoking may be required at regular maintenance intervals. The decision to decoking the H-1101 heater is usually based on a pressure drop or high wall temperature across the furnace becoming greater than allowable design or the thermal efficiency falling below acceptable levels.

The decoking procedure consists of BURNING coke by injecting both steam and air simultaneously while the furnace is fired. Although air is the more essential agent required for this method of coke removal, steam also plays an important role.

Whereas the air causes and supports combustion of the coke, the steam, by acting as a coolant, and as a diluent of the oxygen, helps to prevent excessive burning rates and overheating of the tubes. It thereby also reduces the total time required for decoking because, by acting as a diluent, it permits the use of more air, and as a coolant, it transfers heat from the burning area to the area beyond, where further ignition and burning is desired.

Important note: H-1101 decoking system has not been designed for spalling (only for burning). SEE PRECAUTIONS AT THE END OF THIS POINT.

Important note: H-1101 decoking system has been designed for decoking one pass at a time. SEE PRECAUTIONS AT THE END OF THIS POINT.

Products of combustion from burning are conveyed to a coke knockout drum (D-1117) through a header system, which is cooled by quench water injections.

Internal spray rings of the knockout drum condense part of the steam entering with the decoking effluent and flush all particles to the bottom of the drum. Most of the steam is not condensed and emerges from the drum stack, while the excess quench water and the steam which has been condensed flow to the sewer. This overflow line is cooled to 60 °C by service water.

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8.4. Emergency Shutdown

Refer to the following documents:

- Cause and Effect Chart, 8474L-011-DW-1514-201
- H-1101 safeguarding narratives, 8474L-011-A3501-0110-001-002, issued by Heater Vendor
- A-1101 desalter package operation and maintenance manual, chapter 4, 8474L-011-A5016-0814-001-001 issued by desalter vendor

8.4.1. General Emergency Shutdown

The CDU unit can be completely shut-down by an operator via the emergency push button 011-UXHS-007 which trips the 011-UX-007 “CDU ESD”. The actions triggered by this Interlock are fully described on the Cause and Effect Chart, but in general, this interlock initiates the following actions:

- A-1101-D-01 and 02 Desalters protection, UX-001
- T-1101 main fractionator inventory isolation, UX-002
- T-1103 LGO stripper protection, UX-003
- T-1104 HGO stripper protection, UX-004
- H-1101 crude charge heater protection, UX-005A
- T-1101 main fractionator isolation, UX-017
- P-1102 A/B top p/a pumps protection, UX-020
- P-1104 A/B LGO p/a pumps protection, UX-021
- P-1105 A/B HGO p/a pumps protection, UX-022
- P-1103 A/B kerosene p/a pumps protection, UX-023
- D-1104 inventory isolation, UX-026

Additionally, operator will cut electricity to desalters transformers’ and will stop the M.P. steam to Vacuum package (A-1102).

Operator at field shall verify that all the ESD actions have been executed.

8.4.2. Power Failure

Under a general unit power failure, all electrical motors will stop: oil and water system pumps will fail and air coolers electric motors will also stop. Since the power failure implies a fail of the Heater Air blowers too, B-1101 A or B, that leads to a full Heater shut down via 011-UX-005E ESD interlock.

Regarding the number of systems affected by the loss of power, a general shut down must be carried out via emergency push button, as explained on previous point.

Instrumentation and DCS/ESD are fed from the Uninterruptible Power Supply (UPS) so no impact is expected on them due to the loss of power on the unit.

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Therefore, the actions required after the UX-007 is initiated will be carried out, but in any case, the operator shall check that each action described on the C&E Chart has been completed properly by the system.

8.4.3. Steam Failure

8.4.3.1.L.P. steam failure

A failure on the refinery low pressure steam network implies the loss of stripping steam. As first consequence, products will go out of specification. First action shall be estimate the duration of the steam failure. In case of a long duration failure is expected, unit rate shall be reduced and then the products shall be diverted to slops. If failure is prolonged, a full unit shut-down via ESD interlock 011-UX-007 is required.

If the loss of steam occurs during start-up steaming/purging, open up nitrogen and box-in the plant.

8.4.3.2.M.P. Steam Failure:

The loss of the medium pressure steam implies the loss of motive flow on the steam ejectors of the vacuum package, A-1102. Therefore, vacuum is lost and both LGO and HGO go out of specification. If the steam failure is expected to be prolonged unit feed rate must be reduced and then the products will be routed to crude slops via crude recirculation header.

M.P. Steam is also used to atomize the fuel oil. In the case that H-1101 is working with both Fuel Oil and Fuel Gas, oil connection to burners will be closed, and since unit rate has been reduced, Fuel Gas will compensate the loss of duty. If H-1101 is working only with Fuel Oil, H-1101 must be shut down via 011-UXHS-005A or 005AA push buttons.

8.4.3.3.H.P. Steam failure

The loss of high pressure steam failure implies the loss heat source on the Stabiliser (T-1107) reboiler, E-1121, and therefore, the flow of vapour and liquid within the stabiliser is interrupted.

First measure is to reduce the crude rate to the unit and then divert the stabiliser outlet streams to the following destinations:

- Full Range Naphtha (T-1107 bottoms) to light slops.
- Stabiliser overhead gases shall be routed to the flare.
- LPG off spec from the Stabiliser reflux drum, D-1104, to LPG off spec.

Once the streams have been aligned properly, PL-110080 must be opened and part of the flow returned back to the main fractionator, T-1101.

If failure is prolonged or liquid level on T-1107 reaches high level alarm (011-LAH-042), feed to stabiliser must be cut off and all the liquid from the D-1103 must be send back to the main fractionator. Unit rates shall remain reduced and LGO shall be sent to slops (Heavy slops).

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8.4.3.4. Complete steam failure

A similar procedure will be followed: unit feed rate will be reduced and then products shall be diverted to slops, and if failure is expected to be prolonged a unit shut-down via Emergency Interlock 011-UX-007 is required.

8.4.4. Instrument air failure

Without instrument air, control valves would move to the fail safe condition and the unit will be shut-down via ESD Interlock 011-UX-007.

8.4.5. Cooling water failure

Under cooling water failure scenario, CDU products (Naphtha, Kerosene and Residue) will be too hot for storage tanks, resulting in a dangerous increase of vapour within the tanks. LGO and HGO will run out of spec since without cooling water, vacuum package A-1102 will stop working and vacuum will be lost. Therefore LGO and HGO will be also routed to slops.

In order to minimise the impact on storage, unit throughput will be reduced to allow air coolers to reduce the products outlet temperature. Additionally, tankage responsible personnel shall be adverted that these products could be sent to storage at higher temperatures than required, depending on the Air cooler capacity. If products cannot be cooled down enough, the unit will be shut down.

In the case that the loss of cooling water is expected to last more than a few hours, the lack of cooling on the rotating equipment may cause a mechanical damage. Therefore, the unit shall be shut down via the ESD interlock 011-UX-007.

8.4.6. Fuel gas system failure

The main consequence of a failure on the Fuel gas system is the reduction on the H-1101 duty. Fuel gas failure is detected by the 011-PXALL-094 that activates the Fuel gas Trip, 011-UX-005 B, closing the ESD valves that isolate the Fuel Gas to the Crude Charge Heater H-1101 burners.

Once the fuel gas is lost, the subsequent actions will depend on the H-1101 operating case:

- If H-1101 is working with a mixture of Fuel Oil – Fuel Gas, the firing control system will try to compensate the loss of duty by opening the Fuel Oil control valve: 011-PV-118. If the fuel oil available is not enough to compensate, the crude charge rate will be reduced.
- If H-1101 is working with 100% Fuel Gas, the crude feed rate will be reduced and products will be routed to slops. If this fuel cannot be restored within a short period of time, the unit will be shut-down.

8.4.7. Fuel oil system failure

The failure on the Fuel oil system causes the reduction on the H-1101 duty. Fuel oil failure is detected by the 011-PXALL-122 that activates the Fuel Oil Trip, 011-UX- 005 C, closing the ESD valves that isolate the Fuel oil to the Crude Charge Heater H-1101 burners.

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Once de fuel oil is lost, the subsequent actions will depend on the H-1101 operating case:

- If H-1101 is working with a mixture of Fuel Oil – Fuel Gas, the firing control system will try to compensate the loss of duty by opening the Fuel gas control valve: 011-PV-092. If the fuel gas available is not enough to compensate, the crude charge rate will be reduced.
- If H-1101 is working with 100% Fuel Oil, the crude feed rate will be reduced and products will be routed to slops. If this fuel cannot be restored within a short period of time, the unit will be shut-down.

8.4.8. Pilot gas system failure

The failure of the pilot gas causes the loss of pilot flames. Pilot gas loss is detected by both the 011-PXALL-112 and pilot flame detectors 011-BAL-501 to 516, which activates “Pilot Gas Trip” 011-UX-005 E, closing the emergency valves 011-XV-007 and 011 and opening the depressurizing valve 011-XV-012.

As described on the Cause & Effect Chart, 8474L-011-DW-1514-201, “Pilot Gas Trip” also activates the “Fuel Gas Trip” and the “Fuel Oil Trip”. Heater blower, B-1101 A or B, will continue running to purge the Heater chambers.

Pilot gas system failure requires a complete unit shut down.

8.4.9. Nitrogen failure

Nitrogen is used as blanketing for D-1109 in order to minimise the H₂S presence in gas vented to flare. This happens only in the case of miss operation in SWS unit. Since this is a very unlikely scenario and the flare is able to burn small amounts of H₂S, the loss of nitrogen has not any impact.

Nitrogen is also used to pressurize D-1115. Regarding the purpose of this vessel, and taking into account the level controller 011-LV-070 on this one, which avoid vessel to be empty, no impact on this circuit is expected.

In addition, Nitrogen is used on the sampling systems SC-3B-001, SC-5B-001 to 004, SC-5B-006, SC-5B-008, and SC-5B-012 to 016. In the case of a nitrogen failure, the sampling on these services will not be performed until Nitrogen flow is restored.

Unit general or partial shut down is not required in this scenario.

8.4.10. Mechanical failure

If a spared item fails, first start the standby equipment. Then secure and isolate the failed item and prepare it for maintenance.

If the failed item is not spared, whenever possible, by-pass and isolate the equipment while continuing to operate safely. Throughput will be reduced if necessary, and the equipment will be prepared for maintenance.

The following pumps: P-1101 A/B, P-1104 A/B and P-1106 A/B are protected by a preventive Machine Monitoring System, which control certain parameters (vibration and bearing temperature) and allows the system to warn in advance about a bad performance of the machine.

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SECTION 9 : HSE

9.1. Hazardous Areas

Refer to the following documents:

8474L-011-DW-0051-001 Plot Plan Unit 11 Crude Distillation / Unit 14 Kerosene Treater / Unit 37 Fuel Gas System

8474L-011-DW-1920-002 General Hazardous Area Classification

9.2. Safety Equipment

Refer to:

8474L-011-DW-1933-001 Safety Equipment Layout

9.3. Specific PPE

All personnel involved in chemical handling related work in the SRU must wear the specific PPE required. Moreover, all personnel shall be read and understand the MSDS of the chemicals they will handle before proceeding to work. Such MSDS must be provided by the material supplier and serve as reference.

The following will details the requirements on PPE for specific chemicals used/produced in the CDU. For more information on each of the following materials, refer to the attachment 3 of the operating manual, doc. No. 8474L-011-ML-001.

9.3.1. PPE requirement for Crude Oil

Ventilation Requirement	Use sufficient ventilation to maintain air concentrations below regulatory limits. Select appropriate NIOSH approved respiratory protection where necessary (determined by potential exposure and published respiratory protection factors). For unknown concentration or fire fighting use self-contained breathing apparatus with positive pressure.
Eyes Protection	Safety glasses, chemical goggles or face shield as appropriate
Skin Protection	Gloves: Nitrile, neoprene or other material resistant to crude oil. Flame retardant clothing
Additional Protective Measures	While loading, unloading, tank gauging, etc., remain upwind. Request assistance of safety and industrial hygiene personnel to determine air concentrations. Know the location of eye wash stations and safety showers.

9.3.2. PPE Requirements for Propane

Ventilation Requirement	Use in well-ventilated areas. Use with explosion proof mechanical ventilation in confined spaces or poorly ventilated areas.
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Eyes Protection	Safety glasses or chemical goggles are recommended when transferring product.
Skin Protection	Insulated gloves required if contact with liquid or liquid cooled equipment is expected. Wear gloves and long sleeves when transferring product.
Inhalation	Where concentration in air would reduce the oxygen level below 18% air or exceed occupational exposure limits in section 6, self-contained breathing apparatus is required.

9.3.3. PPE Requirements for Naphta

Respiratory Protection	If engineering controls and ventilation is not sufficient to prevent buildup of aerosols or vapours, appropriate NIOSH/MSHA approved air-purifying respirators or self contained breathing apparatus (SCBA) appropriate for exposure potential should be used. Air supplied breathing apparatus must be used when oxygen concentrations exceed the limits of the air-purifying respirators
Eyes Protection	Wear safety glasses, chemical goggles are recommended if splashing is possible, or to prevent eye irritation from vapours.
Skin Protection	Use impervious gloves when handling product. Wear chemical-resistant safety footwear with good traction to prevent slipping. Work clothing that sufficiently prevents skin contact should be worn, such as coveralls and/or long sleeves shirts and pants. Fire resistant (i.e. Nomex) or natural fibre clothing (i.e. cotton or wool) is recommended. Synthetic clothing can generate static electricity and is not recommended where flammable vapours release may occur.

9.3.4. PPE Requirements for Kerosene

Respiratory Protection	If engineering controls do not maintain airborne contaminant concentrations at a level which is adequate to protect worker health, an approved respirator may be appropriate. Respirator selection, use, and maintenance must be in accordance with regulatory requirements, if applicable. For high airborne concentrations, use an approved supplied-air respirator, operated in positive pressure mode. Supplied air respirators with an escape bottle may be appropriate when oxygen levels are inadequate, gas/vapour warning properties are poor, or if air purifying filter capacity/rating may be exceeded.
Eyes Protection	If contact is likely, safety glasses with side shields are recommended. Chemical / oil resistant clothing if contact with material is likely.
Skin Protection	Chemical resistant gloves are recommended.

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9.3.5. PPE Requirements for Light Gas Oil

Respiratory Protection	Respiratory protection is not required under conditions of normal use. If vapor or mist is generated when the material is heated or handled, use an organic vapor respirator with a dust and mist filter. All respirators must be NIOSH certified.
Eyes Protection	Wear plastic face shield or splash-proof safety goggles
Skin Protection	Use impervious clothing (boots, gloves, aprons, etc.) over parts of the body subject to exposure. Launder soiled clothes. Properly dispose of contaminated leather articles including shoes, which cannot be decontaminated.

9.3.6. PPE Requirements for HGO

Respiratory Protection	If engineering controls and ventilation is not sufficient to prevent buildup of aerosols or vapours, appropriate NIOSH/MSHA approved air-purifying respirators or self-contained breathing apparatus (SCBA) appropriate for exposure potential should be used. Air supplied breathing apparatus must be used when oxygen concentrations are low or if airborne concentrations exceed the limits of the air-purifying respirators.
Eyes Protection	Wear safety glasses; chemical goggles are recommended if splashing is possible or to prevent eye irritation from heated vapors or mists.
Skin Protection	Use chemically resistant gloves when handling product. Wear chemical-resistant safety footwear with good traction to prevent slipping. Work clothing that sufficiently prevents skin contact should be worn, such as coveralls and/or long sleeves and pants. Splashing or contact with liquid material is possible; consider the need for an impervious overcoat. Fire resistant (i.e., Nomex) or natural fiber clothing (i.e., cotton or wool) is recommended. Synthetic clothing can generate static electricity and is not recommended where flammable vapor releases may occur.

9.3.7. PPE Requirements for Fuel Oil

Respiratory Protection	Where concentrations in air may exceed the occupational exposure limits and where engineering, work practices or other means of exposure reduction are not adequate, NIOSH approved respirators may be necessary to prevent overexposure by inhalation.
Eyes Protection	Eye protection (i.e., safety glasses, safety goggles and/or face shield) should be determined based on conditions of use. If product is used in an application where splashing may occur, the use of safety goggles and/or a face shield should be considered.
Skin Protection	Wear appropriate clothing to prevent skin contact. As a minimum long

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	sleeves and trousers should be worn. Wear appropriate chemically protective gloves. When handling hot product ensure gloves are heat resistant and insulated. Wear appropriate footwear to prevent product from coming in contact with feet and skin.
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9.4. Chemical Hazards

Nitrogen introduced for purging and tank and vessel blanketing purposes should be recognized as a non-toxic, asphyxiant.

It is of the utmost importance that all employees involved in this unit read and understand the MSDS of the chemical they will handle before proceeding to work. No work or operation should be allowed to commence before all employees involved in chemical handling related work have demonstrated knowledge of the health hazards they may face.

Note:

Dubai crude oil is even more dangerous than Bach Ho crude due to the amount of H₂S it contains.

9.4.1. Chemical Hazards for Crude Oil

Hazard classification:



flammable

Summary of Hazards:

FLAMMABLE LIQUID AND VAPOR

MAY CONTAIN HYDROGEN SULFIDE – MAY BE FATAL IF INHALED

MAY CAUSE DELAYED LUNG INJURY

CAN CAUSE NERVOUS SYSTEM DEPRESSION

ASPIRATION HAZARD IF SWALLOWED, CAN ENTER LUNGS AND CAUSE DAMAGE

CANCER HAZARD – CONTAINS BENZENE

CAN CAUSE KIDNEY, LIVER AND BLOOD DISORDERS

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Keep away from heat, sparks and flame. Avoid breathing vapor. Use ventilation adequate to keep vapor below recommended exposure limits. Avoid contact with eyes, skin and clothing. Wash thoroughly after handling.

Hazards to health:

Acute Exposure Symptoms:

Skin Contact	Irritation
Inhalation	May cause headache, nasal and respiratory irritation, nausea, drowsiness, breathlessness, fatigue, central nervous system depression, convulsions and loss of consciousness. Hydrogen sulfide paralyzes the respiratory system rapidly causing unconsciousness and death; may cause loss of sense of smell; immediately dangerous to life concentration is 700 ppm
Eye contact	Irritation
Ingestion	Possible burning of mouth and gastrointestinal disturbances. May cause vomiting and diarrhea, depression of the central nervous system, sedation, coma, pneumonitis, pulmonary edema.

Chronic Exposure Symptoms:

Skin Contact	Drying, cracking, redness, itching, burning, or inflammation of skin (Dermatitis).
Inhalation	May cause respiratory tract irritation, central nervous system, kidney, liver and blood disorders. Benzene has been classified as a carcinogen, and may produce blood disorders including anemia and leukemia.

9.4.2. Chemical Hazards for Propane

Hazard Class:



flammable

Hazards to Health:

Inhalation:	Simple asphyxiant. No effect at concentrations of 10,000 ppm (peak exposures). Higher concentrations may cause central nervous system disorder and/or damage. Lack of oxygen may cause dizziness, loss of coordination, weakness, fatigue, euphoria, mental confusion, blurred vision, convulsions, breathing failure, coma and death. Breathing high vapour concentrations (saturated vapours) for a few minutes may be fatal. Saturated vapours may be encountered in confined spaces and/or under conditions of poor ventilation. Avoid breathing vapours or mist.
Skin & Eye Contact	Exposure to vapourizing liquid may cause frostbite (cold burns) and permanent eye damage.
Acute Exposure	Contact with Liquefied Petroleum Gas may cause frostbite or cold burns. Propane acts as a simple asphyxiant as oxygen content in air is displaced by the propane. At increasing concentration levels, propane may cause dizziness, headaches, loss of coordination, fatigue, unconsciousness and death.
Chronic Exposure	No reported effects from long term low level exposure.

9.4.3. Chemical Hazards for Naphta

Hazard class:



flammable

Hazards to health:

Inhalation	Excessive inhalation of this material causes headache, dizziness, nausea and loss of coordination, possible cardiac sensitization, and in extreme conditions coma and possibly death. Liquid aspirated into the lungs may cause chemical pneumonitis. Components of this product are considered cardiogenic. Based on animal testing, a component of this product (xylene) is considered to be a developmental toxicant.
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Eye contact	This product is irritating to the eyes
Skin Contact	Prolonged and/or repeated skin contact with this product may cause irritation, dermatitis and possible chemical blistering. Prolonged or repeated contact with this product may severely dry and/or defat the skin. Material is not a skin sensitizer. Product contains components that may be absorbed through the skin.
Ingestion	Ingestion can cause gastro-intestinal, nausea, vomiting and diarrhea. Ingestion of this product may result in central nervous system effects including headache, irregular heartbeats, nausea, sleepiness, dizziness, slurred speech and blurred vision. Aspiration hazard – swallowing or vomiting liquid may cause aspiration into the lungs. Ingestion may cause liver and kidney damage.

9.4.4. Chemical Hazards for Kerosene

Hazard class:



flammable

Physical/chemical effects:

Combustible. Material can release vapours that readily form flammable mixtures.

Vapour accumulation could flash and/or explode if ignited. Material can accumulate static charges which may cause an incendiary electrical discharge.

Hazards to health:

Inhalation	May be irritating to the nose, throat, and lungs. Breathing of high vapour concentrations may cause dizziness, light-headedness, headache, nausea and loss of co-ordination. Continued inhalation may result in unconsciousness
Eye contact	May be irritating to the eyes
Skin Contact	Irritating to skin High-pressure injection under skin may cause serious damage
Ingestion	If swallowed, may be aspirated and cause lung damage

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9.4.5. Chemical Hazards for LGO

Hazard to Health:

Inhalation	Avoid prolonged inhalation of mist or vapors. Caution should be taken to avoid aerosolization or misting of this products. Overexposure may cause weakness, headache, nausea, confusion, blurred vision, drowsiness, and other nervous system effects; greater exposure may cause dizziness, slurred speech, flushed face, aortic plaques, heart beat irregularities, unconsciousness, or convulsions.
Eye contact	Avoid eye contact. This product is minimally irritating to the eyes upon direct contact. Exposure to high concentrations of vapors may be irritating to the eyes.
Skin Contact	Avoid skin contact. This product may cause slight skin irritation upon direct contact. Prolonged or repeated contact may result in contact dermatitis which is characterized by dryness, chapping, and reddening. This condition may make the skin more susceptible to other irritants, sensitizers, and disease. Pre-existing skin conditions may make the skin more susceptible and facilitate uptake by this route.
Ingestion	Do not ingest. Ingestion of small quantities is usually nonfatal unless aspiration occurs. Aspiration may lead to chemical pneumonitis which is characterized by pulmonary edema and hemorrhage and may be fatal. Signs of lung involvement include increased respiratory rate, increased heart rate and a bluish discoloration of the skin. Coughing, choking, and gagging are often noted at the time of aspiration. Gastrointestinal discomfort may develop, followed by vomiting with a further risk of aspiration. Ingestion may cause gastrointestinal distress. Severe oral intoxication will lead to intense burning of the throat and may result in drowsiness, dullness, numbness, and headache followed by dizziness, weakness, and nausea. Loss of consciousness and convulsions followed by death may result.

9.4.6. Chemical Hazards for HGO

Hazard Classification:

Carcinogen & Skin irritant

Summary of hazards:

This product is flammable when heated to high temperatures. Product is a pale yellow colored, thick liquid with a fuel oil odor. If released, prevent entry into ditches, sewers, and waterways. Small amounts of this product, if aspirated into the lungs, may cause mild to severe pulmonary injury. This product may be irritating to the eyes, skin, and respiratory system. Ingestion or inhalation of heated vapors or mists may result in central nervous system effects including headache,

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sleepiness, dizziness, slurred speech and blurred vision. On prolonged contact, may cause serious dermatitis and possibly skin cancer.

Hazards to health:

Inhalation	Excessive inhalation of this material causes headache, dizziness, nausea and loss of coordination. Inhalation of heated vapors or mists may cause headache, dizziness, nausea and loss of coordination. Excessive inhalation of heated vapors over time may cause lung, kidney and liver damage. Aspiration of liquid into the lungs can cause mild to severe pulmonary injury
Eye contact	This product may cause irritation to the eyes on contact with heated vapors or liquid
Skin Contact	Skin contact with this product may cause irritation/dermatitis. Prolonged and / or repeated contact may cause severe irritation/dermatitis, chemical blistering and possible skin cancer
Ingestion	Ingestion can cause gastrointestinal irritation, nausea, vomiting and diarrhea. Ingestion of this product may result in central nervous system effects including headache, sleepiness, dizziness, slurred speech and blurred vision. Ingestion may cause kidney and liver damage. Swallowing or vomiting of liquid may cause aspiration into the lungs, and subsequent mild to severe pulmonary injury.

9.4.7. Chemical Hazards for Fuel Oil

Hazard Classification:

Animal carcinogen, flammable.



flammable

Hazards to Health

Inhalation	Inhalation of this product may cause respiratory tract irritation and Central Nervous System (CNS) Depression, symptoms of which may include; weakness, dizziness, slurred speech, drowsiness, unconsciousness and in cases of severe overexposure; coma and death.
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Eye contact	Contact with this product may cause eye irritation
Skin Contact	Contact with this product may cause skin irritation Prolonged or repeated contact may cause skin irritation, defatting, drying and dermatitis
Ingestion	Ingestion of this product may cause gastro-intestinal irritation. Aspiration of this product may result in severe irritation or burns to the respiratory tract.

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10.1. Operating Manual/ Licensor Documentation

8474L-011-ML-001-A CDU Operating Manual

10.2. Arrangement Drawings, Layouts and Plot Plans

8474L-011-DW-0051-001 Plot Plan

8474L-011-A2001-0110-001-001 H-1101 Heater Arrangement Drawing

8474L-011-A2001-0110-001-002 H-1101 Heater Arrangement Drawing

8474L-011-A2001-0510-001-001 T-1101 Main Fractionator Arrangement Drawing

8474L-011-A2001-0510-002-001 Kerosene Stripper / LGO Stripper / LGO Drier Arrangement Drawing

8474L-011-A2001-0510-002-002 HGO Stripper / HGO Drier Arrangement Drawing

8474L-011-A2001-0510-002-003 Stabilizer Arrangement Drawing

8474L-011-A2001-0570-001-001 TR-T-1101 Arrangement Drawing

8474L-011-A2001-0570-002-001 TR-T-1102 Kerosene Stripper Arrangement Drawing

8474L-011-A2001-0570-002-002 TR-T-1103 LGO Stripper Arrangement Drawing

8474L-011-A2001-0570-002-003 TR-T-1104 HGO Stripper Arrangement Drawing

8474L-011-A2001-0570-002-004 TR-T-1105 LGO Drier Arrangement Drawing

8474L-011-A2001-0570-002-005 TR-T-1106 HGO Drier General Arrangement

8474L-011-A2001-0570-002-006 TR-T-1107 Stabilizer Arrangement Drawing

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8474L-011-A2001-0610-001-002 Crude / Kerosene pump around exchanger Arrangement Drawing (2/2)

8474L-011-A2001-0610-001-003 Crude / LGO exchanger Arrangement Drawing

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8474L-011-A2001-0610-001-005 Warm Crude / HGO Exchanger Arrangement Drawing

8474L-011-A2001-0610-001-006 Crude / LGO Pumparound Exchanger Arrangement Drawing

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8474L-011-A2001-0610-002-002 Stabilizer feed/bottoms exchanger Arrangement Drawing

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8474L-011-A2001-0610-002-004 Residue / Tempered Water Cooler Arrangement Drawing

8474L-011-A2001-0610-002-005 Stabilizer Reboiler Arrangement Drawing

8474L-011-A2001-0610-002-006 Full Naphta Range Water Cooler Arrangement Drawing

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8474L-011-A2001-0710-001-001 E-1111 Arrangement Drawing

8474L-011-A2001-0710-001-002 E-1112 / E-1116 / E-1126 Arrangement Drawing

8474L-011-A2001-0710-001-003 E-1114 / E-1117 Arrangement Drawing

8474L-011-A2001-0811-001-001 Main Fractionator Accumulator Drum Assembly

8474L-011-A2001-0811-001-002 Stabilizer Reflux Drum Assembly

8474L-011-A2001-0811-001-003 Drier Oil/Water Separator Drum Assembly

8474L-011-A2001-0811-001-004 Drier Off Gas Seal Pot Assembly

8474L-011-A2001-0811-001-005 Desalter Water Surge Drum Assembly

8474L-011-A2001-0811-001-006 Fuel Gas KO Drum Assembly

8474L-011-A2001-0811-001-007 Tempered Water Drum Assembly

8474L-011-A2001-0811-001-008 Closed Drain Vessel Assembly

8474L-011-A2001-0811-001-014 Decoking Knock Out Drum Assembly

8474L-011-A2001-0814-001-003 Desalters Arrangement Drawing

8474L-011-A2001-0850-001-001 Skid Mounted Vacuum Package Arrangement Drawing

8474L-011-A2001-0850-001-002 Pre-condenser type BJU Arrangement Drawing

8474L-011-A2001-0850-001-003 Inter-condenser type BEU Arrangement Drawing

8474L-011-A2001-0850-001-004 After-condenser type BEU Arrangement Drawing

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8474L-011-A2001-0850-001-005 Pre-condenser drain cooling loop seal pipework Arrangement Drawing

8474L-011-A2001-0850-001-006 Inter-condenser drain cooling loop seal pipework Arrangement Drawing

8474L-011-A2001-0850-001-007 After-condenser drain cooling loop seal pipework Arrangement Drawing

8474L-011-A2001-0850-001-008 A-1102-E-30 Pre-condenser Assembly

8474L-011-A2001-0850-001-009 A-1102-E-31 Inter-condenser Assembly

8474L-011-A2001-0850-001-010 A-1102-E-32 After-condenser Assembly

8474L-011-A2001-0850-001-011 Vacuum Package condenser drain cooling loop seal pipework Arrangement Drawing

8474L-011-A2001-0910-001-001 P-1106A/B Arrangement Drawing

8474L-011-A2001-0910-002-002 P-1102 A/B Arrangement Drawing

8474L-011-A2001-0910-002-003 P-1102 A/B Arrangement Drawing

8474L-011-A2001-0910-002-004 P-1104 A/B Arrangement Drawing

8474L-011-A2001-0910-002-005 P-1105 A/B Arrangement Drawing

8474L-011-A2001-0910-002-006 P-1107 A/B Arrangement Drawing

8474L-011-A2001-0910-002-007 P-1110 A/B Arrangement Drawing

8474L-011-A2001-0910-002-008 P-1112 A/B Arrangement Drawing

8474L-011-A2001-0910-002-009 P-1113 A/B Arrangement Drawing

8474L-011-A2001-0910-002-010 P-1114 A/B Arrangement Drawing

8474L-011-A2001-0910-002-011 P-1118 A/B Arrangement Drawing

8474L-011-A2001-0910-002-013 P-1122 A/B Arrangement Drawing

8474L-011-A2001-0910-003-001 P-1120 A/B Arrangement Drawing

8474L-011-A2001-0910-003-002 P-1121 A/B Arrangement Drawing

8474L-011-A2001-0910-004-001 P-1115 A/B Arrangement Drawing

8474L-011-A1102-0920-001-003 P-1123 A/B Arrangement Drawing

8474L-011-A1102-0920-001-004 P-1124 A/B Arrangement Drawing

8474L-011-A1102-0920-001-002 P-1125 A/B Arrangement Drawing

8474L-011-A1102-0920-001-001 P-1126 A/B Arrangement Drawing

8474L-011-A2001-0910-005-007 P-1127 A/B Arrangement Drawing

8474L-011-A2001-0920-001-002 Neutralizer Injection Package Arrangement Drawing

8474L-011-A2001-0920-001-003 Corrosion Package Arrangement Drawing

8474L-011-A2001-0920-001-004 Demulsifier Injection Package Arrangement Drawing

8474L-011-A2001-0920-001-005 Antifoulant Injection Package Arrangement Drawing

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10.3. Process Flow Diagrams

8474L-011-PFD-0010-001 COLD PREHEAT AND DESALTERS (BACH HO)
8474L-011-PFD-0010-002 HOT PREHEAT AND HEATER (BACH HO)
8474L-011-PFD-0010-003 MAIN FRACTIONATOR/STRIPPER (BACH HO)
8474L-011-PFD-0010-004 DRIERS/ EJECTOR SYSTEM (BACH HO)
8474L-011-PFD-0010-005 OVERHEAD/RUNDOWN SYSTEMS (BACH HO)
8474L-011-PFD-0010-006 NAPHTHA STABILISER (BACH HO)
8474L-011-PFD-0010-007 SCHEMATIC- CDU PREHEAT (BACH HO)
8474L-011-PFD-0010-008 MASS BALANCE (BACH HO)
8474L-011-PFD-0010-009 COLD PREHEAT AND DESALTERS (DUBAI)
8474L-011-PFD-0010-010 HOT PREHEAT AND HEATER (DUBAI)
8474L-011-PFD-0010-011 MAIN FRACTIONATOR/STRIPPER (DUBAI)
8474L-011-PFD-0010-012 DRIERS/ EJECTOR SYSTEM (DUBAI)
8474L-011-PFD-0010-013 OVERHEAD/RUNDOWN SYSTEMS (DUBAI)
8474L-011-PFD-0010-014 NAPHTHA STABILISER (DUBAI)
8474L-011-PFD-0010-015 SCHEMATIC- CDU PREHEAT (DUBAI)
8474L-011-PFD-0010-016 MASS BALANCE (DUBAI)

10.4. Piping and Instrumentation Diagrams

8474L-011-PID-0021-101 Cold preheat train (1/3)
8474L-011-PID-0021-102 Cold preheat train (2/3)
8474L-011-PID-0021-103 Cold preheat train (3/3)
8474L-011-PID-0021-104 Crude oil desalting (1/2)
8474L-011-PID-0021-105 Crude oil desalting (2/2)
8474L-011-PID-0021-106 Hot preheat train (1/4)
8474L-011-PID-0021-107 Hot preheat train (2/4)
8474L-011-PID-0021-108 Hot preheat train (3/4)
8474L-011-PID-0021-109 Hot preheat train (4/4)
8474L-011-PID-0021-110 Crude charge heater convection section
8474L-011-PID-0021-111 Crude charge heater radiant section
8474L-011-PID-0021-112 Main fractionator
8474L-011-PID-0021-113 Top pumparound
8474L-011-PID-0021-114 Kerosene stripper
8474L-011-PID-0021-115 LGO stripper section
8474L-011-PID-0021-116 HGO stripper section
8474L-011-PID-0021-117 LGO dryer section

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8474L-011-PID-0021-118 HGO dryer section
 8474L-011-PID-0021-119 Desalting water (1/2)
 8474L-011-PID-0021-120 Desalting water (2/2)
 8474L-011-PID-0021-121 Ejector system (1/2)
 8474L-011-PID-0021-122 Ejector system (2/2)
 8474L-011-PID-0021-123 Dryer accumulator (1/2)
 8474L-011-PID-0021-124 Dryer accumulator (2/2)
 8474L-011-PID-0021-125 Atmospheric residue rundown system
 8474L-011-PID-0021-126 Tempered water system
 8474L-011-PID-0021-127 HGO rundown system
 8474L-011-PID-0021-128 Overhead system
 8474L-011-PID-0021-129 Kerosene rundown system
 8474L-011-PID-0021-130 Naphtha stabiliser
 8474L-011-PID-0021-131 Stabiliser condenser system
 8474L-011-PID-0021-132 Full range Naphtha rundown
 8474L-011-PID-0021-133 Combustion air system
 8474L-011-PID-0021-134 H-1101 Fuel Gas system
 8474L-011-PID-0021-135 H-1101 Fuel Oil system
 8474L-011-PID-0021-136 Chemical injection (1/2)
 8474L-011-PID-0021-137 Chemical injection (2/2)
 8474L-011-PID-0021-138 Crude charge heater / H-1101 heater firing
 8474L-011-PID-0021-139 Crude recirculation, LSO and HSO headers
 8474L-011-PID-0021-140 Decoking station (1/3)
 8474L-011-PID-0021-141 Decoking station (2/3)
 8474L-011-PID-0021-142 Decoking station (3/3)
 8474L-011-PID-0021-143 Sootblowers System
 8474L-011-PID-0021-201 Process area Battery limits (1/2)
 8474L-011-PID-0021-202 Process area Battery limits (2/2)
 8474L-011-PID-0021-301 Sampling systems details
 8474L-011-PID-0021-302 Level instruments details (1/4)
 8474L-011-PID-0021-303 Level instruments details (2/4)
 8474L-011-PID-0021-304 Level instruments details (3/4)
 8474L-011-PID-0021-305 Level instruments details (4/4)
 8474L-011-PID-0021-306 LP steam connection details
 8474L-011-PID-0021-307 FLS and nitrogen connection details
 8474L-011-PID-0021-308 CD connection details

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- 8474L-011-PID-0021-309 Desalter instrumentation details (1/2)
- 8474L-011-PID-0021-310 Desalter instrumentation details (2/2)
- 8474L-011-PID-0021-401 Air cooler details (1/4)
- 8474L-011-PID-0021-402 Air cooler details (2/4)
- 8474L-011-PID-0021-403 Air cooler details (3/4)
- 8474L-011-PID-0021-404 Air cooler details (4/4)

- 8474L-011-PID-0031-203 Utility area battery limits
- 8474L-011-PID-0031-501 Steam, condensate and boiler feed water distribution (1/5)
- 8474L-011-PID-0031-502 Steam, condensate and boiler feed water distribution (2/5)
- 8474L-011-PID-0031-503 Steam, condensate and boiler feed water distribution (3/5)
- 8474L-011-PID-0031-504 Steam, condensate and boiler feed water distribution (4/5)
- 8474L-011-PID-0031-505 Steam, condensate and boiler feed water distribution (5/5)
- 8474L-011-PID-0031-511 Cooling water, demineralized water, service water and potable Water distribution (1/4)
- 8474L-011-PID-0031-512 Cooling water, demineralized water, service water and potable Water distribution (2/4)
- 8474L-011-PID-0031-513 Cooling water, demineralized water, service water and potable Water distribution (3/4)
- 8474L-011-PID-0031-514 Cooling water, demineralized water, service water and potable Water distribution (4/4)
- 8474L-011-PID-0031-521 Nitrogen, instrument air and plant air distribution (1/3)
- 8474L-011-PID-0031-522 Nitrogen, instrument air and plant air distribution (2/3)
- 8474L-011-PID-0031-523 Nitrogen, instrument air and plant air distribution (3/3)
- 8474L-011-PID-0031-531 Fuel oil, fuel gas and pilot gas distribution
- 8474L-011-PID-0031-541 Flare system (1/3)
- 8474L-011-PID-0031-542 Flare system (2/3)
- 8474L-011-PID-0031-543 Flare system (3/3)
- 8474L-011-PID-0031-551 Caustic system
- 8474L-011-PID-0031-561 Flushing oil distribution (1/3)
- 8474L-011-PID-0031-562 Flushing oil distribution (2/3)
- 8474L-011-PID-0031-563 Flushing oil distribution (3/3)
- 8474L-011-PID-0031-571 Closed drain system (1/7)
- 8474L-011-PID-0031-572 Closed drain system (2/7)
- 8474L-011-PID-0031-573 Closed drain system (3/7)
- 8474L-011-PID-0031-574 Closed drain system (4/7)
- 8474L-011-PID-0031-575 Closed drain system (5/7)

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8474L-011-PID-0031-576 Closed drain system (6/7)
 8474L-011-PID-0031-577 Closed drain system (7/7)
 8474L-011-PID-0031-578 Closed drain vessel
 8474L-011-PID-0031-581 Oily water pit (not issued)
 8474L-011-PID-0031-591 Oil mist distribution (1/2)
 8474L-011-PID-0031-592 Oil mist distribution (2/2)
 8474L-011-PID-1931-001(1/3) Fire Water Distribution (1/3)
 8474L-011-PID-1931-001(2/3) Water Spray Systems (2/3)
 8474L-011-PID-1931-001 (3/3) Water Spray Systems Controls (3/3)

8474L-011-PID-0041-601 Pump details (1/5)
 8474L-011-PID-0041-602 Pump details (2/5)
 8474L-011-PID-0041-603 Pump details (3/5)
 8474L-011-PID-0041-604 Pump details (4/5)
 8474L-011-PID-0041-605 Pump details (5/5)

10.5. Equipment list

Refer to the attached equipment list of unit 011: Unit 011 Extracted Equipment List

10.6. Main Equipment Data Sheet

8474L-011-A1001-0110-001-001 H-1101 Heater Data Sheet
 8474L-011-A1001-0110-001-001 Burner Data Sheet
 8474L-011-A1106-0110-001-002 Blowers Data Sheet
 8474L-011-A1005-0110-001-001 Blower Performance Curve

8474L-011-PDS-T-1101-001 T-1101 Data Sheet
 8474L-011-PDS-T-1102-002 T-1102 Data Sheet
 8474L-011-PDS-T-1103-003 T-1103 Data Sheet
 8474L-011-PDS-T-1104-004 T-1104 Data Sheet
 8474L-011-PDS-T-1105-005 T-1105 Data Sheet
 8474L-011-PDS-T-1106-006 T-1106 Data Sheet
 8474L-011-PDS-T-1107-007 T-1107 Data Sheet

8474L-011-A1001-0910-002-001 P-1101 Filled-In Data Sheet
 8474L-011-A1001-0910-002-002 P-1102 Filled-In Data Sheet
 8474L-011-A1001-0910-002-003 P-1103 Filled-In Data Sheet

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8474L-011-A1001-0910-002-004 P-1104 Filled-In Data Sheet
 8474L-011-A1001-0910-002-005 P-1105 Filled-In Data Sheet
 8474L-011-A1001-0910-001-001 P-1106 Filled-In Data Sheet
 8474L-011-A1001-0910-002-006 P-1107 Filled-In Data Sheet
 8474L-011-A1001-0910-002-007 P-1110 Filled-In Data Sheet
 8474L-011-A1001-0910-002-008 P-1112 Filled-In Data Sheet
 8474L-011-A1001-0910-002-009 P-1113 Filled-In Data Sheet
 8474L-011-A1001-0910-002-XXX P-1114 Filled-In Data Sheet

8474L-011-A1001-0910-004-001 P-1115 Filled-In Data Sheet
 8474L-011-A1001-0910-002-010 P-1118 Filled-In Data Sheet
 8474L-011-A1001-0910-002-012 P-1119 Filled-In Data Sheet
 8474L-011-A1001-0910-003-001 P-1120 Filled-In Data Sheet
 8474L-011-A1001-0910-003-002 P-1121 Filled-In Data Sheet
 8474L-011-A1001-0910-002-011 P-1122 Filled-In Data Sheet
 8474L-011-A1001-0910-005-007 P-1127 Filled-In Data Sheet

8474L-011-A1002-0910-002-001 P-1101 Performance Curve
 8474L-011-A1002-0910-002-002 P-1102 Performance Curve
 8474L-011-A1002-0910-002-003 P-1103 Performance Curve
 8474L-011-A1002-0910-002-004 P-1104 Performance Curve
 8474L-011-A1002-0910-002-005 P-1105 Performance Curve
 8474L-011-A1002-0910-001-001 P-1106 Performance Curve
 8474L-011-A1002-0910-002-006 P-1107 Performance Curve
 8474L-011-A1002-0910-002-007 P-1110 Performance Curve
 8474L-011-A1002-0910-002-008 P-1112 Performance Curve
 8474L-011-A1002-0910-002-009 P-1113 Performance Curve
 8474L-011-A1002-0910-002-012 P-1114 Performance Curve
 8474L-011-A1002-0910-004-001 P-1115 Performance Curve
 8474L-011-A1002-0910-002-010 P-1118 Performance Curve
 XXX P-1119 Performance Curve

8474L-011-A1002-0910-003-001 P-1120 Performance Curve
 8474L-011-A1002-0910-003-002 P-1121 Performance Curve
 8474L-011-A1002-0910-002-011 P-1122 Performance Curve
 8474L-011-A1002-0910-005-007 P-1127 Performance Curve

8474L-011-A1002-0920-001-002 Performance Curve A-1104 (1/4)

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8474L-011-A1002-0920-001-003 Performance Curve A-1104 (2/4)

8474L-011-A1002-0920-001-004 Performance Curve A-1104 (3/4)

8474L-011-A1002-0920-001-005 Performance Curve A-1104 (4/4)

8474L-011-A1001-0814-001-002 A-1101-D-01/02 Desalters Vessel Data Sheet

8474L-011-A1001-0814-001-003 / 004 Static Mixer Data Sheet

8474L-011-A1006-0814-001-001 / 002 Emulsifying Valves Data Sheet

8474L-011-A1001-0850-001-002 A-1102 Vacuum Package equipment Data Sheet

10.7. Instrument List

Refer to attached extracted instrument list of unit 011: Unit 011 Extracted Instrument List.xls

10.8. Cause & Effect Matrix

8474L-011-DW-1514-201

10.9. Safety Logic diagram

8474L-XX-XXXX-XXX

10.10. Fire & Gas Cause & Effect Chart

8474L-011-DW-1514-301

10.11. Fire & Gas Detectors Layout

8474L-011-DW-1950-001

10.12. Fire Protection Layout

8474L-011-DW-1933-002

10.13. Hazardous Area Classification

8474L-011-DW-1920-001

10.14. MSDS

Refer to the MSDS listed at the end of the operating manual, doc. No. 8474L-011-ML-001-A

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10.15. Vendors Documentation

10.15.1. A-1101-D-01/02 Desalters

8474L-011-A3501-0814-001-001	Process Description
8474L-011-A1001-0814-001-002	Desalters Vessel Data Sheet
8474L-011-A1001-0814-001-003/004	Static Mixer Data Sheet
8474L-011-A1006-0814-001-001/002	Emulsifying Valves Data Sheet
8474L-011-A0102-0814-001-001	Process Flow Diagram with Heat and Material Balance

10.15.2. H-1101 Crude Heater

8474L-011-A1001-0110-001-001	H-1101 Data Sheet
8474L-011-A1006-0110-001-001	Burner Data Sheet
8474L-011-A1106-0110-001-002	Blowers Data Sheet
8474L-011-A1005-0110-001-002	Blower Performance Curve
8474L-011-A2001-0110-001-001	General Arrangement

10.15.3. A-1102 Vacuum Package

8474L-011-A1001-0850-001-001	Ejector Data Sheet
8474L-011-A3102-0850-001-001	Control Valve Calculation Sheet
8474L-011-A2001-0850-001-001	General Arrangement
8474L-011-A0102-0850-001-001	Process Flow Diagram (Flexibility design case)
8474L-011-A0102-0850-001-003	Process Flow Diagram (Normal case)
8474L-011-A0102-0850-001-002	Process Flow Diagram (Flexibility intermediate case)